

Exam 7

Study Guide

ADVANCED ESTIMATION OF CLAIMS LIABILITIES

Comprehensive study guide
with original and past CAS problems

Exam 7 Study Guide

Fall 2026 Sitting

Rising Fellow



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Introduction

How to Use This Guide

This guide is intended to **supplement** the Content Outline readings. Although we believe it provides a thorough review of the exam material, the readings provide additional context that is invaluable. Please do NOT skip the Content Outline readings.

Past CAS Exam Problems

Past CAS exam problems & solutions are included for each paper. Note that these questions are solely owned by the CAS. They are included in the online course for student convenience. All past CAS problems are Excel-based and can be downloaded from the online course.

Rounding

Numerous examples are provided throughout the guide. We did not round any of the intermediate steps in the examples. All calculations were performed with full precision to ensure accuracy, and only the final answers were rounded when necessary. To exactly reproduce the examples, we recommend working them out in Microsoft Excel.

Feedback

We always working to improve the Exam 7 Study Guide and the rest of the Rising Fellow study material. Please send us an email at exam7@risingfellow.com if you have feedback about any of the following:

- Sections that are confusing or could be improved
- Errors (ex. formatting, spelling, calculations, grammar, etc.)

Note that errata will be posted on the Rising Fellow website on an as-needed basis.

Blank Pages

Since many students want a printed copy of the study guide, blank pages have been inserted throughout the guide to ensure that all outlines start on odd pages.

This paper focuses on the Benktander method and its relationship to the chain-ladder & Bornhuetter/Ferguson methods.

I. General Relationship Between Reserve & Ultimate Loss Estimates

Suppose that C_k is the actual claims amount paid after k years of development. Given a reserve estimate $\hat{R}$ and ultimate loss estimate $\hat{U}$, we have the following general relationship:

$$\hat{U} = C_k + \hat{R}$$

This relationship always holds. If we have a reserve estimate and add the claims paid to date, we always obtain the corresponding ultimate loss estimate.

II. Bornhuetter/Ferguson (BF) Method

The BF method estimates reserves based on an a priori expectation of ultimate losses.

Mathematically:

$$R_{BF} = q_k U_0$$

where:

- R_{BF} is the BF reserve
- $q_k = 1 - \frac{1}{CDF}$ is the proportion of the ultimate claims amount which is expected to remain unpaid after k years of development
- U_0 is the a priori expectation of ultimate losses

Since R_{BF} uses U_0 , it assumes the current claims amount C_k is NOT predictive of future claims.

Using the general relationship described above, we obtain the **BF ultimate loss**:

$$U_{BF} = C_k + R_{BF}$$

III. Chain-Ladder (CL) Method

The **CL method** estimates ultimate losses and reserves based on claims to date. In other words, it assumes that the current claims amount C_k is **fully predictive of future claims**. Mathematically:

$$U_{CL} = \frac{C_k}{p_k} \quad R_{CL} = q_k U_{CL}$$

where:

- U_{CL} is the **chain-ladder ultimate loss**
- $p_k = 1 - q_k$ is the proportion of the ultimate claims amount which is expected to be paid after k years of development

The **main advantage of the CL method** over the BF method is different actuaries should obtain similar results when running the chain-ladder method. This is not the case with the BF method due to differences in the selection of U_0 .

IV. Benktander Method

Since the CL and BF methods represent extreme positions, where the CL method fully believes C_k and the BF method does not rely on C_k at all, Gunnar Benktander replaced U_0 with a credibility mixture:

$$U_c = cU_{CL} + (1 - c)U_0$$

where c is the credibility weight.

As the claims C_k develop, credibility should increase, Benktander proposed the following:

- Set $c = p_k$

- Set $R_{GB} = R_{BF} \cdot \frac{U_{p_k}}{U_0} = q_k U_0 \cdot \frac{U_{p_k}}{U_0} = q_k U_{BF}$, where R_{GB} is the Benktander reserve

Benktander's proposals lead to some important results.

BF Method as a Credibility-Weighted Average

Using our credibility mixture with $c = p_k$, we can show the following:

$$\begin{aligned} U_{p_k} &= p_k U_{CL} + (1 - p_k) U_0 \\ &= p_k U_{CL} + q_k U_0 \\ &= C_k + R_{BF} \\ &= U_{BF} \end{aligned}$$

Hence, the **BF method** is a credibility-weighted average of the CL method and the a priori expectation.

Benktander Method as a Credibility-Weighted Average

The **Benktander method** is a credibility-weighted average of the CL and BF methods:

$$\begin{aligned} U_{GB} &= C_k + R_{GB} \\ &= p_k U_{CL} + q_k U_{BF} \\ &= (1 - q_k) U_{CL} + q_k U_{BF} \end{aligned}$$

The **Benktander reserve** is also a credibility-weighted average of the CL and BF methods:

$$R_{GB} = (1 - q_k) R_{CL} + q_k R_{BF}$$

We can also express the **Benktander method** as a credibility-weighted average of the CL method and the a priori expectation:

$$\begin{aligned} U_{GB} &= C_k + R_{GB} \\ &= (1 - q_k^2) U_{CL} + q_k^2 U_0 \end{aligned}$$

Iterative Relationships

As shown above, the Benktander reserve is obtained by applying the BF method twice, first to U_0 (which produces R_{BF}), and then to U_{BF} (which produces R_{GB}). Hence, the Benktander method is also called the **iterated BF method**.

Using a starting point of $U^{(0)} = U_0$ and iteration rules $R^{(m)} = q_k U^{(m)}$ and $U^{(m+1)} = C_k + R^{(m)}$, we obtain the following **iterative relationships**:

$$\begin{aligned} U^{(m)} &= (1 - q_k^m) U_{CL} + q_k^m U_0 \\ R^{(m)} &= (1 - q_k^m) R_{CL} + q_k^m R_{BF} \end{aligned}$$

These relationships lead to the following:

- $U^{(0)} = (1 - q_k^0) U_{CL} + q_k^0 U_0 = U_0$
- $R^{(0)} = (1 - q_k^0) R_{CL} + q_k^0 R_{BF} = R_{BF}$
- $U^{(1)} = (1 - q_k) U_{CL} + q_k U_0 = C_k + R_{BF} = U_{BF}$ ← Iteration 1 Ultimate
- $R^{(1)} = (1 - q_k) R_{CL} + q_k R_{BF} = R_{GB}$ ← Iteration 2 Reserve
- $U^{(2)} = (1 - q_k^2) U_{CL} + q_k^2 U_0 = U_{GB}$ ← Iteration 2 Ultimate
- ...
- $U^{(\infty)} = (1 - q_k^\infty) U_{CL} + q_k^\infty U_0 = U_{CL}$
- $R^{(\infty)} = (1 - q_k^\infty) R_{CL} + q_k^\infty R_{BF} = R_{CL}$

As shown above, the BF and Benktander methods are iterations 1 and 2, respectively. If we **infinitely iterate between reserves and ultimate losses**, we will eventually obtain the **CL method**.

V. Benktander vs. BF vs. CL

The Benktander method is superior to the BF and CL methods for the following reasons:

- The mean squared error (MSE) is almost always smaller than the BF or CL methods
- Better approximation of the exact Bayesian procedure

- Superior to the CL method since it gives more weight to the a priori expectation of ultimate losses
- Superior to the BF method since it gives more weight to actual loss experience

Example: Benktander Method

An actuary wants to use the Benktander method to estimate ultimate losses for accident year 2021. Given the following information as of December 31, 2023:

AY	<u>Cumulative Paid Losses (\$) as of XX Months:</u>			
	12 mo.	24 mo.	36 mo.	48 mo.
2020	7,000	10,500	12,600	13,860
2021	8,000	12,000	14,400	
2022	9,000	13,500		
2023	10,000			

- The 2021 earned premium is \$25,000
- The expected loss ratio for each year is 75%
- The 48-ultimate loss development factor is 1.05

First, let's calculate the BF ultimate losses:

- Since AY 2021 is 36 months old (i.e., $k = 3$), we need the 36-ultimate CDF
 - The 36-48 LDF is $\frac{13,860}{12,600} = 1.10$
 - The 36-ultimate CDF is $1.10(1.05) = 1.155$
- $U_0 = EP \cdot ELR = 25,000(0.75) = 18,750$
- $R_{BF} = q_3 U_0 = \left(1 - \frac{1}{1.155}\right) (18,750) = 2,516$
- $U_{BF} = C_3 + R_{BF} = 14,400 + 2,516 = 16,916$

Second, let's calculate the Benktander ultimate losses:

- $R_{GB} = q_3 U_{BF} = \left(1 - \frac{1}{1.155}\right) (16,916) = 2,270$
- $U_{GB} = C_3 + R_{GB} = 14,400 + 2,270 = 16,670$

The Benktander ultimate losses for AY 2021 are **\$16,670**.

Note that there are a number of other ways to calculate this quantity. As long as you reproduce the same figure using any of the applicable formulas, you should be good to go.

The methods shown in this paper are inspired by those shown in the Mack Benktander paper.

The main difference is that this paper focuses on loss ratios rather than raw losses.

I. The Collective & Individual Loss Ratio Claims Reserves

This section of the paper defines a number of quantities and builds out the collective and individual loss ratio claims reserves. When first reviewing the formulas below, don't stress about the notation. Once you look at an example and start working problems, it will all come together.

Total Ultimate Claims

Let i represent the origin period (typically an accident year) and n be the total number of origin periods. Assuming that all claims incurred in an origin period are known and closed after n development periods, the **total ultimate claims**, U_i , from origin period i are as follows:

$$U_i = \sum_{k=1}^n S_{ik}$$

where S_{ik} are the incremental paid claims from origin period i .

Cumulative Paid Claims

Cumulative paid claims from origin period i as of k years of development, C_{ik} , are defined as follows:

$$C_{ik} = \sum_{j=1}^k S_{ij}$$

i -th Period Claims Reserve

The i -th period claims reserve, R_i , is the required amount for the incurred but unpaid claims of period i . It is defined as follows:

$$R_i = \sum_{k=n-i+2}^n S_{ik}$$

where $i = 2, \dots, n$. This means that we assume that the oldest origin period is already at ultimate.

Total Claims Reserve

The total amount of incurred but unpaid claims over all periods, R , is defined as follows:

$$R = \sum_{i=2}^n R_i$$

Incremental Expected Loss Ratio

Let V_i be the **premium** for origin period i . The incremental amount of expected paid claims per unit of premium in each development period, m_k , is defined as follows:

$$m_k = \frac{E[\sum_{i=1}^{n-k+1} S_{ik}]}{\sum_{i=1}^{n-k+1} V_i}$$

Based on the definition above, m_k is the incremental expected loss ratio for each development period. If we sum m_k across development periods, we obtain the **total expected loss ratio**.

Expected Value of the Burning Cost

The expected value of the burning cost of the total ultimate claims is similar to the a priori estimate U_0 from the Mack Benktander paper. It is defined as follows:

$$E[U_i^{BC}] = V_i \cdot \sum_{k=1}^n m_k$$

The formula above is multiplying the premium for origin period i by the total expected loss ratio to arrive at the expected value of the burning cost (i.e., expected loss).

Loss Ratio Payout & Reserve Factors

The **loss ratio payout factor**, p_i , represents the percentage of losses paid to date for each origin period. It is defined as follows:

$$p_i = \frac{V_i \cdot \sum_{k=1}^{n-i+1} m_k}{E[U_i^{BC}]} = \frac{\sum_{k=1}^{n-i+1} m_k}{\sum_{k=1}^n m_k}$$

The **loss ratio reserve factor**, q_i , represents the percentage of losses yet to be paid for each origin period. It is defined as follows:

$$q_i = 1 - p_i$$

These quantities are similar to p_k and q_k from the Mack Benktander paper.

Individual Total Ultimate Claims & Loss Ratio Claims Reserve

The **individual total ultimate claims** for origin period i , U_i^{ind} , are defined as follows:

$$U_i^{ind} = \frac{C_{i,n-i+1}}{p_i}$$

The **individual loss ratio claims reserve** for origin period i , R_i^{ind} , is defined as follows:

$$\begin{aligned} R_i^{ind} &= U_i^{ind} - C_{i,n-i+1} \\ &= q_i \cdot U_i^{ind} \\ &= \frac{q_i}{p_i} \cdot C_{i,n-i+1} \end{aligned}$$

These quantities are similar to the CL ultimate loss & reserve from the Mack Benktander paper.

Collective Total Ultimate Claims & Loss Ratio Claims Reserve

The **collective loss ratio claims reserve** for origin period i , R_i^{coll} , is defined as follows:

$$R_i^{coll} = q_i \cdot U_i^{BC}$$

The **collective total ultimate claims** for origin period i , U_i^{coll} , is defined as follows:

$$U_i^{coll} = R_i^{coll} + C_{i,n-i+1}$$

These quantities are similar to the BF ultimate loss & reserve from the Mack Benktander paper.

An **advantage** of the collective loss ratio claims reserve over the standard BF reserve is that different actuaries always come to the same results provided they use the same premiums. This is because the expected loss ratio is based on the actual claims data rather than a judgmental selection.

II. Credible Loss Ratio Claims Reserve

Similar to the CL and BF reserves, the individual and collective loss ratio claims reserves represent extreme positions (fully believe the data vs. do not believe the data at all). Thus, it's sensible to define a **credibility-weighted** average of the individual and collective loss ratio reserves as follows:

$$R_i^c = Z_i \cdot R_i^{ind} + (1 - Z_i) \cdot R_i^{coll}$$

where Z_i is the credibility weight.

Benktander (GB) Loss Ratio Claims Reserve

If we set $Z_i = Z_i^{GB} = p_i$, we obtain the Benktander loss ratio claims reserve, R_i^{GB} :

$$R_i^{GB} = p_i \cdot R_i^{ind} + q_i \cdot R_i^{coll}$$

Neuhaus (WN) Loss Ratio Claims Reserve

If we set $Z_i = Z_i^{WN} = \sum_{k=1}^{n-i+1} m_k = p_i \cdot \sum_{k=1}^n m_k$, we obtain the Neuhaus loss ratio claims reserve, R_i^{WN} :

$$R_i^{WN} = Z_i^{WN} \cdot R_i^{ind} + (1 - Z_i^{WN}) \cdot R_i^{coll}$$

Iterative Relationships

Using a starting point of $U_i^{(0)} = U_i^0 = U_i^{BC}$ and iteration rules $R_i^{(m)} = q_i \cdot U_i^{(m)}$ and $U_i^{(m+1)} = C_{i,n-i+1} + R_i^{(m)}$, we obtain the following **iterative relationships**:

$$\begin{aligned} U_i^{(m)} &= (1 - q_i^m) \cdot U_i^{ind} + q_i^m \cdot U_i^{BC} \\ R_i^{(m)} &= (1 - q_i^m) \cdot R_i^{ind} + q_i^m \cdot R_i^{coll} \end{aligned}$$

Note that in the paper, the reserve relationship is as defined as $R_i^{(m)} = (1 - q_i^m) \cdot R_i^{ind} + q_i^m \cdot R_i^0$. Since $R_i^{coll} = q_i \cdot U_i^{BC}$, it must be the case that $R_i^{coll} = R_i^0$. Thus, these iteration rules are similar to those shown in the Mack Benktander paper.

These relationships lead to the following:

- $U_i^{(0)} = (1 - q_i^0) \cdot U_i^{ind} + q_i^0 \cdot U_i^{BC} = U_i^{BC}$
- $R_i^{(0)} = (1 - q_i^0) \cdot R_i^{ind} + q_i^0 \cdot R_i^{coll} = R_i^{coll}$
- $U_i^{(1)} = (1 - q_i) \cdot U_i^{ind} + q_i \cdot U_i^{BC} = U_i^{coll}$ ← Iteration 1 Ultimate
- $R_i^{(1)} = (1 - q_i) \cdot R_i^{ind} + q_i \cdot R_i^{coll} = R_i^{GB}$ ← Iteration 2 Reserve

- $U_i^{(2)} = (1 - q_i^2) \cdot U_i^{ind} + q_i^2 \cdot U_i^{BC} = U_i^{GB} \longleftarrow \text{Iteration 2 Ultimate}$
- ...
- $U_i^{(\infty)} = (1 - q_i^\infty) \cdot U_i^{ind} + q_i^\infty \cdot U_i^{BC} = U_i^{ind}$
- $R_i^{(\infty)} = (1 - q_i^\infty) \cdot R_i^{ind} + q_i^\infty \cdot R_i^{coll} = R_i^{ind}$

As shown above, the collective and Benktander methods are iterations 1 and 2, respectively. The labeling is a bit strange because $R_i^{(1)}$ is the Benktander reserve and uses a “1” instead of a “2.” But the CAS made it clear that the Benktander method is considered the second-iteration reserve in the official solution for 2016 #1.

Once again, if we **infinitely iterate between reserves and ultimate losses**, we will eventually obtain the **individual method**.

III. The Optimal Credibility Weights & the Mean Squared Error

In this section, we summarize the key formulas from Sections 4 – 6 of the paper regarding the optimal credibility weights.

General Formula for the Optimal Credibility Weights

Assume that $Var(U_i) = f_i \cdot Var(U_i^{BC})$ for some constant $f_i \geq 1$. This implies that U_i is at least as volatile as the burning cost estimate.

The optimal credibility weights Z_i^* which minimize the mean squared error $mse(R_i^c) = E[(R_i^c - R_i)^2]$ and the variance $Var(R_i^c)$ are as follows:

$$Z_i^* = \frac{p_i}{p_i + t_i^*}$$

where $t_i^* = \frac{f_i - 1 + \sqrt{(f_i + 1) \cdot (f_i - 1 + 2 \cdot p_i)}}{2}$.

Special Case of the Optimal Credibility Weights

The **minimum variance optimal credible claims reserve** is obtained when $f_i = 1$. In other words, of all possible values for f_i , $f_i = 1$ yields the lowest possible reserve variance. In this special case, $Var(U_i) = Var(U_i^{BC})$ and $t_i^* = \sqrt{p_i}$. Then, the optimal credibility weights are as follows:

$$Z_i^* = \frac{p_i}{p_i + \sqrt{p_i}}$$

Since $t_i^* = \sqrt{p_i} \leq 1$, it must be the case that $Z_i^* \leq \frac{1}{2}$. This clearly demonstrates that the Neuhaus and Benktander methods are not optimal since they often have weights greater than 0.5.

On the exam, the default formula for the optimal credibility weights is $Z_i^* = \frac{p_i}{p_i + \sqrt{p_i}}$. This applies in the following situations:

- $f_i = 1$
- $Var(U_i) = Var(U_i^{BC})$

If you are told that $f_i > 1$ or that $Var(U_i) = f_i \cdot Var(U_i^{BC})$, then use the general formula.

Mean Squared Error Formula

The mean squared error for the credible loss ratio claims reserve is as follows:

$$mse(R_i^c) = E[\alpha_i^2(U_i)] \cdot \left(\frac{Z_i^2}{p_i} + \frac{1}{q_i} + \frac{(1 - Z_i)^2}{t_i^*} \right) \cdot q_i^2$$

Notice that we have a new term, $E[\alpha_i^2(U_i)]$. This comes from the beginning of Section 4 where Hürlimann lays out the theory for the optimal credibility weights. For the purposes of the exam, just know that it's a constant that must be incorporated into the MSE calculation. It has only shown up one time on released exams (see Spring 2018 #3).

The formula above can also be used to obtain the MSEs of the **collective and individual methods**:

- When $Z_i = 0$, we obtain $mse(R_i^{coll})$
- When $Z_i = 1$, we obtain $mse(R_i^{ind})$

Example: Credible Loss Ratio Claims Reserves

Given the following information as of December 31, 2023:

AY	EP (\$)	<u>Incremental Paid Losses (\$) as of XX Months:</u>					
		12 mo.	24 mo.	36 mo.	48 mo.	60 mo.	72 mo.
2018	13,085	4,370	1,923	3,999	2,168	1,200	647
2019	14,258	2,701	2,590	1,871	1,783	393	
2020	16,114	4,483	2,246	3,345	1,068		
2021	15,142	3,254	2,550	2,547			
2022	16,905	8,010	4,108				
2023	20,224	5,582					

Assuming that $Var(U_i) = Var(U_i^{BC})$, let's calculate the various reserves covered in the paper.

First, we need the applicable parameters:

- $m_k = \frac{E[\sum_{i=1}^{n-k+1} S_{ik}]}{\sum_{i=1}^{n-k+1} V_i}$
 - $m_1 = \frac{4,370+2,701+\dots+5,582}{13,085+14,258+\dots+20,224} = 0.2967$
 - $m_2 = \frac{1,923+2,590+\dots+4,108}{13,085+14,258+\dots+16,905} = 0.1777$
 - ...
 - $m_6 = \frac{647}{13,085} = 0.0494$
- $p_i = \frac{\sum_{k=1}^{n-i+1} m_k}{\sum_{k=1}^n m_k} = Z_i^{GB}$
 - $p_1 = \frac{0.2967+0.1777+\dots+0.0494}{0.2967+0.1777+\dots+0.0494} = 1.0000$
 - $p_2 = \frac{0.2967+0.1777+\dots+0.0583}{0.2967+0.1777+\dots+0.0494} = 0.9450$
 - ...

- $p_6 = \frac{0.2967}{0.2967+0.1777+\dots+0.0494} = 0.3303$
- $q_i = 1 - p_i$
 - $q_1 = 1 - p_1 = 1 - 1 = 0.0000$
 - $q_2 = 1 - p_2 = 1 - 0.9450 = 0.0550$
 - ...
 - $q_6 = 1 - p_6 = 1 - 0.3303 = 0.6697$
- $Z_i^{WN} = \sum_{k=1}^{n-i+1} m_k$
 - $Z_1^{WN} = 0.2967 + 0.1777 + \dots + 0.0494 = 0.8983$
 - $Z_2^{WN} = 0.2967 + 0.1777 + \dots + 0.0583 = 0.8488$
 - ...
 - $Z_6^{WN} = 0.2967$
- $Z_i^* = \frac{p_i}{p_i + \sqrt{p_i}}$
 - $Z_1^* = \frac{p_1}{p_1 + \sqrt{p_1}} = \frac{1.0000}{1.0000 + \sqrt{1.0000}} = 0.5000$
 - $Z_2^* = \frac{p_2}{p_2 + \sqrt{p_2}} = \frac{0.9450}{0.9450 + \sqrt{0.9450}} = 0.4929$
 - ...
 - $Z_6^* = \frac{p_6}{p_6 + \sqrt{p_6}} = \frac{0.3303}{0.3303 + \sqrt{0.3303}} = 0.3650$

The applicable parameters are summarized as follows:

AY	<u>Parameters</u>				
	m_k	$p_i = Z_i^{GB}$	q_i	Z_i^{WN}	Z_i^*
2018 ($i = 1$)	0.2967	1.0000	0.0000	0.8983	0.5000
2019 ($i = 2$)	0.1777	0.9450	0.0550	0.8488	0.4929
2020 ($i = 3$)	0.2007	0.8801	0.1199	0.7906	0.4840
2021 ($i = 4$)	0.1155	0.7515	0.2485	0.6751	0.4644
2022 ($i = 5$)	0.0583	0.5281	0.4719	0.4744	0.4209
2023 ($i = 6$)	0.0494	0.3303	0.6697	0.2967	0.3650

Second, let's calculate the collective, individual, Neuhaus, Benktander, and optimal claims reserves:

AY	<u>Reserves (\$)</u>				
	Collective	Individual	Neuhaus	Benktander	Optimal
2019	705	544	568	553	626
2020	1,736	1,518	1,564	1,544	1,630
2021	3,380	2,761	2,962	2,915	3,092
2022	7,166	10,829	8,904	9,101	8,708
2023	12,167	11,320	11,916	11,887	11,858
Total	25,154	26,972	25,913	25,999	25,914

Notice that there are no reserves for AY 2018 since we assume that the oldest origin period is fully developed.

The calculations for AY 2019 ($i = 2$) are as follows:

- $R_2^{coll} = q_2 \cdot U_2^{BC} = q_2 \cdot V_2 \cdot \sum_{k=1}^n m_k = 0.0550(14,258)(0.2967 + 0.1777 + \dots + 0.0494) = 705$
- $R_2^{ind} = \frac{q_2}{p_2} \cdot C_{2,5} = \frac{0.0550}{0.9450} (2,701 + 2,590 + \dots + 393) = 544$
- $R_2^{WN} = Z_2^{WN} \cdot R_2^{ind} + (1 - Z_2^{WN}) \cdot R_2^{coll} = 0.8488(544) + (1 - 0.8488)(705) = 568$
- $R_2^{GB} = Z_2^{GB} \cdot R_2^{ind} + (1 - Z_2^{GB}) \cdot R_2^{coll} = 0.9450(544) + (1 - 0.9450)(705) = 553$
- $R_2^{opt} = Z_2^* \cdot R_2^{ind} + (1 - Z_2^*) \cdot R_2^{coll} = 0.4929(544) + (1 - 0.4929)(705) = 626$

Now that we have the reserves by AY for each method, we can easily calculate the ultimate losses by AY for each method by adding the AY paid claims to date to each reserve figure.

Third, let's calculate the MSEs for each method relative to the MSE of the optimal reserve:

AY	<u>Relative MSE</u>				
	Collective	Individual	Neuhaus	Benktander	Optimal
2019	1.0271	1.0287	1.0141	1.0228	1.0000
2020	1.0580	1.0659	1.0233	1.0389	1.0000
2021	1.1154	1.1535	1.0238	1.0441	1.0000

2022	1.1986	1.3761	1.0032	1.0129	1.0000
2023	1.2444	1.7401	1.0086	1.0022	1.0000

Recall that $mse(R_i^c) = E[\alpha_i^2(U_i)] \cdot \left(\frac{z_i^2}{p_i} + \frac{1}{q_i} + \frac{(1-z_i)^2}{t_i^*} \right) \cdot q_i^2$. The calculations for AY 2019 ($i = 2$) are as follows:

- Collective Relative MSE = $\frac{E[\alpha_2^2(U_2)] \cdot \left(\frac{0^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-0)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2}{E[\alpha_2^2(U_2)] \cdot \left(\frac{0.4929^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-0.4929)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2} = 1.0271$
- Individual Relative MSE = $\frac{E[\alpha_2^2(U_2)] \cdot \left(\frac{1^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-1)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2}{E[\alpha_2^2(U_2)] \cdot \left(\frac{0.4929^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-0.4929)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2} = 1.0287$
- Neuhaus Relative MSE = $\frac{E[\alpha_2^2(U_2)] \cdot \left(\frac{0.8488^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-0.8488)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2}{E[\alpha_2^2(U_2)] \cdot \left(\frac{0.4929^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-0.4929)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2} = 1.0141$
- Benktander Relative MSE = $\frac{E[\alpha_2^2(U_2)] \cdot \left(\frac{0.9450^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-0.9450)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2}{E[\alpha_2^2(U_2)] \cdot \left(\frac{0.4929^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-0.4929)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2} = 1.0228$
- Optimal Relative MSE = $\frac{E[\alpha_2^2(U_2)] \cdot \left(\frac{0.4929^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-0.4929)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2}{E[\alpha_2^2(U_2)] \cdot \left(\frac{0.4929^2}{0.9450} + \frac{1}{0.0550} + \frac{(1-0.4929)^2}{\sqrt{0.9450}} \right) \cdot 0.0550^2} = 1.0000$
- Notice that the only quantities being swapped out for each calculation are the credibility weights in the numerator. In addition, notice that we don't need $E[\alpha_2^2(U_2)]$ since it cancels out

Using the relative MSE table above, it's clear that the Neuhaus reserve best matches the optimal reserve since the relative MSEs are the closest to 1.0000.

IV. Reinterpreting Traditional Reserving Methods

At the end of Section 6, Hürlimann reinterprets traditional reserving methods using the approaches shown in this paper.

Chain-Ladder Method

The chain-ladder method is similar to the individual loss ratio method with the loss ratio payout factors replaced with the standard chain-ladder lag factors, defined as $p_i^{CL} = \frac{1}{CDF}$, where the CDF is based on volume-weighted development factors.

Hence, the chain-ladder reserve is as follows:

$$R_i^{CL} = \frac{q_i^{CL}}{p_i^{CL}} \cdot C_{i,n-i+1}$$

Cape Cod Method

Hürlimann defines the Cape Cod method as a Benktander-type credibility mixture with the following components:

- $R_i^{ind} = \frac{q_i^{CL}}{p_i^{CL}} \cdot C_{i,n-i+1}$
- $R_i^{coll} = q_i^{CL} \cdot ELR \cdot V_i$, where $ELR = \frac{\sum_{i=1}^n C_{i,n-i+1}}{\sum_{i=1}^n p_i^{CL} \cdot V_i}$
- $Z_i = p_i^{CL}$

There are a couple of things to note in the formulas above:

- 1) The loss ratio payout factors have been replaced with the standard chain-ladder lag factors
- 2) The definition of R_i^{coll} above is the standard Cape Cod reserve. So technically, Hürlimann's "Cape Cod" method is a credibility-weighted average of the standard chain-ladder and Cape Cod methods

Optimal Cape Cod Method

The optimal Cape Cod method is identical to the Cape Cod method above, but with the following credibility weights:

$$Z_i = \frac{p_i^{CL}}{p_i^{CL} + \sqrt{p_i^{CL}}}$$

BF Method

Hürlimann defines the BF method as a Benktander-type credibility mixture with the following components:

- $R_i^{ind} = \frac{q_i^{CL}}{p_i^{CL}} \cdot C_{i,n-i+1}$
- $R_i^{coll} = q_i^{CL} \cdot ELR_i \cdot V_i$, where ELR_i is some selected initial loss ratio for each origin period
- $Z_i = p_i^{CL}$

Similar to the Cape Cod method, Hürlimann's "BF" method is a credibility-weighted average of the standard chain-ladder and BF methods.

Optimal BF Method

The optimal BF method is identical to the BF method above, but with the following credibility weights:

$$Z_i = \frac{p_i^{CL}}{p_i^{CL} + \sqrt{p_i^{CL}}}$$

I. Introduction

Real-world loss data is subject to the following:

- Random fluctuations
- Systematic distortions

The **least squares method** should be considered whenever **random year-to-year fluctuations** in loss experience are significant.

II. Least Squares Method

Before we dive in, we need to define a few things:

- x = losses to date
- y = losses at a future evaluation

Assuming we have a historical loss triangle, we should have a number of historical (x, y) pairs. The **goal is to predict y** based on x . Let $L(x)$ be the estimate of y , given that we have already observed x .

Link Ratio Method

The link ratio method (i.e., chain-ladder method) **estimates y** as follows:

$$L(x) = cx$$

where c is the selected link ratio. In this paper, $c = \frac{\bar{y}}{\bar{x}}$, which is equal to the volume-weighted average LDF.

Budgeted Loss Method

In some cases, it might make sense to ignore the losses to date. For example:

- When fluctuation in loss experience is extreme
- When past data is not available

In these cases, we could **estimate** y using the budgeted loss method as follows:

$$L(x) = k$$

where k is a constant.

The constant k could be chosen by averaging y over several years, or by multiplying earned premium by an expected loss ratio. For the purposes of this paper, we will assume that k is based on averaging y over several years.

Least Squares Method

The least squares method **estimates** y by fitting a line to the points (x, y) that minimizes the sum of the squares of the residuals. Mathematically:

$$L(x) = a + bx$$

where $b = \frac{\bar{xy} - \bar{x}\bar{y}}{\bar{x}^2 - \bar{x}^2}$ and $a = \bar{y} - b\bar{x}$.

Many common methods are **special cases** of the least squares method:

- When $a = 0$, then $L(x) = bx$ (link ratio method)
- When $b = 0$, then $L(x) = a$ (budgeted loss method)
- When $b = 1$, then $L(x) = a + x$ (BF method)

The ability to flex to other methods is a **major advantage** of the least squares method. Later on, we will show that the least squares method is a credibility-weighted average between the link

ratio method and the budgeted loss method, highlighting its ability to give more or less weight to the observed value of x as appropriate.

Example: Basic Least Squares Method

An actuary would like to predict AY 2023 losses at 27 months. Given the following information for a small state as of December 31, 2023:

AY	<u>Incurred Losses (\$) as of XX Months:</u>		15-27 Link Ratios
	15 mo.	27 mo.	
2017	19,039	23,279	1.223
2018	33,040	41,560	1.258
2019	14,637	18,937	1.294
2020	2,785	5,185	1.862
2021	51,606	54,206	1.050
2022	5,726	15,726	2.746
2023	40,490		

Due to the volatility in the data and link ratios, we may not want to give full credibility to the high observed loss for 2023 by applying a large link ratio to it. Since the data fluctuations do not appear to be systematic, we should consider using the least squares method.

First, let's calculate the least squares parameters:

Using the *LINEST* Function

- The known x -values are (19039, 33040, ..., 5726)
- The known y -values are (23279, 41560, ..., 15726)
- We can use the **LINEST** function in Excel to find the parameters as follows:
$$\text{LINEST}(\text{known } y \text{ values, known } x \text{ values}) = \text{LINEST}((23279, 41560, \dots, 15726), (19039, 33040, \dots, 5726)) = (0.96781, 6023.70787)$$
- The first value output by Excel is b and the second value output is a
- Hence, **$b = 0.96781$** and **$a = 6023.70787$**

Using the Least Squares Formulas

On the *exam*, we recommend using the LINEST function to calculate the parameters. However, to demonstrate the least squares formulas, here's how to calculate the parameters by hand:

- $\overline{xy} = \frac{19,039(23,279) + 33,040(41,560) + \dots + 5,726(15,726)}{6} = 832,562,381$
- $\bar{x} = \frac{19,039 + 33,040 + \dots + 5,726}{6} = 21,139$
- $\bar{y} = \frac{23,279 + 41,560 + \dots + 15,726}{6} = 26,482$
- $\overline{x^2} = \frac{19,039^2 + 33,040^2 + \dots + 5,726^2}{6} = 728,681,571$
- $b = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2} = \frac{832,562,381 - 21,139(26,482)}{728,681,571 - 21,139^2} = \mathbf{0.96781}$
- $a = \bar{y} - b\bar{x} = 26,482 - 0.96781(21,139) = \mathbf{6023.70787}$
- These are the same parameters we obtained using the LINEST function

Second, let's estimate the AY 2023 incurred losses at 27 months using the least squares method:

Using the FORECAST Function

- We can use the **FORECAST** function in Excel to estimate y as follows:
 $\text{FORECAST}(x, \text{known } y \text{ values}, \text{known } x \text{ values}) = \text{FORECAST}(40490, (23279, 41560, \dots, 15726), (19039, 33040, \dots, 5726)) = \mathbf{45210.4966}$
- Thus, the estimated AY 2023 incurred losses at 27 months using the least squares method are \$45,210.50
- If the only goal is estimate y using the least squares method, then the FORECAST function is useful because it doesn't require the least squares parameters as inputs. However, we recommend calculating the parameters to make sure they are sensible (more on this later)

Using the Least Squares Formula

- Recall that $L(x) = a + bx$
- Thus, $L(40,490) = 6023.70787 + 0.96781(40,490) = \mathbf{45210.4966}$

- This is the same prediction we obtained using the FORECAST function

Parameter Estimation Errors

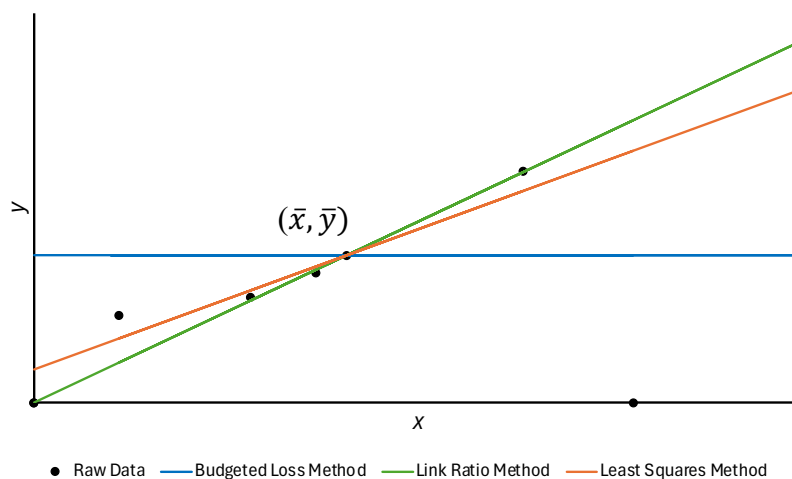
In the example above, there was nothing unusual about the least squares parameters. However, both **significant changes in the nature of the loss experience** and **sampling error** can lead to values of a and b that do not reflect reality:

- When $a < 0$, our estimate of y will be negative for small values of x . In this case, use the **link ratio method**
- When $b < 0$, our estimate of y decreases as x increases. In this case, use the **budgeted loss method**

The parameter estimation errors above are why we recommend always calculating the least squares parameters on the *exam*. In the event that $a < 0$ or $b < 0$, you should estimate y using either the link ratio method or budgeted loss method, respectively.

Comparing the Methods Graphically

The following plot compares the link ratio, budgeted loss, and least squares methods for the basic least squares example above:



We can summarize the plot as follows:

- The **link ratio method** fits a line through the origin with a slope equal to the selected link ratio
- The **budgeted loss method** fits a horizontal line equal to $\bar{y}$
- The **least squares method** yields a “best fit” line with an intercept of a and a slope of b
- The lines intersect at the point $(\bar{x}, \bar{y})$

III. Hugh White’s Question

Question: You are trying to establish the reserve for commercial auto bodily injury and the reported proportion of expected losses as of the statement date for the current accident year period is 8% higher than it should be. Do you:

- Reduce the bulk reserve by a corresponding amount (**budgeted loss method**)
- Leave the bulk reserve at the same percentage level of expected losses (**BF method**)
- Increase the bulk reserve in proportion to the increase of actual reported over expected reported (**link ratio method**)

The point of this question is to show that the three different options might all be reasonable answers to the question. In fact, these three options lie on a continuum of options implied by the least squares estimate $L(x) = a + bx$.

IV. Testing the Least Squares Method Using Loss & Loss Reporting Distributions

The plot of the link ratio, budgeted loss, and least squares methods for the basic least squares example showed a strong fit for the least squares method. However, that could have been pure luck. To truly test the effectiveness of the least squares method, we must know the form of the underlying loss & loss reporting distributions. In this section, we will look at a number of theoretical models.

Simple Model

Given the following information:

- The number of claims incurred each year is a random variable Y which is either 0 or 1 with equal probability
- The number of claims reported by year-end is a random variable X . If there is a claim, there is a 50% chance it will be reported by year-end. Hence, X is a binomial random variable with $n = y$ and $p = 0.5$

Let $Q(x)$ represent the expected total number of claims, and $R(x)$ represent the expected number of claims outstanding, both given that $X = x$. Mathematically:

$$\begin{aligned} Q(x) &= E[Y|X = x] \\ R(x) &= E[Y - X|X = x] \\ &= Q(x) - x \end{aligned}$$

To calculate $Q(x)$ and $R(x)$, we need $P(Y = y|X = x) = \frac{P(X = x|Y = y)P(Y = y)}{P(X = x)}$.

The following table calculates the numerator and denominator of the probability above:

x/y	$\frac{P(X = x Y = y)P(Y = y)}{P(X = x)}$		$P(X = x)$
	0	1	
0	0.50	0.25	0.75
1		0.25	0.25

As an **example calculation**, consider the case where $X = 0$ (zero claims reported by year-end) and $Y = 0$ (zero claims incurred for the year):

- $P(X = 0|Y = 0)P(Y = 0) = \left[\binom{0}{0}(0.5)^0(1 - 0.5)^{(0-0)} \right](0.50) = 0.50$

At this point, we have what we need to calculate $P(Y = y|X = x)$, $Q(x)$, and $R(x)$:

x/y	$P(Y = y X = x)$		$Q(x)$	$R(x)$
	0	1		
0	$0.67 = \frac{0.50}{0.75}$	0.33	$0.33 = 0(0.67) + 1(0.33)$	$0.33 = 0.33 - 0$
1		1.00	1.00	0.00

In this case, $Q(x)$ and $R(x)$ can be written as linear equations:

- $Q(x) = 0.67x + 0.33$
- $R(x) = -0.33x + 0.33$

The best (Bayesian) estimate of y (the number of claims incurred each year), given x (the number of claims reported by year-end), is a line with slope $b = 0.67$ and $a = 0.33$. This relationship is not described by the link ratio, budgeted loss, or BF methods. It is **best described by the least squares method**.

Poisson-Binomial Case

Assume that Y is Poisson distributed with mean μ , and where any given claim has probability d of being reported by year-end. In this case, we have the following for $Q(x)$ and $R(x)$:

$$\begin{aligned} Q(x) &= x + \mu(1 - d) \\ R(x) &= \mu(1 - d) \end{aligned}$$

Which method aligns best with the $Q(x)$ and $R(x)$ shown above?

- The link ratio method is NOT optimal since there is no c such that $x + \mu(1 - d) = cx$
- The budgeted loss method is NOT optimal since $Q(x)$ is not a constant
- The **BF method is optimal** since the expected number of outstanding claims, $R(x)$, does not depend on the number of claims already reported

Negative Binomial-Binomial Case

Assume Y is negative binomial distributed with parameters (r, p) , and where any given claim has probability d of being reported by year-end. In this case, we have the following for $R(x)$:

$$R(x) = \frac{(1-d)(1-p)}{1-(1-d)(1-p)}(x+r)$$

Except in the trivial case where $d = 1$, this is an **increasing linear function** of x , meaning an **increase in reported claims** leads to an **increase in our estimate of outstanding claims**.

This **does NOT** correspond to any of Hugh White's answers:

- Since $R(x)$ is an increasing function of x , the budgeted loss and BF methods are not optimal
- Since the relationship is not proportional, the link ratio method is also not optimal

Question: Why does an increasing function make intuitive sense in this case?

- **Consider a Poisson** distribution with mean 2. This means that the variance of the Poisson distribution is also 2
- Now **consider a negative binomial** distribution with mean 2. Based on the variance formula for the negative binomial distribution, the variance **MUST** be greater than 2
- Hence, the negative binomial distribution has more variance than the Poisson distribution with the same mean!
- This means we have less confidence in our prior estimate of expected losses and are more willing to increase our estimated ultimate claim count. Since a Poisson distribution corresponds to “no dependence between outstanding claims and claims already reported,” an increase in the estimated ultimate claim count implies an increasing function of x

Fixed Prior Case

Suppose Y is not random, which means there is some value k such that Y is sure to equal to k (ex. single-premium whole life policies). In this case, we have the following for $Q(x)$ and $R(x)$:

$$\begin{aligned} Q(x) &= k \\ R(x) &= k - x \end{aligned}$$

This corresponds to the **budgeted loss method**.

Fixed Reporting Case

Suppose there is a number $d \neq 0$ such that the percentage of claims reported by year-end is always d . In this case, we have the following for $Q(x)$ and $R(x)$:

$$\begin{aligned} Q(x) &= \frac{x}{d} \\ R(x) &= \frac{x}{d} - x \end{aligned}$$

This corresponds to the **link ratio method**.

Non-Linear Case

In each of the previous models, $Q(x)$ was linear in x and was of the form $Q(x) = a + bx$. The following example moves away from linearity.

Given the following information:

- The number of claims incurred each year is a random variable Y that is uniformly distributed on the discrete set, $\{2,3,4,5,6\}$
- The number of claims reported by year-end is a random variable X . If there is a claim, there is a 50% chance it will be reported by year-end. Hence, X is a binomial random variable with $n = y$ and $p = 0.5$

Using the same process we used for the **simple model**, we obtain the following table:

x/y	$P(Y = y X = x)$					$Q(x)$	$R(x)$
	2	3	4	5	6		
0	$\frac{16}{31}$	$\frac{8}{31}$	$\frac{4}{31}$	$\frac{2}{31}$	$\frac{1}{31}$	2.839	2.839
1	$\frac{32}{88}$	$\frac{24}{88}$	$\frac{16}{88}$	$\frac{10}{88}$	$\frac{6}{88}$	3.250	2.250
2	$\frac{16}{99}$	$\frac{24}{99}$	$\frac{24}{99}$	$\frac{20}{99}$	$\frac{15}{99}$	3.939	1.939
3		$\frac{8}{64}$	$\frac{16}{64}$	$\frac{20}{64}$	$\frac{20}{64}$	4.813	1.813
4			$\frac{4}{29}$	$\frac{10}{29}$	$\frac{15}{29}$	5.379	1.379
5				$\frac{2}{8}$	$\frac{6}{8}$	5.750	0.750
6					1	6.000	0.000

As the table above shows, $R(x)$ is not linear. However, it is monotonic (the text incorrectly states that it is not monotonic).

Intuitively, it makes sense that $R(x)$ should decrease since Y has less variance than a Poisson distribution with the same mean (hence, we have more confidence in our prior estimate of expected losses and are less willing to revise our estimated ultimate claim count).

V. The Linear Approximation (Bayesian Credibility)

In general, it is difficult to compute a pure Bayesian estimate Q since it requires knowledge of the loss and loss reporting processes. This makes it difficult to make assumptions.

As a replacement for the Bayesian estimate, we can use the best linear approximation, which has the following advantages:

- Simpler to compute

- Easier to understand and explain
- Less dependent upon the underlying distribution

Let L be the best linear approximation to Q . Mathematically, L is the linear function that minimizes $E_X[(Q(X) - L(X))^2]$.

Assuming $L(x) = a + bx$, we must minimize $E_X[(Q(X) - a - bX)^2]$.

Development Formula 1

Given random variables Y describing ultimate losses and X describing reported losses, the best linear approximation to Q is as follows:

$$L(x) = (x - E[X]) \frac{Cov(X, Y)}{Var(X)} + E[Y]$$

This formula provides us with an answer to Mr. White's question:

- If $Cov(X, Y) < Var(X)$, a large reported amount should lead to a **decrease** in the reserve
- If $Cov(X, Y) = Var(X)$, a large reported amount should **not affect** the reserve
- If $Cov(X, Y) > Var(X)$, a large reported amount should lead to an **increase** in the reserve

Example: Development Formula 1

Suppose that $E[Y] = 1,000$ (expected ultimate loss) and $E[X] = 500$ (the amount expected to be reported by the end of the year). Then, the expected reserve at the end of the year is $1,000 - 500 = 500$.

Now, assume that $x = 750$ (actual amount reported by the end of the year). The following table shows how the reserve changes based on the ratio $\frac{Cov(X, Y)}{Var(X)}$:

$\frac{Cov(X, Y)}{Var(X)}$	0.90	1.00	1.10
Expected Ultimate	1,000	1,000	1,000
Expected Reserve	500	500	500
Revised Ultimate, $L(x)$	1,225	1,250	1,275
Revised Reserve, $L(x) - x$	475	500	525

As an **example calculation**, let $\frac{Cov(X, Y)}{Var(X)} = 0.90$. Then, $L(750) = (750 - 500)(0.90) + 1,000 = 1,225$ and $L(750) - 750 = 1,225 - 750 = 475$. As expected, the higher reported amount led to a decrease in the reserve.

As we can see in the formula for $L(x)$ above, we still have random variables. Instead of making assumptions about the distributions of these random variables, we can use **empirical data** to estimate the means, variance, and covariance from the data:

$$L(x) = (x - \bar{X}) \frac{\overline{XY} - \bar{X}\bar{Y}}{\overline{X^2} - \bar{X}^2} + \bar{Y}$$

This formula should look familiar. It's a reconfigured version of the least squares formula!

When is Least Squares Development Appropriate?

As mentioned earlier, the least squares fit **may be appropriate** if year-to-year changes are due largely to **random chance**.

The least squares fit **does NOT make sense** if year-to-year changes in loss experience are due largely to **systematic shifts or distortions** in the book of business.

In the case of systematic distortions, if the data can be adequately adjusted, **we may still be able** to apply the least squares method. Here are a couple of **examples**:

- If studying **incurred loss data**, we can correct for inflation by putting the years on a constant-dollar basis before fitting a line

- If the business expands, we can divide each year's losses by an exposure base to eliminate the distortion

VI. A Credibility Form of the Development Formula

The goal is to reconfigure Development Formula 1 into a credibility weighting system. Let's start with some definitions:

- $E_Y[Var(X|Y)]$ = Expected Value of the Process Variance (*EVPI*). This represents the variability resulting from the loss reporting process
- $Var_Y(E[X|Y])$ = Variance of the Hypothetical Mean (*VHM*). This represents the variability resulting from the loss occurrence process

Development Formula 2

In order to reconfigure Development Formula 1 into a credibility weighting system, we need an additional assumption regarding d . Suppose there is a real number $d \neq 0$ such that $E[X|Y = y] = dy$ for all y . Then, the best linear approximation to Q is as follows:

$$L(x) = Z \frac{x}{d} + (1 - Z)E[Y]$$

where $Z = \frac{VHM}{VHM + EVPI}$.

As we can see from the formula above, L is a credibility weighting of the link ratio estimate $\frac{x}{d}$ and the budgeted loss estimate $E[Y]$:

- If $EVPI = 0$, we give full weight to the link ratio estimate (fixed reporting case)
- If $VHM = 0$, we give full weight to the budgeted loss estimate (fixed prior case)

When using the least squares version of Development Formula 1, the credibility weight is as follows:

$$Z = bd = \frac{b}{c}$$

As defined earlier, $c = \frac{\bar{y}}{\bar{x}}$, which is equal to the volume-weighted average.

Example: Least Squares Method for a Growing Book of Business

An actuary wants to estimate AY 2020 & 2021 ultimate losses using the least squares method. Given the following information for a small book of business as of December 31, 2023:

AY	EP (\$000)	<u>Reported Loss (\$000) as of XX Months:</u>				
		12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2017	4,260	102	104	209	650	847
2018	5,563	0	543	1,309	2,443	3,003
2019	7,777	412	2,310	3,083	3,358	4,099
2020	8,871	219	763	1,637	1,423	
2021	10,465	969	4,090	3,801		
2022	11,986	0	3,467			
2023	12,873	932				

- 60-Ult. Tail Factor = 1.10

AY 2020

First, let's divide the reported losses by the earned premium for each year to adjust for the systematic increase in the losses:

AY	<u>Reported Loss Ratio as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2017	$0.024 = \frac{102}{4,260}$	$0.024 = \frac{104}{4,260}$	0.049	0.153	0.199

2018	$0.000 = \frac{0}{5,563}$	0.098	0.235	0.439	0.540
2019	0.053	0.297	0.396	0.432	0.527
2020	0.025	0.086	0.185	0.160	
2021	0.093	0.391	0.363		
2022	0.000	0.289			
2023	0.072				

By dividing the losses by the earned premium for each year, the AYs are on a more apples-to-apples basis.

Second, let's account for the tail factor:

AY	<u>Reported Loss Ratio as of XX Months:</u>					Ult.
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.	
2017	0.024	0.024	0.049	0.153	0.199	0.219 = 0.199(1.10)
2018	0.000	0.098	0.235	0.439	0.540	0.594 = 0.540(1.10)
2019	0.053	0.297	0.396	0.432	0.527	0.580 = 0.527(1.10)
2020	0.025	0.086	0.185	0.160		
2021	0.093	0.391	0.363			
2022	0.000	0.289				
2023	0.072					

Third, let's calculate the least squares parameters for AY 2020 to ensure there are no parameter estimation errors:

- Since AY 2020 is 48 months old, the known x -values are (0.153, 0.439, 0.432)
- Since we want to estimate the AY 2020 ultimate loss ratio, the known y -values are (0.219, 0.594, 0.580)
- $\text{LINEST}((0.219, 0.594, 0.580), (0.153, 0.439, 0.432)) = (1.30145, 0.02007)$
- Thus, $b = 1.30145$ and $a = 0.02007$. No parameter estimation errors are present

Fourth, let's estimate the AY 2020 ultimate loss ratio using the least squares method:

- $\text{FORECAST}(0.160, (0.219, 0.594, 0.580), (0.153, 0.439, 0.432)) = 0.2288$

Fifth, let's convert the AY 2020 ultimate loss ratio into an ultimate loss:

- AY 2020 ultimate losses = (AY 2020 ultimate loss ratio)(AY 2020 EP) =
(0.2288)(8,871,000) = 2,030,032
- Thus, the AY 2020 ultimate losses are **\$2,030,032**

Lastly, let's demonstrate that the least squares method is a **credibility weighting** of the link ratio method and the budgeted loss method:

- $c = \frac{\bar{y}}{\bar{x}} = \frac{(0.219+0.594+0.580)/3}{(0.153+0.439+0.432)/3} = 1.3603$. This is equal to the 48-ultimate volume-weighted average, where the "weights" are the loss ratios at 48 months
- $Z = \frac{b}{c} = \frac{1.30145}{1.3603} = 0.9567$
- Link ratio ultimate loss ratio = $1.3603(0.160) = 0.2182$
- Budgeted loss ultimate loss ratio = $\bar{y} = \frac{0.219+0.594+0.580}{3} = 0.464$
- $L(0.160) = 0.9567(0.2182) + (1 - 0.9567)(0.464) = 0.2288$. This is **equal to the ultimate loss ratio** we calculated in step 4 above

AY 2021

To calculate the AY 2021 ultimate loss ratio, we re-run the least squares method with the AY 2020 estimate included in the y-values.

First, let's calculate the least squares parameters for AY 2021 to ensure there are no parameter estimation errors:

- Since AY 2021 is 36 months old, the known x -values are (0.049, 0.235, 0.396, 0.185)
- Since we want to estimate the AY 2021 ultimate loss ratio, the known y -values are (0.219, 0.594, 0.580, **0.229**). Notice that the AY 2020 estimate of 0.229 has been added to the list of y -values
- $\text{LINEST}((0.219, 0.594, 0.580, 0.229), (0.049, 0.235, 0.396, 0.185)) = (1.16244, 0.15381)$

- Thus, $b = 1.16244$ and $a = 0.15381$. No parameter estimation errors are present

Second, let's estimate the AY 2021 ultimate loss ratio using the least squares method:

- $\text{FORECAST}(0.363, (0.219, 0.594, 0.580, 0.229), (0.049, 0.235, 0.396, 0.185)) = 0.5760$

Third, let's convert the AY 2021 ultimate loss ratio into an ultimate loss:

- AY 2021 ultimate losses = (AY 2021 ultimate loss ratio)(AY 2021 EP) =
(0.5760)(10,465,000) = 6,028,028
- Thus, the AY 2021 ultimate losses are **\$6,028,028**

To calculate subsequent ultimate loss ratios, we keep adding the new estimates to the known y -values and re-running the least squares method.

In the prior example, we were able to adjust for systematic distortions and apply the least squares method. How do we handle situations where we **CANNOT correct** for year-to-year changes in the loss & loss reporting distributions? Let's look at another example.

Example: Law Change

We wish to develop AY 2023 personal auto losses for a state which has just instituted a strict verbal tort threshold. Given the following information as of December 31, 2023:

- Under the old system, expected losses were \$20M. Industry estimates show the reform should save 40% in the first year
- Under the old system, about 62% of incurred losses were reported by year-end. Under the new system, this is expected to rise to 75%

Based on the information above, we expect the following under the new system:

- $E[Y] = 20(1 - 0.40) = 12$ (expected ultimate losses are \$12M)

- $E\left[\frac{X}{Y}\right] = 0.75$
- $E[X] = E[Y] \cdot E\left[\frac{X}{Y}\right] = 12(0.75) = 9$ (expected losses reported at year-end are \$9M).

Note that Brosius assumes that $E\left[\frac{X}{Y}\right]$ is independent of Y

Suppose the year-end reported loss is only \$6M.

Are the savings from the reform greater than expected (in which case we should reduce our estimate of ultimate loss), or are there temporary reporting delays (in which case we should not reduce our estimate of ultimate loss)?

Neither the least squares method nor the link ratio method works here since they rely on past loss experience, which was based on a different tort system. It may not be appropriate to use the budgeted loss method due to uncertainty surrounding the estimated savings.

In this case, it is best to estimate *EVPPV* and *VHM* using Bayesian credibility.

First, let's estimate the means and standard deviations of the loss Y and the reporting ratio $\frac{X}{Y}$:

- We already have estimates of the means:
 - $E[Y] = 12$
 - $E\left[\frac{X}{Y}\right] = 0.75$
- Discussions with the underwriting and claims team lead to the following estimates of the standard deviations:
 - $\sigma(Y) = 3$
 - $\sigma\left(\frac{X}{Y}\right) = 0.14$

Second, let's calculate *EVPPV* and *VHM*:

- Brosius does not provide the derivations of *EVPPV* and *VHM* in the paper. Here are the required formulas:

$$\begin{aligned}
 EVPV &= \text{Var}\left(\frac{X}{Y}\right) (\text{Var}(Y) + E[Y]^2) \\
 VHM &= \left(E\left[\frac{X}{Y}\right]\right)^2 (\text{Var}(Y))
 \end{aligned}$$

- $EVPV = 0.14^2(3^2 + 12^2) = 3$
- $VHM = 0.75^2(3^2) = 5.06$

Third, let's calculate the credibility weight Z :

- $Z = \frac{VHM}{VHM + EVPV} = \frac{5.06}{5.06 + 3} = 0.628$

Fourth, let's estimate AY 2023 ultimate losses using Bayesian credibility:

- Recall from Development Formula 2 that $L(x) = Z \frac{x}{d} + (1 - Z)E[Y]$
- $L(6) = 0.628\left(\frac{6}{0.75}\right) + (1 - 0.628)(12) = 9.5$
- Thus, the estimate of the AY 2023 ultimate losses is \$9.5M

Lastly, let's compare the estimate to the other methods:

- It is larger than the link ratio estimate of $\frac{6}{0.75} = \$8\text{M}$
- It is smaller than the budgeted loss estimate of $E[Y] = \$12\text{M}$
- It is slightly larger than the BF estimate of $x + E[Y]\left(1 - E\left[\frac{X}{Y}\right]\right) = 6 + 12(1 - 0.75) = \9M . This makes sense because $b = \frac{0.628}{0.75} < 1$. Conceptually, this means we placed more confidence in the prior estimate $E[Y]$ than if we had used the BF method

VII. The Caseload Effect

Development Formula 2 assumes that the expected number of claims reported is proportional to the number of claims incurred. Since a claim is more likely to be reported quickly when the

caseload is low, we expect the development ratio $\frac{E[X|Y = y]}{y}$ to be a decreasing function of y , not a constant. Development Formula 3 addresses this.

Development Formula 3

Suppose there are real numbers $d \neq 0$ and x_0 such that $E[X|Y = y] = dy + x_0$ for all y . Then, we have the following:

$$L(x) = Z \left(\frac{x - x_0}{d} \right) + (1 - Z)E[Y]$$

where $Z = \frac{VHM}{VHM + EVPV}$.

Regarding the formula above:

- It gives a development ratio of $d + \frac{x_0}{y}$, which decreases as y increases
- It gives $E[X|Y = 0] = x_0 > 0$, which doesn't make much sense
- When $x_0 = 0$, we obtain Development Formula 2 as a special case

Example: Caseload Effect

Given the following information for an accident year:

- Expected losses are \$15M
- When the caseload is low ($y = \$5M$), 60% of the claims are expected to be reported by year-end
- When the caseload is high ($y = \$20M$), 37.5% of the claims are expected to be reported by year-end
- The actual amounts of losses reported by year-end are \$7.5M
- $Z = 0.25$

Since the amount of losses expected to be reported by year-end depends on y , we should use Development Formula 3.

First, let's set up a system of equations to solve for d and x_0 :

$$d + \frac{x_0}{5,000,000} = 0.600 \quad \longrightarrow \quad 5,000,000d + x_0 = 3,000,000$$

$$d + \frac{x_0}{20,000,000} = 0.375 \quad \longrightarrow \quad 20,000,000d + x_0 = 7,500,000$$

This yields $d = 0.3$ and $x_0 = 1,500,000$.

Second, let's estimate the ultimate losses for the AY:

- $L(7,500,000) = 0.25 \left(\frac{7,500,000 - 1,500,000}{0.3} \right) + (1 - 0.25)(15,000,000) = 16,250,000$
 - Thus, the estimated ultimate losses are \$16.25M
-

VIII. Conclusion

Conclusions regarding the least squares method are as follows:

- It is easy to implement and uses easily accessible data
- It works well for developing losses for small states or lines that are subject to serious fluctuations
- It can lead you astray if corrections are not made to account for significant exposure changes or other shifts in loss history
- It is subject to sampling error since parameters are estimated from observed data

I. Introduction

Sound estimates of unpaid losses are important for reinsurers and their stakeholders. Reasons include:

- **Internal management** of a reinsurer requires sound reserves because they impact pricing, underwriting, strategic planning and financial decision making
- **Investors** require sound reserves in order to appropriately evaluate a reinsurer's balance sheet and income statement when making investment decisions
- **Regulators** require sound reserves to appropriately evaluate a reinsurer's financial position
- **Rating agencies** require sound reserves in order to issue strong financial ratings. If a reinsurer experiences significant adverse development, it risks a rating downgrade. This may make the reinsurer look less attractive to primary insurers

Although much of this will be review for the majority of students, let's cover some basic **reinsurance terms**:

- **Reinsurance** – insurance for primary insurers
- **Reinsured** (i.e. cedant, ceding company, primary insurer) – the insurer that cedes its business to a reinsurer
- **Reinsurer** – the insurer that accepts or assumes business from the cedant for a stated premium
- **Retrocession** – insurance for reinsurers
- **Retrocedant** – the reinsurer that cedes its business to another reinsurer
- **Retrocessionaire** – the reinsurer that accepts or assumes business from the retrocedant for a stated premium
- **Gross experience** – for a reinsurer, this refers to assumed business

- Ceded experience – for a reinsurer, this refers to business transferred through retrocessions
- Net experience – gross less ceded experience
- Retention – amount of loss that the cedant retains after consideration of reinsurance
- Attachment point – point at which reinsurance begins to apply
- Working layer – dollar range in which the insurer or reinsurer expects relatively predictable loss experience with a fairly high level of loss frequency. The working layer for the cedant is usually different from the working layer for a reinsurer
- Bordereau – a detailed report of insurance premiums or losses affected by reinsurance. Reinsurers often receive data by bordereau from cedants or the brokers of their cedants
- Counterparty default risk – risk that the reinsurer is unable to meet its contractual obligations. When this occurs, the liability falls back to the ceding insurer
- Subject loss – the losses that are relevant to the reinsurance coverage

There are **five primary reasons** that ceding companies purchase reinsurance:

1) Promote Stability

- Reinsurance stabilizes the loss experience of ceding companies from year to year by retaining smaller, more predictable claims and ceding larger, more unpredictable claims

2) Increase Capacity

- Reinsurance allows ceding companies to assume more risk by ceding a portion of all of their policies or by ceding their larger policies
- As an example, suppose a primary insurer is asked to insure a commercial building for \$200 million. Further assume that the primary insurer has an internally defined maximum retention of \$10 million. By securing reinsurance equal to \$190 million, the primary insurer could write the policy
- By increasing its capacity through reinsurance, cedants may be able to write quality accounts that are otherwise unattainable. It also allows smaller insurers compete with larger insurers who are able to retain more risk

3) Protect Against Catastrophes

- Reinsurance protects ceding companies from a single catastrophic loss event (ex. single hurricane) as well as multiple large loss events (ex. multiple hurricanes within a single year)

4) Manage Capital & Solvency Margin

- Reinsurance passes risk from the cedant to the reinsurer. With less risk present, less capital is needed
- The smaller risk level shows up in the cedant's financial results. For example, most reinsurance contracts involve a ceding commission paid by the reinsurer to the cedant, which effectively transfers surplus from the reinsurer to the cedant. In addition, the ceded premium reduces the cedant's net premium-to-surplus ratio. Companies with lower net premium-to-surplus ratios are able to write more business

5) Access Technical Expertise

- Reinsurers can lend their expertise in the areas of underwriting, marketing, claims, and pricing to insurers seeking to enter new lines of business or new regions

Unlike many personal insurance policies, reinsurance contracts can change dramatically depending on the needs of the cedant. However, reinsurance can be categorized into broad buckets: **treaty vs. facultative** and **proportional vs. non-proportional**.

Treaty Reinsurance

Under treaty reinsurance, the cedant enters into a contract with one or more reinsurers to cede all business from certain lines of business, subject to the retentions, limits, and attachment points specified in the treaty.

An **important aspect** of treaty reinsurance is that the reinsurer is not involved in the underwriting of the underlying policies written by the cedant. Treaty reinsurance effectively transfers underwriting risk from the cedant to the reinsurer.

Facultative Reinsurance

Unlike treaty reinsurance, reinsurers have the option of rejecting cession submissions from the cedant under facultative reinsurance. Furthermore, a submission, acceptance, and agreement are required for each individual risk or group of risks that the cedant wants to reinsure.

The **primary purpose** of facultative reinsurance is to increase a cedant's capacity by ceding large risks (often high-value commercial risks).

Underwriting risk for the reinsurer is reduced under facultative reinsurance (all else being equal) since the reinsurer can conduct its own underwriting.

Example: Combining Treaty & Facultative Reinsurance

As an example of treaty and facultative reinsurance, a cedant may maintain property treaty reinsurance for policyholders with total insured values (TIV) less than \$25 million, while policyholders with TIV of \$25 million or more may be placed through the facultative market.

For the treaty reinsurance portion, all ceded risks would be subject to the terms and limits of the treaty.

For the facultative portion, terms and conditions would be unique to the ceded risks.

Hybrid Contracts

Some contracts include both treaty and facultative characteristics and are known as **hybrid contracts**. Here are two **examples**:

- 1) **Facultative automatic** – a bordereau of risks ceded is submitted to the reinsurer, which has limited rights to decline individual risks
- 2) **Facultative obligatory treaty** – a treaty under which the cedant has the option to cede or not cede individual risks. The reinsurer must accept any risks that are ceded

Proportional (i.e. Pro-Rata) Reinsurance

Both treaty reinsurance and facultative reinsurance can be written on a proportional or non-proportional basis.

Under proportional reinsurance, the reinsurer typically pays a ceding commission to the cedant to reimburse for acquisition and underwriting expenses associated with issuing the underlying policies.

Two common types of proportional reinsurance are quota share and surplus share reinsurance.

Example: Quota Share Reinsurance

As an example of quota share reinsurance, assume the following:

- A quota share treaty applies to a single line of business with a cession percentage of 60%
- There are two underlying policies written by the cedant

The premiums, losses, and loss ratios for the cedant are summarized below:

Insured	<u>Gross</u>		<u>Net</u>		<u>Ceded</u>	
	EP (\$M)	Ult. Loss (\$M)	EP (\$M)	Ult. Loss (\$M)	EP (\$M)	Ult. Loss (\$M)
#1	1,000	600	400	240	600	360
#2	1,000	3,000	400	1,200	600	1,800
Total	2,000	3,600	800	1,440	1,200	2,160

As we can see in the table, the cedant retains 40% of the experience and cedes 60% of the experience to the reinsurer. It's also important to note that the ultimate loss ratios are always the same for the cedant and the reinsurer under quota share reinsurance:

Insured	Ultimate Loss Ratios		
	Gross	Net	Ceded
#1	60%	60%	60%
#2	300%	300%	300%
Total	180%	180%	180%

Unlike quota share reinsurance, surplus share reinsurance does not have a uniform cession percentage across risks. Instead, the cedant cedes the surplus amount of risk above its retained line subject to a maximum ceded percentage and limit:

- The retained line is the amount the cedant is willing to retain per risk
- The reinsurer's share is expressed as a multiple of the cedant's retained line. Thus, a three-line surplus share treaty provides reinsurance for three times the cedant's retained line, which means the cedant can write four times as much coverage as possible before reinsurance

Example: Surplus Share Reinsurance

As an example of surplus share reinsurance, assume the following:

- A cedant wants to write commercial auto policies to a maximum limit of \$10 million per policy
- The cedant's retained line is \$2.5 million per policy
- The cedant purchases a three-line surplus share treaty that provides an additional \$7.5 million in coverage per policy ($7.5 = 2.5(3)$)
- The proportion ceded per policy is calculated as follows: $\frac{\text{Policy Limit} - \text{Retained Line}}{\text{Policy Limit}}$
- There are three underlying policies written by the cedant

The losses for the cedant are summarized below:

Insured	Policy Limit (\$M)	Ult. Loss (\$M)	Proportion Ceded	Retained Ult. Loss (\$M)	Ceded Ult. Loss (\$M)
#1	2.5	2.5	0%	2.5	0
#2	5.0	2.5	$50\% = \frac{5 - 2.5}{5}$	$1.25 = 2.5(1 - 0.50)$	1.25
#3	10.0	2.5	$75\% = \frac{10 - 2.5}{10}$	$0.625 = 2.5(1 - 0.75)$	1.875

In the table above, the ceded column is equal to the ultimate loss column minus the retained ultimate loss column. As we can see, the cession percentage differs depending on the policy limit of the underlying policy. In this sense, we can think of surplus share reinsurance as variable quota share reinsurance.

So, what is proportional reinsurance typically used for?

- To manage capital and solvency margins (net premium-to-surplus ratios are typically improved)
- To increase capacity
- To protect against catastrophes

Non-Proportional (i.e. Excess of loss) Reinsurance

Non-proportional reinsurance can be written on a treaty or facultative basis.

Unlike proportional reinsurance, the premium paid to the reinsurer is not proportional to the limits of coverage. Also, the reinsurer or reinsurers pay for all losses above a specified retention, subject to any specified limits.

Common types of non-proportional reinsurance are excess per risk, excess per occurrence, catastrophe, annual aggregate excess of loss and clash reinsurance.

Excess Per Risk Reinsurance

This reinsurance indemnifies the cedant against the amount of loss in excess of a specified retention subject to specified limits on a per risk basis. In this case, each “risk” would be defined in the contract.

Excess per risk reinsurance is **typically used** to protect property exposures and increase capacity.

Although excess per risk reinsurance often includes ceding commissions, those commissions provide less surplus relief to the cedant because the reinsurance premium tends to be significantly less.

Example: Excess Per Risk Reinsurance

A cedant sells property policies with a \$10 million limit and maintains excess per risk reinsurance with a \$3 million attachment point and reinsurance limit of \$7 million.

For a loss of \$3 million, reinsurance is not triggered and the cedant retains the full loss.

For a loss of \$6.5 million, the cedant retains \$3 million and the reinsurer assumes losses of \$3.5 million.

Excess Per Occurrence Reinsurance

This reinsurance operates very similarly to excess per risk reinsurance. The main difference is that excess per occurrence reinsurance protects a cedant from an accumulation of losses due to a single occurrence. The subject loss under excess per occurrence reinsurance is the sum of all losses arising from an insured event for all subject policies. This means the insurer’s retention only comes into play once rather than for each individual risk.

Catastrophe Reinsurance

This reinsurance indemnifies the cedant for the accumulation of losses in excess of a specified retention arising from a single catastrophic event or a series of events, subject to a specified limit. Thus, it is essentially a **special case of excess per occurrence** reinsurance.

There are **two important terms** to understand when discussing catastrophe reinsurance:

- In the event of a full limit loss or some other amount specified in the reinsurance contract, cedants can obtain a **reinstatement** of the reinsurance policy limit
- Depending on the reinsurance contract terms, the reinstatement may or may not require an additional premium to be paid by the cedant to the reinsurer. If such a premium is required, it is known as a **reinstatement premium**

Example: Catastrophe Reinsurance

A cedant maintains unlimited catastrophe reinsurance with a retention of \$35 million. A flood results in total property losses of \$42 million. The cedant would retain \$35 million in losses and the reinsurer would assume \$7 million in losses.

Before we dive into the other non-proportional contracts, let's consider how multiple reinsurance contracts might *inure to the benefit* of another. When one reinsurance contract is applied first and reduces the loss subject to other reinsurance contracts, then we say that the first contract inures to the benefit of the other contracts. If all other reinsurance is ignored when considering a reinsurance contract's impact on losses, then we say that the reinsurance contracts inure to the benefit of the reinsured.

Example: Inuring Contracts

As an example of inuring contracts, assume the following:

- The cedant writes 200 personal property policies. Each policy has a \$2 million limit
- The cedant purchases excess per risk reinsurance with a retention of \$1 million and a reinsurance policy limit of \$1 million
- The cedant also purchases catastrophe reinsurance with a retention of \$20 million and a reinsurance policy limit of \$150 million
- The excess per risk reinsurance inures to the benefit of the catastrophe reinsurance

A wildfire occurs with the following loss profile:

Individual Losses Per Policy (\$M)	# Insureds Suffering Losses	Total Insured Losses (\$M)	Losses Ceded to Excess Per Risk Reinsurance (\$M)
0.2	35	7	0
1.0	10	10	0
2.0	5	10	5

In this case, the excess per risk reinsurance does NOT apply to the 35 individuals with losses of \$200,000 each since \$200,000 is below the \$1 million retention. The same logic applies to the 10 individuals with losses of \$1 million each. For the remaining 5 individuals with losses of \$2 million each, the cedant retains the first \$1 million of each loss and the reinsurer covers the remaining \$1 million. Note that this also happens to be the per risk reinsurance policy limit. At this point, the cedant's retained losses are \$22 million ($27 - 5 = 22$). Now, the catastrophe reinsurance cover kicks in. The cedant retains the first \$20 million and the catastrophe reinsurer covers the remaining \$2 million.

Annual Aggregate Excess of Loss (i.e., Aggregate Stop-Loss) Reinsurance

Under this reinsurance, the total losses to the cedant cannot exceed a specified annual threshold (expressed as either a percent of premium or fixed dollar amount).

This type of reinsurance is typically **used to protect net results** (i.e. this reinsurance kicks in after other contracts have already been applied) and an insurer's capital base.

The **issue with this type of reinsurance** is that it is often unavailable or very expensive.

Example: Aggregate Stop-Loss Reinsurance

A captive insurer seeks aggregate excess of loss reinsurance. The reinsurance contract is stated as follows: "20% loss ratio excess of the captive's retention of a 90% loss ratio." Assuming that the captive has a subject premium of \$10 million, the aggregate excess of loss contract provides coverage of \$2 million (10 x 20%) excess of \$9 million (10 x 90%).

Clash Reinsurance

This reinsurance is a casualty reinsurance contract that attaches above all other policy limits.

Clash coverage is meant to cover exceptional events where traditional reinsurance contracts will not fully reimburse a cedant's claims. This can occur when a cedant receives multiple claims from multiple insureds arising out of the same catastrophe.

In order for clash coverage to kick in, a clash event must occur. A clash event has **three components**:

- 1) The loss must arise out of multiple policies held by one insured or similar policies held by multiple insureds
- 2) All damages must be traceable to a specific event
- 3) The event must take place in its entirety within a specific timeframe

Example: Clash Coverage

Suppose a reinsurance contract covers losses from a single event up to a \$10 million policy limit regardless of the number of claims. If a massive hurricane occurs and total losses are well above \$10 million, then the cedant will be stuck with a number of un-reinsured claims. Clash coverage can be used to fill this gap.

Outside of the broad categories of reinsurance we have discussed so far, there are a few other types of special reinsurance. These include finite risk reinsurance, loss portfolio transfers, and adverse development covers.

Finite Risk Reinsurance

Typical finite risk reinsurance contracts are multi-year contracts that spread risk over time and consider the investment income generated over the period.

Finite risk reinsurance typically has the following features:

- Risk transfer and risk financing combined in a multi-year contract
- Emphasis on the time value of money with investment income explicitly included in the contract
- Limited assumption of risk by the reinsurer
- Sharing of the results with the cedant

Loss Portfolio Transfers (LPTs)

An LPT is a form of reinsurance that transfers all or a portion of a cedant's loss reserves present at a specific accounting date to a reinsurer.

When determining the premium paid to the reinsurer to assume the reserves, the time value of money must be considered. Thus, the premium is less than the reserves.

The cedant's statutory surplus increases by the difference between the premium and the loss reserves.

LPTs are often used by cedants to withdraw from a specific line of business while meeting their obligations to policies they already wrote. The lines of business involved in LPTs are often long-tailed (ex. medical malpractice, asbestos) where there are significant delays in the reporting of claims and the losses may not be settled for years.

Reinsurers could lose money on an LPT contract in the event that claims are settled faster than expected, resulting in less than anticipated investment income.

Adverse Development Cover

This reinsurance cover is an alternative to an LPT where the cedant receives reimbursement from the reinsurer for losses in excess of a pre-agreed retention level.

In its basic form, an adverse development cover would attach at the level of the cedant's current carried reserves and cover any development beyond that point.

Unlike an LPT, reserves are not transferred under an adverse development cover.

These covers are often used for Mergers & Acquisitions to transfer the risks of timing and adverse reserve development.

Miscellaneous Topics

The last portion of the introduction of this paper covers a few topics that influence the estimation of unpaid losses for reinsurers. These are the business **covered clause**, **subscription percentages**, and **commutation clauses**.

Business Covered Clause

The **business covered clause** describes which claims are covered under the reinsurance contract.

There are two ways in which reinsurance covers claims:

- **Losses-occurring-during** coverage – provides reinsurance coverage for all losses that occur between the contract inception and expiration date **REGARDLESS** of when the cedant issued the underlying policy that resulted in the loss
- **Risks-attaching** coverage – provides reinsurance coverage only for those policies that began during the reinsurance contract effective period. Note that the underlying policies can have expiration dates later than the expiration date of the reinsurance contract as long as the underlying policies began during the reinsurance contract effective period

Subscription Percentages

When the amount of coverage sought by the primary insurer is more than a single reinsurer wants to provide, the primary insurer can issue a **subscription policy**.

Subscription policies are policies in which multiple reinsurers share a risk subject to corresponding subscription percentages.

For **example**, if the total reinsured loss is \$5 million and Reinsurer A's percentage is 20%, then Reinsurer A would provide \$1 million of the total \$5 million reinsured loss.

Subscription policies are also useful ways for primary insurers to reduce credit risk by diversifying their reinsurer pools.

Commutation Clauses

Under a commutation, the reinsurer pays the present value of reinsurance recoveries not yet due to the cedant in exchange for full termination of all future obligations related to the reinsurance contract.

If present in the reinsurance contract, the commutation clause lays out the terms and conditions for the estimation, payment, and discharge of all obligations of the parties to a reinsurance contract for the purposes of a commutation.

Cedants may commute a reinsurance contract for the following reasons:

- To exit a line of business or geographic region
- To manage reserves for transfer or sale
- To avoid the credit risk associated with its reinsurer, especially if the reinsurer has experienced a ratings downgrade
- To better manage claims and claims-related expenses

Reinsurers may commute a reinsurance contract for the following reasons:

- To end a relationship with a cedant that is in run-off or one with which it no longer conducts business
- To protect itself from the potential insolvency of the cedant
- To avoid disputes when there are significant differences of opinion with respect to future loss development of subject losses

Understanding commutations is important for the **reserving actuary** for the following reasons:

- Actuaries are often involved in the analysis of reinsurance contracts that are subject to commutation
- Commutations affect the estimation of unpaid ceded losses. Thus, a ceding company's actuary should be aware of commuted contracts
- Commutations eliminate the corresponding liability to the reinsurer. Thus, a reinsurer's actuary should also be aware of commuted contracts
- Loss development patterns for commuted contracts could be different from contracts that remain in-force

II. Data Requirements

Sufficiency

Data are **sufficient** if they include the needed information for the work.

The loss development method is one of the primary methods for projecting ultimate losses. The loss development method depends on consistent historical data to produce accurate projections. If there have been significant changes in the mix of business, attachment points, policy limits, or claims processing throughout the experience period, then the loss development method may not work well.

Given the key assumption around consistency for the loss development method, ensuring the sufficiency of the data can be challenging for reinsurers for the following reasons:

- Contract terms can differ from one cedant to another and from year to year
- Operational and strategic changes implemented at the cedant and the reinsurer can cause significant changes in mix of business, attachment points, policy limits, and claims processing

Reliability

Data are **reliable** if they are sufficiently complete, consistent, and accurate for the purposes of the work.

To ensure the data are accurate, the actuary must validate the data. **Data validation** consists of the following steps:

- Reconcile data against audited financial statements, trial balances, or other relevant records (if available)
- Test the data for reasonableness against external or independent data
- Test the data for internal consistency and consistency with other relevant information
- Compare the data to those for a prior period or periods

Data validation for actuaries working for reinsurers *may be more difficult* than data validation for actuaries for primary insurers for the following reasons:

- For each cedant and broker, different IT systems capture different types of data and use different terms for similar types of data
- Bordereau reporting may differ by cedant and broker in the types of data reported, data labels, and the frequency of submission to the reinsurer
- There may be lags in reporting to the reinsurer due to the following:
 - 1) Claims must first be reported to the primary insurer before being reported to the reinsurer
 - 2) The long-tailed nature of certain types of reinsurance
 - 3) The use of bordereau reporting, where losses are only reported on a quarterly or more infrequent basis
- There may be gaps in reporting critical claims information from the cedants (ex. loss payments, case reserves, etc.)
- The unique nature of reinsurance policies can lead to different coverage for similar loss events with different cedants
- There may be issues related to data coding for the reinsurer

Homogeneity

Homogeneous reinsurance obligations, known as **homogeneous risk groups (HRGs)**, are managed together and have similar risk characteristics such as underwriting policies, claims settlement patterns, product features, etc.

When separating data into HRGs for the purposes of estimating unpaid losses, actuaries should consider the following (listing a few):

- Consistency of the coverage triggered by the losses in the group
- Length of time to report the claim once an insured event has occurred (i.e. reporting patterns)

- Length of time to settle the claim once it is reported (i.e. payment patterns)
- Average settlement value (i.e. severity)
- Volume of losses in the group

Credibility

Credibility is a measure of the predictive value in a given application that the actuary attaches to a particular set of data.

When estimating unpaid losses, the goal is to divide the data into homogeneous groupings without compromising credibility. If the actuary divides the data into too many groupings, the volume of data in each group may lead to low credibility and unreliable analyses.

Differences in Considerations Related to Homogeneity and Credibility for Primary Insurers vs. Reinsurers

Primary insureds tend to segment data by LOB since claims within a LOB are typically subject to similar laws, policy terms, claims-management expenses, etc. However, further segmentation is often needed by LOB for reinsurers since significant differences can exist between reinsurance contracts within a single LOB.

Example: Segmenting Data

An actuary working with a large primary insurer may segment auto insurance as follows:

- Personal lines auto separate from commercial lines auto
- Jurisdiction, such as state or region
- Sub-coverage, such as third-party liability, no-fault benefits, and physical damage

On the other hand, an actuary working for a large reinsurer may segment auto insurance as follows:

- Personal lines separate from commercial lines auto

- Treaty separate from facultative
- Pro-rata separate from excess
- Aggregate stop-loss and finite risk covers separate from all other segments

In this example, notice that jurisdiction is not generally used to segment reinsurance data. In addition, a reinsurance actuary may choose to further segment excess of loss reinsurance at various attachments whereas a primary insurance actuary may choose to segment losses at various policy limits.

In the Patrik paper that used to be on the syllabus, he provided a list of important variables that affect the pattern of claim report lags to the reinsurer and the development of individual case amounts. The list (in priority order) is as follows:

- Line of business: property, casualty, ocean marine, etc.
- Type of contract: facultative, treaty, finite
- Type of reinsurance cover: quota share, surplus share, excess per risk, excess per-occurrence, aggregate excess, etc.
- Primary line of business for casualty
- Attachment points for casualty
- Contract terms: flat-rated, retro-rated, sunset clause, etc.
- Type of cedant: small, larger, or excess and surplus
- Broker

This is essentially a more detailed version of the list shown earlier in the Homogeneity section. For the exam, you should at least know the first few items on the list. It's probably also worth looking at old exam problems from the Patrik paper that cover this topic.

Accident Year Aggregation for Estimating Unpaid Losses

Accident year data refers to losses grouped according to the date of occurrence.

Calendar year earned premiums are used with accident year losses to provide an approximate matching of premiums and losses.

Advantages to using accident year data are as follows:

- Easy to achieve
- Easy to understand
- Become reliably estimable sooner than underwriting year data since accident year data represents losses occurring over a shorter time frame
- Industry benchmarks, such as Reinsurance Association of America data, are based on accident year losses
- Tracking losses by accident year is valuable when there are changes due to economic or regulatory forces (ex. inflation or law amendments) or major loss events (ex. hurricane) that can influence loss experience

A **disadvantage** to using accident year data is the potential mismatch between losses and premiums (the matching of calendar year earned premiums and accident year losses is only approximate).

Underwriting (i.e. Treaty) Year Aggregation for Estimating Unpaid Losses

Underwriting year data refers to losses grouped according to the date in which the reinsurance contract was inceptioned.

Underwriting year data takes much longer to mature than accident year data. In fact, losses arising from an underwriting year can extend over three calendar years.

The main **advantage** of underwriting year aggregation is an exact match of losses and premiums. This is especially important when underwriting or pricing changes occur, such as:

- A shift in attachment points or limits
- A new emphasis on certain classes of business or regions
- A change in the types of ceding company

- An increase or decrease in the price

All of the changes above can lead to changes in expected loss ratios and loss development patterns. If we were using accident year aggregation, changes such as the ones above could lead to significant heterogeneity in the data.

The **main disadvantage** of underwriting year aggregation is that data takes longer to mature. This is particularly challenging in the most immature underwriting years where development factors are highly leveraged, and the written premium is not fully earned.

Another disadvantage of underwriting year aggregation is that it can make it difficult to isolate the effect of a single large event (ex. major court ruling that changes how insurance contracts are interpreted).

Example: Underwriting Year Length to Maturity

Suppose a reinsurer writes annual risks-attaching reinsurance contracts with effective dates between January 1, 2020 and December 31, 2020:

Reinsurance Treaty Effective Date	Primary Policy Effective Dates	Primary Policy Loss Dates
1/1/20	1/1/20 – 12/31/20	1/1/20 – 12/30/21
12/31/20	12/31/20 – 12/30/21	12/31/20 – 12/29/22

Notice that the underlying primary policies can have losses occurring from 1/1/20 – 12/29/22, which is effectively three calendar years. Those losses would need time to develop to ultimate, further lengthening the time to maturity.

Allocation to Accident Year From Underwriting Year

It is common for reinsurers to use underwriting year data for the purposes of developing unpaid estimates. Once the estimates are created, results are allocated to accident year for financial reporting.

Underwriting year results are typically allocated to accident year based on how premium is earned over the contract period. An exception to this general rule is when the reinsurer receives detailed loss data including loss dates and policy effective dates.

In this case, the reinsurer can allocate underwriting year results to accident year more precisely than using earned premium.

Quality & Quantity of Reinsurance Data

The quality and quantity of data available to actuaries working with reinsurers is considerably less than the data available to actuaries working with primary insurers. Reserving actuaries working with reinsurers typically rely on the following:

- Paid losses
- Case reserves (set by the cedant)
- Additional case reserves (set by the reinsurer)
- Written premium
- Earned premium

Notice that the list above is missing detailed claim count and exposure information. This means reserving actuaries working with reinsurers are often unable to calculate triangles of average claim values or count-based ratio triangles for the purposes of running diagnostics. Instead, the actuary must rely on interviews with management of the reinsurer and cedant to understand changes in the underlying experience.

Bordereau Reporting

There are a **number of issues** associated with bordereau reporting:

- How data are cumulated by the cedant and absorbed by the reinsurer
- Potential infrequency of bordereau reporting. The more infrequent the reporting, the greater the lag in reporting and the longer the payment pattern
- The creation of a bordereau is often manual and time intensive
- The loss detail on a bordereau is not nearly as detailed as the claim files of the cedant

Loss Adjustment Expenses (LAE)

LAE consists of allocated loss adjustment expenses (ALAE) and unallocated loss adjustment expenses (ULAE). ALAE can be assigned to a specific claim whereas ULAE cannot.

In many cases, **ULAE are excluded** from reinsurance coverage. For **ALAE**, there are three common approaches:

- 1) Included with the claim amount when determining excess of loss coverage
- 2) Included on a pro-rata basis. In other words, the ratio of the excess portion of the loss to the total loss is used to determine the reinsurance coverage for ALAE
- 3) Not included in the reinsurance coverage

Example: Dealing with ALAE

A cedant issues underlying liability policies with limits of \$5 million and maintains liability excess per occurrence reinsurance with a retention of \$2 million and a limit of \$3 million. The following table shows the total ceded loss and ALAE under various loss scenarios and ALAE treatment scenarios:

Loss Scenario	Gross of Reinsurance (in \$M)			Ceded Loss & ALAE (in \$M)		
	Loss	ALAE	Total	ALAE Included	ALAE Pro-Rata	ALAE Not Included
#1	2	2	4	2	0	0
#2	3	2	5	3	1.67	1
#3	0	3	3	1	0	0

Let's discuss the table above:

- For the “ALAE Included” column, we subtract the retention from the “Total = Loss + ALAE” column to compute the ceded loss & ALAE, assuming that the total loss & ALAE is greater than the retention. For loss scenario #1, this is $2 = 4 - 2$
- For the “ALAE Pro-Rata” column, the formula for the ceded loss & ALAE is $(Total\ Loss - Retained\ Loss) + \frac{Total\ Loss - Retained\ Loss}{Total\ Loss} (Total\ ALAE)$, assuming that the losses are greater than the retention. For loss scenario #1, this is $0 = (2 - 2) + \frac{2-2}{2} (2)$
- For the “ALAE Not Included” column, we subtract the retention from the “Loss” column to compute the ceded loss & ALAE, assuming that the loss is greater than the retention. For loss scenario #1, this is $0 = 2 - 2$

Multiple Currencies

Loss data for cedants may exist in multiple currencies. To deal with multiple currencies, actuaries working with reinsurance might consider separating the data by currency, converting each current to a common currency using “point-in-time” exchange rates, and then combining the common currency data.

The **advantage** of converting the data to a “point-in-time” common currency is that it avoids the influence of changes in exchange rates over time.

Another approach is to aggregate losses based on the cedant's currency of origin (used when writing catastrophe reinsurance in a region with numerous countries and currencies).

Large Losses

Since reinsurance is often used to cover large claims, large loss adjustments are particularly important for reinsurers.

One approach to dealing with large losses when projecting ultimate losses is as follows:

- Exclude the large losses from the initial ultimate loss projection
- Once the initial projection is complete, add a case-specific projection for the reported portion of large losses and smoothed provision for the IBNR portion of large losses

When losses are unusually large, reinsurers often rely on the expertise of claims adjusters and input from catastrophe models to supplement basic loss development techniques.

Recoveries

Generally, recoveries such as deductibles, salvage, and subrogation on the underlying primary policies are applied *before* ceding losses to the reinsurer.

Example: Recoveries

For calendar year 2019, a primary insurer wrote \$10 million limit commercial property policies and maintained commercial property excess per risk reinsurance with a retention of \$2 million and a limit of \$8 million. Suppose the following loss timeline occurs:

- On January 2, 2019, an insured experienced a major fire due to a boiler explosion. This resulted in total losses of \$7 million gross of salvage and subrogation recoveries
- The primary insurer paid losses of \$2 million in 2019, \$3 million in 2020, and the final \$2 million in 2021
- During 2019, the expected salvage recoveries of \$0.25 million were received

- During 2022, the cedant received an unexpected subrogation recovery from the boiler manufacturer of \$1.5 million
- For year-end 2019, the reported losses net of salvage before reinsurance are equal to $\$7 - \$0.25 = \$6.75$ million. The cedant would report losses net of reinsurance and salvage of \$2 million (i.e. the retention) and ceded losses of \$4.75 million
- In 2022, after receiving the subrogation payment, the reported losses net of salvage and subrogation before reinsurance are equal to $\$7 - \$0.25 - \$1.5 = \5.25 million. Since this is still above the \$2 million retention, there is no benefit to the cedant on a net of reinsurance basis. Instead, the full \$1.5 million transfers to the reinsurer and reduces the ceded losses from \$4.75 million to \$3.25 million

What would happen if the reported losses net of salvage before reinsurance were \$2.75 million instead of \$6.75 million?

In this case, the subrogation payment of \$1.5 million would drop the reported losses below the \$2 million retention ($2.75 - 1.5 = 1.25$). Since reinsurance no longer applies here, any payments made by the reinsurer would be returned, and the subrogation payment would accrue to the benefit of the cedant.

In terms of accounting, this revised loss scenario would look as follows:

- For year-end 2019, the reported losses net of salvage before reinsurance are equal to \$2.75 million. The cedant would report losses net of reinsurance and salvage of \$2 million (i.e. the retention) and ceded losses of \$0.75 million
- In 2022, the reinsurer would receive reimbursement of \$0.75 million (i.e. the amount the reinsurer already paid), and the cedant would report favorable development of \$0.75 million

Influence of Change in Operations & the Environment

An actuary working for a reinsurer may experience greater challenges in understanding the impact of operational changes on the estimation of unpaid claims. A big reason for this is that the actuary must understand operational changes occurring at cedants AND reinsurers.

Potential operational changes include:

- Systems modernization efforts such as the implementation of new policy administration systems and claims administration systems
- The increased use of analytics and big data to influence pricing, marketing, and underwriting
- Acquisitions and divestitures made by ceding companies

Actuaries working with reinsurers must also understand legal and economic environment changes that might impact the loss experience of cedants (ex. tort reform).

Reporting Lags

As mentioned earlier, loss data for reinsurers lags those of cedants because the losses must first be reported and investigated by the cedant before being reported to the reinsurer.

Reporting lags can be especially bad for excess of loss reinsurance where it can take time to realize that a claim will exceed the retention threshold.

To lessen the impact of reporting lags, reinsurers often put reporting requirements in reinsurance contracts. For example, a contract might state that a claim must be reported to the reinsurer once it reaches a specified dollar amount.

Sources of Data

Large reinsurers typically rely on detailed loss and premium data from internal IT systems. Due to IT limitations, small reinsurers often rely heavily on external data sources when analyzing development factors, trend rates, and expected loss ratios.

Reinsurance Association of America (RAA)

The RAA publishes a biannual study of loss development triangles. The study:

- 1) Includes historical loss development patterns by accident year for reinsurers writing casualty excess reinsurance for auto liability, general liability, and medical malpractice
- 2) Organizes patterns separately by treaty and facultative business and five ranges of attachment points
- 3) Discusses how loss development patterns have changed over the last few years and provides possible reasons for those changes
- 4) Discusses how loss development has varied depending on the business being considered

Best's Aggregates & Averages

This external data is based on the non-proportional reinsurance loss data found in Schedule P.

Unlike primary insurance, Schedule P – Part 1 does not include the number of reported claims and the number of claims outstanding for the three non-proportional reinsurance lines.

The non-proportional reinsurance triangles found in Schedule P – Parts 2 through 4 include incurred (i.e. paid + case outstanding + IBNR), paid, and IBNR losses for ten accident years.

For actuaries working with reinsurers, Schedule P data has the following limitations:

- An experience period of only ten years, which is not typically long enough for excess of loss reinsurance
- The segmentation is not granular enough
- The combined experience may not reflect the experience of an individual reinsurer

Internet Searches

Internet searches can turn up publicly available loss triangles that are presented on a worldwide basis and highly aggregated by major line of business.

It's unlikely that highly aggregated data like this would be sufficient when estimating unpaid losses for a single reinsurer unless some sort of data modification was performed prior to the analysis.

Shortcomings of External Data

External data may not be useful for a reinsurer due to differences in the following:

- Reinsurance contract terms and conditions
- Mix of assumed business
- Types of reinsurance (ex. treaty, facultative, proportional, non-proportional)
- Underwriting processes
- Claims management processes
- IT systems

III. Methods Frequently Used to Estimate Unpaid Losses for Reinsurance

There are **two basic approaches** for determining unpaid losses:

- 1) Project ultimate losses and unpaid losses on a gross of reinsurance basis and net of reinsurance basis. Then, subtract the two to estimate ceded unpaid losses
- 2) Project ultimate losses and unpaid losses on a gross of reinsurance basis and ceded basis. Then, subtract the two to estimate net of reinsurance unpaid losses

Actuaries typically use the first approach since ceded data often have limited credibility due to the following:

- Higher volatility from large claims and catastrophe events
- Frequent changes in terms and conditions (ex. attachment points, limits, etc.) that result in less homogenous data

Actuaries typically use the **following methods to estimate unpaid losses**:

- Development method

- Expected method
- Bornhuetter-Ferguson (BF) method

Development Method

Key assumptions of the development method are as follows:

- Losses recorded to date will continue to develop in a similar manner in the future
- For an immature year, the losses or premiums observed thus far tell the actuary something about the losses or premiums yet to be observed
- Claims processing, mix of business, policy limits, and reinsurance coverage is consistent throughout the experience period

A **major difference** in projecting ultimate losses for primary insurance and reinsurance is that reinsurance data tends to be far less credible due to volume, volatility, and heterogeneity of the data. This makes sense given the nature of the losses associated with excess of loss reinsurance.

When **selecting age-to-age factors**, the following considerations should be made:

- Smooth progression of individual age-to-age factors and average factors across development periods
 - Ideally, there should be steadily decreasing incremental development from valuation to valuation
- Stability of age-to-age factors for the same development period
 - Ideally, there should be a relatively small range of factors within each development interval
- Credibility of the experience
 - Benchmark development factors should be considered for use when loss experience has low credibility

- Changes in patterns and applicability of the historical experience
 - When relying on historical age-to-age factors, actuaries should consider changes in the book of business or insurer operations over time to ensure the past is indicative of the future
 - Historical experience may not be appropriate due to things such as:
 - Changes in claims or premium volume
 - The presence of large claims that distort historical experience
 - Changes in claims processing that affect how claims are paid or reserved

Expected Method

The expected method is **often used in the following situations**:

- Entering a new line of business or new region
- Changes in strategy, operations, or the environment make historical data no longer indicative of the future
- Immature years where development factors are highly leveraged, in which case the development method is not appropriate
- Data are unavailable for other method

The **key assumption of the expected method** is that the actuary can better estimate unpaid losses based on an “a priori” estimate than from loss experience observed to date.

For reinsurance, the **most common expected method** is to multiply an expected loss ratio by earned premium. The selected expected loss ratio is based on pricing information, industry data, and historical experience adjusted to the conditions of the year under review. To aid in the selection, the actuary seeks input from management and considers changes in market conditions, pricing, terms and conditions, underwriting claims emergence, etc.

BF Method

The BF method is a **blend** of the development and expected methods.

The BF method addresses disadvantages of the development and expected method:

- The development method can produce volatile and unreliable estimates in immature accident years with large cumulative development factors since a small change in reported or paid losses can lead to a large change in projected ultimate losses
- The expected method ignores actual reported or paid losses to date
- The BF blends the two methods by adding actual reported or paid losses to expected unreported or unpaid losses to project ultimate losses

Key Differences in Assumptions Between Primary Insurance & Reinsurance

When running any of the three methods above, there are **key differences in assumptions between primary insurance and reinsurance**:

- For a similar LOB, loss development factors in the earlier maturity age intervals are often higher for reinsurance than for primary insurance due to reporting lags
- For a similar LOB, tail factors are often higher due to reporting lags and the nature of large claims associated with excess of loss reinsurance
- Loss trend factors tend to be higher for excess of loss reinsurance than primary insurance
- There is less precision in premium on-level factors that adjust for rate changes. Detailed rate change information is often difficult to obtain for reinsurers due to the unique nature of reinsurance arrangements
- Adjustments for changes such as tort and product reform are less common in reinsurance reserving

Effect of Changes in Currency Exchange Rates

As mentioned earlier, when aggregating multiple currencies to a common currency, it's best to use a single "point-in-time" exchange rate. Let's look at an example.

Example: Changes in Currency Exchange Rates

Given the following USD reported loss triangle, Euro reported loss triangle, and USD-Euro exchange rates as of December 31, 2019:

<u>USD Reported Losses (in M) as of XX Months:</u>						
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.	72 mo.
2014	103	309	463.5	515	515	515
2015	108	324	486	540	540	
2016	114	342	513	570		
2017	119	357	535.5			
2018	125	375				
2019	132					
Age-to-Age	3.000	1.500	1.111	1.000	1.000	
Age-to-Ult.	5.000	1.667	1.111	1.000	1.000	
% Reptd.	20%	60%	90%	100%	100%	

<u>EUR Reported Losses (in M) as of XX Months:</u>						
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.	72 mo.
2014	52	156	234	260	260	260
2015	54	162	243	270	270	
2016	57	171	256.5	285		
2017	60	180	270			
2018	63	189				
2019	66					
Age-to-Age	3.000	1.500	1.111	1.000	1.000	
Age-to-Ult.	5.000	1.667	1.111	1.000	1.000	
% Reptd.	20%	60%	90%	100%	100%	

USD-EUR Exchange Rate as of XX Months:						
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.	72 mo.
2014	1.21100	1.08660	1.05225	1.19990	1.14550	1.12270
2015	1.08660	1.05225	1.19990	1.14550	1.12270	
2016	1.05225	1.19990	1.14550	1.12270		
2017	1.19990	1.14550	1.12270			
2018	1.14550	1.12270				
2019	1.12270					

Now, let's aggregate the two currencies to a single USD triangle using the 12/31/19 exchange rate of 1.12270 (notice that the latest diagonal of the exchange rate triangle has the same rate since this represents 12/31/19 for each AY):

Aggregated USD Reported Losses (in M) as of XX Months:						
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.	72 mo.
2014	161.38 = 103 + 52(1.1227)	484.14	726.21	806.90	806.90	806.90
2015	168.63 = 108 + 54(1.1227)	505.88	758.82	843.13	843.13	
2016	177.99	533.98	800.97	889.97		
2017	186.36	559.09	838.63			
2018	195.73	587.19				
2019	206.10					
Age-to-Age	3.000	1.500	1.111	1.000	1.000	
Age-to-Ult.	5.000	1.667	1.111	1.000	1.000	
% Reptd.	20%	60%	90%	100%	100%	

Next, let's aggregate the two currencies to a single USD triangle using the exchange rate in effect at each cell on the triangle:

Aggregated USD Reported Losses (in M) as of XX Months:						
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.	72 mo.
2014	165.97 = 103 + 52(1.2110)	478.51	709.73	826.97	812.83	806.90
2015	166.68 = 108 + 54(1.0866)	494.46	777.58	849.29	843.13	
2016	173.98	547.18	806.82	889.97		
2017	190.99	563.19	838.63			

2018	197.17	587.19			
2019	206.10				
Age-to-Age	2.985	1.504	1.119	0.988	0.993
Age-to-Ult.	4.923	1.650	1.097	0.981	0.993
% Reptd.	20.3%	60.6%	91.2%	102.0%	100.7%

We see from the tables above that the first approach retains the true reporting pattern of 20%, 60%, 90%, and 100% reported at 12, 24, 36, and 48 months, respectively.

The remainder of the paper provides a number of examples. The first three examples highlight differences in development. The final example highlights the interaction between various types of reinsurance.

Since the first three examples are meant to highlight differences in development, simplifying assumptions have been made for the purposes of developing tail factors and expected loss ratios.

Tail Factors

The tail factors for reported losses are equal to $Max(1.00, latest\ observed\ factor)$.

The tail factors for paid losses are based on a review of the estimated ultimate losses based on the development method for reported losses for mature years.

In practice, actuaries would use other techniques as well to estimate tail factors. **Examples** include using industry benchmark factors or fitting a curve to the observed development factors and extrapolating the tail factors.

Expected Loss Ratios

The initial ultimate losses used to derive expected loss ratios are based solely on the reported losses.

Friedland makes no adjustment for loss trend, premium trend, rate level, etc. when developing the expected loss ratios.

In practice, expected loss ratios might also consider paid ultimate losses, pricing loss ratios, or planning loss ratios. In addition, any significant trend or rate level adjustments would be made when deriving expected loss ratios.

Example: Primary Insurance & Reinsurance for a Similar Type of Business

This example considers loss triangle data for the professional lines of an insurer that writes primary insurance and reinsurance.

Professional lines include sub-lines such as Director & Officers (D&O) Liability, Employment Practices Liability, Errors & Omissions Liability, etc.

For this particular insurer, D&O liability is a larger proportion of the reinsurance business than the primary insurance business

The reinsurance contracts are summarized as follows:

- Includes non-proportional and proportional treaties. The majority of exposures are non-proportional excess treaties
- D&O exposures typically attach at higher levels than the rest of the portfolio
- The reinsurance is primarily written on claims-made and risks-attaching bases

The following table highlights the volatility of the paid and reported age-to-age factors for the primary insurance business and reinsurance business:

	Age-to-Age Interval					
	12-24	24-36	36-48	48-60	60-72	72-84
	Std. Dev. – Reported Age-to-Age Factors					
Primary	0.50	0.06	0.05	0.07	0.04	0.03
Reinsurance	0.84	0.16	0.14	0.10	0.08	0.12

	Std. Dev. – Paid Age-to-Age Factors					
Primary	0.73	0.17	0.18	0.10	0.07	0.03
Reinsurance	2.91	0.46	0.19	0.12	0.07	0.04

	Abs. Diff. – Reported Age-to-Age Factors					
Primary	1.763	0.177	0.163	0.189	0.093	0.081
Reinsurance	2.181	0.528	0.379	0.257	0.214	0.263

	Abs. Diff. – Paid Age-to-Age Factors					
Primary	2.167	0.516	0.539	0.274	0.180	0.062
Reinsurance	7.643	1.179	0.568	0.331	0.179	0.080

What can we **conclude from the table** above?

- There is greater volatility at the earlier maturity ages with paid losses than with reported losses
 - Occurs for both primary insurance and reinsurance
 - This is due to the longer time frame for claims settlement, and as a result, the lower volume of paid loss data
- There is greater volatility in the age-to-age factors for reinsurance than with primary insurance
 - The standard deviations and absolute differences for paid and reported age- to-age factors are consistently higher for reinsurance. This makes sense given that a **majority of the reinsurance is non-proportional excess of loss coverage**, which will be less stable than primary insurance losses
 - Note that the **absolute difference** is the difference between the smallest and largest age-to-age factors in that age-to-age interval
 - Greater volatility in development factors leads to greater volatility in projections of ultimate losses from the development method

If we examine the actual loss data in the exhibits included with this paper, we see that the **reporting and payment patterns are longer** (i.e. slower) for reinsurance than for primary insurance. Reasons for this include the following:

- The claim must first be recognized by the cedant before it can be reported to the reinsurer
 - It takes time for the claim to develop beyond the cedant's retention and become eligible for reinsurance
 - Delays in bordereau reporting
-

Example: Proportional & Non-Proportional Reinsurance for the Same Line of Business

This example considers loss triangle data for the same line of business for 1) proportional treaty reinsurance and 2) non-proportional treaty and facultative reinsurance.

If we review the same types of tables discussed in the prior example, we **conclude the following**:

- There is greater volatility in the age-to-age factors for non-proportional treaty and facultative reinsurance than for proportional treaty reinsurance
- This makes sense given that **proportional reinsurance attaches on a ground-up basis, whereas non-proportional reinsurance is excess of loss coverage**. Ground-up losses are more stable than excess losses
- The cumulative development factors (CDFs) are greater for non-proportional treaty and facultative reinsurance than for proportional treaty reinsurance (i.e. the development patterns are longer/slower). This makes sense given the delays associated with non-proportional reinsurance

At this point, Friedland presents a **diagnostic triangle** that displays the **ratio of paid-to-reported losses** for each age-to-age interval:

- Distortions in these ratios might be caused by changes in case outstanding adequacy or changes in settlement patterns
- If distortions are observed, further investigation is required to determine whether it is coming from the paid losses (numerator) or the reported losses (denominator)
- It's also worth noting that changes in settlement patterns might offset changes in case outstanding adequacy, resulting in no distortions to these ratios. Thus, a lack of distortions does NOT mean changes are not occurring

A review of the ratio of paid-to-reported losses triangle shows greater volatility in ratios for non-proportional treaty and facultative reinsurance than for proportional reinsurance. The reasons for this are the same as the reasons for the greater volatility in the age-to-age factors for non-proportional and facultative reinsurance.

As we saw in the prior example, greater volatility in age-to-age factors and ratios for non-proportional and facultative reinsurance leads to greater volatility in projections of ultimate losses from the development method.

Example: Property Reinsurance Excluding Catastrophe & Property Reinsurance Catastrophe

This example considers loss triangle data for 1) property reinsurance excluding catastrophe and 2) property reinsurance catastrophe.

When developing reserves for large catastrophe events, actuaries often estimate reserves for specific events individually using ground-up exposure assessments provided by the claims team. If an event has just occurred and access to the affected area is limited, catastrophe models may be used to estimate losses in the interim. In either case, the catastrophe-specific reserves are added to the estimated reserves for other losses, excluding those from large catastrophes.

Loss development for catastrophe loss triangles can be significantly distorted by the timing of the event. If a catastrophe occurs early in the year, then development may be complete by the end of the year. If a catastrophe occurs later in the year, then development could extend well past 12

months. Loss development could be further extended if the contract is on a risks-attaching basis vs. an occurrence basis.

There is greater volatility in the age-to-age factors for property reinsurance catastrophe than for property reinsurance excluding catastrophe. This makes sense given the inherent uncertainty present in catastrophic events.

Due to the additional uncertainty, the development method is typically not used for catastrophic losses. Instead, catastrophe models and/or other expert-driven estimation methods are often used to estimate unpaid catastrophe losses.

Example: Dealing with Catastrophe Losses When Projecting Ultimate Losses

Given the following property reinsurance catastrophe data for a reinsurer as of December 31, 2005:

Treaty Year (1)	<u>Cat Losses at 12/31/05</u>		<u>Specific Cat Losses at 12/31/05</u>		
	Reported (2)	Paid (3)	Reported (4)	Paid (5)	Est. Ult. (6)
01	59,903	54,576	50,000	49,000	50,500
02	16,540	12,558	0	0	0
03	4,211	3,167	0	0	0
04	18,677	15,577	13,000	8,900	20,000
05	920	179	0	0	0

<u>CDF Incl. Cats at 12/31/05</u>		<u>Proj. Ult. With Cat Adj.</u>		<u>Proj. Ult. Without Cat Adj.</u>	
Reported (7)	Paid (8)	Reported (9)	Paid (10)	Reported (11)	Paid (12)
1.007	1.042	60,469	56,309	60,299	56,853
1.032	1.100	17,062	13,811	17,062	13,811
1.076	1.297	4,530	4,108	4,530	4,108
1.244	1.898	27,065	32,670	23,242	29,558
2.988	6.626	2,749	1,186	2,749	1,186

The following formulas apply to the table above:

- $(9) = [(2) - (4)] \times (7) + (6)$
- $(10) = [(3) - (5)] \times (8) + (6)$
- $(11) = (2) \times (7)$
- $(12) = (3) \times (8)$

In this example, the reinsurer's actuary begins by removing losses associated with specific, large catastrophes. Next, the actuary develops the remaining (non-specific) catastrophes to ultimate using standard development techniques. Finally, the actuary adds back the estimated ultimate losses for the large catastrophes to arrive at a total catastrophe projected ultimate loss.

Notice that the projected ultimate losses differ significantly with and without the cat adjustment for Treaty Year 4. It appears that the traditional development method is likely to understate the total reserve need.

In the next section of the paper, Friedland points out the development method, the expected method, and the BF method are all impacted by volatility in age-to-age factors:

- The development method uses the age-to-age factors directly
- Estimates of ultimate losses for mature years are often used to select expected loss ratios used in the expected method and the BF method
- The BF method also uses the age-to-age factors directly when calculating the percentage unpaid

The **main takeaway from this section of the paper** is that greater volatility in age-to-age factors and projected ultimate losses leads to a greater range in indicated IBNR and total unpaid for:

- Reinsurance compared to primary insurance for a similar line of business
- Non-proportional treaty and facultative compared to proportional treaty reinsurance for the same line of business

- Property reinsurance catastrophe compared to property reinsurance excluding catastrophe

Example: Interaction Between Quota Share (QS) Reinsurance, Stop-Loss (SL) Reinsurance, & Excess-of-Loss Reinsurance

For this example, we will examine the impacts various reinsurance treaties have on a ceding company. Let's start by applying the QS treaties to gross losses:

AY (1)	Sel. Ult. Losses (2)	<u>Gross of QS Reinsurance at 12/31/05</u>			Indicated IBNR (5)
		Paid (3)	Case O/S (4)		
01	1,892	1,600	200		92
02	1,583	1,050	350		183
03	2,986	900	1,500		586
04	2,424	525	975		924
05	2,328	750	450		1,128

QS % Ceded (6)	<u>Ceded of QS Reinsurance at 12/31/05</u>					Retained Ult. Losses After Q/S (12)
	Paid (7)	Case O/S (8)	IBNR (9)	Unpaid (10)	Ult. (11)	
50%	800	100	46	146	946	946
35%	368	123	64	187	554	1,029
30%	270	450	176	626	896	2,090
20%	105	195	185	380	485	1,939
20%	150	90	226	316	466	1,862

In the tables above, items (7) through (9) are found by applying QS % Ceded to the corresponding gross items (i.e. (3) – (5)). Column (12) is equal to (2) – (11).

Next, we will examine the impact of incorporating stop-loss reinsurance in this same example. The important thing to remember is that stop-loss provisions are applied after the QS percentages. Here is the table after applying stop-loss limits:

AY (13)	Ret. Ult. Losses After QS (14)	SL Limit (15)	Ret. Ult. Losses After QS & SL (16)	Losses at 12/31/05 Net of QS & SL			
				Reported (17)	Paid (18)	Case O/S (19)	IBNR (20)
01	946	750	750	750	750	0	0
02	1,029	1,500	1,029	910	683	228	119
03	2,090	1,500	1,500	1,500	630	870	0
04	1,939	3,000	1,939	1,200	420	780	739
05	1,862	3,000	1,862	960	600	360	902

The following formulas apply to the table above:

- (16) = $\text{Min}[(14), (15)]$. In other words, we cap the retained ultimate losses after QS at the stop-loss limit
- (17) = $\text{Min}[(3) + (4) - (7) - (8), (15)]$. In other words, we cap the retained reported losses after QS at the stop-loss limit
- (18) = $\text{Min}[(3) - (7), (15)]$. In other words, we cap the retained paid losses after QS at the stop-loss limit
- (19) = (17) – (18)
- (20) = (16) – (17)

Lastly, we will examine the impact of incorporating excess-of-loss reinsurance in this same example. The reported losses for AY 03 are equal to column (3) + column (4) = 900 + 1,500 = 2,400. Suppose that there is a single large loss in AY 03 with an estimated ultimate value of 500. Further suppose that the ceding company has an excess per occurrence treaty with an attachment point of 100 that inures to the benefit of the QS and SL coverages. How does this impact the calculation of the final retained ultimate losses?

- Selected ultimate losses gross of reinsurance = 2,986
- Single large loss = 500

- Excess per occurrence attachment point = 100
- Ceded losses to excess per occurrence reinsurer = $500 - 100 = 400$
- Ultimate losses net of excess per occurrence reinsurance = $2,986 - 400 = 2,586$
- QS ceded percentage = 30%
- Ultimate losses net of excess per occurrence and QS reinsurance = $2,586(1 - 0.30) = 1,810$
- SL limit = 1,500
- Ultimate losses net of all reinsurance = $\text{Min}(\text{Ultimate losses net of excess per occurrence and QS reinsurance, SL limit}) = \text{Min}(1,810, 1,500) = 1,500$

We have now seen how multiple reinsurance treaties impact retained losses for a ceding company. The last thing we should note is that once a company breaches a stop-loss limit, it is **common for the reinsurer to increase the price, or the attachment point of the stop-loss coverage**. Furthermore, multiple breaches of a stop-loss limit may make it challenging for a ceding company to secure stop-loss coverage in the future.

I. Introduction

A statistical loss reserving model has two key elements:

- 1) The expected amount of loss to emerge in some time period
- 2) The distribution of actual emergence around the expected value

In this paper, Clark explores the elements above using emergence curves and common reserving methods.

II. Expected Loss Emergence

Clark's model estimates the expected amount of loss to emerge based on:

- An estimate of the ultimate loss by year
- An estimate of the pattern of loss emergence

Let $G(x) = \frac{1}{LDF_x}$ be the cumulative % of loss paid (or reported) as of time x , where x represents the time (in months) from the “average” accident date to the evaluation date.

Assume that the loss emergence pattern is described by one of the following curves with scale θ and shape ω :

- Loglogistic

$$G(x|\omega, \theta) = \frac{x^\omega}{x^\omega + \theta^\omega}$$

- Weibull

$$G(x|\omega, \theta) = 1 - \exp(-(x/\theta)^\omega)$$

Note that with these curves, we assume a strictly increasing pattern. If there is real expected negative development (salvage recoveries), different models should be used.

Advantages to using parameterized curves to describe the emergence pattern are as follows:

- Estimation is simple since we only have to estimate two parameters
- We can use data that is not from a triangle with evenly spaced evaluation dates (ex. latest diagonal is only a 9-month period)
- The final pattern is smooth and does not follow random movements in the historical age-to-age factors

In order to estimate the loss emergence amount, we require an estimate of the ultimate loss by AY. **Two methods** are as follows:

- 1) LDF method
- 2) Cape Cod method

LDF Method

Let $\mu_{AY;x,y}$ = expected incremental loss dollars in accident year AY between ages x and y . For the LDF method, we have the following formula for $\mu_{AY;x,y}$:

$$\mu_{AY;x,y} = ULT_{AY} \cdot [G(y|\omega, \theta) - G(x|\omega, \theta)]$$

As mentioned in the Mack Chain-Ladder and Venter Factors papers, the LDF method assumes the loss amount in each AY is independent from all other years.

Cape Cod Method

For the Cape Cod method, we have the following formula for $\mu_{AY;x,y}$:

$$\mu_{AY;x,y} = \text{Premium}_{AY} \cdot ELR \cdot [G(y|\omega, \theta) - G(x|\omega, \theta)]$$

In general, the Cape Cod method is preferred since data is summarized into a loss triangle with relatively few data points. Since the LDF method requires an estimation of a number of parameters (one for each AY ultimate loss, as well as θ and ω), it tends to be over-parameterized when few data points exist.

Comparing the Variance Between the Two Methods

The variance of the actual loss emergence can be estimated in two pieces:

- 1) Process variance – the “random” amount
- 2) Parameter variance – the uncertainty in the estimator, also known as the estimation error

Due to the additional information given by the exposure base (as well as fewer parameters), the Cape Cod method has a smaller parameter variance. The process variance can be higher or lower than the LDF method. In general, the Cape Cod method produces a lower total variance than the LDF method.

III. The Distribution of Actual Loss Emergence & Maximum Likelihood

As mentioned above, the variance of the actual loss emergence is comprised of process variance and parameter variance. We will consider each separately.

Process Variance

Assume that the loss in any period has a constant ratio of variance/mean:

$$\boxed{\frac{\text{Variance}}{\text{Mean}} = \sigma^2 \approx \frac{1}{n-p} \sum_{AY,t}^n \frac{(c_{AY,t} - \mu_{AY,t})^2}{\mu_{AY,t}}}$$

where n = # of data points in the triangle, p = # of parameters, $c_{AY,t}$ = actual incremental loss emergence, and $\mu_{AY,t}$ = expected incremental loss emergence.

The number of parameters for the LDF and Cape Cod methods are as follows:

Method	p	Parameters
LDF	# of AYs + 2	ULT_{AY}, θ, ω
Cape Cod	3	ELR, θ, ω

For estimating the parameters of our model, let's assume that the actual loss emergence c follows an over-dispersed Poisson distribution with scaling factor (i.e., dispersion parameter) σ^2 .

Assuming λ represents the mean of a standard Poisson random variable, the mean and variance of an over-dispersed Poisson are as follows:

$$\begin{aligned} E[c] &= \lambda \sigma^2 = \mu \\ Var(c) &= \lambda \sigma^4 = \mu \sigma^2 \end{aligned}$$

The $Var(c)$ quantity above is the process variance. It is equal to the mean multiplied by the variance/mean ratio.

Key **advantages** of using the over-dispersed Poisson distribution are as follows:

- Inclusion of scaling factors allows us to match the first and second moments of any distribution, allowing high flexibility
- Maximum likelihood estimation produces the LDF and Cape Cod estimates of ultimate losses, so the results can be presented in a familiar format

The Likelihood Function

Clark uses maximum likelihood estimation to parameterize the model. For an over-dispersed Poisson distribution, $\Pr(c) = \frac{\lambda^{c/\sigma^2} e^{-\lambda}}{(c/\sigma^2)!}$. Hence, for independent losses c_i , we have the following likelihood function:

$$\prod_i \Pr(c_i) = \prod_i \frac{\lambda_i^{c_i/\sigma^2} e^{-\lambda_i}}{(c_i/\sigma^2)!} = \prod_i \frac{(\mu_i/\sigma^2)^{c_i/\sigma^2} e^{-(\mu_i/\sigma^2)}}{(c_i/\sigma^2)!}$$

Since the log-likelihood function is easier to maximize than the likelihood function, we take the natural log of the likelihood function to obtain the log-likelihood. Given that c_i is observed and σ^2 is assumed to be known, maximizing the log-likelihood is equivalent to maximizing the following:

$$l = \sum_i c_i \cdot \ln(\mu_i) - \mu_i$$

Before we summarize the MLE estimates for the LDF and Cape Cod methods, let's define some notation:

- $c_{i,t}$ = actual loss in AY i and development period t
- P_i = premium for AY i
- x_{t-1} = beginning age for development period t
- x_t = ending age for development period t

If we set the derivative of l to zero and solve for μ_i , we obtain the following MLE estimates for our two methods:

- **LDF Method** ($\mu_i = ULT_i \cdot [G(x_t|\omega, \theta) - G(x_{t-1}|\omega, \theta)]$)

$$\widehat{ULT}_i = \frac{\sum_t c_{i,t}}{\sum_t [G(x_t) - G(x_{t-1})]}$$

where $\widehat{ULT}_i$ is the MLE estimate of ULT_i and is equivalent to the LDF ultimate.

- **Cape Cod Method** ($\mu_i = P_i \cdot ELR \cdot [G(x_t|\omega, \theta) - G(x_{t-1}|\omega, \theta)]$)

$$\widehat{ELR} = \frac{\sum_{i,t} c_{i,t}}{\sum_{i,t} P_i \cdot [G(x_t) - G(x_{t-1})]}$$

where $\widehat{ELR}$ is the MLE estimate of ELR and is equivalent to the Cape Cod expected loss ratio. The premium for each AY is assumed to be known.

An **advantage** of maximizing the log-likelihood function is that it works in the presence of negative or zero incremental losses since we never actually take the log of $c_{i,t}$.

Parameter Variance

Since the parameters were estimated using maximum likelihood, the parameter variance of each estimator can be found using the information matrix (known as the *Delta Method*).

For each method, if we take the inverse of the information matrix, we obtain the covariance matrix. The diagonals of the covariance matrices provide the variances of each parameter.

- **LDF Method**

$$\Sigma = \begin{bmatrix} Var(Ult_1) & \cdots & Cov(Ult_1, \theta) \\ \vdots & \ddots & \vdots \\ Cov(\theta, Ult_1) & \cdots & Var(\theta) \end{bmatrix}$$

- **Cape Cod Method**

$$\Sigma = \begin{bmatrix} Var(ELR) & Cov(ELR, \omega) & Cov(ELR, \theta) \\ Cov(\omega, ELR) & Var(\omega) & Cov(\omega, \theta) \\ Cov(\theta, ELR) & Cov(\theta, \omega) & Var(\theta) \end{bmatrix}$$

Once we have the covariance matrices, we can use them to calculate the parameter variance of the estimated reserves.

Variance of the Reserves

At the end of the day, our goal is to calculate the variance of the estimated reserves $\hat{R}$. We estimate this variance as follows:

$$\begin{aligned} \widehat{Var}(\hat{R}) &= \text{Process Variance} + \text{Parameter Variance} \\ &= \sigma^2 \cdot \sum \hat{\mu}_{AY;x,y} + \text{Parameter Variance} \\ &= \sigma^2 \cdot \text{Estimated Reserves} + \text{Parameter Variance} \end{aligned}$$

For the exam, don't worry about knowing how to calculate the parameter variance in a general setting. It involves matrix multiplication and seems to be outside of the scope of the exam. In nearly every past exam problem, the parameter variance was provided to you.

Later on, we will look at a specific case where you can easily calculate the parameter variance. This special case has shown up on a past exam.

IV. Key Assumptions of the Model

Clark's model has three key assumptions that could potentially understate the variance of future losses:

- 1) Incremental losses are independent and identically distributed (i.i.d.)
- 2) The variance/mean scale parameter σ^2 is fixed and known
- 3) Variance estimates are based on an approximation to the Rao-Cramer lower bound

Assumption 1 (i.i.d.)

Independence means that one period does not affect the surrounding periods:

- Can be tested using residual analysis
- Positive correlation could exist if all periods are equally impacted by a change in loss inflation
- Negative correlation could exist if a large settlement in one period replaces a stream of payments in later periods

Identically distributed assumes that the emergence pattern is the same for all accident years:

- This is clearly over-simplified since different risks and a different mix of business would have been written in each historical period, each subject to different claims handling and settlement practices

Assumption 2 (Variance/Mean Ratio)

Technically, σ^2 should be estimated simultaneously with the other model parameters, with the variance around its estimate included in the covariance matrix. However, doing so results in messy mathematics. For convenience and simplicity, we assume that σ^2 is fixed and known.

Assumption 3 (Rao-Cramer Lower Bound)

The estimate of variance based on the information matrix is only exact when we are using linear functions. Since our model is non-linear, the variance estimate is a Rao-Cramer lower bound (i.e., the variance estimate is as low as it possibly can be).

V. A Practical Example

Before we into an example, let's briefly discuss growth curve extrapolation:

- The growth curve extrapolates losses to ultimate
- For curves with “heavy” tails (such as loglogistic), it may be necessary to truncate the growth curve at a finite point in time to reduce reliance on the extrapolation
- An alternative to truncation is using a growth curve with a “lighter” tail (such as Weibull)
- Clark only truncates growth curves when estimating reserves (and consequently, ultimate losses). The untruncated growth curve should be used when calculating fitted values for the estimate of the variance/mean ratio

Example: LDF Method

Given the following cumulative paid losses as of December 31, 2023:

AY	<u>Cumulative Paid Losses (\$000) as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	500	1,500	2,250	2,590	2,720
2020	550	1,700	2,400	2,725	
2021	450	1,200	2,000		

2022	600	1,750
2023	575	

Assume that expected loss emergence is described by a loglogistic curve. Model parameters are as follows:

Parameters	
θ	20.92
ω	1.44
σ^2 (in 000s)	27.884

Assume that the parameter standard deviation of the total estimated reserves is \$2,100,000.

Untruncated Growth Curve

First, let's estimate the reserves by AY and in total:

AY	Paid Losses (\$000)	Average Age (x)	Growth Curve $G(x)$	LDF	Ultimate Loss (\$000)	Estimated Reserves (\$000)
2019	2,720	54	0.797	1.2552	3,414	694
2020	2,725	42	0.732	1.3665	3,724	999
2021	2,000	30	0.627	1.5951	3,190	1,190
2022	1,750	18	0.446	2.2417	3,923	2,173
2023	575	6	0.142	7.0405	4,048	3,473
Total	9,770				18,299	8,529

Let's discuss the table above:

- The paid losses are from the latest diagonal of the triangle
- Losses are assumed to occur evenly throughout the AY. Hence, the average age of the losses in each AY is the age of the AY – 6
- The growth curve column is calculated using $G(x|\omega, \theta) = \frac{x^\omega}{x^\omega + \theta^\omega}$. For example, for AY 2019, $G(54) = \frac{54^{1.44}}{54^{1.44} + 20.92^{1.44}} = 0.797$

- The LDFs are the age-to-ultimate factors for each AY. They are equal to $\frac{1}{G(x)}$
- The ultimate losses are found by multiplying the paid losses for each AY by the LDF.
For example, for AY 2019, ultimate losses = $2,720(1.2552) = 3,414$
- The estimated reserves are found by subtracting the paid losses from the ultimate losses.
For example, for AY 2019, estimated reserves = $3,414 - 2,720 = 694$

Second, let's calculate the process standard deviation (SD) by AY and in total:

AY	Estimated Reserves (\$000)	Process SD (\$000)
2019	694	139
2020	999	167
2021	1,190	182
2022	2,173	246
2023	3,473	311
Total	8,529	488

The process variance for each AY and in total is equal to $(\sigma^2 \cdot \text{Estimated Reserves})$. Hence, the process SD is equal to $\sqrt{\sigma^2 \cdot \text{Estimated Reserves}}$. For example, for AY 2019, the process SD is $\sqrt{27.884(694)} = 139$.

Third, let's calculate the standard deviation and coefficient of variation (CV) of the total estimated reserves:

- The standard deviation is $\sqrt{\text{Process Variance} + \text{Parameter Variance}} = \sqrt{488^2 + 2,100^2} = 2,156$
- The total CV is $\frac{\text{Total SD}}{\text{Total Estimated Reserves}} = \frac{2,156}{8,529} = 25.3\%$

Truncated Growth Curve

An actuary has determined that the growth curve should be truncated at ten years for the purposes of computing the reserve estimate. How does this change things?

Let's estimate the reserves by AY and in total:

AY	Paid Losses (\$000)	Average Age (x)	Growth Curve $G(x)$	Trunc. LDF	Ultimate Loss (\$000)	Estimated Reserves (\$000)
Trunc.		114	0.920			
2019	2,720	54	0.797	1.1548	3,141	421
2020	2,725	42	0.732	1.2571	3,426	701
2021	2,000	30	0.627	1.4673	2,935	935
2022	1,750	18	0.446	2.0622	3,609	1,859
2023	575	6	0.142	6.4768	3,724	3,149
Total	9,770				16,834	7,064

Let's discuss the table above:

- We added a row to represent the ten-year truncation point
- Since the ten-year truncation point occurs at 120 months, the average age at the truncation point is 114 and the growth curve is 0.920
- The truncated LDF for each AY is equal to $\frac{0.920}{G(x)}$ since we assume that the curve stops at 120 months
- The ultimate losses and estimated reserves are calculated based on the truncated LDF. As expected, they are both smaller than the untruncated scenario

The variance components are calculated in the same manner as before. We simply replace the untruncated reserves with the truncated reserves in the calculations.

Example: Cape Cod Method

Given the cumulative losses from the prior example, along with the following on-level earned premium (OLP) and model parameters:

<u>Cumulative Paid Losses (\$000) as of XX Months:</u>						
AY	OLP (\$000)	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	5,000	500	1,500	2,250	2,590	2,720
2020	5,200	550	1,700	2,400	2,725	
2021	5,400	450	1,200	2,000		
2022	5,600	600	1,750			
2023	5,800	575				

<u>Parameters</u>	
θ	21.30
ω	1.42
σ^2 (in 000s)	21.353

Assume that expected loss emergence is described by a loglogistic curve.

Untruncated Growth Curve

First, let's calculate the Cape Cod ELR:

AY	OLP (\$000)	Paid Losses (\$000)	Average Age (x)	Growth Curve $G(x)$	Used-Up Premium (\$000)
2019	5,000	2,720	54	0.789	3,947
2020	5,200	2,725	42	0.724	3,765
2021	5,400	2,000	30	0.619	3,344
2022	5,600	1,750	18	0.441	2,467
2023	5,800	575	6	0.142	823

where *Used – Up Premium* = $OLP \times G(x)$. The idea is to put the premium on the same basis

as the paid losses. The **Cape Cod ELR** is equal to $\frac{\text{Total Paid Losses}}{\text{Total Used-Up Premium}} =$

$$\frac{2,720+2,725+2,000+1,750+575}{3,947+3,765+3,344+2,467+823} = 0.681.$$

Before moving on, here are a **couple of important notes** on the Cape Cod ELR:

- 1) The premium used should be on-leveled first
- 2) The ELR should be calculated before truncation. Remember that truncation is only used when estimating reserves. If we truncate the growth curve before calculating the ELR, the fitted values underlying the MLE process will not be correct

Second, let's estimate the reserves by AY and in total:

AY	OLP (\$000)	Paid Losses (\$000)	Average Age (x)	Growth Curve $G(x)$	% Unpaid	Expected Loss (\$000)	Est. Reserves (\$000)	Ult. Loss (\$000)
2019	5,000	2,720	54	0.789	0.211	3,405	717	3,437
2020	5,200	2,725	42	0.724	0.276	3,541	978	3,703
2021	5,400	2,000	30	0.619	0.381	3,678	1,400	3,400
2022	5,600	1,750	18	0.441	0.559	3,814	2,134	3,884
2023	5,800	575	6	0.142	0.858	3,950	3,389	3,964
Total		9,770					8,618	

Let's discuss the table above:

- The % Unpaid column is equal to $1 - G(x)$
- The Expected Loss column is equal to $OLP(ELR)$. For example, the Expected Loss for AY 2019 is $5,000(0.681) = 3,405$
- The Est. Reserves column is equal to $Expected\ Loss(\% \text{ Unpaid})$. For example, the Est. Reserves for AY 2019 are $3,405(0.211) = 717$
- The Ult. Loss column is equal to $Paid\ Loss + Est.\ Reserves$. For example, the Ult. Loss for AY 2019 is $2,720 + 717 = 3,437$
- The ultimate loss is NOT the same as the expected loss! This is super important! The expected loss is used to calculate the reserves, similar to the BF method. The ultimate loss for any method is always equal to the paid loss + estimated reserves

To calculate the variance components, we follow the same procedure used for the LDF method. Just make sure to use the correct scale parameter!

Truncated Growth Curve

As stated above, the Cape Cod ELR is calculated before truncation. Hence, the ELR is still 0.681.

Thus, we can go ahead and calculate the reserve by AY and in total:

AY	OLP (\$000)	Paid Losses (\$000)	Average Age (x)	Growth Curve $G(x)$	$0.915 - G(x)$	Expected Loss (\$000)	Est. Reserves (\$000)	Ult. Loss (\$000)
Trunc.			114	0.915				
2019	5,000	2,720	54	0.789	0.126	3,405	429	3,149
2020	5,200	2,725	42	0.724	0.191	3,541	678	3,403
2021	5,400	2,000	30	0.619	0.296	3,678	1,089	3,089
2022	5,600	1,750	18	0.441	0.475	3,814	1,811	3,561
2023	5,800	575	6	0.142	0.773	3,950	3,055	3,630
Total		9,770					7,064	

Let's discuss the table above:

- We added a row to represent the ten-year truncation point
- Since the ten-year truncation point occurs at 120 months, the average age at the truncation point is 114 and the growth curve is 0.915
- The % Unpaid column has been replaced with the $0.915 - G(x)$ column since we assume that the curve stops at 120 months. This is the truncated version of the % Unpaid amount. For example, for AY 2019, the truncated % Unpaid is $0.915 - 0.789 = 0.126$
- You may be tempted to calculate the truncated LDF and convert that to a truncated % Unpaid. Don't do that! Follow the procedure outlined above as this aligns with the paper

Once again, the variance components are calculated in the same manner as before.

Reviewing Residuals

Similar to the Mack Chain-Ladder and Venter Factors papers, we can plot residuals to test model assumptions. In this case, the normalized residuals, $r_{AY;x,y}$, are defined as follows:

$$r_{AY;x,y} = \frac{c_{AY;x,y} - \hat{\mu}_{AY;x,y}}{\sqrt{\sigma^2 \cdot \hat{\mu}_{AY;x,y}}}$$

These are similar to the weighted residuals from the Mack paper.

We can plot the normalized residuals against a variety of things:

- Increment age (i.e., AY age)
- Expected loss in each increment – useful for testing if the variance/mean ratio is constant
- AY
- CY – useful for testing for diagonal effects

In all of the cases above, we want the residuals to be randomly scattered around $y = 0$.

Example: Reviewing Residuals

Given the following information from the Cape Cod example:

<u>Cumulative Paid Losses (\$000) as of XX Months:</u>					
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	500	1,500	2,250	2,590	2,720
2020	550	1,700	2,400	2,725	
2021	450	1,200	2,000		
2022	600	1,750			
2023	575				

Avg. Age x	Growth Curve $G(x)$	AY	Expected Loss (\$000)
54	0.789	2019	3,405
42	0.724	2020	3,541
30	0.619	2021	3,678
18	0.441	2022	3,814
6	0.142	2023	3,950

- $\sigma^2 = 21.353$

First, let's calculate the actual incremental paid loss triangle:

AY	<u>Incremental Paid Losses (\$000) as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	500	1,000	750	340	130
2020	550	1,150	700	325	
2021	450	750	800		
2022	600	1,150			
2023	575				

This triangle is found by differencing the cells in the cumulative paid loss triangle.

Second, let's calculate the fitted incremental paid loss triangle:

AY	<u>Fitted Incremental Paid Losses (\$000) as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	482	1,018	610	357	223
2020	501	1,058	634	371	
2021	521	1,099	658		
2022	540	1,140			
2023	559				

The fitted incremental losses, $\hat{\mu}_i$, for the Cape Cod method are calculated as $\text{Premium}_i \cdot ELR \cdot [G(x_t|\omega, \theta) - G(x_{t-1}|\omega, \theta)] = \text{Expected Losses}_i \cdot [G(x_t|\omega, \theta) - G(x_{t-1}|\omega, \theta)]$. For example:

- AY 2019 at 12 mo. = $3,405(0.142) = 482$
- AY 2019 at 24 mo. = $3,405(0.441 - 0.142) = 1,018$

As mentioned earlier, we always use the untruncated growth curve to calculate fitted values.

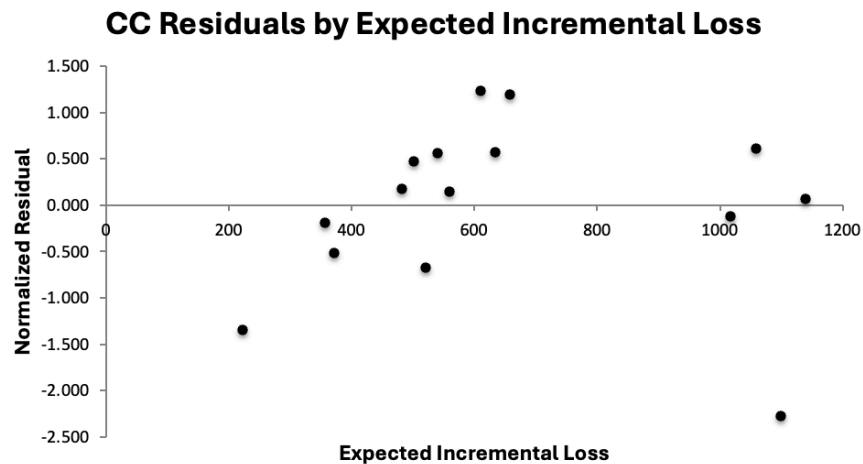
Third, let's calculate the normalized residuals for each cell in the triangle:

AY	Normalized Residuals				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	0.177	-0.119	1.231	-0.194	-1.346
2020	0.471	0.611	0.568	-0.519	
2021	-0.669	-2.278	1.195		
2022	0.560	0.067			
2023	0.145				

The normalized residuals are calculated as $r_{AY;x,y} = \frac{c_{AY;x,y} - \hat{\mu}_{AY;x,y}}{\sqrt{\sigma^2 \cdot \hat{\mu}_{AY;x,y}}}$. For example:

- AY 2019 at 12 mo. = $\frac{500 - 482}{\sqrt{21.353(482)}} = 0.177$
- AY 2019 at 24 mo. = $\frac{1,000 - 1,018}{\sqrt{21.353(1,018)}} = -0.119$

Fourth, let's plot the normalized residuals against the expected loss in each increment (i.e., the fitted incremental loss for each cell of the triangle):



In the plot above, the normalized residuals do NOT appear to be randomly scattered around $y = 0$. Since we are plotting them against the expected incremental loss, we conclude that the variance/mean ratio is not constant. In other words, the data does not support a constant σ^2 across the triangle.

How does this conclusion compare to the chain-ladder method as shown in the **Mack Chain-Ladder paper**? Assumption 3 (i.e., variance assumption) of the chain-ladder method assumes that the scale parameter differs by development period (i.e., not constant across the triangle).

Finally, let's calculate the scale parameter and show that it equals the given value of $\sigma^2 =$

21.353:

- Recall that $\sigma^2 \approx \frac{1}{n-p} \sum_{AY,t}^n \frac{(c_{AY,t} - \mu_{AY,t})^2}{\mu_{AY,t}}$
- In this example, $n = 15$ (15 cells in the triangle) and $p = 3$ (ELR, θ, ω)
- $\sigma^2 \approx \frac{1}{15-3} \left[\frac{(500-482)^2}{482} + \frac{(1,000-1,018)^2}{1,018} + \dots + \frac{(575-559)^2}{559} \right] = 21.353$

As expected, $\sigma^2 = \mathbf{21.353}$.

Testing the Constant Cape Cod ELR Assumption

The Cape Cod method assumes that there is a constant ELR across AYs. This is one of the reasons why we should always use on-leveled premium when calculating the ELR. Ideally, we would also adjust the losses for loss trend net of exposure trend as well to ensure all years are at the same cost level.

To test the constant ELR assumption, Clark suggests graphing the Cape Cod expected ultimate loss ratios by AY. These are defined as follows:

$$\boxed{\text{Cape Cod Expected Ult. LR}_{AY} = \frac{\text{Loss}_{AY}}{\text{Used} - \text{Up Premium}_{AY}}}$$

Remember, these are “expected” ultimate loss ratios. These are NOT the same as the ultimate loss ratios one would achieve by dividing “Paid Loss + Cape Cod Reserves” by “OLP.”

If the expected ultimate loss ratios follow some sort of increasing or decreasing trend, then it suggests that a constant ELR across AYs is not appropriate.

Example: Testing the Constant ELR Assumption

Recall this table from the Cape Cod example we looked at before:

AY	OLP (\$000)	Paid Losses (\$000)	Average Age (x)	Growth Curve $G(x)$	Used-Up Premium (\$000)
2019	5,000	2,720	54	0.789	3,947
2020	5,200	2,725	42	0.724	3,765
2021	5,400	2,000	30	0.619	3,344
2022	5,600	1,750	18	0.441	2,467
2023	5,800	575	6	0.142	823

The Cape Cod expected ultimate loss ratios are as follows:

AY	Paid Losses (\$000)	Used-Up Prem. (\$000)	Expected Ult. LR
2019	2,720	3,947	0.689
2020	2,725	3,765	0.724
2021	2,000	3,344	0.598
2022	1,750	2,467	0.709
2023	575	823	0.698

We don’t need a plot to see that a constant ELR assumption seems reasonable. Excluding AY 2021, the loss ratios are very similar. One “off” year is not enough to establish a trend.

Variance of the Prospective Losses

In addition to estimating reserves and their variability, we can also estimate prospective losses and their variability using the Cape Cod method. Let's look at an example.

Example: Variance of the Prospective Losses

Given the following from the Cape Cod example we looked at before:

- Estimated ELR = 68.1%
- $\sigma^2 = 21.353$

Suppose the AY 2024 expected premium (in 000s) is \$6,000. In addition, suppose we are given the following covariance matrix:

$$\begin{array}{c} \begin{array}{ccc} & ELR & \omega & \theta \\ \begin{array}{c} ELR \\ \omega \\ \theta \end{array} & \begin{pmatrix} 0.002421 & -0.002997 & 0.242396 \\ -0.002997 & 0.007853 & -0.401000 \\ 0.242396 & -0.401000 & 33.021994 \end{pmatrix} \end{array}$$

First, let's calculate the AY 2024 prospective losses:

- AY 2024 prospective losses = $OLP(ELR) = 6,000(0.681) = 4,086$

Second, let's calculate the process variance of the prospective losses:

- Process variance = $\sigma^2(Prospective Losses) = 21.353(4,086) = 87,248$

Third, let's calculate the parameter variance of the prospective losses:

- Remember earlier when we mentioned a simple example where it's possible to calculate the parameter variance? This is it! Since we are looking at prospective losses, we do NOT need to consider a growth curve. The variance of the *ELR* informs the entire parameter variance
- Parameter variance = $Var(ELR) \cdot OLP^2 = 0.002421(6,000^2) = 87,156$

Finally, let's calculate the total variance and CV of the prospective losses:

- Total variance = $87,248 + 87,156 = 174,404$
 - Total CV = $\frac{\sqrt{174,404}}{4,086} = 0.102$
-

Calendar Year Development

Rather than estimating the reserves for each accident year, we can estimate development for the next calendar year period beyond the latest diagonal:

To estimate development for the next 12-month calendar period, we take the difference in growth functions at the two evaluation ages and multiply it by:

- The estimated ultimate losses for the LDF method
- $OLP \cdot ELR$ for the Cape Cod method

The process variance and parameter variance are calculated as usual.

A **major reason** for calculating the 12-month development is that the estimate is testable within a short timeframe. One year later, we can compare it to the actual development and see if it was within the forecast range.

Example: Calendar Year Development

Let's revisit the Cape Cod example once again. Given the following information:

Avg. Age x	Growth Curve $G(x)$	AY	Expected Loss (\$000)
54	0.789	2019	3,405
42	0.724	2020	3,541
30	0.619	2021	3,678
18	0.441	2022	3,814
6	0.142	2023	3,950

We have nearly everything we need to calculate the CY 2024 development. Since AY 2019 is 60 months old (i.e., average age of $x = 54$ months), we need the growth curve at $x = 66$ months in order to calculate its development over the next 12 months. Here are the model parameters:

Parameters	
θ	21.30
ω	1.42
σ^2 (in 000s)	21.353

Thus, $G(66) = \frac{66^{1.42}}{66^{1.42} + 21.30^{1.42}} = 0.833$. The CY 2024 development (along with its process standard deviation) is calculated as follows:

AY	CY 2024 Development (\$000)	CY 2024 Process SD (in \$000)
2019	$148 = 3,405(0.833 - 0.789)$	$56 = \sqrt{21.353(148)}$
2020	$232 = 3,541(0.789 - 0.724)$	$70 = \sqrt{21.353(232)}$
2021	385	91
2022	682	121
2023	1,179	159
Total	2,626	237

Variability in Discounted Reserves

In Appendix C, Clark considers discounted reserves. Depending on the accounting rules or the line of business, it might make sense to calculate discounted reserves.

To estimate the total discounted reserve, $\hat{R}_d$, we simply “discount” the incremental losses at a specified interest rate, i . Mathematically:

$$\hat{R}_d = \sum_{AY} \sum_{k=1}^{y-x} \overline{ULT}_{AY} \cdot v^{k-0.5} \cdot (G(x+k) - G(x+k-1))$$

where $v = \frac{1}{1+i}$.

The “ $k - 0.5$ ” exponent indicates that we are discounting from the mid-point of each AY. The “ $\widehat{ULT}_{AY}$ ” term refers to either the estimated ultimate loss from the LDF method or the “expected” ultimate loss from the Cape Cod method (i.e., $OLP \times \widehat{ELR}$).

The process variance of the total discounted reserve is as follows:

$$Process Var(\hat{R}_d) = \sigma^2 \cdot \sum_{AY} \sum_{k=1}^{y-x} \widehat{ULT}_{AY} \cdot v^{2k-1} \cdot (G(x+k) - G(x+k-1))$$

This formula looks very similar to the process variance for the undiscounted reserve. The only difference is the inclusion of the discount factor. Since $Var(cX) = c^2 Var(X)$ for a constant c , the “ $k - 0.5$ ” exponent becomes “ $2k - 1$ ” after taking the variance of the discounted reserve.

Clark also provides the formula for the parameter variance of the total discounted reserve, but it seems outside of the scope of the exam.

Example: Discounted Reserves

Given the following expected losses and model parameters for the Cape Cod method for AY 2020:

- Expected losses (in \$000s) = 3,541
- $\theta = 21.30$
- $\omega = 1.42$
- σ^2 (in 000s) = 21.353

Assume that the interest rate is 3% and that there is a truncation point of ten years.

First, let's calculate the discounted reserves for AY 2020 by decomposing it into its CY pieces:

Age	Average Age (x)	Growth Curve $G(x)$	Undisc. Inc. Paid (\$000)	Disc. Inc. Paid (\$000)
Trunc.	114	0.915	46	39
108	102	0.902	60	52
96	90	0.886	79	71
84	78	0.863	108	100
72	66	0.833	154	147
60	54	0.789	232	228
48	42	0.724		
Total			678	638

Let's discuss the table above:

- Since AY 2020 is 48 months old, the table runs between 48 months and the truncation point of 120 months
- The undiscounted incremental paid column is calculated by multiplying the AY 2020 expected losses by the difference in the growth curve at adjacent evaluation points. **For example**, the incremental payment between 108 months and 120 months is $3,541(0.915 - 0.902) = 46$
- The discounted incremental paid column is calculated by multiplying the incremental paid column by $v^{k-0.5}$. **For example**, the discounted incremental payment between 108 months and 120 months is $46 \cdot \left(\frac{1}{1.03}\right)^{6-0.5} = 39$. We set $k = 6$ since the 108-120 development period is six calendar years into the future
- Notice that the total undiscounted incremental paid loss is 678. This is equal to the estimated AY 2020 reserves from the truncated version of the original Cape Cod example we showed earlier! This is because the sum of the future incremental payments for an AY should always equal its reserve!
- The discounted reserve for AY 2020 is 638

Second, let's calculate the process variance for the AY 2020 discounted reserve:

Age	Average Age (x)	Growth Curve $G(x)$	Process Var. Excl. σ^2
Trunc.	114	0.915	$33 = \frac{3,541(0.915 - 0.902)}{(1.03)^{2(6)} - 1}$
108	102	0.902	$46 = \frac{3,541(0.902 - 0.886)}{(1.03)^{2(5)} - 1}$
96	90	0.886	64
84	78	0.863	93
72	66	0.833	141
60	54	0.789	225
48	42	0.724	
Total			602

The process variance of the AY 2020 discounted reserve is $21.353(602) = 12,856$.

In this example, we only considered AY 2020. To calculate the total discounted reserve and its process variance, we simply repeat this process for the remaining AYs and sum across them.

VI. Comments & Conclusion

Abandon Your Triangles

Clark points out that the MLE model works best when using a tabular format of data rather than a triangular format (see exhibits in the paper for an example).

The CV Goes with the Mean

If we select a carried reserve other than the maximum likelihood estimate, can we still use the CV from the model?

Technically, the answer is “no.” The estimate of the standard deviation in the MLE model is directly tied to the maximum likelihood estimate.

However, for practical purposes, the answer is “yes.” Since the final carried reserve is a selection based on a number of factors (some of which are not captured in the model), it stands to reason that the standard deviation should also be a selection. The output from the MLE model is a reasonable basis for that selection.

Other Curve Forms

This paper focused on the loglogistic and Weibull growth curves for a few reasons:

- Smoothly move from 0% to 100%
- Closely match the empirical data
- First and second derivatives are calculable

Clark’s method is not limited to these forms. Other curves could be used if desired!

Final Conclusion

The main conclusion of the paper is that parameter variance is generally larger than the process variance, implying that our need for more complete data (such as the exposure information in the Cape Cod method) outweighs the need for more sophisticated models.

VII. Appendix B: Adjustments for Different Exposure Periods

In every example we have looked at so far, the exposure periods have all been annual. However, in practice, there are many times where that is not the case. For example, suppose we want to estimate AY 2024 reserves as of 03/31/24. In this case, the booked reserve should only reflect the first quarter of the AY 2024 exposure period.

Clark develops formulas for partial periods on an AY and PY basis. We will consider each separately.

Accident Year Partial Periods

Let $G^*(Avg. Age(t)|\omega, \theta)$ represent the cumulative percent of ultimate for the earned portion of the AY at time t .

The final cumulative percent of ultimate curve, including annualization, is given by the following:

$$G_{AY}(t|\omega, \theta) = Expos(t) \cdot G^*(Avg. Age(t)|\omega, \theta)$$

where $Expos(t)$ and $Avg. Age(t)$ are defined as follows:

$$Expos(t) = \begin{cases} \frac{t}{12} & t \leq 12 \\ 1 & t > 12 \end{cases}$$

$$Avg. Age(t) = \begin{cases} \frac{t}{2} & t \leq 12 \\ t - 6 & t > 12 \end{cases}$$

Example: AY Partial Period

Given the following as of September 30, 2023:

AY	Cumulative Paid Losses (\$000)
2021	8,000
2022	6,000
2023	3,000

Expected loss emergence is described by a loglogistic curve with the following LDF method parameters:

Parameters	MLE
ω	1.40
θ	5.00

The annualized reserves for each AY are calculated as follows:

AY	Paid Losses (\$000)	$Expos(t)$	Age	$Avg. Age(t)$	G^*	G	LDF	Annualized Reserves (\$000)
2021	8,000	1	33	27	0.914	0.914	1.0943	755
2022	6,000	1	21	15	0.823	0.823	1.2148	1,289
2023	3,000	0.75	9	4.5	0.463	0.347	2.8786	5,636

Let's discuss the table above:

- The age column represents t
- AYs 2021 & 2022 are fully earned ($Expos(t) = 1$). However, only 75% of AY 2023 is earned ($Expos(t) = 0.75$)
- For AY 2021 & 2022, the average age at time t is $t - 6$. For AY 2023, the average age at time t is $\frac{t}{2}$
- G^* is the growth curve evaluated at the average age. For **example**, for AY 2023, $G^* = \frac{4.5^{1.4}}{4.5^{1.4} + 5.00^{1.4}} = 0.463$. This means that the paid losses related to the first 9 months of AY 2023 are 46.3% emerged
- G is the annualized growth curve. For **example**, for AY 2023, $G = 0.75(0.463) = 0.347$. This means that the paid losses related to the entire AY are 34.7% emerged. This makes sense as the full AY should be less emerged than the exposures related only to the first 9 months
- The LDF is equal to $\frac{1}{G}$
- The annualized reserves are found by multiplying the cumulative paid losses for each AY by the LDF - 1

Keep in mind that we would NOT book the annualized reserves in practice. We only book reserves related to the earned portion of an AY. Thus, for AY 2023, the booked reserves as of September 30, 2023 would be $3,000 \left(\frac{1}{G^*} \right) - 3000 = 3,000 \left(\frac{1}{0.463} \right) - 3000 = 3,477$.

Policy Year Partial Periods

The policy year formulas are a bit more complicated. The final cumulative percent of ultimate curve, including annualization, is given by the following:

$$G_{AY}(t|\omega, \theta) = Expos(t) \cdot G^*(Avg.Age(t)|\omega, \theta)$$

where $Expos(t)$ and $Avg.Age(t)$ are defined as follows:

$$Expos(t) = \begin{cases} \frac{1}{2} \cdot \left(\frac{t}{12}\right)^2, & t \leq 12 \\ 1 - \frac{1}{2} \cdot \max\left(2 - \frac{t}{12}, 0\right)^2, & t > 12 \end{cases}$$

$$Avg.Age(t) = \begin{cases} \frac{t}{3}, & t \leq 12 \\ \frac{(t - 12) + \frac{1}{3} \cdot (24 - t) \cdot (1 - Expos(t))}{Expos(t)}, & t > 12 \end{cases}$$

Are these formulas worth memorizing? Probably not. For that reason, we are not including an example in the study guide.

However, we recommend knowing the following:

- Assuming annual policies, PYs take two years to fully earn out vs. one year for an AY
- Unlike an AY, the earning pattern for a PY is not linear. Instead, it looks more like a logistic curve (i.e., S shape). Take a look at page 87 of the source for an example graph

I. Introduction & Overview

On the surface, the chain-ladder method appears to be based on almost no assumptions.

However, that's not the case.

This paper discusses a number of implicit assumptions underlying the chain-ladder method.

Based on those assumptions, Mack develops the mean and mean squared error (MSE) of the chain-ladder reserves, giving us the ability to construct confidence intervals for estimated reserves & estimated ultimate losses.

Confidence intervals are **appealing** for the following reasons:

- Estimated ultimate losses are not an exact forecast of the true ultimate losses
- They allow the inclusion of business policy (i.e., management philosophy) by using a specific confidence probability
- They allow comparison between the chain-ladder method and other reserving procedures

II. Notations & First Analysis of the Chain-Ladder Method

Given the following:

- C_{ik} = cumulative loss amount of accident year i ($1 \leq i \leq I$) through development year k ($1 \leq k \leq I$)
- C_{iI} = ultimate loss amount for accident year i
- $R_i = C_{iI} - C_{i,I+1-i}$ = outstanding loss reserve for accident year i
- f_k = age-to-age factor

The ultimate loss amount, C_{il} , is **estimated** as follows:

$$\hat{C}_{il} = C_{i,l+1-i} \hat{f}_{l+1-i} \dots \hat{f}_{l-1}$$

where $\hat{f}_k = \frac{\sum_{j=1}^{l-k} C_{j,k+1}}{\sum_{j=1}^{l-k} C_{jk}}$.

In other words, under the chain-ladder method, we apply volume-weighted LDFs to the latest diagonal of cumulative losses to estimate ultimate losses.

Notice the “hat” above f_k . This is because each “increase from C_{ik} to $C_{i,k+1}$ is considered a **random disturbance** of an expected increase from C_{ik} to $C_{ik}f_k$, where f_k is an unknown true factor of increase which is the same for all accident years.”

First Implicit Assumption of the Chain-Ladder Method

Considering the chain-ladder method stochastically, we obtain the **first implicit assumption** of the chain-ladder method:

$$E[C_{i,k+1} | C_{i1}, \dots, C_{ik}] = C_{ik} f_k$$

In words, expected losses in the next development period are proportional to losses-to-date.

Notice that the loss projection uses the most recent losses-to-date, ignoring losses as of earlier development periods.

A **major consequence** of Assumption 1 is that it **implies that subsequent development factors are uncorrelated**:

- $E \left[\frac{C_{i,k+1}}{C_{ik}} \mid C_{i1}, \dots, C_{ik} \right] = f_k$ regardless of the prior development $C_{i1}, \dots, C_{ik}$
- This means that the expected size of $\frac{C_{i,k+1}}{C_{ik}}$ is the same after a high value of $\frac{C_{ik}}{C_{i,k-1}}$ or a low value
- We should **NOT** apply the chain-ladder method to a book of business where we usually observe a smaller-than-average increase between development periods k and $k + 1$ (i.e.,

$\frac{C_{i,k+1}}{C_{ik}} < f_k$) following a larger-than-average increases between development periods $k - 1$ and k (i.e., $\frac{C_{ik}}{C_{i,k-1}} > f_{k-1}$)

III. Analysis of the Age-to-Age Factor Formula

Second Implicit Assumption of the Chain-Ladder Method

As shown in the prior section, we do not know the true values of f_k . Instead, we must estimate them. Since a good estimator is unbiased, we want $E[\hat{f}_k] = f_k$. This is only the case if we assume that $\{C_{i1}, \dots, C_{iI}\}$ and $\{C_{j1}, \dots, C_{jI}\}$ of different accident years $i \neq j$ are independent.

Thus, the **second implicit assumption** of the chain-ladder method is that **losses are independent between accident years**.

A **major consequence** of Assumption 2 is that it cannot be used for triangles where calendar year effects, such as a change in claims handling or case reserving practices, affect several accident years in the same way.

Third Implicit Assumption of the Chain-Ladder Method

Recall that $\hat{f}_k = \frac{\sum_{j=1}^{I-k} C_{j,k+1}}{\sum_{j=1}^{I-k} C_{jk}}$. This is equivalent to the weighted average of the individual development factors across all accident years, where the weights are proportional to C_{jk} . Hence, it is the volume-weighted LDF.

Mack points out that the volume-weighted average is **used instead of** the simple average since it has a smaller variance. Basic statistical theory says that among several unbiased estimators, we should choose the one with the smallest variance.

Under the volume-weighted average LDF, the following can be shown:

$$\text{Var}\left(\frac{C_{j,k+1}}{C_{jk}} \mid C_{j1}, \dots, C_{jk}\right) = \frac{\alpha_k^2}{C_{jk}}$$

where α_k^2 is a proportionality constant that varies by age.

Using the fact that $Var\left(\frac{X}{c}\right) = \frac{Var(X)}{c^2}$ for any constant c , we can re-write the variance formula above as follows:

$$Var(C_{j,k+1} | C_{j1}, \dots, C_{jk}) = C_{jk} \alpha_k^2$$

This is **third implicit assumption** of the chain-ladder method.

In words, the variance of the losses in the next development period is proportional to losses-to-date with proportionality constant α_k^2 that varies by age.

In summary, the **three implicit assumptions** of the chain-ladder method are as follows:

- 1) Expected losses in the next development period are proportional to losses-to-date
 - $E[C_{i,k+1} | C_{i1}, \dots, C_{ik}] = C_{ik} f_k$
- 2) Losses are independent between accident years
- 3) The variance of the losses in the next development period is proportional to losses-to-date with proportionality constant α_k^2 that varies by age
 - $Var(C_{j,k+1} | C_{j1}, \dots, C_{jk}) = C_{jk} \alpha_k^2$

IV. Quantifying the Variability of the Ultimate Loss Amount

Mack shows that the MSE of the estimate of the ultimate loss amount for accident year i is as follows:

$$MSE(\hat{C}_{il}) = Var(C_{il} | D) + (E[C_{il} | D] - \hat{C}_{il})^2$$

where D represents the observed losses. The MSE above is the sum of the following two components:

- 1) Pure random error, $Var(C_{il} | D)$
- 2) Estimation error, $(E[C_{il} | D] - \hat{C}_{il})^2$

Standard Errors for a Single Accident Year

Since $MSE(\hat{C}_{il})$ depends on the unknown parameters f_k and α_k^2 , we must develop an estimator for the MSE. The square root of the estimator of the MSE is called the **standard error**.

The standard error of $\hat{C}_{il}$ is calculated as follows:

$$\left(s.e.(\hat{C}_{il}) \right)^2 = MSE(\hat{C}_{il}) = \hat{C}_{il}^2 \sum_{k=I+1-i}^{I-1} \frac{\hat{\alpha}_k^2}{\hat{f}_k^2} \left(\frac{1}{\hat{C}_{ik}} + \frac{1}{\sum_{j=1}^{I-k} C_{jk}} \right)$$

where $\hat{\alpha}_k^2 = \frac{1}{I-k-1} \sum_{j=1}^{I-k} C_{jk} \left(\frac{C_{j,k+1}}{C_{jk}} - \hat{f}_k \right)^2$.

An issue with the $\hat{\alpha}_k^2$ formula is that it **does NOT** provide an estimator for α_{I-1}^2 . Instead, we can do one of the following:

- Set $\hat{\alpha}_{I-1}^2 = 0$
 - Only do this if $\hat{f}_{I-1} = 1$ and development is expected to be finished after $I - 1$ years
- Extrapolate the series of $\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_{I-2}$ by one additional member using log-linear regression
 - To do this, take the natural log of the alphas
 - Fit a line to the natural logs to predict $\ln(\hat{\alpha}_{I-1})$
 - Exponentiate and square $\ln(\hat{\alpha}_{I-1})$ to obtain $\hat{\alpha}_{I-1}^2$
- Set $\hat{\alpha}_{I-1}^2 = \min \left(\frac{\hat{\alpha}_{I-2}^4}{\hat{\alpha}_{I-3}^2}, \min(\hat{\alpha}_{I-3}^2, \hat{\alpha}_{I-2}^2) \right)$
 - Do this if you cannot nicely extrapolate the series using log-linear regression
 - This formula ensures the $\hat{\alpha}_k^2$ values decrease as k increases

Finally, Mack points out that the standard error of the estimated ultimate losses for accident year i is equal to the standard error of the estimated reserves for accident year i . Mathematically:

$$s.e.(\hat{C}_{il}) = s.e.(\hat{R}_i)$$

Confidence Intervals for a Single Accident Year

Let's start with the confidence interval for the true reserves for accident year i , R_i . In order to build a confidence interval for R_i , we must assume a distribution for R_i .

Normal Distribution

Assuming a normal distribution for R_i , the confidence interval for R_i is as follows:

$$\boxed{\hat{R}_i \pm z \cdot s.e.(\hat{R}_i)}$$

where z is the z -score at the $\left(1 - \frac{\alpha}{2}\right)$ percentile of the standard normal distribution.

For **example**, for a 95% confidence interval, $\alpha = 0.05$ and $z_{1-\frac{0.05}{2}} = z_{0.975} = 1.96$. Hence, the confidence interval for R_i is $\hat{R}_i \pm 1.96 \cdot s.e.(\hat{R}_i)$. Note that in the paper, Mack rounds 1.96 to 2 for simplicity.

There are **two potential problems** when using the normal distribution as an approximation to the true distribution of R_i :

- 1) If data is highly skewed, it is a poor approximation because it's symmetric
- 2) If $\frac{s.e.(\hat{R}_i)}{\hat{R}_i}$ is large (greater than 50%), we can have a negative lower limit for the confidence interval

For these reasons, a lognormal distribution is preferred.

Lognormal Distribution

Assuming a lognormal distribution for R_i , the confidence interval for R_i is as follows:

$$\boxed{\hat{R}_i \cdot e^{\pm z \cdot \sigma_i - \frac{\sigma_i^2}{2}}}$$

where:

$$\bullet \sigma_i^2 = \ln \left[1 + \left(\frac{s.e.(\hat{R}_i)}{\hat{R}_i} \right)^2 \right]$$

Regardless of the chosen distribution, we obtain the **confidence interval for the true ultimate losses**, C_{il} , by shifting the confidence interval for R_i by the losses-to-date, $C_{i,l+1-i}$. We can do this because $C_{i,l+1-i}$ is known (i.e., not random).

Empirical Limits for a Single Accident Year

In the example at the end of the paper, Mack also computes confidence intervals for each C_{il} using **empirical limits**:

- The lower empirical limit results from applying the minimum age-to-age factors for each development period to the losses-to-date
- The upper empirical limit results from applying the maximum age-to-age factors for each development period to the losses-to-date

Standard Error of the Overall Reserve

The overall reserve, $R = R_1 + R_2 + \dots + R_I$, is estimated as follows:

$$\boxed{\hat{R} = \hat{R}_1 + \hat{R}_2 + \dots + \hat{R}_I}$$

In order to estimate $MSE(\hat{R})$, it's tempting to simply add the squares $(s.e.(\hat{R}_i))^2$ of the individual accident year standard errors based on the assumption of independence.

However, although the true R_i values are independent, the **estimators $\hat{R}_i$ are NOT independent** because they are all influenced by the same age-to-age factors, $\hat{f}_k$, resulting in positive correlation between accident year reserve estimates.

This correlation causes the **overall MSE to be greater than the sum of the individual accident year MSEs**.

The standard error of $\hat{R}$ is calculated as follows:

$$\left(s.e.(\hat{R}) \right)^2 = \sum_{i=2}^I \left[\left(s.e.(\hat{R}_i) \right)^2 + \hat{C}_{iI} \left(\sum_{j=i+1}^I \hat{C}_{jI} \right) \sum_{k=I+1-i}^{I-1} \frac{2\hat{\alpha}_k^2 / \hat{f}_k^2}{\sum_{n=1}^{I-k} C_{nk}} \right]$$

The formula above is especially horrific. *In our opinion*, it's unlikely that you would need to execute this formula on the exam. The **most important thing to remember** is the positive correlation between accident year reserve estimates and its impact on the overall MSE. The correlation is captured in the second term, $\hat{C}_{iI} \left(\sum_{j=i+1}^I \hat{C}_{jI} \right) \sum_{k=I+1-i}^{I-1} \frac{2\hat{\alpha}_k^2 / \hat{f}_k^2}{\sum_{n=1}^{I-k} C_{nk}}$.

Example: Standard Errors & Confidence Intervals

Given the following cumulative reported losses as of December 31, 2023:

Cumulative Reported Losses (\$000) as of XX Months:				
AY	12 mo.	24 mo.	36 mo.	48 mo.
2020	100	200	260	275
2021	150	305	350	
2022	200	250		
2023	150			

First, let's calculate the individual age-to-age and volume-weighted LDFs:

AY	Age-to-Age Factors		
	12-24 ($k = 1$)	24-36 ($k = 2$)	36-48 ($k = 3$)
2020 ($i = 1$)	$2.000 = \frac{200}{100}$	$1.300 = \frac{260}{200}$	1.058
2021 ($i = 2$)	$2.033 = \frac{305}{150}$	1.148	
2022 ($i = 3$)	1.250		
$\hat{f}_k$	1.678	1.208	1.058

The volume-weighted LDFs, $\hat{f}_k$, are found by dividing the sum of the losses in each column by the sum of the corresponding cells in the previous column. For example:

- The 12-24 volume-weighted LDF is $\frac{200+305+250}{100+150+200} = 1.678$
- Alternatively, we can calculate the 12-24 volume-weighted LDF as the C_{ik} -weighted average of the individual age-to-age factors: $\frac{2.000(100)+2.033(150)+1.250(200)}{100+150+200} = 1.678$

Second, let's square the triangle using the volume-weighted LDFs calculated in the prior step:

<u>Cumulative Reported Losses (\$000) as of k Years:</u>				
AY i	1	2	3	4
1	100	200	260	275
2	150	305	350	370
3	200	250	302 = 250(1.208)	319
4	150	252 = 150(1.678)	304 = 252(1.208)	322 = 304(1.058)

- Notice that we have switched out the raw AYs and development months for i and k . This will make it easier to follow along with the formulas for the next few steps.

Third, let's calculate the $\hat{\alpha}_k^2 = \frac{1}{I-k-1} \sum_{j=1}^{I-k} C_{jk} \left(\frac{C_{j,k+1}}{C_{jk}} - \hat{f}_k \right)^2$ parameters:

- $I = 4$ since losses are assumed to be at ultimate at 48 months
- $\hat{\alpha}_1^2 = \frac{1}{4-1-1} [100(2.000 - 1.678)^2 + 150(2.033 - 1.678)^2 + 200(1.250 - 1.678)^2] = 32.972$
- $\hat{\alpha}_2^2 = \frac{1}{4-2-1} [200(1.300 - 1.208)^2 + 305(1.148 - 1.208)^2] = 2.808$
- For $\hat{\alpha}_3^2$, we cannot use log-linear regression since there are only two available data points.

Instead, let's use $\hat{\alpha}_{I-1}^2 = \min \left(\frac{\hat{\alpha}_{I-2}^4}{\hat{\alpha}_{I-3}^2}, \min(\hat{\alpha}_{I-3}^2, \hat{\alpha}_{I-2}^2) \right)$

- Thus, $\hat{\alpha}_3^2 = \min \left(\frac{\hat{\alpha}_2^4}{\hat{\alpha}_1^2}, \min(\hat{\alpha}_1^2, \hat{\alpha}_2^2) \right) = \min \left(\frac{2.808^2}{32.972}, \min(32.972, 2.808) \right) = 0.239$

As we see from the calculations above, the $\hat{\alpha}_k^2$ essentially describe the spread of the individual age-to-age factors around the overall chain-ladder factor $\hat{f}_k$.

To summarize:

k	1	2	3
$\hat{\alpha}_k^2$	32.972	2.808	0.239

Fourth, let's calculate the standard errors of the accident year reserves:

- Recall that $(s.e.(\hat{C}_{il}))^2 = (s.e.(\hat{R}_i))^2 = \hat{C}_{il}^2 \sum_{k=l+1-i}^{l-1} \frac{\hat{\alpha}_k^2}{\hat{f}_k^2} \left(\frac{1}{\hat{C}_{ik}} + \frac{1}{\sum_{j=1}^{l-k} C_{jk}} \right)$. To help

identify the pattern for easily calculating these quantities, start with the following diagram:

AY i	$k = 1$	$k = 2$	$k = 3$	$k = 4$
1	A	B	C	
2	A	B		
3	A			
4				

In the diagram above, the letters lie above the latest diagonal. For bigger triangles, we simply add more letters. For example, if the original loss triangle is 5 x 5, then we would have 4 As, 3 Bs, 2 Cs, and 1 D.

Now, let's use the squared triangle from step 2 and the diagram above to calculate the standard errors for each accident year. Note that we will skip AY 2020 since it's assumed to be at ultimate.

AY 2021

$$\begin{aligned}
 (s.e.(\hat{R}_2))^2 &= \hat{C}_{2,4}^2 \left[\frac{\hat{\alpha}_3^2}{\hat{f}_3^2} \left(\frac{1}{C_{2,3}} + \frac{1}{C} \right) \right] \\
 &= 370^2 \left[\frac{0.239}{1.058^2} \left(\frac{1}{350} + \frac{1}{260} \right) \right] \\
 &= 196
 \end{aligned}$$

Thus, $s.e.(\hat{R}_2) = \sqrt{196} = 14$.

AY 2022

$$\begin{aligned}(s.e.(\hat{R}_3))^2 &= \hat{C}_{3,4}^2 \left[\frac{\hat{\alpha}_2^2}{\hat{f}_2^2} \left(\frac{1}{C_{3,2}} + \frac{1}{B} \right) + \frac{\hat{\alpha}_3^2}{\hat{f}_3^2} \left(\frac{1}{\hat{C}_{3,3}} + \frac{1}{C} \right) \right] \\ &= 319^2 \left[\frac{2.808}{1.208^2} \left(\frac{1}{250} + \frac{1}{505} \right) + \frac{0.239}{1.058^2} \left(\frac{1}{302} + \frac{1}{260} \right) \right] \\ &= 1,330\end{aligned}$$

Thus, $s.e.(\hat{R}_3) = \sqrt{1,330} = 36$.

AY 2023

$$\begin{aligned}(s.e.(\hat{R}_4))^2 &= \hat{C}_{4,4}^2 \left[\frac{\hat{\alpha}_1^2}{\hat{f}_1^2} \left(\frac{1}{C_{4,1}} + \frac{1}{A} \right) + \frac{\hat{\alpha}_2^2}{\hat{f}_2^2} \left(\frac{1}{C_{4,2}} + \frac{1}{B} \right) + \frac{\hat{\alpha}_3^2}{\hat{f}_3^2} \left(\frac{1}{\hat{C}_{4,3}} + \frac{1}{C} \right) \right] \\ &= 322^2 \left[\frac{32.972}{1.678^2} \left(\frac{1}{150} + \frac{1}{450} \right) + \frac{2.808}{1.208^2} \left(\frac{1}{252} + \frac{1}{505} \right) + \frac{0.239}{1.058^2} \left(\frac{1}{304} + \frac{1}{260} \right) \right] \\ &= 12,106\end{aligned}$$

Thus, $s.e.(\hat{R}_4) = \sqrt{12,106} = 110$.

Summary

Let's summarize the estimated reserves and standard error for each accident year:

AY	Reported Losses	Estimated Ultimate Losses	Estimated Reserves	$(s.e.(\hat{R}_i))^2$	$s.e.(\hat{R}_i)$	$\frac{s.e.(\hat{R}_i)}{\hat{R}_i}$
2020	275	275	0	0	0	
2021	350	370	20	196	14	69.4%
2022	250	319	69	1,330	36	52.5%
2023	150	322	172	12,106	110	64.1%

Fifth, let's build 80% confidence intervals for the accident year reserves:

- Since $\frac{s.e.(\hat{R}_i)}{\hat{R}_i} > 50\%$ for each AY, we will use a lognormal distribution
- For a lognormal distribution, the confidence interval limit is as follows: $\hat{R}_i \cdot e^{\pm z \cdot \sigma_i - \frac{\sigma_i^2}{2}}$,
where $\sigma_i^2 = \ln \left[1 + \left(\frac{s.e.(\hat{R}_i)}{\hat{R}_i} \right)^2 \right]$
- The following table displays the confidence intervals for each AY:

AY	Estimated Reserves	σ_i^2	80% CI for R_i
2021	20	0.393	(7, 37)
2022	69	0.244	(33, 116)
2023	172	0.345	(68, 306)

As an **example** calculation, consider AY 2021:

- $\sigma_2^2 = \ln[1 + (0.694)^2] = 0.393$
- Since we are calculating an 80% confidence interval, we need the 10th and 90th percentiles of the lognormal distribution ($0.90 - 0.10 = 0.80$). Hence, we need the z-score at the 90th percentile of the standard normal distribution. Using NORMSINV(0.90) in Excel, we obtain $z = 1.282$
- The 80% confidence interval is $20 \cdot e^{\pm 1.282 \cdot \sqrt{0.393} - \frac{0.393}{2}} = (7, 37)$

Sixth, let's build 80% confidence intervals for the accident year ultimate losses:

- To construct confidence intervals for ultimate losses, we simply shift the confidence intervals for the accident year reserves by the cumulative losses-to-date:

AY	Reported Losses	$\hat{C}_{i,4}$	80% CI for R_i	80% CI for $C_{i,4}$
2021	350	370	(7, 37)	(357, 387)
2022	250	319	(33, 116)	(283, 366)
2023	150	322	(68, 306)	(218, 456)

As an **example** calculation, consider AY 2021:

- The 80% confidence interval is $(7 + 350, 37 + 350) = (357, 387)$

Seventh, let's build confidence intervals for the accident year ultimate losses using empirical limits:

- To construct the empirical confidence interval for each ultimate loss, we must determine the minimum and maximum age-to-age factors for each development period. In this case, we have the following:

	$k = 1$	$k = 2$	$k = 3$
Min. Age-to-Age	1.250	1.148	1.058
Max. Age-to-Age	2.033	1.300	1.058

- The empirical confidence intervals are found by applying the minimum and maximum factors above to the losses-to-date for each AY:

AY	Reported Losses	Empirical CI for $C_{i,4}$
2021	350	(370, 370)
2022	250	(303, 344)
2023	150	(228, 419)

- As an **example** calculation, the empirical confidence interval is $(250(1.148)(1.058), 250(1.300)(1.058)) = (303, 344)$
- The empirical confidence intervals are far too narrow for AYs 2021 & 2022. This is because there is little variation in the age-to-age factors for later development periods
- Hence, empirical limits do NOT seem like a good approach for building confidence intervals

Eighth, let's calculate the overall reserve standard error:

- Recall that $(s.e.(\hat{R}))^2 = \sum_{i=2}^I \left[(s.e.(\hat{R}_i))^2 + \hat{C}_{iI} (\sum_{j=i+1}^I \hat{C}_{jI}) \sum_{k=I+1-i}^{I-1} \frac{2\hat{\alpha}_k^2 / \hat{f}_k^2}{\sum_{n=1}^{I-k} C_{nk}} \right]$

- Since we already have $(s.e.(\hat{R}_i))^2$ values, we only need to deal with

$\hat{C}_{il}(\sum_{j=i+1}^I \hat{C}_{jl}) \sum_{k=l+1-i}^{l-1} \frac{2\hat{\alpha}_k^2/\hat{f}_k^2}{\sum_{n=1}^{l-k} C_{nk}}$. Since this term sums over the accident years, we will

look at each one separately, starting with AY 2021 ($i = 2$):

AY 2021

- Let's start with $\sum_{k=l+1-i}^{l-1} \frac{2\hat{\alpha}_k^2/\hat{f}_k^2}{\sum_{n=1}^{l-k} C_{nk}}$. This quantity sums over the development periods:

$$\begin{aligned} \sum_{k=4+1-2}^{4-1} \frac{2\hat{\alpha}_k^2/\hat{f}_k^2}{\sum_{n=1}^{4-k} C_{nk}} &= \frac{2\hat{\alpha}_3^2/\hat{f}_3^2}{\sum_{n=1}^{4-3} C_{n,3}} \\ &= \frac{2\hat{\alpha}_3^2/\hat{f}_3^2}{C} \\ &= \frac{2(0.239)/1.058^2}{260} \\ &= 0.0016 \end{aligned}$$

- Next, let's calculate $\hat{C}_{il}(\sum_{j=i+1}^I \hat{C}_{jl})$:

$$\begin{aligned} \hat{C}_{il} \left(\sum_{j=i+1}^I \hat{C}_{jl} \right) &= \hat{C}_{2,4}(\hat{C}_{3,4} + \hat{C}_{4,4}) \\ &= 370(319 + 322) \\ &= 237,269 \end{aligned}$$

- Thus, $\hat{C}_{il}(\sum_{j=i+1}^I \hat{C}_{jl}) \sum_{k=l+1-i}^{l-1} \frac{2\hat{\alpha}_k^2/\hat{f}_k^2}{\sum_{n=1}^{l-k} C_{nk}} = 0.0016(237,269) = 390$

AY 2022

- Starting with $\sum_{k=l+1-i}^{l-1} \frac{2\hat{\alpha}_k^2/\hat{f}_k^2}{\sum_{n=1}^{l-k} C_{nk}}$:

$$\begin{aligned}
\sum_{k=4+1-3}^{4-1} \frac{2\hat{\alpha}_k^2/\hat{f}_k^2}{\sum_{n=1}^{4-k} C_{nk}} &= \frac{2\hat{\alpha}_2^2/\hat{f}_2^2}{\sum_{n=1}^{4-2} C_{n,2}} + \frac{2\hat{\alpha}_3^2/\hat{f}_3^2}{\sum_{n=1}^{4-3} C_{n,3}} \\
&= \frac{2\hat{\alpha}_2^2/\hat{f}_2^2}{B} + \frac{2\hat{\alpha}_3^2/\hat{f}_3^2}{C} \\
&= \frac{2(2.808)/1.208^2}{505} + \frac{2(0.239)/1.058^2}{260} \\
&= 0.0093
\end{aligned}$$

- Next, let's calculate $\hat{C}_{il}(\sum_{j=i+1}^I \hat{C}_{jl})$:

$$\begin{aligned}
\hat{C}_{il} \left(\sum_{j=i+1}^I \hat{C}_{jl} \right) &= \hat{C}_{3,4}(\hat{C}_{4,4}) \\
&= 319(322) \\
&= 102,698
\end{aligned}$$

- Thus, $\hat{C}_{il}(\sum_{j=i+1}^I \hat{C}_{jl}) \sum_{k=l+1-i}^{l-1} \frac{2\hat{\alpha}_k^2/\hat{f}_k^2}{\sum_{n=1}^{l-k} C_{nk}} = 0.0093(102,698) = 951$

AY 2023

- Since $\hat{C}_{il}(\sum_{j=i+1}^I \hat{C}_{jl}) = 0$ for the latest AY, $\hat{C}_{il}(\sum_{j=i+1}^I \hat{C}_{jl}) \sum_{k=l+1-i}^{l-1} \frac{2\hat{\alpha}_k^2/\hat{f}_k^2}{\sum_{n=1}^{l-k} C_{nk}} = 0$

Finally, we calculate the overall reserve standard error as follows:

$$\begin{aligned}
(s.e.(\hat{R}))^2 &= \sum_{i=2}^I \left[(s.e.(\hat{R}_i))^2 + \hat{C}_{il} \left(\sum_{j=i+1}^I \hat{C}_{jl} \right) \sum_{k=l+1-i}^{l-1} \frac{2\hat{\alpha}_k^2/\hat{f}_k^2}{\sum_{n=1}^{l-k} C_{nk}} \right] \\
&= [(196 + 1,330 + 12,106) + (390 + 951 + 0)] \\
&= 14,974
\end{aligned}$$

The overall reserve standard error is $\sqrt{14,974} = 122$.

Revisiting the Summary

Let's summarize the estimated reserves and standard error for each accident year and in total:

AY	Reported Losses	Estimated Ultimate Losses	Estimated Reserves	$(s.e.(\hat{R}_i))^2$	$s.e.(\hat{R}_i)$	$\frac{s.e.(\hat{R}_i)}{\hat{R}_i}$
2020	275	275	0	0	0	
2021	350	370	20	196	14	69.4%
2022	250	319	69	1,330	36	52.5%
2023	150	322	172	12,106	110	64.1%
Total	1,025		261	14,974	122	46.9%

Finally, let's build an 80% confidence interval for the overall reserves:

- $\sigma_{Total}^2 = \ln[1 + (0.469)^2] = 0.199$
- The 80% confidence interval is $261 \cdot e^{\pm 1.282 \cdot \sqrt{0.199} - \frac{0.199}{2}} = (134, 419)$

Example: Allocating Limits of the Overall Reserve Confidence Interval

An actuary analyzing the data from the prior example would like to determine the level of confidence for each AY that results in lower & upper limits that sum up to the limits of the overall reserve. Given the following summary from the prior example:

AY	Estimated Reserves	$s.e.(\hat{R}_i)$	σ_i^2	80% CI for R_i
2021	20	14	0.393	(7, 37)
2022	69	36	0.244	(33, 116)
2023	172	110	0.345	(68, 306)
Total	261	122	0.199	(134, 419)

Notice that the sum of the lower & upper limits of the 80% confidence intervals do NOT equal the lower & upper limit of the 80% confidence interval for the overall reserve:

- Lower: $7 + 33 + 68 \neq 134$
- Upper: $37 + 116 + 306 \neq 419$

To answer the actuary's question, we must consider each limit separately.

Lower Limit

We need to find the t such that $\sum_{i=2}^4 \hat{R}_i \cdot e^{t \cdot \sigma_i - \frac{\sigma_i^2}{2}} = 134$. Using **trial and error**, we find that $t = -0.905$, which corresponds to the 18.3% percentile of the standard normal distribution.

When $t = -0.905$, the lower limit for each AY is as follows:

- AY 2021: $\hat{R}_2 \cdot e^{t \cdot \sigma_2 - \frac{\sigma_2^2}{2}} = 20 \cdot e^{-0.905 \cdot \sqrt{0.393} - \frac{0.393}{2}} = 9$
- AY 2022: $\hat{R}_3 \cdot e^{t \cdot \sigma_3 - \frac{\sigma_3^2}{2}} = 69 \cdot e^{-0.905 \cdot \sqrt{0.244} - \frac{0.244}{2}} = 39$
- AY 2023: $\hat{R}_4 \cdot e^{t \cdot \sigma_4 - \frac{\sigma_4^2}{2}} = 172 \cdot e^{-0.905 \cdot \sqrt{0.345} - \frac{0.345}{2}} = 85$

As expected, $9 + 39 + 85 = 134$.

Upper Limit

We need to find the t such that $\sum_{i=2}^4 \hat{R}_i \cdot e^{t \cdot \sigma_i - \frac{\sigma_i^2}{2}} = 419$. Using **trial and error**, we find that $t = 1.119$, which corresponds to the 86.8% percentile of the standard normal distribution.

When $t = 1.119$, the upper limit for each AY is as follows:

- AY 2021: $\hat{R}_2 \cdot e^{t \cdot \sigma_2 - \frac{\sigma_2^2}{2}} = 20 \cdot e^{1.119 \cdot \sqrt{0.393} - \frac{0.393}{2}} = 33$
- AY 2022: $\hat{R}_3 \cdot e^{t \cdot \sigma_3 - \frac{\sigma_3^2}{2}} = 69 \cdot e^{1.119 \cdot \sqrt{0.244} - \frac{0.244}{2}} = 107$
- AY 2023: $\hat{R}_4 \cdot e^{t \cdot \sigma_4 - \frac{\sigma_4^2}{2}} = 172 \cdot e^{1.119 \cdot \sqrt{0.345} - \frac{0.345}{2}} = 278$

As expected, $33 + 107 + 278 = 419$.

This analysis shows that an $86.8\% - 18.3\% = 68.5\%$ confidence interval for each AY leads to an 80% confidence interval for the overall reserve.

V. Checking the Chain-Ladder Assumptions Against the Data

Once again, the three implicit assumptions underlying the chain-ladder method are as follows:

- 1) Expected losses in the next development period are proportional to losses-to-date
 - $E[C_{i,k+1} | C_{i1}, \dots, C_{ik}] = C_{ik} f_k$
- 2) Losses are independent between accident years
- 3) The variance of the losses in the next development period is proportional to losses-to-date with proportionality constant α_k^2 that varies by age
 - $Var(C_{j,k+1} | C_{j1}, \dots, C_{jk}) = C_{jk} \alpha_k^2$

In this section, we will look at tests for the **first** and **third** assumptions. We will look at a test for the second assumption in Appendix H.

Mack uses a regression framework to compare **three separate attempts** at finding the best f_k parameter. We want the f_k that minimizes the weighted sum of squared differences between actual results and fitted results. The three attempts are as follows:

- 1) $f_{k0} = \frac{\sum_{i=1}^{I-k} C_{ik} C_{i,k+1}}{\sum_{i=1}^{I-k} C_{ik}^2}$
 - This is the C_{ik}^2 -weighted average of the individual development factors. It is equal to the standard linear regression slope found using ordinary least squares for a no-intercept model
- 2) $f_{k1} = \frac{\sum_{i=1}^{I-k} C_{i,k+1}}{\sum_{i=1}^{I-k} C_{ik}}$
 - This is the C_{ik} -weighted average of the individual development factors (i.e., the usual chain-ladder age-to-age factor $\hat{f}_k$)
- 3) $f_{k2} = \frac{1}{I-k} \sum_{i=1}^{I-k} \frac{C_{i,k+1}}{C_{ik}}$
 - This is the unweighted average (simple average) of the individual development factors

The variance assumptions for each f_k are as follows:

- f_{k0} assumes that $\text{Var}(C_{j,k+1} | C_{j1}, \dots, C_{jk})$ is proportional to 1
- f_{k1} assumes that $\text{Var}(C_{j,k+1} | C_{j1}, \dots, C_{jk})$ is proportional to C_{jk} (i.e., third chain-ladder assumption)
- f_{k2} assumes that $\text{Var}(C_{j,k+1} | C_{j1}, \dots, C_{jk})$ is proportional to C_{jk}^2

Now that we have laid out the details of for each LDF, we can start testing assumptions.

Testing Assumption 1 (Linearity)

To test Assumption 1, we plot $C_{i,k+1}$ against C_{ik} to see if the points cluster closely around a straight line through the origin with a slope equal to the chain-ladder factor. If not, then the chain-ladder assumption is violated.

Testing Assumption 3 (Variance)

To test Assumption 3, we plot weighted residuals against C_{ik} to see if the residuals appear random. **Three separate residual plots** should be analyzed to ensure that the usual age-to-age factors $\hat{f}_k$ are appropriate:

- 1) Plot 0: $(C_{i,k+1} - C_{ik}f_{k0})$ against C_{ik}
- 2) Plot 1: $\frac{(C_{i,k+1} - C_{ik}f_{k1})}{\sqrt{C_{ik}}}$ against C_{ik}
- 3) Plot 2: $\frac{(C_{i,k+1} - C_{ik}f_{k2})}{C_{ik}}$ against C_{ik}

The three plots correspond to f_{k0} , f_{k1} , and f_{k2} , respectively. Notice that we divide by the square root of the “quantity that the variance is proportional to” when calculating the weighted residuals.

In general, we want weighted residuals to be **randomly scattered around $y = 0$** . If Plot 0 or Plot 2 shows a more random pattern than Plot 1 (and if this holds true for several values of k), we should consider replacing $f_{k1} = \hat{f}_k$ with f_{k0} or f_{k2} .

Example: Testing Assumptions 1 & 3

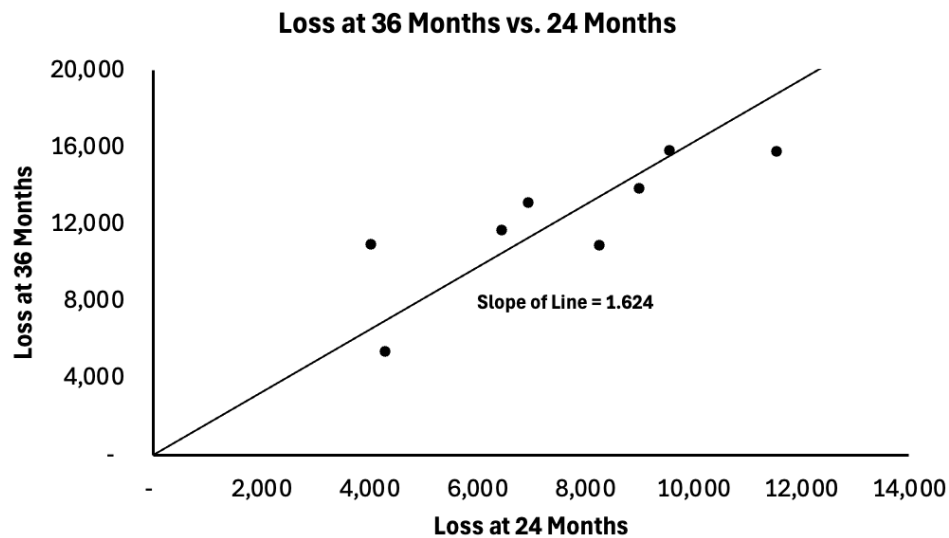
Given the following cumulative losses as of December 31, 2023:

AY	Losses at 24 Months (\$)	Losses at 36 Months (\$)
2014	8,269	10,907
2015	4,285	5,396
2016	8,992	13,873
2017	11,555	15,766
2018	9,565	15,836
2019	6,445	11,702
2020	4,020	10,946
2021	6,947	13,112
2022	5,395	

First, let's calculate the overall chain-ladder factor:

- 24-36 LDF = $\hat{f}_2 = \frac{10,907+5,396+\dots+13,112}{8,269+4,285+\dots+6,947} = 1.624$

Second, let's test Assumption 1 using a linearity plot:

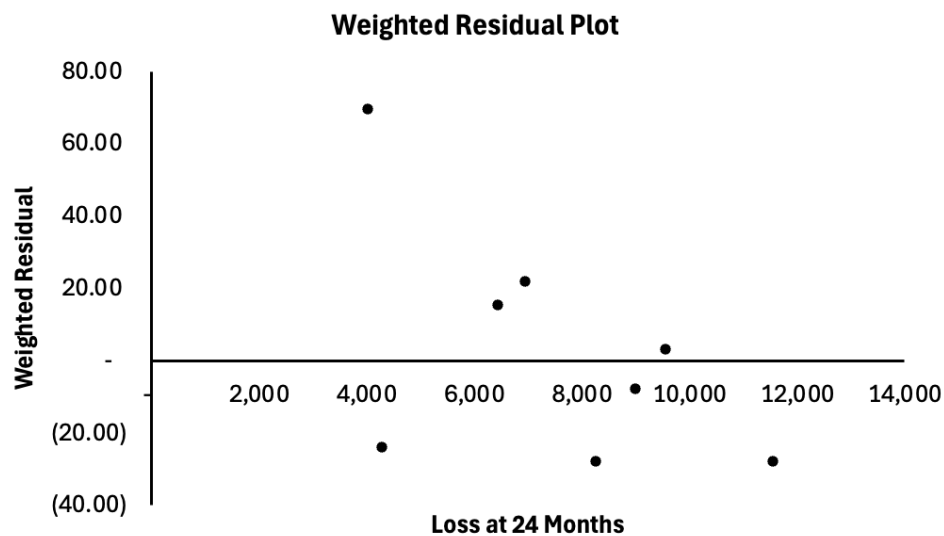


Since the data appears to revolve around a straight line through the origin with a slope equal to the chain-ladder factor, **Assumption 1 appears to be met.**

Third, let's calculate the weighted residuals:

AY	Losses at 24 Months (\$)	Losses at 36 Months (\$)	Weighted Residuals
2014	8,269	10,907	$-27.69 = \frac{10,907 - 8,269(1.624)}{\sqrt{8,269}}$
2015	4,285	5,396	-23.84
2016	8,992	13,873	-7.65
2017	11,555	15,766	-27.85
2018	9,565	15,836	3.14
2019	6,445	11,702	15.43
2020	4,020	10,946	69.70
2021	6,947	13,112	22.00

Fourth, let's test Assumption 3 by creating a weighted residual plot:



In the plot above, the weighted residuals are not randomly scattered around $y = 0$. Instead, they appear to show a downward trend. Thus, **Assumption 3 appears to be violated.**

In this case, we should create weighted residual plots for the alternative LDFs to see if the residuals are more random. If the other plots do not show an improvement in randomness, then it's reasonable to move forward with the chain-ladder method.

VI. Appendix G: Testing for Correlations Between Subsequent Development Factors

Recall that a major consequence of Assumption 1 is that it **implies that subsequent development factors are uncorrelated**. To check this, Mack describes a test procedure based on Spearman's rank correlation coefficient.

The test is based on the *table of development factors* corresponding to the cumulative losses. The procedure is as follows:

- 1) Rank each development period k along with the rank of the prior development period $k - 1$
- 2) Calculate S_k , the sum of squared rank differences for each applicable development period k
- 3) Calculate T_k , Spearman's rank correlation coefficient for each applicable development period k as follows:

$$T_k = 1 - 6 \left(\frac{S_k}{n_k^3 - n_k} \right)$$

where n_k is the number of individual development factors in development period k . A value of T_k close to 0 indicates that the development factors between development periods $k - 1$ and k and those between periods k and $k + 1$ are not correlated

- 4) Calculate the weighted average of the individual T_k s as follows:

$$T = \frac{\sum_{k=2}^{I-2} (n_k - 1) T_k}{\sum_{k=2}^{I-2} (n_k - 1)}$$

T is our test statistic. It is a global quantity that allows us to test the entire triangle for correlation

- 5) Construct a confidence interval for T as follows:

$$\pm z \cdot \sqrt{\frac{2}{(I-2)(I-3)}}$$

where z is the z -score at the $\left(1 - \frac{\alpha}{2}\right)$ percentile of the standard normal distribution. This assumes the following:

- T is normally distributed
- $E[T] = 0$ since the null hypothesis is that subsequent development factors are uncorrelated
- $Var(T) = \frac{2}{(I-2)(I-3)}$

- 6) Interpret the confidence interval as follows:

- If T lies outside of the confidence interval, we reject the null hypothesis that subsequent development factors are uncorrelated
- If T lies inside of the confidence interval, we fail to reject the null hypothesis that subsequent development factors are uncorrelated

We use T as the test statistic for **two reasons**:

- 1) Some correlation will occur in subsequent LDFs purely due to random chance
- 2) It's more important to know whether correlations prevail globally than to find a small part of the triangle with correlations

Example: Testing for Correlation Between Subsequent Development Factors

Given the following reported losses as of December 31, 2023:

AY	<u>Reported Losses (\$) as of XX Months:</u>					
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.	72 mo.
2018	1,092	9,565	15,836	22,169	25,955	26,180
2019	1,513	6,445	11,702	12,935	15,852	
2020	557	4,020	10,946	12,314		
2021	1,351	6,947	13,112			
2022	3,133	5,395				
2023	2,063					

First, let's create the table of individual development factors:

AY	<u>Development Factors</u>				
	12-24	24-36	36-48	48-60	60-72
2018	8.759	1.656	1.400	1.171	1.009
2019	4.260	1.816	1.105	1.226	
2020	7.217	2.723	1.125		
2021	5.142	1.887			
2022	1.722				

Second, let's calculate S_k , the sum of squared rank differences for each applicable development period k :

<u>$k = 2$ (24-36)</u>			<u>$k = 3$ (36-48)</u>			<u>$k = 4$ (48-60)</u>		
$Rank_1$	$Rank_2$	$Diff^2$	$Rank_2$	$Rank_3$	$Diff^2$	$Rank_3$	$Rank_4$	$Diff^2$
4	1	9	1	3	4	2	1	1
1	2	1	2	1	1	1	2	1
3	4	1	3	2	1			
2	3	1						
S_2		12	S_3		6	S_4		2

Let's discuss the $k = 2$ table:

- $k = 2$ refers to the second development factor (i.e., 24-36)
- $Rank_2$ is the ranking of the individual development factors in the 24-36 development factor column
- $Rank_1$ is the ranking of the individual development factors in the 12-24 development factor column, excluding the final factor of 1.722. We exclude this factor since there is not a corresponding factor in the 24-36 column
- $Dif f^2$ is the squared difference of the rank columns. For example, the first row has a squared difference of $(4 - 1)^2 = 9$
- S_2 is the sum of the squared rank differences. Hence, $S_2 = 9 + 1 + 1 + 1 = 12$
- The $k = 3$ and $k = 4$ tables work in the same way

Notice that we don't have $k = 1$ or $k = 5$ tables. For $k = 1$, there is no prior column for comparison. For $k = 5$, there is only one factor, so the rank difference is always zero.

Also, when calculating the ranks, feel free to utilize the RANK() function in the Pearson environment. See the Cookbook for an example.

Third, let's calculate T_k , Spearman's rank correlation coefficient for each applicable development period k :

k	n_k	S_k	T_k
2	4	12	-0.20
3	3	6	-0.50
4	2	2	-1.00

As an example calculation, consider $k = 2$:

- $n_2 = 4$ since there are 4 development factors in the 24-36 column
- $S_2 = 12$ as calculated above
- $T_2 = 1 - 6 \left(\frac{S_2}{n_2^3 - n_2} \right) = 1 - 6 \left(\frac{12}{4^3 - 4} \right) = -0.20$

Fourth, let's calculate the test statistic T as follows:

$$\bullet \quad T = \frac{\sum_{k=2}^{I-2} (n_k - 1) T_k}{\sum_{k=2}^{I-2} (n_k - 1)} = \frac{(4-1)(-0.20) + (3-1)(-0.50) + (2-1)(-1.00)}{(4-1) + (3-1) + (2-1)} = -0.433$$

Fifth, let's construct a 50% confidence interval for T :

- To construct a 50% confidence interval for T , we need the z-score at the 75th percentile. Using NORMSINV(0.75), we find that $z = 0.674$
- $Var(T) = \frac{2}{(I-2)(I-3)} = \frac{2}{(6-2)(6-3)} = 0.167$. Recall that I is the number of AYs
- The 50% confidence for T is $\pm z \cdot \sqrt{Var(T)} = (-0.674 \cdot \sqrt{0.167}, 0.674 \cdot \sqrt{0.167}) = (-0.275, 0.275)$
- Note that the 50% confidence level was not chosen at random. Mack defaults to a 50% confidence interval due to the approximate nature of the test AND because we want to detect correlations already in a substantial part of the triangle
- If you are not provided with a confidence level on the exam, we recommend defaulting to 50%

Lastly, let's interpret the confidence interval:

- Since $T = -0.433$ lies outside of the confidence interval, we reject the null hypothesis that subsequent development factors are uncorrelated
- In this case, we should be concerned about applying the chain-ladder method

Appendix H: Testing for Calendar Year Effects

The second implicit assumption of the chain-ladder method is that losses are independent between accident years. In general, this assumption is violated when we have calendar year influences that affect multiple accident years.

Examples of calendar year influences are as follows:

- Reserve strengthening or weakening
- Changes in payment processes
- Changes in inflation

Notice that the last bullet above is “changes” in inflation. Constant inflation is not an issue for the chain-ladder method. Inflation only becomes a problem when it is changing over time.

Mack developed a test to check for calendar year effects. Once again, the test is based on a table of development factors corresponding to the cumulative losses. The procedure is as follows:

- 1) Rank the development factors for each development period as “small (S)” or “large (L)” relative to the median. If a development factor is the median, then label it with an asterisk
- 2) For each diagonal j starting with the second diagonal from the top left:
 - Count the number of S_j and L_j values
 - Calculate $Z_j = \min(S_j, L_j)$
 - Calculate $E[Z_j] = \frac{n}{2} - \binom{n-1}{m} \frac{n}{2^n}$, where $n = S_j + L_j$ and $m = \frac{n-1}{2}$ (truncate if fraction)
 - Calculate $Var(Z_j) = \frac{n(n-1)}{4} - \binom{n-1}{m} \frac{n(n-1)}{2^n} + E[Z_j] - (E[Z_j])^2$

- 3) Calculate the test statistic Z as follows:

$$Z = Z_2 + Z_3 + \cdots + Z_{I-1}$$

- 4) Calculate $E[Z]$ as follows:

$$E[Z] = E[Z_2] + E[Z_3] + \cdots + E[Z_{I-1}]$$

5) Calculate $Var(Z)$ as follows:

$$Var(Z) = Var(Z_2) + Var(Z_3) + \cdots + Var(Z_{I-1})$$

6) Construct a confidence interval for Z as follows:

$$E[Z] \pm z \cdot \sqrt{Var(Z)}$$

where z is the z -score at the $\left(1 - \frac{\alpha}{2}\right)$ percentile of the standard normal distribution.

Similar to the test for correlated development factors, this assumes the test statistic Z is normally distributed

7) Interpret the confidence interval as follows:

- If Z lies outside of the confidence interval, we reject the null hypothesis that losses are independent between accident years
- If Z lies inside of the confidence interval, we fail to reject the null hypothesis that losses are independent between accident years

Example: Testing for Calendar Year Effects

Given the development factor triangle from the prior example:

AY	<u>Development Factors</u>				
	12-24	24-36	36-48	48-60	60-72
2018	8.759	1.656	1.400	1.171	1.009
2019	4.260	1.816	1.105	1.226	
2020	7.217	2.723	1.125		
2021	5.142	1.887			
2022	1.722				

First, let's rank the development factors for each development period as "small (S)" or "large (L)" relative to the median:

AY	<u>Development Factor Ranks</u>				
	12-24	24-36	36-48	48-60	60-72
2018	L	S	L	S	*
2019	S	S	S	L	
2020	L	L	*		
2021	*	L			
2022	S				

Let's discuss the table above:

- For the **12-24 column**, the median is 5.142. Thus, all factors greater than 5.142 are labelled as "L" and all factors less than 5.142 are labelled as "S". The median is labelled with an asterisk
- For the **24-36 column**, the median is between 1.816 and 1.887. Thus, all factors greater than or equal to 1.887 are labelled as "L" and all factors less than or equal to 1.816 are labelled as "S." No factors are labelled with an asterisk
- The other columns work in the same way

Second, let's calculate the test statistic components for each diagonal $j \geq 2$:

j	S_j	L_j	Z_j	n	m	$E[Z_j]$	$Var(Z_j)$
2	2	0	0	2	0	0.50	0.25
3	1	2	1	3	1	0.75	0.1875
4	2	1	1	3	1	0.75	0.1875
5	1	2	1	3	1	0.75	0.1875
Total			3			2.75	0.8125

As an example calculation, consider $j = 2$:

- $S_2 = 2$ since there are two Ss on the second diagonal
- $L_2 = 0$ since there are zero Ls on the second diagonal

- $Z_j = \min(S_2, L_2) = \min(2, 0) = 0$
- $n = S_2 + L_2 = 2 + 0 = 2$
- $m = \frac{n-1}{2} = \frac{2-1}{2} = 0$ (recall that we truncate if this is a fraction)
- $E[Z_j] = \frac{n}{2} - \binom{n-1}{m} \frac{n}{2^n} = \frac{2}{2} - \binom{2-1}{0} \frac{2}{2^2} = 0.50$. To calculate $\binom{n-1}{m}$, use the COMBIN() function in the Pearson environment. See the Cookbook for an example
- $Var(Z_j) = \frac{n(n-1)}{4} - \binom{n-1}{m} \frac{n(n-1)}{2^n} + E[Z_j] - (E[Z_j])^2 = \frac{2(2-1)}{4} - \binom{2-1}{0} \frac{2(2-1)}{2^2} + 0.50 - (0.50)^2 = 0.25$

The table above shows that $Z = 3$, $E[Z] = 2.75$, and $Var(Z) = 0.8125$.

Third, let's construct a 95% confidence interval for the test statistic Z :

- To construct a 95% confidence interval for Z , we need the z-score at the 97.5th percentile. Using NORMSINV(0.975), we find that $z = 1.96$
- $E[Z] = 2.75$
- $Var(Z) = 0.8125$
- The 95% confidence interval for Z is $E[Z] \pm z \cdot \sqrt{Var(Z)} = (2.75 - 1.96 \cdot \sqrt{0.8125}, 1.96 \cdot \sqrt{0.8125}) = (0.983, 4.517)$
- Mack does not provide a default confidence level for this test

Lastly, let's interpret the confidence interval:

- Since $Z = 3$ lies inside of the confidence interval, we fail to reject the null hypothesis that losses are independent between accident years
- Thus, Assumption 2 of the chain-ladder method appears to be met

This paper is a follow-up to the Mack Chain-Ladder paper. It revisits the three implicit assumptions of the chain-ladder method, and then introduces six testable implications of those assumptions.

I. Preliminaries

Venter begins by defining the following quantities:

- $c(w, d)$ = cumulative loss from AY w as of age d
- $c(w, \infty)$ = total loss from AY w when end of triangle is reached
- $q(w, d)$ = incremental loss for AY w from $d - 1$ to d
- $f(d)$ = factor applied to $c(w, d)$ to estimate $q(w, d + 1)$
- $F(d)$ = factor applied to $c(w, d)$ to estimate $c(w, \infty)$

Unlike other papers on the Content Outline, Venter assumes the first accident year in a loss triangle is year $w = 0$ and the first development period is age $d = 0$. In addition, Venter's version of the chain-ladder method predicts next period's incremental losses rather than next period's cumulative losses.

Although it is important to understand how Venter is defining terms when reading through the paper, notation is not important for the exam. Once you review examples and work a few problems, you will start to feel comfortable with the procedures discussed in the paper.

Mack's Chain-Ladder Assumptions

Venter restates the chain-ladder assumptions from the Mack Chain-Ladder paper as follows:

- 1) Expected incremental losses in the next development period are proportional to cumulative losses-to-date

- $E[q(w, d + 1) | \text{data to } w + d] = f(d)c(w, d)$
- 2) Losses are independent between accident years
- 3) The variance of the incremental losses in the next development period is a function of age and cumulative losses-to-date
 - $\text{Var}[q(w, d + 1) | \text{data to } w + d] = a[d, c(w, d)]$

These assumptions are identical to the ones found in the Mack Chain-Ladder except for the fact that Venter focuses on next period's incremental losses rather than next period's cumulative losses. If you are asked to provide the chain-ladder assumptions on the exam, both Mack's and Venter's versions are acceptable.

Mack's Results

In the Mack Chain-Ladder paper, it is shown that the chain-ladder LDF is as follows (remember, this is the incremental version):

$$f(d) = \frac{\sum_w q(w, d + 1)}{\sum_w c(w, d)}$$

Assuming $\text{Var}[q(w, d + 1) | \text{data to } w + d] = k(d)c(w, d)$, the $f(d)$ above is the minimum variance unbiased linear estimator of future emergence. In other words, the traditional chain-ladder method assumes that the variance of incremental losses in the next development period is proportional to cumulative losses-to-date with a proportionality factor that varies by age.

However, if we alter the variance assumption, we obtain different minimum variance unbiased $f(d)$ s:

- If $\text{Var}[q(w, d + 1) | \text{data to } w + d] = k(d)$, then $f(d) = \frac{\sum_w c(w, d)q(w, d + 1)}{\sum_w c(w, d)^2}$
- If $\text{Var}[q(w, d + 1) | \text{data to } w + d] = k(d)c(w, d)^2$, then $f(d)$ is the simple average of the individual AY d to $d + 1$ ratios

The two $f(d)$ values above are equivalent to f_{k0} and f_{k2} from the Mack Chain-Ladder paper.

II. Testable Implications of Assumptions

Testable implications of the chain-ladder assumptions include the following:

- 1) Significance of factor $f(d)$
- 2) Superiority of factor assumption compared to alternative emergence patterns such as:
 - **Linear with constant:** $E[q(w, d + 1)|\text{data to } w + d] = f(d)c(w, d) + g(d)$
 - **Factor times parameter:** $E[q(w, d)|\text{data to } w + d - 1] = f(d)h(w)$
 - **Including CY effect:** $E[q(w, d)|\text{data to } w + d - 1] = f(d)h(w)g(w + d)$
- 3) Linearity of model: look at residuals as a function of $c(w, d)$
- 4) Stability of factor: look at residuals as a function of time
- 5) No correlation among columns
- 6) No high or low diagonals

Note that we slightly modified the “factor times parameter” and “including CY effect” emergence patterns from what is shown in the paper since Venter’s formulas are a bit off here (note that the modified versions apply when $d > 0$).

Lastly, please keep in mind that Venter uses a standard linear regression framework to test many of the implications above, which assumes that the variance of the next incremental loss is constant. In a no-intercept model, this leads to the $c(w, d)^2$ -weighted average LDF. In contrast, the traditional chain-ladder method assumes that the variance of the next incremental loss is proportional to $c(w, d)$, resulting in the $c(w, d)$ -weighted average LDF. While these estimators differ, Venter’s regression-based approach still provides useful tests of the general form of the chain-ladder assumptions.

III. Testing Loss Emergence – Implications 1 & 2

Implication 1: Significance of Factors

In a regression analysis, the absolute value of a parameter is required to be at least twice its standard deviation for the factor to be considered significantly different from zero. This is based

on the fact that *approximately* 95% of normally distributed data is within ± 2 standard deviations from the mean. Although the normal distribution is a fine distribution to use, positive skewness is often observed, which implies that a lognormal distribution may be more appropriate.

Recall that Venter focuses on incremental losses instead of cumulative losses. If the factor $f(d)$ predicts cumulative losses rather than incremental losses, we need to test the significance of the *difference of the factor from 1*. This can be done by comparing that difference to the standard deviation of the factor.

Example: Regression Analysis

Given the following incremental incurred losses as of December 31, 2023:

AY	<u>Incremental Incurred Losses (\$000) as of XX Months:</u>						
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.	72 mo.	84 mo.
2017	100	116	51	36	15	16	6
2018	150	179	81	66	34	10	
2019	200	206	128	42	46		
2020	175	159	77	66			
2021	180	180	90				
2022	190	162					
2023	165						

Regression Analysis

To test implication 1 (i.e., the significance of the factors), we need to run a regression analysis to compute the estimate of each factor along with its standard error. To accomplish this, we can use the LINEST() function in Excel. The function is as follows:

- LINEST(known y values, known x values, FALSE, TRUE)

The “FALSE” indicates “no intercept.” Recall that the chain-ladder method assumes a line through the origin. The “TRUE” indicates that we want the associated statistics. To limit the output to the factor estimates and standard errors, we can use the INDEX() function as follows:

- Factor estimates: Use INDEX(LINEST(known y values, known x values, FALSE, TRUE), 1, 1). This pulls cell (1, 1) from the output array, which is the slope (i.e., factor estimate)
- Standard errors: Use INDEX(LINEST(known y values, known x values, FALSE, TRUE), 2, 1). This pulls cell (2, 1) from the output array, which is the standard error of the slope

If we run the regression separately for each incremental column (excluding the first column), we obtain the following summary:

	12-24 ($d = 0$)	24-36 ($d = 1$)	36-48 ($d = 2$)	48-60 ($d = 3$)	60-72 ($d = 4$)	72-84 ($d = 5$)
b	0.994	0.263	0.124	0.073	0.028	0.018
$s.e.(b)$	0.050	0.017	0.022	0.007	0.014	0
$abs(b)/s.e.(b)$	19.785	15.429	5.533	10.136	2.046	

Let’s discuss the table above:

- The factor estimates are labelled as b based on the standard linear regression estimate of $a + bx$. Since this is a “no intercept” model, $a = 0$
- Using the 48-60 factor as an example, we compute b , $s.e.(b)$, and $b/s.e.(b)$ as follows:
 - The known y values are (15, 34, 46). These are the incremental losses that emerge between 48-60 months
 - The known x values are (303, 476, 576). These are the cumulative losses as of 48 months
 - b : INDEX(LINEST((15, 34, 46), (303, 476, 576), FALSE, TRUE), 1, 1) = 0.073. Note that this is equal to the $c(w, d)^2$ -weighted average of the individual incremental age-to-age factors

- $s.e.(b)$: INDEX(LINEST((15, 34, 46), (303, 476, 576), FALSE, TRUE), 2, 1)
= 0.007
- $abs(b)/s.e.(b) = 0.073/0.007 = 10.136$
- The regression analysis does not predict the first column. Venter assumes that the first column is always known

Interpreting the Regression Analysis

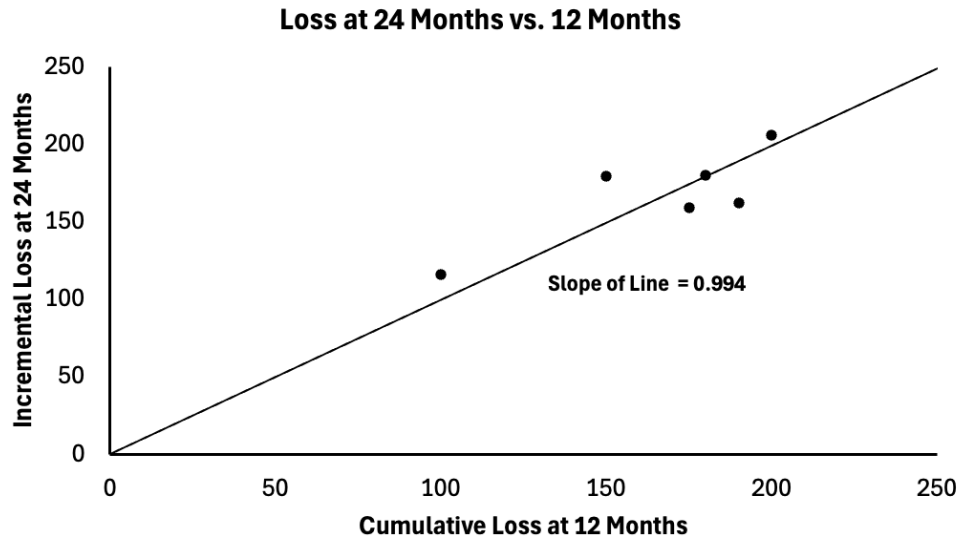
As shown in the table above, the absolute value of each factor is at least twice its standard error (other than the final column which only has one factor). Thus, the **first testable implication** passes (i.e., the factors are significant).

Note that it won't always be the case that each factor passes the "two times its standard deviation" test. As long as the majority of the factors pass the test (especially the ones earlier in the triangle), then it's reasonable to use the chain-ladder method.

If we wanted to perform a full test of Assumption 1 of the Chain-Ladder method, we should fit models for each period with non-zero intercepts (i.e., Linear Plus Constant models) and test the significance of the intercepts as well. If the intercepts are significant, then Assumption 1 is violated.

Supporting Our Conclusion Graphically

If the plot of incremental losses against the prior cumulative losses clusters closely around a no-intercept linear regression line, then Assumption 1 of the Chain-Ladder method is met. For example, here is the plot of the incremental losses at 24 months against the cumulative losses at 12 months:



As we can see, a straight line through the origin with a slope equal to 0.994 provides a nice fit to the data. Remember that the slope of 0.994 is equal to the $c(w, d)^2$ -weighted average.

Implication 2: Superiority to Alternative Emergence Patterns

In addition to regression analyses, we can compare emergence patterns using goodness-of-fit tests. Venter discusses the following test statistics:

- Adjusted SSE
- Akaike Information Criteria (AIC)
- Bayesian Information Criteria (BIC)

The **adjusted SSE** is computed as follows:

$$\boxed{Adjusted\ SSE = \frac{SSE}{(n - p)^2}}$$

where n is the number of predicted points and p is the number of parameters.

Akaike Information AIC and BIC are computed as follows:

$$AIC \approx SSE \cdot e^{\frac{2p}{n}}$$

$$BIC \approx SSE \cdot n^{\frac{p}{n}}$$

Using more parameters often gives an *advantage in fitting* but a *disadvantage in prediction*. Each of the test statistics above incorporates a penalty for an increasing number of parameters.

In Venter's world, the number of predicted points is equal to the number of cells in the loss triangle, *excluding the first column*. During the model fit, we do not produce fitted values for the first column since they are assumed to be given.

For the **chain-ladder method**, the **number of parameters** is equal to the $m - 1$, where m is the number of AYs.

Alternative Emergence Pattern 1: Linear with Constant

For this pattern, $E[q(w, d + 1)|\text{data to } w + d] = f(d)c(w, d) + g(d)$.

The constant is often significant (i.e., absolute value of the constant is at least twice its standard deviation) in the age 0 to age 1 stage, especially for highly variable and slowly reporting lines (ex. excess reinsurance).

For this pattern, the **number of parameters** is $2m - 2$ (twice the chain-ladder method) due to the inclusion of the $g(d)$ parameters.

Alternative Emergence Pattern 2: Factor Times Parameter

For this pattern, $E[q(w, d)|\text{data to } w + d - 1] = f(d)h(w)$. In words, this pattern states that the next period's expected emerged loss is a lag factor $f(d)$ times the expected ultimate loss amount $h(w)$ for an AY (or at least something proportional to the expected ultimate loss).

Venter refers to this as the **parameterized BF model**.

For this pattern, the **number of parameters** is $2m - 2$ (twice the chain-ladder method). To understand this, consider a 4 x 4 triangle:

- We need a parameter for each accident period $(h(0), h(1), h(2), h(3))$
- We need a parameter for each development period $(f(0), f(1), f(2), f(3))$
- We can subtract one parameter because the $f(d)$ s typically sum to 1, so we only need three of them to calculate the remaining $f(d)$
- We can also subtract the $f(0)$ parameter since we are not predicting any losses for the first column
- This results in $6 = 2(4) - 2$ parameters, which is the twice the number of parameters needed for the chain-ladder method

Since the BF model has so many parameters, we can use some of the information in the triangle to create a reduced parameter BF model. Three ways to create a reduce parameter BF model are as follows:

- 1) We can reduce parameters by grouping AYs using apparent jumps in loss levels and fitting a single h parameter to each group
- 2) We can assume that subsequent periods all have the same expected percentage development
- 3) We can also fit a trend line through the BF ultimate loss parameters

Alternative Emergence Pattern 3: Cape Cod Method

The Cape Cod method is a special case of the parameterized BF model where $h(w) = h$. Hence, $E[q(w, d) | \text{data to } w + d - 1] = f(d)h$.

For this pattern, the **number of parameters** is $m - 1$ (same as the chain-ladder method). To understand this, consider a 4 x 4 triangle:

- On the surface, it appears that we have an accident year parameter h and four development period parameters $(f(0), f(1), f(2), f(3))$

- However, notice that we can multiply/divide h by any number and divide/multiply all of the $f(d)$ factors by the same number and still maintain the same fitted values $f(d)h$. Thus, the model is actually over-parameterized by one parameter
- For example, if we divide h by h and multiply all of the $f(d)$ factors by h , we are left with four $f(d)$ parameters and $h = 1$. The fitted values would remain the same, but they would now be defined as $f(d)(1) = f(d)$. Since the first column is given, we do not need $f(0)$. This leaves us with $(f(1), f(2), f(3))$, which is the same number of parameters as the chain-ladder method

Venter points out that the Cape Cod method works **better for loss ratio triangles** than for loss triangles, since a single target ultimate loss ratio makes more sense than a single target ultimate loss.

Alternative Emergence Pattern 4: Additive Chain-Ladder Method

This is a special case of the “Linear with Constant” model, where the constant is significant, and the factor is not.

Under this pattern, $E[q(w, d + 1)|\text{data to } w + d] = g(d)$.

When using least squares regression, the additive chain-ladder method and the Cape Cod method **always produce the same fitted values**. This also means they **always produce the goodness-of-fit statistics**.

Why do the two methods produce the same fitted values?

- If we fit the Cape Cod model, we can define $g(d) = f(d)h$, where $g(d)$ is the additive development amount for age d
- If we fit the additive chain-ladder model, we can define $f(d)h = g(d)$
- As long as models are fit using least squares regression, the equations in the two preceding bullets will always hold

Summary of Emergence Patterns

The emergence patterns discussed in this section are summarized as follows:

	$E[q(w, d)]$	Number of Parameters p
Chain-Ladder	$f(d)c(w, d)$	$m - 1$
Linear with Constant	$f(d)c(w, d) + g(d)$	$2m - 2$
Parameterized BF	$f(d)h(w)$	$2m - 2$
Cape Cod	$f(d)h$	$m - 1$
Additive Chain-Ladder	$g(d)$	$m - 1$

Using a Regression Analysis to Compare Emergence Patterns

Before we look at an example of performing goodness-of-fit tests, we should note that the chain-ladder method, linear with constant model, and additive chain-ladder method can all be compared using a regression analysis. This is because the chain-ladder method and additive chain-ladder method are special cases of the linear with constant model:

- Factors and constants are significant
 - Use linear with constant model
 - Data should revolve around a straight line with a slope and intercept equal to b and a from the regression output, respectively
- Only factors are significant
 - Use chain-ladder method
 - Data should revolve around a straight line through the origin with a slope equal to b from the regression output
- Only constants are significant
 - Use additive chain-ladder method
 - Data should revolve around a horizontal line equal to a from the regression output (i.e., the additive chain-ladder parameter $g(d)$)

Example: Goodness-of-Fit Tests

Various emergence methods were used on a 5 x 5 loss triangle. The resulting SSEs are as follows:

Method	SSE
Chain-Ladder	200,000
Parameterized BF	50,000
Cape Cod	120,000

First, let's determine the number of predicted points n :

- Excluding the first column, a 5 x 5 loss triangle has $4 + 3 + 2 + 1 = 10$ data points
- Hence, $n = 10$

Second, let's determine the number of parameters p for each method:

- Chain-Ladder: $p = 5 - 1 = 4$ (number of AYs - 1)
- Parameterized BF: $p = 2(4) = 8$ (twice the chain-ladder method)
- Cape Cod: $p = 4$ (same as the chain-ladder method)

Third, let's calculate the test statistics for each method:

Method	Adjusted SSE	AIC	BIC
Chain-Ladder	5,556	445,108	502,337
Parameterized BF	12,500	247,652	315,479
Cape Cod	3,333	267,065	301,426

Using the chain-ladder method as an example:

- $Adjusted\ SSE = \frac{SSE}{(n-p)^2} = \frac{200,000}{(10-4)^2} = 5,556$
- $AIC \approx SSE \cdot e^{\frac{2p}{n}} = 200,000 \left(e^{\frac{2(4)}{10}} \right) = 445,108$
- $BIC \approx SSE \cdot n^{\frac{p}{n}} = 200,000 \left(10^{\frac{4}{10}} \right) = 502,337$

In general, the tests agree that the Cape Cod method is the better model. However, the AIC statistic favors the BF model. This is due to the fact that the AIC penalizes less for the number of parameters.

Iterative Method for Fitting a Parameterized BF Model

In a linear regression, we “directly” find the parameters that minimize the sum of the squared residuals (i.e., SSE). Since the parameterized BF model is a non-linear model, it’s not that simple. Instead, we must employ an *iterative* method to minimize the sum of the squared residuals, where the (w, d) residual is $[q(w, d) - f(d)h(w)]$.

The method procedure depends on the variances of the residuals:

- If the *variances of the residuals are constant* over the triangle, we use the standard *least squares* procedure
- If the *variances of the residuals are NOT constant* over the triangle, we use the *weighted least squares* procedure

Variances of the Residuals are Constant Over the Triangle

To begin the iteration, we need a starting point for the $f(d)$ s:

- Under the parameterized BF model, the $f(d)$ s represent incremental lag factors or emergence
- We can use the chain-ladder method to produce good starting values for the $f(d)$ s:
 - Calculate the age-to-ultimate factors using the chain-ladder method
 - Take the reciprocal of the age-to-ultimate factors to create cumulative percents emerged
 - Difference the cumulative percentages to create the incremental lags

Once we have our starting $f(d)$ s, we can calculate the corresponding $h(w)$ s as follows:

$$h(w) = \frac{\sum_d q(w, d) f(d)}{\sum_d f(d)^2}$$

Then, we run another regression to calculate the next iteration of $f(d)$ s:

$$f(d) = \frac{\sum_w q(w, d) h(w)}{\sum_w h(w)^2}$$

We repeat this process until convergence occurs and the $f(d)$ s and $h(w)$ s no longer change.

Variances of the Residuals are NOT Constant Over the Triangle

In this case, we need to use weighted least squares. Venter assumes that the residual variances are proportional to $f(d)^p h(w)^q$. This means that the regression weights are equal to $\frac{1}{f(d)^p h(w)^q}$.

If $p = q = 1$, we obtain the following iterative formulas:

$$h(w)^2 = \frac{\sum_d q(w, d)^2 / f(d)}{\sum_d f(d)}$$

$$f(d)^2 = \frac{\sum_w q(w, d)^2 / h(w)}{\sum_w h(w)}$$

Just as before, we repeat this process until convergence occurs and the $f(d)$ s and $h(w)$ s no longer change.

Iterative Method for Fitting a Cape Cod Model

Given that the Cape Cod model is a special case of the parameterized BF model, the fitting process is largely the same. Assuming residual variances are constant over the triangle, we obtain the following iterative formulas:

$$h = \frac{\sum_{w,d} q(w, d) f(d)}{\sum_{w,d} f(d)^2}$$

$$f(d) = \frac{\sum_w q(w, d)h}{\sum_w h^2}$$

Notice that the $h(w)$ s have been replaced with a single h that is found by summing over w and d . The $f(d)$ s are calculated in the same way, except every $h(w)$ is replaced with the single h parameter. Once we calculate the h parameter and corresponding $f(d)$ s, we no longer need to iterate. This is because any further iterations will produce the same h parameter.

Adjusting $f(d)$ Values

For the parameterized BF and Cape Cod models, the iterative procedures often produce $f(d)$ s that do not sum to 1. This is fine for estimating emergence since the $h(w)$ or h values will adjust accordingly, but it may be a good idea to multiply the $f(d)$ s by a factor (and correspondingly, multiply the $h(w)$ or h values by the reciprocal of the factor) to force them to sum to 1. Then, the $f(d)$ s can be interpreted as incremental emergence and the $h(w)$ or h values can be interpreted as expected losses.

Example: Fitting Parameterized BF & Cape Cod Models

Given the following incremental incurred losses as of December 31, 2023:

AY	<u>Incremental Incurred Losses (\$000) as of XX Months:</u>			
	12 mo.	24 mo.	36 mo.	48 mo.
2020	400	200	100	50
2021	450	300	200	
2022	425	100		
2023	500			

Let's fit the parameterized BF and Cape Cod methods through one full iteration.

Parameterized BF – Variances of Residuals Constant Over Triangle

First, let's calculate the starting $f(d)$ s using the chain-ladder method (i.e., volume-weighted LDFs):

	<u>Development Period</u>		
	12-24	24-36	36-48
Age-to-Age	1.471	1.222	1.071
Age-to-Ultimate	1.926	1.310	1.071

Then, the starting $f(d)$ s are as follows:

- $f(0) = \frac{1}{1.926} = 0.519$
- $f(1) = \frac{1}{1.310} - \frac{1}{1.926} = 0.244$
- $f(2) = \frac{1}{1.071} - \frac{1}{1.310} = 0.170$
- $f(3) = 1 - \frac{1}{1.071} = 0.067$

Second, let's calculate the first iteration of $h(w)$ s (in \$000s):

- Recall that $h(w) = \frac{\sum_d q(w,d)f(d)}{\sum_d f(d)^2}$
- $h(0) = \frac{400(0.519)+200(0.244)+100(0.170)+50(0.067)}{0.519^2+0.244^2+0.170^2+0.067^2} = 763.611$
- $h(1) = \frac{450(0.519)+300(0.244)+200(0.170)}{0.519^2+0.244^2+0.170^2} = 951.882$
- $h(2) = \frac{425(0.519)+100(0.244)}{0.519^2+0.244^2} = 744.258$
- $h(3) = \frac{500(0.519)}{0.519^2} = 962.885$

Third, let's calculate the first iteration of $f(d)$ s:

- Recall that $f(d) = \frac{\sum_w q(w,d)h(w)}{\sum_w h(w)^2}$
- $f(0) = \frac{400(763.611)+450(951.882)+425(744.258)+500(962.885)}{763.611^2+951.882^2+744.258^2+962.885^2} = 0.516$
- $f(1) = \frac{200(763.611)+300(951.882)+100(744.258)}{763.611^2+951.882^2+744.258^2} = 0.251$

- $f(2) = \frac{100(763.611)+200(951.882)}{763.611^2+951.882^2} = 0.179$
- $f(3) = \frac{50(763.611)}{763.611^2} = 0.065$

The $h(w)$ and $f(d)$ values after the first iteration are as follows:

Year w or Age d	0	1	2	3
$h(w)$	763.611	951.882	744.258	962.885
$f(d)$	0.516	0.251	0.179	0.065

In practice, we would continue the iterative process until the $h(w)$ and $f(d)$ values no longer change.

Parameterized BF – Variances of Residuals Proportional to $f(d)h(w)$

This is the weighted least squares setup with $p = q = 1$. We can use the same starting $f(d)$ s we calculated above.

First, let's calculate the first iteration of $h(w)$ s (in \$000s):

- Recall that $h(w)^2 = \frac{\sum_d q(w,d)^2 / f(d)}{\sum_d f(d)}$
- $h(0) = \sqrt{\frac{400^2/0.519+200^2/0.244+100^2/0.170+50^2/0.067}{0.519+0.244+0.170+0.067}} = 753.818$
- $h(1) = \sqrt{\frac{450^2/0.519+300^2/0.244+200^2/0.170}{0.519+0.244+0.170}} = 1,031.981$
- $h(2) = \sqrt{\frac{425^2/0.519+100^2/0.244}{0.519+0.244}} = 713.510$
- $h(3) = \sqrt{\frac{500^2/0.519}{0.519}} = 962.885$

Second, let's calculate the first iteration of $f(d)$ s:

- Recall that $f(d)^2 = \frac{\sum_w q(w,d)^2 / h(w)}{\sum_w h(w)}$
- $f(0) = \sqrt{\frac{400^2/753.818+450^2/1,031.981+425^2/713.510+500^2/962.885}{753.818+1,031.981+713.510+962.885}} = 0.516$

- $f(1) = \sqrt{\frac{200^2/753.818+300^2/1,031.981+100^2/713.510}{753.818+1,031.981+713.510}} = 0.248$
- $f(2) = \sqrt{\frac{100^2/753.818+200^2/1,031.981}{753.818+1,031.981}} = 0.171$
- $f(3) = \sqrt{\frac{50^2/753.818}{753.818}} = 0.066$

The $h(w)$ and $f(d)$ values after the first iteration are as follows:

Year w or Age d	0	1	2	3
$h(w)$	753.818	1,031.981	713.510	962.885
$f(d)$	0.516	0.248	0.171	0.066

Cape Cod Method

Once again, we can use the same starting $f(d)$ s we calculated above.

First, let's calculate the h parameter (in \$000s):

- Recall that $h = \frac{\sum_{w,d} q(w,d)f(d)}{\sum_{w,d} f(d)^2}$
- $h = \frac{[400(0.519)+200(0.244)+100(0.170)+50(0.067)]+\dots+[500(0.519)]}{[0.519^2+0.244^2+0.170^2+0.067^2]+\dots+[0.519^2]} = 850.589$

Second, let's calculate the $f(d)$ s:

- Recall that $f(d) = \frac{\sum_w q(w,d)h}{\sum_w h^2}$
- $f(0) = \frac{400(850.589)+450(850.589)+425(850.589)+500(850.589)}{850.589^2+850.589^2+850.589^2+850.589^2} = 0.522$
- $f(1) = \frac{200(850.589)+300(850.589)+100(850.589)}{850.589^2+850.589^2+850.589^2} = 0.235$
- $f(2) = \frac{100(850.589)+200(850.589)}{850.589^2+850.589^2} = 0.176$
- $f(3) = \frac{50(850.589)}{850.589^2} = 0.059$

If we calculated an h parameter from these $f(d)$ s, we would get same one as before. Hence, these are the final h and $f(d)$ values.

For presentation purposes and increased intuition, it's useful to adjust the final $f(d)$ s such that they sum up to 1:

- The $f(d)$ s sum up to $0.522 + 0.235 + 0.176 + 0.059 = 0.992$
- Thus, we need to multiply each $f(d)$ by $\frac{1}{0.992} = 1.008$. We also need to multiply h by 0.992 to ensure that the same fitted values are produced before and after the adjustment

The final h and $f(d)$ values after the adjustment are as follows:

Year w or Age d	0	1	2	3
h	$843.750 = 850.589(0.992)$	843.750	843.750	843.750
$f(d)$	$0.526 = 0.522(1.008)$	0.237	0.178	0.059

After the adjustment, the h parameter can be interpreted as the expected loss for each AY and the $f(d)$ parameters can be interpreted as the percent to emerge in each development period. Since we have the final set of parameters, we can calculate fitted and/or forecasted values using the Cape Cod emergence pattern. **For example**, suppose we wanted to calculate AY 2022 reserves as of December 21, 2023. This is equal to $843.750(0.178 + 0.059) = 200$.

Also, suppose we wanted to compare the Cape Cod method to the chain-ladder method using one of the goodness-of-fit statistics covered earlier. Then, for each emergence pattern:

- Calculate the fitted value for each cell in the loss triangle
- Subtract the fitted value from the observed value to obtain the residual for each cell in the loss triangle
- Square the residuals and sum them up to obtain the SSE

Once we have the SSE for each emergence pattern, we calculate the desired test statistic to compare each method.

IV. Residual Analysis – Testing Implications 3 & 4

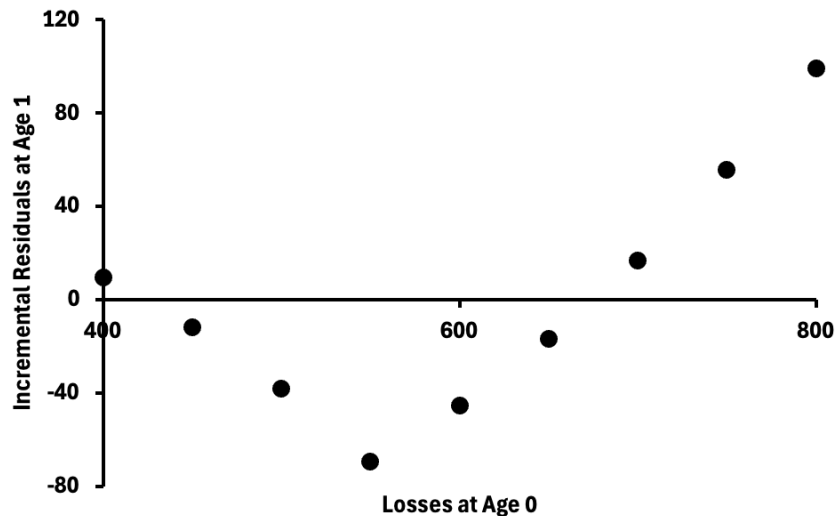
Implication 3: Test of Linearity

To test linearity (i.e., Assumption 1), Venter uses a scatter plot of the incremental residuals for a development period against the prior cumulative losses.

Since Venter's regression framework assumes that $Var[q(w, d + 1) | \text{data to } w + d] = k(d)$, the “weighted” residuals he uses are simply the raw residuals (similar to the weighted residuals for f_{k0} from the Mack Chain-Ladder paper).

For this test, we don't require a random scattering around zero. Instead, we want to ensure that there are not strings of positive and negative residuals in a row. If there are strings of positive and negative residuals in a row, then a non-linear process may be indicated.

As an **example**, consider the following residual plot:



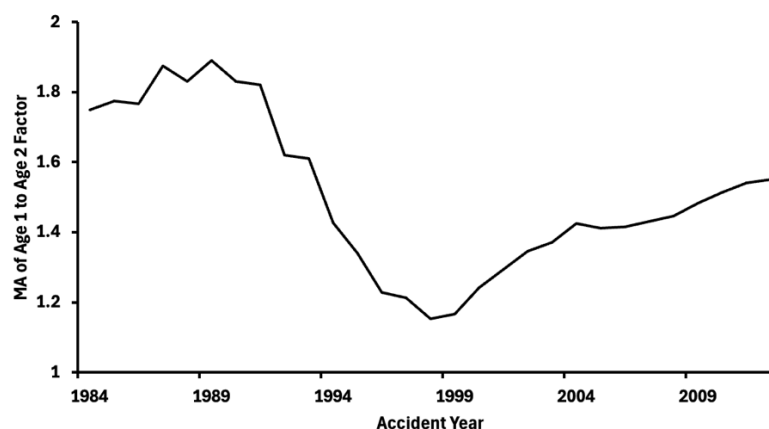
Since there are strings of positive and negative residuals in a row, we can conclude that the age 1 incremental losses are NOT a linear function of the age 0 cumulative losses.

Implication 4: Tests of Stability

Assumption 1 uses the same development factor for all accident years, so we would expect that the appropriate development factor is stable over time. We can test the stability of the development factor in a couple of different ways:

- 1) Plot the incremental residuals over time (i.e., AY)
 - If there are strings of positive and negative residuals in a row, then the development factors may not be stable. Hence, we may not want to use the same development factor for all AYs
- 2) Plot a moving average of a specific age-to-age factor over time
 - If the moving average shows clear shifts over time, then instability exists, and we may want to use a weighted average of the factors
 - If the moving average shows large fluctuations around a fixed level, this does NOT mean we should focus only on recent data. In this case, a broader range of data is actually better. We are really looking for cases where the fixed level moves over time (i.e. are there groupings of accident years revolving around different means)

As an **example**, consider the following moving average plot:



In general, the age 1 to age 2 development factor is NOT stable over time. However, there are temporary periods of stability.

When *development factors are unstable*, two options to consider are as follows:

- 1) Use a weighted average of the available factors, with more weight going to the more recent years (ex. only use the last 5 diagonals)
 - Keep in mind that this type of weighted average will increase estimation errors by over-emphasizing some observations and under-emphasizing others
- 2) Adjust the triangle for instability
 - For example, if there are changes in the settlement rate of claims, we can adjust the triangle to the latest pattern (Berquist-Sherman)

V. Independence – Testing Implications 5 & 6

Implication 5: Correlation of Development Factors

Similar to Mack, Venter created a correlation test for development factors. However, unlike Mack, Venter's test does NOT require the factors to be from adjacent columns.

In the Mack Chain-Ladder paper, the correlation test was a test of Assumption 1. Venter considers the correlation test to be a test of *Assumption 2*. Keep this in mind when answering questions on the exam involving Venter's correlation test.

Testing for Correlation for a Single Pair of Development Factor Columns

Venter's correlation test is based on the *table of development factors*. The procedure for a single pair of development factor columns is as follows:

- 1) Calculate the sample correlation coefficient, $r = \frac{\overline{XY} - \bar{X}\bar{Y}}{\sigma_X \sigma_Y}$, for any pair of development factor columns with at least three elements. Since this is the standard Pearson correlation coefficient, use the CORREL(array 1, array 2) function to calculate r
- 2) Calculate the test statistic, T , as follows:

$$T = r \sqrt{\frac{n-2}{1-r^2}}$$

where n is the number of elements in each development factor column and T is t -distributed with $n - 2$ degrees of freedom. The $n - 2$ comes from the fact that at least three elements are needed in each development column to run the test

3) Interpret T as follows:

- If $|T|$ is greater than the critical t -value at the $1 - \frac{\alpha}{2}$ confidence level with $n - 2$ degrees of freedom, then r is significant at the α level. Hence, we reject the null hypothesis that the development factors are independent
- If $|T|$ is less than the critical t -value at the $1 - \frac{\alpha}{2}$ confidence level with $n - 2$ degrees of freedom, then r is not significant at the α level. Hence, we fail to reject the null hypothesis that the development factors are independent
- The critical t -value is calculated using the $T.INV(1 - \frac{\alpha}{2}, n - 2)$ function
- Notice that this is a two-tailed test since correlation can be negative or positive

Testing for Correlation Across the Triangle

A 10% significance level means that 10% of the pairs of columns could show a false positive, so a single correlation doesn't mean much. Instead, we need a large number of correlations to determine if a correlation problem exists globally:

- Let $m = \binom{\text{number of AYs}-3}{2}$ be the number of testable pairs of columns in the triangle
 - We use “number of AYs – 3” because T cannot be calculated for column pairs with less than three elements
 - We use “choose 2” to ensure we include all column pairs, not just consecutive ones
- The number of pairs that display significant correlation at the α level is a binomial random variable with parameters m and α
- The mean is $\mu = m\alpha$ and the standard deviation is $\sigma = \sqrt{m\alpha(1 - \alpha)}$

- This can then be translated into an upper bound for the number of allowable significant correlations by picking the number of standard deviations we are comfortable with. For example:
 - For 3.33 standard deviations, the upper bound for allowable significant correlations is equal to $\mu + 3.33\sigma = m\alpha + 3.33\sqrt{m\alpha(1-\alpha)}$
 - Any more than this would imply correlation

Example: Test for Correlation

Given the following cumulative losses as of December 31, 2023:

AY	<u>Cumulative Incurred Losses (\$000) as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	75	135	160	170	175
2020	100	200	260	275	
2021	150	280	350		
2022	200	420			
2023	150				

An actuary wants to test for correlation at the 10% level between the 12-24 and 24-36 development factors.

First, let's create the table of development factors:

AY	<u>Development Factors</u>			
	12-24	24-36	36-48	48-60
2019	1.800	1.185	1.063	1.029
2020	2.000	1.300	1.058	
2021	1.867	1.250		
2022	2.100			

In the paper, Venter uses incremental development factors instead of cumulative development factors. The correlation is the same either way and both solutions have been accepted in the past.

Second, let's calculate the correlation between the 12-24 and 24-36 columns:

- $\text{CORREL}((1.800, 2.000, 1.867), (1.185, 1.300, 1.250)) = 0.965$

Third, let's calculate the test statistic, T :

- $T = r \sqrt{\frac{n-2}{1-r^2}} = 0.965 \sqrt{\frac{3-2}{1-0.965^2}} = 3.692$

Fourth, let's interpret T :

- The critical t -value at the $1 - \frac{\alpha}{2}$ confidence level with $n - 2$ degrees of freedom is
 $\text{T.INV}(1 - 0.10/2, 3 - 2) = 6.314$
- Since $|T| = 3.692 < 6.314$, we fail to reject the null hypothesis that the 12-24 and 24-36 development factors are independent

As mentioned earlier, we should test for correlation cross the entire triangle before drawing any conclusions. See the Cookbook for an example of testing for correlation globally.

Implication 6: Significantly High or Low Diagonals

Venter mentions that CY effects can be multiplicative or additive.

Multiplicative Diagonal Effects

As an example of multiplicative effects, suppose that diagonal $w + d = 7$ has been estimated to be 10% higher than normal. Then, an adjusted BF estimate of a cell is as follows:

$$q(w, d) = \begin{cases} 1.1f(d)h(w), & w + d = 7 \\ f(d)h(w), & \text{otherwise} \end{cases}$$

Additive Diagonal Effects (Standard Chain-Ladder Method)

Additive diagonal effects can be estimated using regression. To do this, we re-organize the loss triangle into columns and add dummy columns for the diagonals. A regression is run on the re-organized data to determine which columns have significant coefficients.

If the coefficients for the dummy columns are significant, we assume that CY effects exist.

Example: Testing for Additive Diagonal Effects

Given the following incremental incurred losses as of December 31, 2023:

AY	<u>Incremental Incurred Losses (\$000) as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	75	60	25	10	5
2020	100	100	60	15	
2021	150	130	70		
2022	200	220			
2023	150				

First, let's re-organize the loss triangle into columns and add the diagonal dummies:

Inc. 1 – 4	C0	C1	C2	C3	D1	D2	D3
60	75	0	0	0	0	0	0
100	100	0	0	0	1	0	0
130	150	0	0	0	0	1	0
220	200	0	0	0	0	0	1
25	0	135	0	0	1	0	0
60	0	200	0	0	0	1	0
70	0	280	0	0	0	0	1
10	0	0	160	0	0	1	0
15	0	0	260	0	0	0	1
5	0	0	0	170	0	0	1

Let's discuss the table above:

- This is the design matrix for the multiple regression
- The Inc. 1 – 4 column represents the incremental losses for ages 1 – 4
- The C0 – C3 columns represents the prior cumulative losses corresponding to the incremental losses in the Inc. 1 – 4 column (C0 predicts I1, C1 predicts I2, etc.)
- The D1 column represents the incremental losses on the third latest diagonal (100, 25)
- The D2 column represents the incremental losses on the second latest diagonal (130, 60, 10)
- The D3 column represents the incremental losses on the latest diagonal (220, 70, 15, 5)

If the dummy variables are significant, then it suggests that CY effects are present. In that case, the incremental losses are predicted by the prior cumulative losses AND the diagonal.

Second, let's use the LINEST() function to run a regression in Excel:

	$f(0)$	$f(1)$	$f(2)$	$f(3)$	D1	D2	D3
Coeff.	0.770	0.077	-0.151	-0.302	18.811	31.117	56.282
Std. Er.	0.194	0.152	0.172	0.261	22.248	29.541	41.838

By re-organizing the data into a design matrix, we can easily incorporate all of the predictors into the LINEST() function:

- The known y values are in the Inc. 1 – 4 column
- The known x values are all other columns. We simply highlight the entire array in Excel
- The final LINEST() function needed is LINEST(known y values, known x values, FALSE, TRUE)

Be careful when summarizing the coefficients! Excel reverses the order of the estimates. From left to right, the output order is “D3, D2, D1, ..., $f(0)$.”

Third, let's interpret the coefficients:

- Since the absolute values of the dummy coefficients are less than twice their standard deviations, the coefficients are insignificant
- Thus, we conclude that CY effects do NOT exist

Additive Diagonal Effects (Additive Chain-Ladder Method)

In the prior example, we assumed a standard chain-ladder method was being used and tested for additive diagonal effects. How would the design matrix change if the additive chain-ladder method was being used instead?

Since the additive chain-ladder method assumes next period's incremental losses are equal to a constant amount, we simply replace the cumulative losses in the design matrix with 1s. Then, the columns become dummy variables, resulting in additive amounts in the regression analysis. The design matrix would look as follows:

Inc. 1 – 4	A0	A1	A2	A3	D1	D2	D3
60	1	0	0	0	0	0	0
100	1	0	0	0	1	0	0
130	1	0	0	0	0	1	0
220	1	0	0	0	0	0	1
25	0	1	0	0	1	0	0
60	0	1	0	0	0	1	0
70	0	1	0	0	0	0	1
10	0	0	1	0	0	1	0
15	0	0	1	0	0	0	1
5	0	0	0	1	0	0	1

The A0 – A3 columns represent the dummy variables for the prior cumulative losses. If we dropped the diagonal variables D1 – D3, we would obtain the additive chain-ladder without diagonal effects (which is the same as the Cape Cod method)!

Diagonal Effects as a Measure of Inflation

One way to interpret diagonal effects is as a measure of inflation. An emergence model for this is as follows:

$$E[q(w, d) | \text{data to } w + d - 1] = f(d)g(w + d)$$

where $f(d)$ is the additive age term for age d , and $g(w + d)$ measures the cumulative claims inflation applicable to that diagonal since the first accident year.

In addition to CY effects, we can include AY effects as well:

$$E[q(w, d) | \text{data to } w + d - 1] = f(d)h(w)g(w + d)$$

where $h(w)$ is the AY effect.

The emergence model above requires far too many parameters (f , h and g parameters). We can reduce these substantially by **writing them as trends**:

- For the CY effect associated with cumulative inflation, set $g(w + d) = (1 + j)^{w+d}$
- For the AY effect, set $h(w) = h(1 + k)^w$
- For the age parameters, set $f(d) = (1 + i)^d$

We have reduced the model to four parameters: h, i, j and k . The model for the (w, d) cell is as follows:

$$h(1 + i)^d(1 + j)^{w+d}(1 + k)^w$$

Assuming these trends are constant for all periods, we can reduce the model further. To demonstrate this, let's expand the terms:

$$h(1 + i + j + ij)^d(1 + k + j + jk)^w$$

We now only have three parameters, as i becomes $i + j + ij$ and k becomes $k + j + jk$.

Lastly, Venter points out that we should remove the AY trend if the triangle has already been adjusted for AY changes. In that case, the model for the (w, d) cell would then be $h(1 + i)^d(1 + j)^{w+d}$.

VI. Appendix

The main takeaways from the Appendix are as follows:

- Since the Cape Cod method is the same as the additive chain-ladder method, it can be expressed as a linear model
- For forecasting, diagonal effects can be difficult to deal with. If there are diagonal effects, the assumption is that including them in the model fit will produce better estimates of the age effects (which are typically the dominant effect). Then, when forecasting losses, only the “better estimated” age effects would be used
- If significant additive diagonal effects are found using multiple regression, then it might be more useful to convert them into multiplicative effects using non-linear regression. Then, they could be used in the inflation models we discussed above

I. Introduction

This paper addresses the bootstrapping method for constructing a distribution of reserves. At the highest level, bootstrapping is “sampling with replacement.” As you can imagine, there are a variety of ways to implement this. The method shown in this paper samples from the model residuals.

II. Notation

As always, we need to define the quantities used in the model:

- w = the accident year, $1 \leq w \leq n$
- d = the development age, $1 \leq d \leq n - w + 1$
- $k = w + d$ = the diagonal representing the loss information for each accident year w as of accounting year k
- $c(w, d)$ = the cumulative loss from accident year w as of age d
- $c(w, n) = U(w)$ = the total loss from accident year w when claims are at ultimate values
- $q(w, d)$ = the incremental loss for accident year w from $d - 1$ to d
- $f(d)$ = the factor applied to $c(w, d)$ to estimate $q(w, d + 1)$
- $F(d)$ = the factor applied to $c(w, d)$ to estimate $c(w, d + 1)$

III. The Bootstrap Model

The bootstrap model deployed in this paper is the **ODP bootstrap model**. This one is useful in this context because it reproduces the standard chain-ladder (CL) method.

Assuming that each accident year has the same development factor, development factors can be defined as follows:

$$\hat{F}(d) = \frac{\sum_{w=1}^{n-d} c(w, d+1)}{\sum_{w=1}^{n-d} c(w, d)}$$

Assuming that each accident year has a parameter representing its relative level, ultimate claims can be estimated as follows:

$$\hat{c}(w, n) = c(w, d) \prod_{i=d+1}^n \hat{F}(i)$$

where the level parameters are the current cumulative values for each accident year (i.e., $c(w, n - w + 1)$).

The ODP Model

Under the ODP model, we use a GLM to model the incremental claims directly. The GLM uses a log link function and an ODP distribution. Under this model, the linear predictor is as follows:

$$\eta_{w,d} = c + \alpha_w + \sum_{r=2}^d \beta_r$$

where $\alpha_1 = \beta_1 = 0$

Note that the linear predictor shown in the source does not “sum” the β parameters. However, if you review the Excel exhibits provided by the author, the model is clearly parameterized such that the β parameters must be summed. In addition, solutions for past exam problems always sum the β parameters.

In the formulation above, the α parameters act as adjustments to the constant, c , and the β parameters adjust for the development trends after the first development period.

Alternatively, we can remove the constant, c , which causes the α parameters to act as individual AY level parameters:

$$\eta_{w,d} = \alpha_w + \sum_{r=2}^d \beta_r$$

where $\alpha_1 > 0$ and $\beta_1 = 0$. This is the linear predictor used for the remainder of the paper.

The **expected value** and **variance** of incremental claims are as follows:

$$\begin{aligned} E[q(w, d)] &= m_{w,d} \\ \text{Var}(q(w, d)) &= \phi E[q(w, d)]^z = \phi m_{w,d}^z \end{aligned}$$

where $m_{w,d} = e^{\eta_{w,d}}$ and ϕ is the scale parameter (i.e., dispersion parameter).

Notice that the variance is equal to a “power” of the mean times the scale parameter. The power, z , is used to specify the error distribution:

- $z = 0$: Normal distribution
- $z = 1$: ODP distribution
- $z = 2$: Gamma distribution
- $z = 3$: Inverse Gaussian distribution

Thus, under the ODP distribution, $\text{Var}(q(w, d)) = \phi m_{w,d}$.

Estimating the α & β Parameters Using a GLM

Shapland describes the following process for estimating the α & β parameters using a GLM:

- Apply the log-link function to the incremental loss triangle
- Specify the GLM using a system of equations with α & β parameters
- Convert the system of equations into matrix notation, $Y = XA$, where Y is the matrix of log-incremental data, X is the design matrix, and A is the matrix of α & β parameters

- Use a numerical technique (ex. iteratively weighted least squares, Newton-Raphson) to solve for the model parameters in the A matrix that minimize the squared difference between the Y matrix and the solution matrix, $\hat{Y}$. The solution matrix is the column vector whose entries are the applicable $\ln(m_{w,d})$ values

The method above for estimating the α & β parameters is a bit different from the one described in Taylor & McGuire. Furthermore, Shapland's description is not exactly correct (although the Excel exhibits provided with the paper correctly solve for the parameters). For the purposes of the exam, we recommend treating Shapland's description as another formulation for solving for the model parameters.

Example: Setting Up the ODP GLM

A GLM with a log link function and an ODP error distribution is fit to a 4 x 4 incremental loss triangle.

The system of equations underlying the GLM are as follows:

- $\ln[q(1,1)] = \alpha_1$
- $\ln[q(2,1)] = \alpha_2$
- $\ln[q(3,1)] = \alpha_3$
- $\ln[q(4,1)] = \alpha_4$
- $\ln[q(1,2)] = \alpha_1 + \beta_2$
- $\ln[q(2,2)] = \alpha_2 + \beta_2$
- $\ln[q(3,2)] = \alpha_3 + \beta_2$
- $\ln[q(1,3)] = \alpha_1 + \beta_2 + \beta_3$
- $\ln[q(2,3)] = \alpha_2 + \beta_2 + \beta_3$
- $\ln[q(1,4)] = \alpha_1 + \beta_2 + \beta_3 + \beta_4$

We convert the system of equations into matrix notation, $Y = XA$, as follows:

$$Y = \begin{bmatrix} \ln[q(1,1)] \\ \ln[q(2,1)] \\ \ln[q(3,1)] \\ \ln[q(4,1)] \\ \ln[q(1,2)] \\ \ln[q(2,2)] \\ \ln[q(3,2)] \\ \ln[q(1,3)] \\ \ln[q(2,3)] \\ \ln[q(1,4)] \end{bmatrix} \quad X = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \quad A = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix}$$

The model parameters are those that minimize the squared difference between the following two matrices:

$$Y = \begin{bmatrix} \ln[q(1,1)] \\ \ln[q(2,1)] \\ \ln[q(3,1)] \\ \ln[q(4,1)] \\ \ln[q(1,2)] \\ \ln[q(2,2)] \\ \ln[q(3,2)] \\ \ln[q(1,3)] \\ \ln[q(2,3)] \\ \ln[q(1,4)] \end{bmatrix} \quad \hat{Y} = \begin{bmatrix} \ln[m_{1,1}] \\ \ln[m_{2,1}] \\ \ln[m_{3,1}] \\ \ln[m_{4,1}] \\ \ln[m_{1,2}] \\ \ln[m_{2,2}] \\ \ln[m_{3,2}] \\ \ln[m_{1,3}] \\ \ln[m_{2,3}] \\ \ln[m_{1,4}] \end{bmatrix}$$

The ODP Bootstrap Model

One **important property** of the over-dispersed Poisson model is that the fitted incremental claims will exactly equal the fitted incremental claims derived using the standard CL volume-weighted factors:

- If we start with the latest diagonal and divide backwards successively by each age-to-age factor, we obtain fitted cumulative claims. Using subtraction, the fitted cumulative claims can be used to determine the fitted incremental claims
- These fitted incremental claims exactly match those obtained using the GLM
- This simplified model is known as the ODP bootstrap model

The property above has **three important consequences**:

- A simple link ratio algorithm can be used in place of the more complicated GLM algorithm, while still maintaining an underlying GLM framework
- The use of the age-to-age factors serves as a bridge to the deterministic framework. This allows the model to be more easily explained
- In general, the log link function does not work for negative incremental claims. Using link ratios remedies this problem

Unscaled Pearson Residuals

The bootstrap process involves sampling with replacement from the residuals of the incremental loss data (i.e., actual – fitted). Although the deviance, Pearson, and Anscombe residuals are potential candidates, the **Pearson residuals** are used since they are calculated consistently with the scale parameter, ϕ .

The **unscaled Pearson residuals** and **scale parameter** are calculated as follows:

$$\begin{aligned} r_{w,d} &= \frac{q(w,d) - m_{w,d}}{\sqrt{m_{w,d}^z}} \\ \phi &= \frac{\sum r_{w,d}^2}{N - p} \end{aligned}$$

where N is the number of data cells in the triangle and p is the number of parameters. In general, $p = 2 \cdot (\text{of AYs}) - 1$ since $\beta_1 = 0$.

In the formula above, the numerator is the raw residual. The denominator is the square root of the variance. The variance excludes the scale parameter since it's simply a proportionality constant.

Distribution of the Residuals

Although sampling with replacement assumes the residuals are independent and identically distributed (i.i.d.), it does *not require* the residuals to be normally distributed:

- This is an **advantage** of the ODP bootstrap model since the actual distributional form of the residuals will flow through the simulation process
- This is sometimes referred to as a “semi-parametric” bootstrap model since we are not parameterizing the residuals

Sample Triangles

Sampling with replacement from the residuals can be used to create new sample triangles of incremental claims using the following formula:

$$q^*(w, d) = r^* \sqrt{m_{w,d}^z} + m_{w,d}$$

where r^* is a randomly select residual from the sample.

The new sample triangles can then be cumulated, and age-to-age factors can be calculated and applied to calculate reserve point estimates. This produces a *distribution of point estimates*.

Process Variance in the Future Incremental Losses

The point estimates above do not fully capture the predictive distribution of future losses. Although they incorporate process variance and parameter variance in the simulation of the *historical data* (i.e., each sample triangle is a recreation of history), they fail to incorporate process variance in the simulation of the *future data*. This is because we are producing point estimates for each sample triangle, which is an expected value.

To *incorporate process variance in the simulation of future data*, we can expand the simulation process. Rather than projecting each incremental loss using age-to-age factors, we sample from a Gamma distribution with mean $m_{w,d}$ and variance $\phi m_{w,d}$. In doing so, our distribution of point estimates becomes a *distribution of possible outcomes*, which incorporates process variance and parameter variance in the simulation of the historical and future data.

Degrees of Freedom Adjustment Factor

Historical papers have taken slightly different approaches for implementing the ODP bootstrap model. For example, England & Verrall suggest multiplying the distribution of point estimates by the **degrees of freedom adjustment factor**, f^{DoF} , and the **scale parameter**. The degrees of freedom factor is defined as follows:

$$f^{DoF} = \sqrt{\frac{N}{N-p}}$$

By multiplying the point estimates by the two quantities mentioned above, we allow for over-dispersion of the residuals in the sampling process AND add process variance to future incremental losses.

Scaled Pearson Residuals

Another way to add over-dispersion in the residuals is by multiplying the unscaled Pearson residuals by the degrees of freedom adjustment factor. This produces the **scaled Pearson residuals**, defined as follows:

$$r_{w,d}^S = r_{w,d} \times f^{DoF}$$

We can think of the scaled Pearson residuals as a “bias-corrected” version of the unscaled Pearson residuals. To add process variance to future incremental losses, you would still need to sample from a gamma distribution at the end of the simulation process.

Standardized Residuals

The issue with the scaled Pearson residuals is that the f^{DoF} factor does not produce standardized residuals. Assuming our model is properly specified, standardized residuals should have a constant variance. This is important since the ODP bootstrap model assumes that residuals are i.i.d.

If the model is not properly specified, standardized residuals may not display a constant variance. This is known as **heteroscedasticity**. In this case, it might indicate that something other than a Poisson distribution should be used. Alternatively, it might indicate that more predictors are needed.

In order to calculate the standardized residuals, the hat matrix adjustment factor, $f_{w,d}^H$, is required. Mathematically:

$$f_{w,d}^H = \sqrt{\frac{1}{1 - H_{i,i}}}$$

where $H_{i,i}$ represents the i^{th} point on the diagonal of the hat matrix. Note that the hat matrix is the projection matrix that maps the response vector Y to the solution vector $\hat{Y}$.

As a quick example of the structure of the hat matrix, suppose we are given the following hat matrix for a 3 x 3 loss triangle:

$$\begin{bmatrix} H_{1,1} & H_{1,2} & H_{1,3} & H_{1,4} & H_{1,5} & H_{1,6} \\ H_{2,1} & H_{2,2} & H_{2,3} & H_{2,4} & H_{2,5} & H_{2,6} \\ H_{3,1} & H_{3,2} & H_{3,3} & H_{3,4} & H_{3,5} & H_{3,6} \\ H_{4,1} & H_{4,2} & H_{4,3} & H_{4,4} & H_{4,5} & H_{4,6} \\ H_{5,1} & H_{5,2} & H_{5,3} & H_{5,4} & H_{5,5} & H_{5,6} \\ H_{6,1} & H_{6,2} & H_{6,3} & H_{6,4} & H_{6,5} & H_{6,6} \end{bmatrix}$$

The hat matrix adjustment factors are as follows:

$$\left[\begin{array}{ccccccccc} f_{1,1}^H = \sqrt{\frac{1}{1-H_{1,1}}} & & & & & & & & \\ & f_{2,1}^H = \dots & & & & & & & \\ & & f_{3,1}^H = \dots & & & & & & \\ & & & f_{1,2}^H = \dots & & & & & \\ & & & & f_{2,2}^H = \dots & & & & \\ & & & & & f_{1,3}^H = \dots & & & \end{array} \right]$$

The hat matrix adjustment factor is considered as a replacement for and an improvement over the degrees of freedom adjustment factor.

The **standardized residuals** are found by multiplying the hat matrix adjustment factors by the unscaled Pearson residuals:

$$r_{w,d}^H = \frac{q(w,d) - m_{w,d}}{\sqrt{m_{w,d}^Z}} \cdot f_{w,d}^H$$

The standardized residuals are the final residuals that should be sampled from when running the ODP bootstrap model.

Approximating the Scale Parameter

Although the standardized residuals serve as the foundation of the bootstrapping process, the scale parameter should still be calculated using the unscaled Pearson residuals.

In the event that the unscaled Pearson residuals are not readily available, we can approximate the scale parameter using the standardized residuals:

$$\phi^H = \frac{\sum (r_{w,d}^H)^2}{N}$$

Zero-Value Residuals

When sampling from residuals, we should not include any zero-value residuals. These are normally found in the “corners” of the triangle (i.e., the $(n, 1)$ and $(1, n)$ cells). A common explanation for excluding these residuals is that these cells contain variance...we just don’t know what that variance is yet!

Example: ODP Bootstrap Model

An actuary is running an ODP bootstrap model. Given the following information for an insurer as of December 31, 2023:

AY	<u>Cumulative Paid Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	95	150	180
2022	115	160	
2023	105		

The hat matrix is as follows:

$$\begin{bmatrix} 0.8335 & 0.1665 & 0.0000 & 0.1665 & -0.1665 & 0.0000 \\ 0.1561 & 0.8439 & 0.0000 & -0.1561 & 0.1561 & 0.0000 \\ 0.0000 & 0.0000 & 1.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.3496 & -0.3496 & 0.0000 & 0.6504 & 0.3496 & 0.0000 \\ -0.3278 & 0.3278 & 0.0000 & 0.3278 & 0.6722 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 \end{bmatrix}$$

- $\phi = 2.58$

First, let’s calculate the standard CL volume-weighted factors:

- 12-24: $\frac{150+160}{95+115} = 1.476$
- 24-36: 1.200

Second, let's calculate the fitted cumulative losses by starting with the latest diagonal and dividing backwards successively by each age-to-age factor:

AY	<u>Fitted Cumulative Paid Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	$101.61 = \frac{150}{1.476}$	$150 = \frac{180}{1.200}$	180
2022	108.39	160	
2023	105		

Third, let's calculate the fitted incremental losses:

AY	<u>Fitted Incremental Paid Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	101.61	$48.39 = 150 - 101.61$	30
2022	108.39	51.61	
2023	105		

Fourth, let's calculate the actual incremental losses:

AY	<u>Actual Incremental Paid Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	95	$55 = 150 - 95$	30
2022	115	45	
2023	105		

Fifth, let's calculate the unscaled Pearson residuals:

AY	<u>Unscaled Pearson Residuals as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	-0.66	$0.95 = \frac{55 - 48.39}{\sqrt{48.39}}$	0.00
2022	0.64	-0.92	
2023	0.00		

Sixth, let's calculate the hat matrix adjustment factors (set to 0 if undefined):

<u>Hat Matrix Adj. Factors as of XX Months:</u>			
AY	12 mo.	24 mo.	36 mo.
2021	$2.451 = \sqrt{\frac{1}{1 - 0.8335}}$	1.691	0.000
2022	$2.531 = \sqrt{\frac{1}{1 - 0.8439}}$	1.750	
2023	0.000		

Seventh, let's calculate the standardized residuals:

<u>Standardized Residuals as of XX Months:</u>			
AY	12 mo.	24 mo.	36 mo.
2021	$-1.61 = -0.66(2.451)$	1.61	0.00
2022	$1.61 = 0.64(2.531)$	-1.61	
2023	0.00		

Eighth, let's randomly sample from the standardized residuals to create the following sample:

<u>Sample Standardized Residuals as of XX Months:</u>			
AY	12 mo.	24 mo.	36 mo.
2021	1.61	1.61	-1.61
2022	-1.61	1.61	
2023	-1.61		

Ninth, let's convert the sample standardized residuals into sample incremental losses:

<u>Sample Incremental Paid Losses (\$) as of XX Months:</u>			
AY	12 mo.	24 mo.	36 mo.
2021	$117.82 = 1.61\sqrt{101.61} + 101.61$	59.57	21.19
2022	91.65	63.16	
2023	88.53		

Tenth, let's use the sample incremental losses to create a triangle of sample cumulative losses:

<u>Sample Cumulative Paid Losses (\$) as of XX Months:</u>			
AY	12 mo.	24 mo.	36 mo.
2021	117.82	177.39	198.58
2022	91.65	154.81	
2023	88.53		
Age-to-Age		1.586	1.119

Eleventh, let's project the sample cumulative losses to ultimate using the age-to-age factors:

<u>Projected Sample Cumulative Paid Losses (\$) as of XX Months:</u>			Reserve Point Estimate
AY	12 mo.	24 mo.	36 mo.
2021	117.82	177.39	198.58
2022	91.65	154.81	173.31
2023	88.53	140.40	157.17
Total			87.14

The reserve point estimates above *do not include process variance in the simulation of the future values*. To include this process variance, we replace the future incremental losses with simulated values from a gamma distribution with mean $m_{w,d}$ and variance $\phi m_{w,d}$.

For example, the future incremental losses including process variance might look like this:

<u>Sample Incremental Paid Losses w/PV (\$) as of XX Months:</u>			Reserve Possible Outcome
AY	12 mo.	24 mo.	36 mo.
2021	117.82	59.57	21.19
2022	91.65	63.16	6.71
2023	88.53	59.13	22.80
Total			88.64

The point estimates are now possible outcomes. To simulate from a gamma distribution, we can use the GAMMAINV(RAND(), α , β) function in Excel. The mean and variance of a gamma distribution are $\alpha\beta$ and $\alpha\beta^2$, where α is the shape parameter and β is the scale parameter. Thus,

$\alpha = \frac{m_{w,d}}{\phi}$ and $\beta = \phi$. Note that the Verrall paper uses an alternative parameterization of the gamma distribution.

As an example:

- AY 2022 at 36 months
 - $m_{2,3} = 173.31 - 154.81 = 18.50$
 - $\phi = 2.58$ (given in problem)
 - $\text{GAMMAINV}(\text{RAND}(), \frac{18.50}{2.58}, 2.58) = 6.71$
- AY 2023 at 24 months
 - $m_{3,2} = 140.40 - 88.53 = 51.87$
 - $\text{GAMMAINV}(\text{RAND}(), \frac{51.87}{2.58}, 2.58) = 59.13$

Keep in mind that the simulated gamma values above depend on the random seed (i.e., the random uniform variable produced by the $\text{RAND}()$ function). If you use the function above in Excel, it's unlikely you will get the same values since it will be based on a different random seed.

At this point, we have completed one pass through the bootstrapping process. The total point estimate (or possible outcome) produced at the end represents one point (or possible outcome) in the distribution of point estimates (or possible outcomes).

Starting with the random sampling of the standardized residuals, we would redo this entire process a number of times until a full distribution is produced.

Bootstrapping the Incurred Loss Triangle

The ODP bootstrap model can be applied to both paid and incurred data. If applied to paid data, the distribution of possible outcomes represents the total unpaid. If applied to incurred data, the distribution of possible outcomes represents IBNR. Thus, they are not directly comparable.

Shapland covers two approaches for modeling an unpaid loss distribution using incurred data.

Approach 1

Under Approach 1, we run a paid data model in conjunction with the incurred data model.

Then, we use the random payment pattern from each iteration of the paid data model to convert the ultimate values from each corresponding incurred model iteration to develop paid losses by accident year.

A **benefit** of this approach is that it allows us to use the case reserves to help predict the ultimate losses, while still focusing on the payment stream for measuring risk.

An **improvement** to this approach would be the inclusion of correlation between the paid and incurred models (possibly in the residual sampling process). For example, if we want to compare iterations showing long payment streams with iterations showing high incurred results, we must consider correlation.

Approach 2

Under this approach, we apply the ODP bootstrap to the Munich Chain-Ladder (MCL) model. The MCL uses the inherent relationship/correlation between paid and incurred losses to predict ultimate losses. When paid losses are low relative to incurred losses, then future paid loss development tends to be higher than average. When paid losses are high relative to incurred losses, then future paid loss development tends to be lower than average.

Two advantages of this approach versus the first approach are as follows:

- It does not require us to model paid losses twice
- It explicitly measures the correlation between paid and incurred losses

Bootstrapping the BF and Cape Cod Models

An **issue** with using the ODP bootstrap process is that iterations for the latest few accident years tend to be more variable than what we would expect given the simulations for earlier accident

years. This is due to the fact that *more* age-to-age factors are used to extrapolate the sampled values to develop point estimates for each iteration.

To **address** this issue, future incremental values can be extrapolated using the BF or Cape Cod method.

Typically, these methods are deterministic. However, they can be made stochastic by converting the deterministic assumptions to stochastic assumptions. For example:

- Instead of simply assuming a priori loss ratios for the BF method, we can add a vector of standard deviations to go with these means
- We can then assume a distribution and simulate a different a priori loss ratio for every iteration of the model

The GLM Bootstrap Model

So far, we have discussed the ODP bootstrap model, where the standard CL factors are used to produce fitted values. Two **limitations** of the ODP bootstrap model are as follows:

- It does not adjust for CY effects
- Since it includes a parameter for each AY and development period beyond the first period, it might over-fit the data

If we operate completely in the GLM framework (known as the *GLM bootstrap model*), we can address these limitations.

The GLM bootstrap model has the following **two drawbacks**:

- The GLM must be solved for each iteration of the bootstrap model, which may slow down the simulation
- The model is no longer directly explainable to others using age-to-age factors

The GLM bootstrap model has the following **benefits**:

- We can specify fewer parameters, which helps avoid over-fitting

- We can add parameters for CY trends. If we include CY trends, the linear predictor changes to the following:

$$\eta_{w,d} = \alpha_w + \sum_{r=2}^d \beta_r + \sum_{k=2}^{w+d-1} \gamma_k$$

where γ_k is the CY trend, $\alpha_1 > 0$, $\beta_1 = 0$, and $\gamma_1 = 0$

- We can model shapes other than triangles
 - If incremental data is missing for the first few diagonals, the remaining values in those rows would not be useful under the CL method since it requires cumulative values
 - Since the GLM bootstrap uses incremental values, the entire trapezoid can be used to fit the model
- We can match the model parameters to the statistical features found in the data
 - For example, modeling with fewer development trend parameters means that the last parameter is assumed to continue past the end of the triangle. This gives us a tail without actually having to specify a tail factor

One issue with the CY trend model is that the system of equations no longer has a unique solution. For example:

- Excluding CY trends, a 3 x 3 triangle has five parameters (three α parameters and two β parameters)
- Including CY trends, a 3 x 3 triangle has seven parameters (three α parameters, two β parameters, and two γ parameters)
- Since six cells of the triangle contain data, we now have six knowns and seven unknowns
- To deal with this issue, we start with a model with one α parameter, one β parameter, and one γ parameter. We then add and remove parameters as needed

How do we calculate expected future incremental losses using the GLM bootstrap model?

- Unlike the ODP bootstrap model, we do not apply age-to-age factors to each sample triangle to produce point estimates. Instead, we fit the same GLM model underlying the residuals to each sample triangle. Then, we use the resulting parameters to produce reserve point estimates (or possible outcomes)
 - One **drawback** of this approach is the additional time required to fit a GLM to each sample triangle
-

Example: GLM Bootstrap Model

An actuary is using the GLM bootstrap model with a log link and an ODP error distribution to produce a reserve distribution. Given the following information for an insurer as of December 31, 2023:

AY	<u>Cumulative Paid Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	50	80	100
2022	60	90	
2023	80		

The fitted parameters are as follows:

- $\hat{\alpha}_1 = 3.947$
- $\hat{\alpha}_2 = 4.064$
- $\hat{\alpha}_3 = 4.382$
- $\hat{\beta}_2 = -0.606$
- $\hat{\beta}_3 = -0.345$

First, let's calculate the fitted incremental paid losses using $m_{w,d} = e^{\alpha_w + \sum_{r=2}^d \beta_r}$

AY	<u>Fitted Incremental Paid Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	$51.76 = e^{3.947}$	$28.24 = e^{3.947-0.606}$	20
2022	$58.24 = e^{4.064}$	31.76	
2023	80		

Second, let's calculate the actual incremental paid losses:

AY	<u>Actual Incremental Paid Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	50	30	20
2022	60	30	
2023	80		

Third, let's calculate the unscaled Pearson residuals:

AY	<u>Unscaled Pearson Residuals as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	$-0.25 = \frac{50 - 51.76}{\sqrt{51.76}}$	0.33	0.00
2022	0.23	-0.31	
2023	0.00		

The remainder of the GLM bootstrap model process is as follows:

- Calculate standardized residuals by multiplying the unscaled Pearson residuals by the hat matrix adjustment factors
- Randomly sample from the standardized residuals to create a sample residual triangle
- Convert the sample residual triangle to a sample incremental paid loss triangle
- Fit the same GLM structure underlying the original model to the sample incremental paid loss triangle

- Calculate future expected incremental values using the fitted parameters from the prior step
- Sample from a gamma distribution with mean $m_{w,d}$ and variance $\phi m_{w,d}$ to incorporate process variance into each future cell of the sample incremental paid loss triangle
- Sum up the expected future incremental paid losses to obtain a possible outcome for each AY
- Starting with the random sampling of the standardized residuals, redo this process thousands of times

Notice that we never need to produce sample cumulative paid losses when using the GLM bootstrap model. This is because the GLM models incremental losses directly.

IV. Practical Issues

Negative Incremental Values – GLM Bootstrap Model

Negative incremental values can be problematic when parameterizing a GLM with a log link function since we cannot take the log of a negative number. Depending on the extent of negative values, we have two options for dealing with them.

Option 1: Modified Log-Link Function

If some of the incremental values in a column are negative, but the **sum of the incremental values in the column is positive**, we can use the following modified log-link function to solve the GLM:

$$\begin{cases} \ln[q(w, d)], & q(w, d) > 0 \\ 0, & q(w, d) = 0 \\ -\ln[\text{abs}(q(w, d))], & q(w, d) < 0 \end{cases}$$

The reason the GLM doesn't break down here is because each β parameter is based on the entire column. As long as the column sum is positive, the β parameter will usually be found.

Example: Modified Log-Link Function

Given the following incremental incurred losses as of December 31, 2023:

AY	<u>Incremental Incurred Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	100	-20	15
2022	110	50	
2023	120		

Using the modified log-link function, the values used to solve the GLM are as follows:

AY	<u>Log-Incremental Incurred Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	4.605	$-2.996 = -\ln[\text{abs}(-20)]$	2.708
2022	4.700	3.912	
2023	4.787		

Option 2: Subtracting the Largest Negative Value

If the **sum of the incremental values in a column is negative**, the iterative algorithm used to solve for the GLM parameters will diverge. Instead, we must use a different approach:

- Find the largest negative value in the triangle
- Set the constant ψ equal to the largest negative value
- Subtract ψ from every incremental value in the triangle to arrive at the following adjusted incremental values:

$$\boxed{q^+(w, d) = q(w, d) - \psi}$$

- Solve for the GLM parameters based on the logs of the adjusted incremental values
- Calculate the fitted incremental values using $m_{w,d}^+ = e^{\text{linear predictor}}$

- Adjust the resulting fitted incremental values as follows:

$$m_{w,d} = m_{w,d}^+ + \psi$$

The reason we add ψ back in the end is to ensure the final fitted values are on the same basis as the original data.

Example: Subtracting the Largest Negative Value

Given the following incremental incurred losses as of December 31, 2023:

AY	<u>Incremental Incurred Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	100	-20	15
2022	110	10	
2023	120		

In this case, the sum of the incremental losses in the second column is negative. Thus, we cannot use the simple modification we used in the previous example. Instead, we must use the second modification.

First, let's calculate the adjusted incremental losses by subtracting the largest value in the triangle:

AY	<u>Adjusted Incremental Incurred Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	120	$0 = -20 - (-20)$	35
2022	130	30	
2023	140		

Second, let's take the log of the adjusted incremental losses (set to 0 if equal to 0):

AY	<u>Log of Adjusted Incremental Incurred Losses (\$) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	4.787	0.000	3.555
2022	4.868	3.401	
2023	4.942		

Once the GLM parameters have been determined, we can calculate the fitted values, $m_{w,d}^+$.

Then, we “adjust them back to normal” by adding the negative value we subtracted earlier (i.e., $m_{w,d} = m_{w,d}^+ + \psi$). This reduces all of the fitted values.

In the second option above, some of the final fitted values will be negative since we add the largest negative value back in at the end. This is a problem when calculating residuals and sampled incremental values because we must take the square root of the fitted value.

To deal with this, we modify the **unscaled Pearson residuals** and **sampled incremental losses** as follows:

$$r_{w,d} = \frac{q(w,d) - m_{w,d}}{\sqrt{\text{abs}(m_{w,d}^z)}}$$

$$q^*(w,d) = r^* \sqrt{\text{abs}(m_{w,d}^z)} + m_{w,d}$$

Negative Incremental Values – ODP Bootstrap Model

Unlike the GLM bootstrap model, the ODP bootstrap is not impacted by negative values when calculating fitted values. The development factors being used to “divide backwards” will simply be less than 1.00.

Since development factors less than 1.00 will produce negative incremental fitted values, we must use the modified unscaled Pearson residuals and sampled incremental losses shown above to run the full bootstrap process.

Negative Values During Simulation

Although we have dealt with the issue of negative incremental values when parameterizing a model, negative values still affect the process variance in the simulation process.

When each future incremental value is sampled from a gamma distribution to add process variance, the gamma parameters must be positive. To simulate negative values from a gamma distribution, we have **two options**:

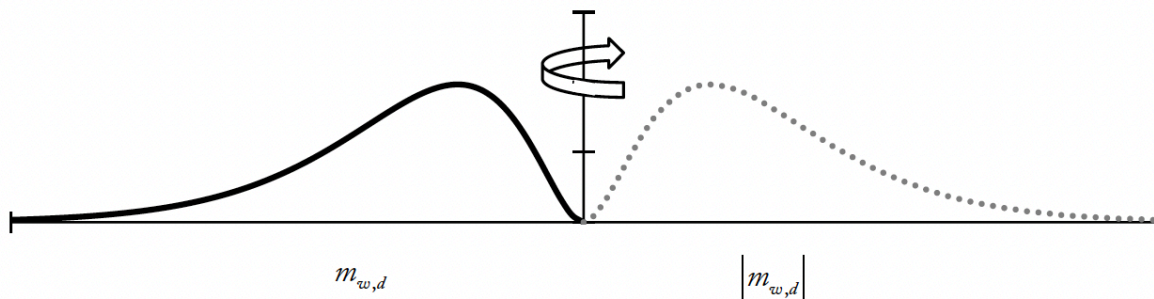
$$\begin{aligned} m_{w,d} &= -\text{Gamma}[\text{abs}(m_{w,d}), \phi \text{abs}(m_{w,d})] \\ m_{w,d} &= \text{Gamma}[\text{abs}(m_{w,d}), \phi \text{abs}(m_{w,d})] + 2m_{w,d} \end{aligned}$$

Although the first option is more intuitive, the second option is preferred because the resulting distribution will be skewed to the right, as it should be for a gamma distribution.

Graphically, the two options have the following densities:

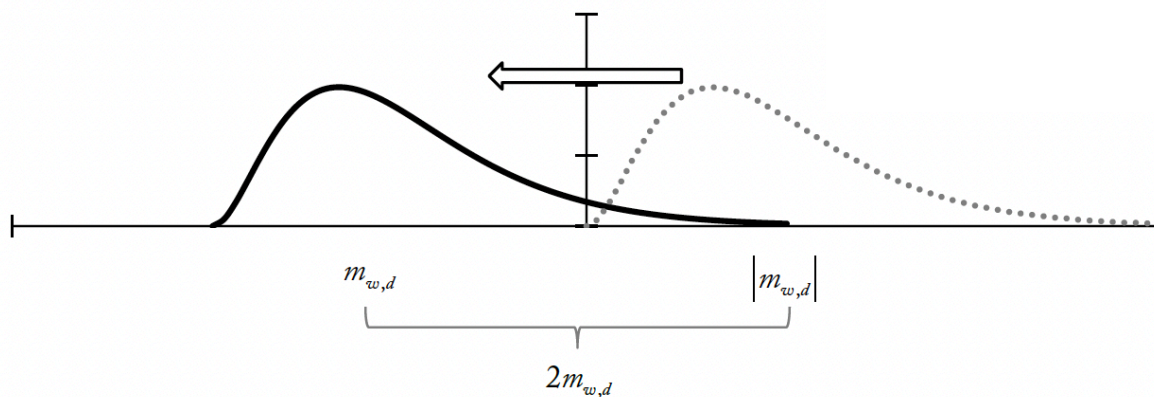
Option 1: Change the sign of the simulated value

$$-\text{Gamma}(\text{mean} = |m_{w,d}|, \text{var} = \phi \cdot |m_{w,d}|)$$



Option 2: Shift the entire distribution to have a mean of $m_{w,d}$

$$\text{Gamma}\left(\text{mean} = |m_{w,d}|, \text{var} = \phi \cdot |m_{w,d}|\right) + 2m_{w,d}$$



Under option 1, the density is skewed to the left and all values are negative. Under option 2, the density is skewed to the right and there is a mixture of positive and negative values. However, the mean is the same in both graphs.

Negative Values & Extreme Outcomes

Negative incremental values can cause extreme outcomes. This is most common when newly sampled triangles have negative incremental losses in the first few development columns. When a column of cumulative values sums close to zero and the next column sums to a large number, extremely large age-to-age factors and reserve estimates are produced.

To deal with extreme outcomes, the author provides the following **three options**:

- 1) Identify the extreme iterations and remove them
 - Be careful to only remove unreasonable extreme iterations so that the probability of extreme outcomes is not understated
- 2) Recalibrate the model
 - Identify the source of the negative incremental losses and remove it if necessary
 - For example, if the first row has negative incremental values due to sparse data, remove it and reparametrize the model

- Alternatively, we could create separate models. For example, if negative values are caused by salvage/subrogation, we can model the gross losses and salvage/subrogation separately. Then, we can combine the iterations assuming 100% correlation
- Limit incremental losses to zero
 - This involves replacing negative incremental values with zeroes in the original triangles, zeroes in the sample triangles, OR zeroes in the projected future incremental losses

Non-Zero Sum of Residuals

Although the residuals are theoretically identically distributed with mean zero, they tend to have a non-zero average in practice.

Should the residuals be adjusted so that their average is zero?

- If the average of the residuals is positive, then resampling from the residuals will add variability to the resampled incremental losses. It may also cause the resampled incremental losses to have an average greater than the fitted loss. In this respect, the residuals should be adjusted
- Others argue that the non-zero average of the residuals is a characteristic of the data set. Thus, they should not be adjusted

If a residual average of zero is desired, we can add a single constant to all residuals such that the sum of the shifted residuals is zero.

Using an N -Year Weighted Average

At times, we may not use all years in the loss triangle when determining development factors. Instead, we may only use N years of data. The method for accomplishing this differs between the GLM bootstrap model and the ODP bootstrap model.

GLM Bootstrap Model

- We use N years of data by excluding the first few diagonals in the triangle, leaving us with $N + 1$ included diagonals
- This changes the shape of the triangle to a trapezoid
- The excluded diagonals are given zero weight in the model and fewer calendar year parameters are required
- When running the bootstrap simulations, we only need to sample residuals for the trapezoid that was used to parameterize the original model. The reason for this is that the GLM models incremental claims directly and can be parameterized using a trapezoid
- Each parameter set is then used to calculate future incremental values

ODP Bootstrap Model

- We calculate N -year weighted-average factors instead of all-year weighted-average factors
- We exclude the first few diagonals when calculating residuals
- Unlike the GLM bootstrap model, we must still sample residuals for the entire triangle when running bootstrap simulations. The reason for this is that the ODP bootstrap model requires cumulative values in order to calculate link ratios
- Once we have cumulative values for each sample triangle, we use N -year average factors to project the sample triangles to ultimate

Why does using N years of data leave us with $N + 1$ included diagonals? As an **example**, assume we have a 5 x 5 triangle, but only want to use a two-year weighted average. The required data in the triangle changes as follows:

AY	12 mo.	24 mo.		AY	12 mo.	24 mo.
1	x_1	y_1		1		
2	x_2	y_2		2		y_2
3	x_3	y_3	$\Rightarrow$	3	x_3	y_3
4	x_4	y_4		4	x_4	y_4

$$5 \quad | \quad x_5$$

$$5 \quad | \quad x_5$$

In this example, the original 12-24 age-to-age factor is $\frac{y_1+y_2+y_3+y_4}{x_1+x_2+x_3+x_4}$. The new 12-24 age-to-age factor for $N = 2$ years of data is $\frac{y_3+y_4}{x_3+x_4}$. The new factor uses $N + 1 = 3$ diagonals:

- The first diagonal is comprised of x_3
- The second diagonal is comprised of x_4 and y_3
- The third diagonal is comprised of y_4

Missing Values

If values are missing from the loss triangle, the following calculations are affected:

- Loss development factors
- Fitted triangle (if the missing value lies on the last diagonal)
- Residuals
- Degrees of freedom

Once again, the method for dealing with missing values differs between the GLM bootstrap model and the ODP bootstrap model.

ODP Bootstrap Model

When using the ODP bootstrap model, there are multiple approaches to managing missing values:

- Estimate the missing value using surrounding values
- Exclude the missing value when calculating the loss development factors
 - If we do this, no corresponding residual will be calculated for the missing value
 - Similar to N -year weighted averages, we must still sample for the entire triangle so that we can calculate cumulative values during the simulation process
 - Once the sample triangles are calculated, we should exclude the cells corresponding to the missing values from the projection process (i.e., when

calculating the age-to-age factors based on the sample triangles, the missing values should be excluded)

- If the missing value lies on the last diagonal, we can either estimate the value or we can use the value in the second to last diagonal to construct the fitted triangle

GLM Bootstrap Model

If using the GLM bootstrap model, the missing data simply reduces the number of observations used in the model. Similar to the ODP bootstrap model, we could use one of the methods described above to estimate the missing data if desired.

Outliers in the Original Dataset

If extreme values exist in the original triangle, and they are not considered representative of the future variability of the data, we can remove their impact from the model.

The method for dealing with outliers differs between the GLM bootstrap model and the ODP bootstrap model.

ODP Bootstrap Model

If using the ODP bootstrap model, approaches to managing outliers include the following:

- Exclude the outliers completely
 - In this case, we would proceed in the same manner as a missing value
- Exclude the outliers when calculating the age-to-age factors and the residuals (similar to missing values), but include the outlier cells during the sample triangle projection process (different from missing values)
 - The idea here is to remove the extreme impact of the incremental cell by excluding the outlier during the fitting process while still including some non-extreme variability by including the cell in the sample triangle projections

When excluding outliers from the calculation of age-to-age factors, there are **three options**:

- 1) Exclude the outlier in the numerator
- 2) Exclude the outlier in the denominator
- 3) Exclude the outlier in the numerator and the denominator

GLM Bootstrap Model

When using the GLM bootstrap model, outliers are treated similarly to missing data. If the data is not considered representative of real variability, the outlier should be excluded, and the model should be parameterized without it.

Example: Dealing with Outliers in the ODP Bootstrap Model

An actuary is running the ODP bootstrap model and has decided to deal with outliers by excluding them from the calculation of the age-to-age factors. Given the following incremental paid losses (in \$) as of December 31, 2023:

AY	12 mo.	24 mo.
2019	100	80
2020	90	120
2021	105	85
2022	100	75

In this case, AY 2020 at 24 months appears to be an outlier. Let's walk through the three options for excluding outliers from the calculation of age-to-age factors:

- 1) Excluding the outlier in the numerator
 - 12-24 age-to-age factor = $\frac{180+190+175}{100+105+100} = 1.787$
- 2) Excluding the outlier in the denominator
 - 12-24 age-to-age factor = $\frac{180+210+190+175}{100+90+105+100} = 1.911$
 - Notice there is no exclusion here since the outlier only exists in the numerator

3) Excluding the outlier in the numerator and the denominator

- 12-24 age-to-age factor = $\frac{180+190+175}{100+105+100} = 1.787$

What do we do if there are a significant number of outliers?

- This might indicate that the model is a poor fit to the data
- For the *GLM bootstrap model*, new parameters could be chosen, or the distribution of the error could be changed (i.e. the z parameter)
- For the *ODP bootstrap model*, an *N*-year weighted average could be used to provide a better model fit

Shapland also discusses *residual outliers*. Since the bootstrap process does not make a distributional assumption about the residuals, a large number of residual outliers may just mean that the residuals are highly skewed. If the skewness is real, then the outliers in the original dataset driving the residual outliers should be included in the fitting process to replicate the true nature of the residuals.

Heteroscedasticity

Since residuals are assumed to be i.i.d., we are able to apply a residual from one development period/accident period to the fitted loss in any other development period/accident period to produce sampled values. This is known as **homoscedasticity** (i.e., residuals have the same variance).

As mentioned before, if model assumptions are not quite met, it's possible for standardized residuals to exhibit more variability in certain development (or accident) periods than in other development (or accident) periods. This non-constant variance of the residuals is known as **heteroscedasticity**.

When assessing if heteroscedasticity is present, we must keep credibility in mind. There are fewer residuals as the development years become older, so comparing development years at the tail-end of the triangle is difficult.

Shapland covers three options to adjust for heteroscedasticity:

- Stratified sampling
- Calculating variance parameters
- Calculating scale parameters

Option 1: Stratified Sampling

Under this option, we do the following:

- Group development (or accident) periods with homogeneous variances
- Sample with replacement from the residuals in each group separately

An **advantage** of this method is that it is straightforward and easy to implement.

A **disadvantage** of this method is that some groups may only have a few residuals in them, which limits the amount of variability in the possible outcomes.

Option 2: Calculating Variance Parameters

Under this option, we do the following:

- Group development (or accident) periods with homogeneous variances
- Calculate the standard deviation of the standardized residuals in each of the “hetero” groups (each group has a different variance)
- Calculate the hetero-adjustment factor, h_i , for each group i as follows:

$$h_i = \frac{\text{stdev}(U_1^j r_{w,d}^H)}{\text{stdev}(U_i r_{w,d}^H)}$$

where $stdev(\cup_1^j r_{w,d}^H)$ is the standard deviation for all standardized residuals combined and $stdev(\cup_i r_{w,d}^H)$ is the standard deviation of the standardized residuals in group i . Notice that there is a total of j groups

- Multiply all residuals in group i by h_i as follows:

$$r_{w,d}^{iH} = \frac{q(w,d) - m_{w,d}}{\sqrt{m_{w,d}^z}} \times f_{w,d}^H \times h_i$$

All groups now have the same standard deviation, and we can sample with replacement from among all $r_{w,d}^{iH}$

- Divide the resampled residuals by the corresponding hetero-adjustment factor as follows:

$$q^{i*}(w,d) = \frac{r^*}{h_i} \times \sqrt{m_{w,d}^z + m_{w,d}}$$

This final adjustment ensures that the resampled triangles are consistent with the original triangle in terms of the residual distribution

Option 3: Calculating Scale Parameters

Under this option, we do the following:

- Group development (or accident) periods with homogeneous variances
- Calculate the total scale parameter using the unscaled Pearson residuals as follows:

$$\begin{aligned} \phi &= \frac{\sum r_{w,d}^2}{N - p} \\ &= \frac{\frac{N}{N - p} \sum r_{w,d}^2}{N} \end{aligned}$$

- Calculate the group i scale parameter using the unscaled Pearson residuals as follows:

$$\phi_i = \frac{\sum_{i=1}^{n_i} \left(\sqrt{\frac{N}{N-p}} \cdot r_{w,d} \right)^2}{n_i} = \frac{\frac{N}{N-p} \sum_{i=1}^{n_i} r_{w,d}^2}{n_i}$$

- Calculate the hetero-adjustment factor, h_i , for each group i as follows:

$$h_i = \frac{\sqrt{\phi}}{\sqrt{\phi_i}} = \frac{\sqrt{\frac{N}{N-p} \sum r_{w,d}^2}}{\sqrt{\frac{N}{N-p} \sum_{i=1}^{n_i} r_{w,d}^2}} = \frac{\sqrt{\frac{\sum r_{w,d}^2}{N}}}{\sqrt{\frac{\sum_{i=1}^{n_i} r_{w,d}^2}{n_i}}}$$

- Multiply all standardized residuals in group i by h_i as shown for the variance parameter option
- Divide the resampled residuals by the corresponding hetero-adjustment factor as shown for the variance parameter option

Example: Adjusting for Heteroscedasticity Using Variance Parameters

Given the following standardized residuals and standard deviations as of December 31, 2023:

AY	<u>Standardized Residuals as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	160	40	-90	-140	0
2020	-45	-30	300	120	
2021	-150	-120	-200		
2022	40	100			
2023	0				

Development Period/Range	Standard Deviation
12	113.71
24	94.65
36	262.74
48	183.85
12-24	99.14
12-36	140.36
24-60	148.68
36-60	185.52
12-60	133.82

First, let's group the development periods with homogeneous variances:

- Based on the individual development period standard deviations, there are two reasonable groupings: 12-24 and 36-60

Second, let's calculate the hetero-adjustment factors for the two groups:

- Recall that $h_i = \frac{stdev(U_1^j r_{w,d}^H)}{stdev(U_i r_{w,d}^H)}$
- $h_1 = \frac{133.82}{99.14} = 1.350$
- $h_2 = \frac{133.82}{185.52} = 0.721$

Third, let's apply the hetero-adjustment factors to the standardized residuals to produce hetero-adjusted standardized residuals:

<u>Hetero-Adjusted Standardized Residuals as of XX Months:</u>					
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	215.98 = 160(1.350)	54.00	-64.92 = -90(0.721)	-100.99	0
2020	-60.74	-40.50	216.40	86.56	
2021	-202.48	-161.99	-144.27		
2022	54.00	134.99			
2023	0				

Now that we have hetero-adjusted standardized residuals, we can randomly sample from the full set of standardized residuals above.

Suppose our sampling mechanism chooses the residual for 2021 at 36 months as the top left cell in the resampled triangle:

AY	<u>Resampled Hetero-Adjusted Standardized Residuals as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	-144.27	xxx	xxx	xxx	xxx
2020	xxx	xxx	xxx	xxx	
2021	xxx	xxx	xxx		
2022	xxx	xxx			
2023	xxx				

Further suppose that the fitted loss for 2019 at 12 months is \$250,000. Then, the resampled incremental loss is $q^{1*}(1,1) = \frac{-144.27}{1.350} \times \sqrt{250,000} + 250,000 = \$196,563$.

One thing students often ask is whether or not the corner zeroes should be included in the calculation of the standard deviations of ranges. In the Excel file provided with the paper, Shapland defaults to including the corner zeroes in the standard deviation calculations. For that reason, we recommend doing the same.

For example, the 36-60 standard deviation given in the problem is the sample standard deviation of -90, 300, -200, -140, 120, and 0. This can be calculated using the STDEV() function in Excel.

Example: Adjusting for Heteroscedasticity Using Scale Parameters

The same actuary that adjusted for heteroscedasticity using variance parameters wants to compare it to the adjustment using scale parameters. Given the following unscaled Pearson residuals as of December 31, 2023:

AY	Unscaled Pearson Residuals as of XX Months:				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	120	30	-50	-95	0
2020	-15	-20	225	90	
2021	-125	-100	-190		
2022	30	80			
2023	0				

- The standardized residuals and corresponding standard deviations are the same as the prior example

First, let's group the development periods with homogeneous variances:

- Based on the individual development period standard deviations, there are two reasonable groupings: 12-24 and 36-60

Second, let's calculate the total scale parameter using the unscaled Pearson residuals:

- Recall that $\phi = \frac{\sum(r_{w,d})^2}{N-p}$
- Determining N
 - $N = 15$ since there are 15 cells in the triangle
- Determining p
 - Since we were not told otherwise, it's reasonable to assume that the model has 5 AY parameters (i.e., $\alpha_1 - \alpha_5$) and 4 development period parameters (i.e., $\beta_2 - \beta_5$)
 - Since hetero-adjustment factors act as parameters, we need to consider them as well. Although there are two hetero-adjustment factors, we only include one as a parameter since we could simply rebase the factors so that one of them is 1.000
 - In general, we need to include "the number of hetero groups - 1" as parameters
 - Thus, $p = 5 + 4 + 1 = 10$

- $\phi = \frac{120^2 + (-15)^2 + \dots + 90^2 + 0^2}{15-10} = 31,040$

Third, let's calculate the scale parameters for each hetero group:

- Recall that $\phi_i = \frac{\frac{N}{N-p} \sum_{i=1}^{n_i} r_{w,d}^2}{n_i}$
- $\phi_1 = \frac{\frac{15}{15-10}(120^2 + (-15)^2 + \dots + (-100)^2 + 80^2)}{9} = 16,283$
- $\phi_2 = \frac{\frac{15}{15-10}((-50)^2 + (225)^2 + \dots + (90)^2 + 0^2)}{6} = 53,175$

Fourth, let's calculate the hetero-adjustment factors for the two groups:

- Recall that $h_i = \frac{\sqrt{\phi}}{\sqrt{\phi_i}}$
- $h_1 = \frac{\sqrt{31,040}}{\sqrt{16,283}} = 1.381$
- $h_2 = \frac{\sqrt{31,040}}{\sqrt{53,175}} = 0.764$
- Alternatively, we could have calculated h_i using the final version of the formula where the $\frac{N}{N-p}$ terms are cancelled out. The nice thing about this version of the formula is that it doesn't require us to calculate the number of parameters

Fifth, let's apply the hetero-adjustment factors to the standardized residuals to produce hetero-adjusted standardized residuals:

<u>Hetero-Adjusted Standardized Residuals as of XX Months:</u>					
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	220.91 = 160(1.381)	55.23	-68.76 = -90(0.764)	-106.96	0
2020	-62.13	-41.42	229.21	91.68	
2021	-207.10	-165.68	-152.80		
2022	55.23	138.07			
2023	0				

Now that we have hetero-adjusted standardized residuals, we can randomly sample from the full set of standardized residuals above.

Suppose our sampling mechanism chooses the residual for 2021 at 36 months as the top left cell in the resampled triangle:

AY	<u>Resampled Hetero-Adjusted Standardized Residuals as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	-152.80	xxx	xxx	xxx	xxx
2020	xxx	xxx	xxx	xxx	
2021	xxx	xxx	xxx		
2022	xxx	xxx			
2023	xxx				

Further suppose that the fitted loss for 2019 at 12 months is \$250,000. Then, the resampled incremental loss is $q^{1*}(1,1) = \frac{-152.80}{1.381} \times \sqrt{250,000} + 250,000 = \$194,663$.

Before we end our discussion on heteroscedasticity, we should discuss its impact on the incorporation of process variance into future incremental values:

- Adjusting for heteroscedasticity using variance parameters
 - The total scale parameter should be adjusted to a group-specific scale parameter using $\phi_i = \frac{\phi}{h_i^2}$
 - Thus, when simulating future incremental losses, we draw from a gamma distribution with mean $m_{w,d}$ and variance $\phi_i m_{w,d}$
- Adjusting for heteroscedasticity using scale parameters
 - The group-specific scale parameters should be used when adding process variance to future incremental losses
 - Thus, when simulating future incremental losses, we draw from a gamma distribution with mean $m_{w,d}$ and variance $\phi_i m_{w,d}$

Heteroecthesious Data

Heteroecthesious data refers to incomplete or uneven exposures at interim evaluation dates. The two most common types of heteroecthesious data triangles are as follows:

- Partial first development period triangles
- Partial last calendar period triangles

Shapland discusses both of these from the viewpoint of the **ODP bootstrap model**.

Partial First Development Period Data

This occurs when the **first development column has a different exposure period** than the rest of the columns. For example, with annual data evaluated as of June 30, partial first development period data would have development periods ending at 6 months, 18 months, 30 months, etc.

How does this impact the bootstrap process?

- Parameterizing the model
 - **Not a problem** since the unscaled Pearson residuals use the square root of the fitted value to make them all “exposure independent”
- Projecting future incremental values
 - For each sample incremental triangle, **we must reduce** the projected future values for the latest AY to correspond to the earned exposure
 - For **example**, if the latest AY is only 6 months old, then the future values for the latest AY must be cut in half. No adjustments are needed for the remaining AYs since they are fully earned (i.e., at least 12 months old)
 - After adjusting the future values for the latest AY, we incorporate process variance using a gamma distribution as discussed before

Partial Last Calendar Period Data

This occurs when the latest diagonal is less than an annual period. For example, partial last calendar period data would have development periods ending at 12 months, 24 months, 36

months, etc. for all of the data in the triangle except the latest diagonal, which might have development periods ending at 6 months, 18 months, 30 months, etc.

How does this impact the bootstrap process?

- Parameterizing the model
 - We must **annualize the exposures in the latest diagonal** to make them consistent with the rest of the triangle
 - The fitted values (and consequently, the residuals) are based on this annualized triangle
- Projecting future incremental values
 - For each annualized sample incremental triangle, age-to-age factors are calculated and **interpolated**
 - The latest diagonal in the sample triangle is **adjusted back** to the partial period present in the original data
 - The cumulative values are then multiplied by the interpolated age-to-age factors to project future cumulative & incremental values
 - Once again, **we must reduce** the projected future values for the latest AY to correspond to the earned exposure
 - After adjusting the future values for the latest AY, we incorporate process variance using a gamma distribution as discussed before

Exposure Adjustment

A common issue in real data is exposures that have changed dramatically over the years.

Examples include LOBs with rapid growth or run-off business.

Shapland discusses exposure adjustments under both the ODP bootstrap and GLM bootstrap models.

ODP Bootstrap Model

To adjust for this, we divide the claim data by earned exposures for each accident year. Then, the entire bootstrap process is run on the exposure-adjusted data.

After the “incorporate process variance step” is completed, we multiply the results by the earned exposures to restate them in terms of total values.

GLM Bootstrap Model

Similar to the ODP bootstrap model, the GLM bootstrap model is fit to the exposure-adjusted losses. However, there are two differences in the model fit as compared to the ODP bootstrap model:

- The fit is **exposure-weighted**, meaning that exposure-adjusted losses with higher exposure are assumed to have a lower variance
- The exposure adjustment could allow fewer AY parameters to be used

Once the fit is complete, the entire bootstrap process is run on the exposure-adjusted data. The GLM fits to the sample incremental triangles would need to be exposure-weighted fits to align with the fit to the original triangle.

After the “incorporate process variance step” is completed, we multiply the results by the earned exposures to restate them in terms of total values.

Tail Factors

Tail factors are commonly used to extrapolate development beyond the end of the triangle. We can incorporate them into the bootstrapping process.

ODP Bootstrap Model

Instead of using a deterministic tail factor, we can assign a distribution (ex. normal or lognormal) to the tail factor parameter. This makes the tail factor stochastic within the ODP bootstrap model.

Since the tail factor represents an “accumulation” of incremental factors, a single factor may not produce accurate incremental values. Thus, it might be useful to split the tail factor into its incremental counterparts using a decay model.

For **example**, suppose that the selected 60-ultimate tail factor is 1.05. This could be extrapolated over three years using a decay model. If the decay percentage is 50%, the 60-ultimate, 72-ultimate, and 84-ultimate tail factors would be 1.05, $1.025 (1 + 0.05(0.50))$, and $1.013 (1 + 0.025(0.50))$. This is what Shapland does in the Excel files provided with the text.

A **rule of thumb** for the tail factor standard deviation is 50% or less of the tail factor minus 1. So, for our example above, the standard deviation should be less than or equal to $0.50(1.05 - 1) = 0.025$.

GLM Bootstrap Model

Earlier in the outline, we mentioned that modeling with fewer development trend parameters means that the last parameter is assumed to continue past the end of the triangle. This effectively gives us a tail without actually having to specify a tail factor.

To implement this, we simply assume that the final development period will continue to apply incrementally until its effect on the future incremental claims is negligible.

If we fit a GLM bootstrap model with development year and calendar year parameters, we would assume both continue past the end of the sample triangle until the effects on the future incremental claims are negligible.

Fitting a Distribution to the Residuals

Due to the limited number of observations in a loss triangle, it's difficult to determine whether the most extreme observation is a 1-in-100 event, 1-in-1000 event, etc.

If we believe that extreme observations are not captured well in the loss triangle, we can parameterize a distribution for the residuals (such as normal) and resample using the distribution. This is known as the **parametric bootstrap**.

V. Diagnostics

In order to assess the quality of a stochastic model, various diagnostic tests should be run. The diagnostic tools are designed for three purposes:

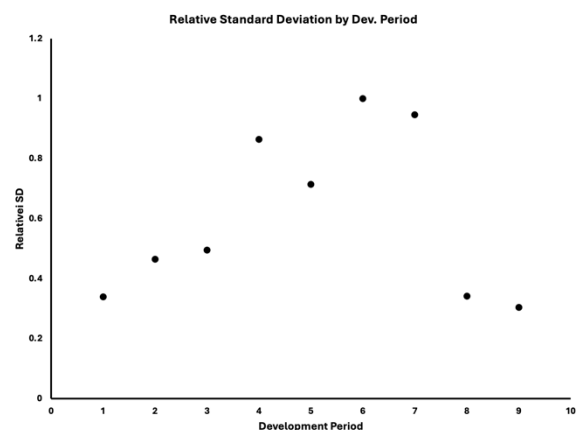
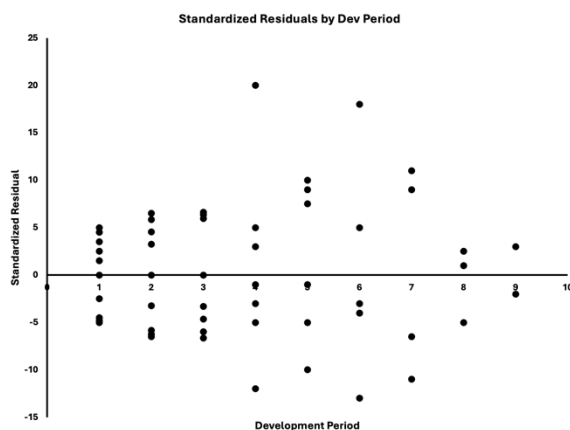
- 1) Test various assumptions in the model
- 2) Gauge the quality of the model fit
- 3) Guide the adjustment of model parameters

Residual Graphs

Residual graphs can be used to test the assumption that residuals are independent and identically distributed. We can graph the standardized residuals by development period, accident period, calendar period, or fitted incremental loss.

In each case, we should be able to draw a **relatively horizontal line** through the residuals. In addition, we want the residuals to show a **constant spread** across the graph. If the spread is non-constant, then heteroscedasticity exists, and hetero adjustments need to be made before bootstrapping the residuals.

To help visualize how residuals should be grouped for hetero adjustments, we can graph relative standard deviations and look for natural groupings. For example, consider the following plots:



In the plots above, there seem to be three groups of residuals (periods 1-3, periods 4-7, and periods 8-9). We can calculate hetero-adjustment factors for these groups and adjust them to a common standard deviation. After adjusting the residuals to a common standard deviation, we can run the graphs again to ensure that the residuals are homoscedastic.

Normality Test

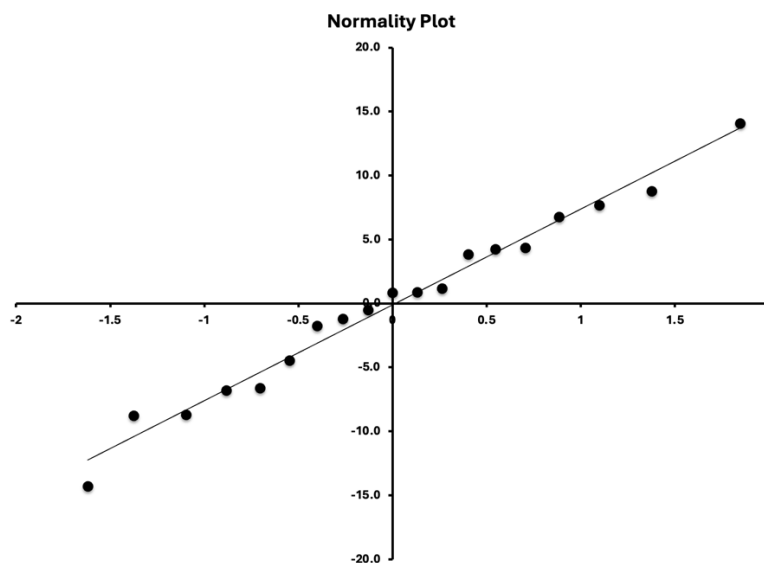
Although we do not require the residuals to be normally distributed, it's still helpful to compare residuals against a normal distribution. This allows us to compare parameter sets and assess the skewness of the residuals.

We can assess normality using plots and test values.

Normality Plot

A normality plot shows the relationship between theoretical quantiles of a standard normal distribution (i.e., the x -axis) and the empirical quantiles of the observed data (i.e., the y -axis). In our case, the observed data are the standardized residuals.

If the data is normally distributed, then the points in the plot will lie along a diagonal line. For **example**, consider the following normality plot for a set of standardized residuals:



In the graph above, the data points are tightly distributed around the diagonal line. This suggests that the standardized residuals are normally distributed.

Test Values

A number of test values can be used to assess normality:

- **p-value** from Shapiro's Test for Normality
 - If the residuals are normally distributed, the p -value should be large (i.e., greater than 5%)
- **R^2**
 - The diagonal line of the normality plot can be thought of as the regression line through the points
 - If the residuals are normally distributed, the R^2 should be close to 1
- **AIC/BIC**
 - p -values and R^2 fail to penalize for the number of parameters used in the model
 - To address this limitation, we can look at AIC and BIC. Both tests use the difference between each residual and its normal counterpart from the normality plot to calculate the Residual Sum Squared (RSS)
 - AIC and BIC are defined as follows:

$$\begin{aligned} AIC &= 2p + n \times \left[\ln \left(\frac{2 \times \pi \times RSS}{n} \right) + 1 \right] \\ BIC &= n \times \ln \left(\frac{RSS}{n} \right) + p \times \ln(n) \end{aligned}$$

- If residuals are normally distributed, the AIC and BIC should be small

For the normality plots and test values, we can run them before and after making heteroscedasticity adjustments. This enables us to see if applying the hetero groupings improved the fit of the model.

Keep in mind that failing the normality assumption does not mean the bootstrap model is poor. It just means that the residuals are not normally distributed.

Outliers

Shapland uses **box-whisker plots** of the standardized residuals to identify outliers in the data:

- The border of the box is formed using the inter-quartile range (the 25th to 75th percentiles)
- The median is graphed within the box
- The whiskers extend to the largest values within three times the inter-quartile range
- Values beyond the whiskers are considered outliers and are identified with a point

Once outliers are identified, judgment must be used to decide if they should be removed from the model fit:

- If the outliers represent extreme events *not expected to happen again*, they can be given *zero weight in the GLM* (or alternatively, we could remove them from the age-to-age factor calculation as described earlier)
- If the outliers represent extreme events that could happen again, they should be kept in the model

When residuals are not normally distributed, outliers tend to be more common. In this case, we don't necessarily want to remove them because they are representative of the shape of the data.

Parameter Adjustment

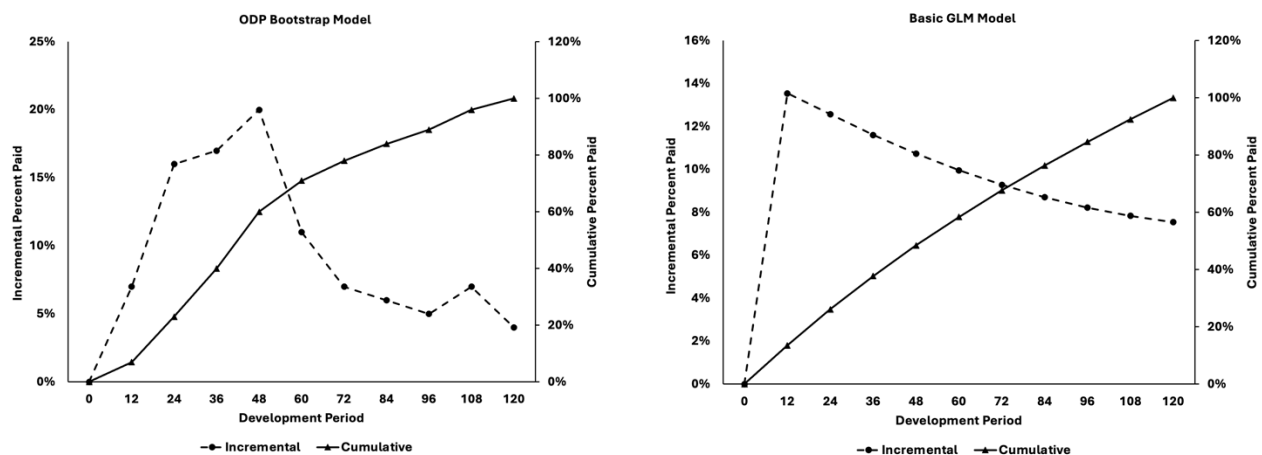
Including parameters for every accident period and development period may lead to an over-parameterized model (i.e., over-fitting the noise in the data). This is effectively what the ODP bootstrap model is doing.

The benefit of the GLM bootstrap model is that we can reduce the number of parameters until we find the optimal mix. The *principle of parsimony* states that a model with fewer parameters is preferred as long as the goodness of fit is not markedly different.

To find the optimal mix of parameters in the GLM bootstrap model, we do the following:

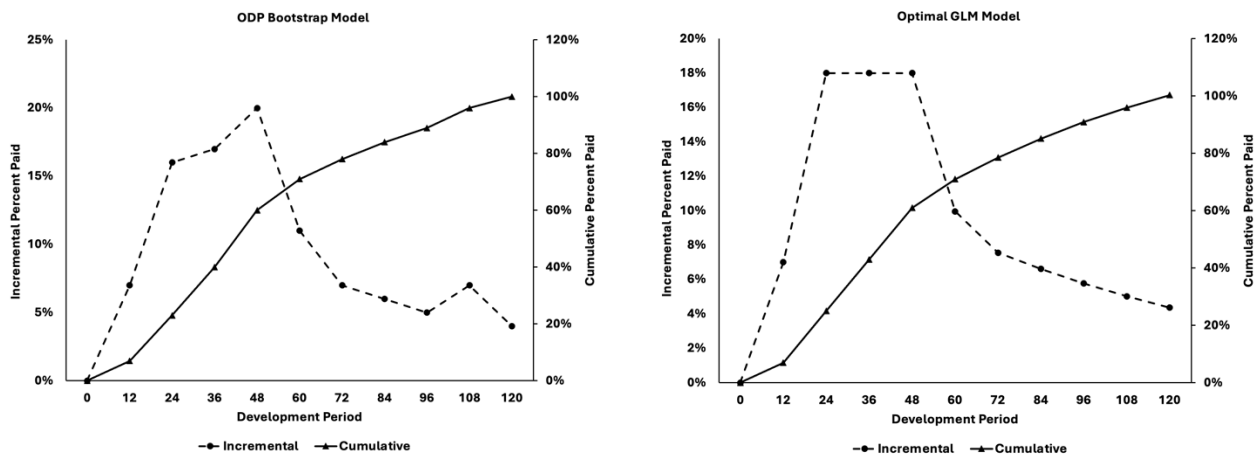
- Start with a “basic” GLM model which includes one parameter for accident, development, and calendar periods
- Check the residual plots for the basic model
 - If the residuals by accident period, development period, and calendar period are not randomly scattered around zero, then we should consider adding parameters
 - In addition, if certain parameters are not statistically significant, we should remove them
 - The implied development pattern should look like a smoothed version of the chain-ladder development pattern
- We keep adding/removing parameters until we see a proper residual plot

The following graphs show the implied development patterns for the ODP bootstrap model and a “basic” GLM model:



In the ODP bootstrap model, we have a different factor for each development period. This produces a jagged pattern in the incremental paid percentages. In the basic GLM model, we have a single development period parameter (i.e., a development period trend). This over-smooths the development pattern resulting in a poor fit.

The following graphs show the implied development patterns for the ODP bootstrap model and an “optimal” GLM model:



In the optimal GLM model, we have a set of parameters that balances parsimony and goodness of fit. In this case, the implied development pattern is a smoothed version of the ODP bootstrap model development pattern that still achieves the overall shape.

As we mentioned earlier, the optimal model should show no pattern in the residuals, indicating that all of the significant development period trends have been captured.

Estimated Unpaid Model Results

For each bootstrap model run, descriptive statistics (ex. mean, percentiles, standard deviation) relating to the bootstrap simulations can provide an additional diagnostic tool.

For the **estimated unpaid model results by AY**, Shapland recommends zooming in on the standard error and coefficient of variation (CoV).

Standard Error

We should see the following trends in the standard error:

- Standard error should *increase* from older to more recent years
 - This is because the standard error should follow the magnitude of the results

- Total standard error should be larger than the standard error for any individual year

CoV

We should see the following trends in the CoV

- CoV should generally decrease from older to more recent years
 - This is because older accident years have fewer payments remaining, which causes all of the variability to be reflected in the coefficient
 - For more recent years, random variations in the remaining payments tend to offset each other, which reduces overall variability
- Total CoV should be less than any individual year

The CoV may rise in the most recent years due to a couple of reasons:

- With an increasing number of parameters in the model, parameter uncertainty increases when moving from the oldest years to the most recent years. This parameter uncertainty may overpower the process uncertainty, causing an increase in variability
- The model may simply be overestimating the variability in the most recent years. In this case, the BF or Cape Cod models may need to be used in place of the CL model

Lastly, since AY are assumed to be independent, we should note that the standard error or CoV for all years combined will be less than the sum of the standard error or CoV for individual years.

These points should not be used as diagnostic tests because they should be true of all models, including ones that are unreasonable.

Mean & Standard Deviation of Incremental Values

The mean and standard deviation of each incremental value (aggregated across all simulations) provide another diagnostic tool.

If the future mean or standard deviation values appear inconsistent, then this may help identify the source of any CoV issues that have surfaced (i.e., is it the mean or standard deviation causing the problem?)

Example: Reviewing the Mean & Standard Deviation of Incremental Values

Suppose we have the following incremental means from the simulated data after running the ODP bootstrap model as of December 31, 2023:

<u>Simulated Incremental Means (\$) as of XX Months:</u>					
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	2,005	1,502	800	402	207
2020	2,113	1,555	753	352	178
2021	2,225	1,409	727	372	220
2022	2,497	1,713	844	368	196
2023	2,365	1,638	776	394	212

In the table above, the means down every development period appear consistent. There are no future means that seem out of place.

Further suppose we have the following incremental standard deviations from the simulated data after running the ODP bootstrap model:

<u>Simulated Incremental SDs (\$) as of XX Months:</u>					
AY	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	1,012	327	161	142	125
2020	1,216	313	202	148	146
2021	985	298	183	164	181
2022	1,165	304	245	205	170
2023	1,106	500	320	231	191

In the table above, the standard deviations down every development period show some inconsistencies. In particular, the standard deviations of the future incremental values appear much higher than the standard deviations for the historical period.

The resulting CoVs are as follows:

AY	<u>Simulated CoVs as of XX Months:</u>				
	12 mo.	24 mo.	36 mo.	48 mo.	60 mo.
2019	50.5%	21.8%	20.1%	35.3%	60.4%
2020	57.5%	20.1%	26.8%	42.0%	82.0%
2021	44.3%	21.1%	25.2%	44.1%	82.3%
2022	46.7%	17.7%	29.0%	55.7%	86.7%
2023	46.8%	30.5%	41.2%	58.6%	90.1%

In the table above, the higher standard deviations are driving higher CoVs for the future incremental values. In addition, the inconsistencies appear to be more severe in the latest two accident years. This is not completely unexpected since the CoV may rise in the latest few accident years due to the reasons mentioned earlier.

VI. Using Multiple Models

Multiple stochastic models should be run when performing a bootstrap analysis. Each model should go through diagnostic testing and all model output should be reviewed. Once the models have been reviewed, weights should be assigned to the models to combine results.

There are two methods for combining the results for multiple models:

- Run models with the same random variables
- Run models with independent random variables

Run Models with the Same Random Variables

Each model is run with the exact same random variables (i.e., random residuals in terms of position).

For **example**, suppose two models are run and each produces 50 residuals. When performing the random sampling portion of the bootstrap process, the first model chooses residual number 40

from the list of possible residuals and uses it in the first cell of the sample triangle. The second model would also choose residual number 40 from its list of possible residuals and use it in the first cell of its sample triangle. This process continues until both sample triangles are built.

Once all of the models have been run, the incremental values for each model are weighted together for each iteration by accident year.

This method for combining models causes correlation in model results since each model is run using the same set of random residuals.

Run Models with Independent Random Variables

Each model is run with its own random variables (i.e., different samples of random residuals in terms of position). In this case, the position of the random residual in the first model has no impact on the position of the random residual in other models. Once all of the models have been run, the weights are used to select a model for each iteration by accident year.

For **example**, suppose we are estimating unpaid losses using CL and BF bootstrap models. Further suppose the weights are 25% and 75%, respectively. For each iteration by accident year, we would draw a uniform random variable on (0, 1). If the drawn uniform random variable is “< 0.25”, then the CL unpaid losses would be used for that iteration/AY combination. Otherwise, the BF unpaid losses would be used.

The result is a weighted mixture of models, where the CL and BF model results represent approximately 25% and 75% of the iterations by AY, respectively.

For detailed examples of each of the “model combining” methods above, see the Cookbook.

Other Notes on Weighting the Results of Different Models

Shapland discusses some other miscellaneous items regarding the weighting of results from different modes:

- The process of weighting the results of different models produces an actuarial best estimate of a distribution
- The weights can be determined using Bayesian methods that account for the quality of each model's forecasts
- The weights may produce results by accident year that do not reconcile with case reserves
 - For **example**, the weighted results for an accident year may produce negative IBNR. If negative IBNR is not reasonable for that particular LOB, then the weighted distributions can be “adjusted”
 - If the actuary believes that the shape and width of the distribution is appropriate, then they can add a fixed amount to the accident year to produce positive IBNR
 - Alternatively, if the actuary believes that the shape of the distribution is appropriate but would like to adjust the width, they can multiply the results by a factor to produce positive IBNR

Fitting Distributions to the Data

Once model results have been combined and tabulated, distributions can be fit to the data to smooth results.

For **example**, we can fit normal, lognormal, and gamma distributions to the total unpaid claim distributions.

The smoothed results can be used to:

- Assess the quality of the fit
- Parameterize a DFA (i.e., dynamic financial analysis) model
- Estimate extreme values
- Estimate TVaR (i.e., Tail Value at Risk)

The benefit of using smoothed results is that some of the random noise is prevented from distorting the calculations of specific metrics.

Estimated Cash Flow Results

Similar to the estimated unpaid model results, we can review a model's output by future calendar year. This is equivalent to reviewing "estimated cash flow results."

The **main difference** between future CYs (cash flows) and historical AYs (unpaid losses) is that the standard errors and CoVs move in opposite directions:

- For historical AYs, standard errors increase and CoVs decrease as we move from older to more recent years
- For future CYs, standard errors decrease and CoVs increase as we move further out into the future
 - Ex. If we are reviewing future cash flows for unpaid claims as of 12/31/23, the first future CY is 1/1/24 – 12/31/24

This makes intuitive sense for CYs. As we move further out into the future, the CY cash flows will decrease, which also leads to a decrease in the absolute standard error. However, relative to its mean, the variability increases substantially.

Estimated Ultimate Loss Ratio Results

The estimated ultimate loss ratios by AY are calculated using all simulated values, not just the values beyond the latest diagonal of the historical triangle:

- Since the simulated values represent additional possibilities of what could have happened in the past, we are able to estimate the complete variability in the loss ratio from day one until all claims are paid and settled
 - If we are **only interested in the future volatility of the loss ratios**, we can add the estimated unpaid claim estimates to the actual cumulative paid values and divide by the premiums
- By using all simulated values, we should have a fairly constant CoV across the accident years (since the standard errors should be proportional to the mean)
 - An increasing CoV would indicate large parameter uncertainty as described earlier

Distribution Graphs

We can create a total unpaid distribution histogram by dividing the range of all values generated from the simulation into equal size buckets and then counting the number of simulations that fall within each bucket (Shapland uses 100 buckets in the paper).

Dividing the number of simulations in each bucket by the total number of simulations results in a frequency/probability for each bucket.

Since simulation results tend to appear jagged, a Kernel density can be fit to the data to provide a smoothed distribution.

We can also compare the individual models used to create the best estimate by graphing each model's Kernel density function on the same graph.

Correlation

Once best estimates have been calculated for each business segment, results can be aggregated to the business unit level. Since business segments tend have specific correlations, results cannot simply be added together.

Shapland described two useful correlation process for the bootstrap model:

- 1) Location mapping
- 2) Re-sorting

Location Mapping

For each iteration:

- Sample residuals for segment 1
- Track the location in the residual triangle where each sampled residual was taken
- For all other segments, sample the residuals from their residuals triangles using the same locations

Advantages of location mapping are as follows:

- Easily implemented
- Does not require an estimated correlation matrix

Disadvantages of location mapping are as follows:

- Requires all business segments to have the same size data triangles with no missing data
- Since the correlation of the original residuals is used, we cannot test other correlation assumptions for stress testing purposes

Re-sorting

Re-sorting relies on algorithms such as Iman-Conover (rank correlation algorithm) or copulas to induce a desired correlation.

For **example**, we could induce correlation among business segments by re-sorting the residuals until the rank correlation between each business segment matches the desired correlation specified by a correlation matrix.

Advantages of re-sorting are as follows:

- Data triangles can be different shapes/sizes by segment
- Can use different correlation assumptions
- Different correlation assumptions may have other beneficial impacts on the aggregate distribution
 - Ex. We can use a copula with a heavy tail distribution to strengthen the correlation between segments in the tails, which is important for risk-based capital modeling

I. Introduction

This paper focuses on developing ultimate excess losses, which can then be used to determine excess IBNR.

In this paper, excess losses refer to losses excess of a per-occurrence deductible or an aggregate limit. As usual, losses below the per-occurrence deductible are retained by the insured and losses above the per-occurrence deductible are paid by the insurer.

Once aggregate deductible losses reach the aggregate limit, the insured stops retaining losses, and the insurer pays for everything else.

Advantages of a high deductible program include:

- Achieves price flexibility while passing additional risk to larger insureds
- Reduces residual market charges and premium taxes
- Gives cash flow advantages to insured (insurer pays claim first and must seek reimbursement from insured)
- Provides incentive for insureds to control losses while protecting them from large losses
- Allows “self-insurance” without subjecting insureds to demanding state requirements

II. Development Approaches

Siewert explores the following development approaches:

- 1) Loss ratio approach
- 2) Implied development approach
- 3) Direct development approach
- 4) BF approach

Loss Ratio Approach

The loss ratio approach should be used in the following situations:

- When no data is available
- For immature years where data is sparse

The loss ratio approach requires a database of individual accounts and pricing elements, including an estimate of the full coverage loss ratio, as well as estimates of excess losses for both occurrence and aggregate limits.

Full Coverage Loss Ratio

To determine the full coverage loss ratio, we can use one of the following approaches:

- Use company experience by state, reflecting the individual account's premium distribution
- Supplement company experience with industry experience if credibility is a problem

Per-Occurrence Excess Losses

To calculate the per-occurrence excess losses, we first need to estimate the per-occurrence charge (i.e., excess ratio):

- **Approaches for estimating** the per-occurrence charge include estimating the excess ratio based upon company experience and/or using an industry excess ratio

The final estimate of per-occurrence excess losses is as follows:

$$\boxed{\text{Per - Occurrence Excess Losses} = P \cdot E \cdot \chi}$$

where P = premium, E = expected loss ratio, and χ = per-occurrence charge.

Per-Aggregate Excess Losses

To calculate the per-aggregate excess losses, we first need to estimate the per-aggregate charge (i.e., aggregate ratio):

- **One approach** is to use the NCCI Table M, which reflects size of the account, deductible, state severity relativities, prospective rating period and rating plan parameters

The final estimate of per-aggregate excess losses is:

$$\boxed{\text{Per} - \text{Aggregate Excess Losses} = P \cdot E \cdot (1 - \chi) \cdot \phi}$$

where ϕ = per-aggregate charge.

Notice that the factor ϕ applies to the expected losses below the per-occurrence deductible. This is because the aggregate limit applies to losses below the deductible (i.e., the ones retained by the insured).

Applying these two formulas to each account and summing them leads to an estimate of ultimate AY losses.

Advantages of the loss ratio approach are as follows:

- Can be used when no data is available or when data is immature
- Loss ratio estimates can be consistently tied to pricing programs
- Relies on a more credible pool of company and industry experience

Disadvantages of the loss ratio approach are as follows:

- Ignores actual emerging experience (not as useful for mature years)
- May not properly reflect account characteristics, since development may emerge differently due to the exposures written

Example: Loss Ratio Approach

Given the following for a book of business:

- Premium = \$1,000,000
- ELR = 0.50
- Excess ratio = 0.10
- Aggregate ratio = 0.02

The total expected losses are equal to $P \cdot E = 1,000,000(0.50) = \$500,000$.

The insurer pays the losses above the per-occurrence deductible and the losses above the aggregate limit:

- The losses above the per-occurrence deductible are $P \cdot E \cdot \chi = 1,000,000(0.50)(0.10) = \$50,000$
- The losses above the aggregate limit are $P \cdot E \cdot (1 - \chi) \cdot \phi = 1,000,000(0.50)(1 - 0.10)(0.02) = \$9,000$
- In total, the insurer pays $\$50,000 + \$9,000 = \$59,000$

The insured pays the rest of the losses, which are equal to $\$500,000 - \$59,000 = \$441,000$.

Before moving on, we want to point out that we will ignore per-aggregate excess losses for the remaining development approaches. So, when you see the words “excess” or “limited,” we are referring to losses “excess” of the per-occurrence deductible or losses “limited” by the per-occurrence deductible. We will revisit per-aggregate excess losses later on.

Implied Development Approach

The implied development approach works as follows:

- Develop full coverage losses to ultimate

- Develop deductible losses (i.e., all losses limited by the deductible) to ultimate by applying development factors that reflect various inflation indexed limits
- Determine ultimate excess losses by subtracting the limited ultimate losses from the full coverage ultimate losses

Consistency of Tail Factors

When **determining tail factors**, make sure the full coverage tail factor is consistent with the limited loss tail factors. We do not want to develop limited losses beyond unlimited losses.

Indexing Limits for Inflation

When calculating development factors for various deductibles, we need to index the limits for inflation. This keeps the proportion of deductible/excess losses constant about the limit from year to year.

When determining development factors, indexing allows us to combine different experience years.

Siewert mentions a couple of ways to determine the index value:

- Fit a line to average severities over a long-term history
- Use an index that reflects the movement in annual severity changes

Example: Indexing Limits for Inflation

An insurer has a deductible of \$250,000 and inflation is equal to 5%. The insurer experienced five losses in AY 2023 and AY 2024. The AY 2024 losses are exactly 5% higher than the AY 2023 losses. Let's see what happens if the deductible is not indexed for inflation. Given the following loss information (in \$00):

Loss ID	AY 2024 - \$250,000 Ded.			AY 2023 - \$250,000 Ded.			AY 2023 - \$238,095 Ded.		
	Total Loss	Ded. Loss	Excess Loss	Total Loss	Ded. Loss	Excess Loss	Total Loss	Ded. Loss	Excess Loss
1	2,100	2,100	0	2,000	2,000	0	2,000	2,000	0
2	3,150	2,500	650	3,000	2,500	500	3,000	2,381	619
3	5,250	2,500	2,750	5,000	2,500	2,500	5,000	2,381	2,619
4	12,600	2,500	10,100	12,000	2,500	9,500	12,000	2,381	9,619
5	315	315	0	300	300	0	300	300	0
Total	23,415	9,915	13,500	22,300	9,800	12,500	22,300	9,443	12,857
Ratio	73.4%			78.4%			73.4%		

For AY 2024, the ratio of deductible to excess losses is 73.4%. When we apply the same \$250,000 deductible to the AY 2023 losses (which have been deflated by 5%), the ratio of deductible to excess losses is 78.4%.

To ensure that the ratio of deductible to excess losses remains constant from year to year, the current deductible needs to be deflated before applying it to the AY 2023 losses. As we see in the table, the ratio of deductible to excess losses for AY 2023 is 73.4% using a deflated deductible of $\frac{250,000}{1.05} = 238,095$. These accident years can now be combined for analysis.

Advantages of the implied development approach are as follows:

- Provides an estimate of excess losses at early maturities even when excess losses have not emerged
- Development factors for limited losses are more stable than those determined for excess losses
- Estimating deductible losses helps determine the asset value represented by service revenue (described in detail later)

A **disadvantage** of the implied development approach is that it does not explicitly recognize excess loss development.

Example: Implied Development Approach

Given the following information for a book of business:

- AY 2024 unlimited (i.e., full coverage) losses as of 12/31/24 = \$1,000,000
- AY 2024 limited (i.e. deductible) losses as of 12/31/24 = \$960,000
- Unlimited 12-ult. factor = 2.40
- Limited 12-ult. factor = 2.30

Using the numbers above, we obtain the following:

- AY 2024 ultimate unlimited losses = $1,000,000(2.40) = \$2,400,000$
- AY 2024 ultimate limited losses = $960,000(2.30) = \$2,208,000$ (*insured* responsible)
- AY 2024 ultimate excess losses = $2,400,000 - 2,208,000 = \$192,000$ (*insurer* responsible)

Thus, the AY 2024 excess IBNR is $192,000 - (1,000,000 - 960,000) = \$152,000$.

Direct Development Approach

The direct development approach focuses on excess development directly. Given development factors for limited and full coverage losses, excess loss development factors are calculated so that excess factors in conjunction with limited development factors *balance back* to full coverage factors.

An **advantage** of the direct development approach is that it explicitly recognizes excess loss development.

Disadvantages of the direct development approach are as follows:

- Excess factors tend to be overly leveraged and extremely volatile
- If excess losses have not yet emerged, we can't estimate IBNR (nothing to apply the factors to!)

In the next section, we will look at a development model that builds out formulas for excess loss development factors that maintain consistency with the limited LDFs. Thus, we will save the Direct Development Approach example for then.

BF Approach

As a credibility-weighting between the direct development method and the expected method, the BF approach relies on weighting indications based upon actual experience (i.e., direct development approach) with expected values (i.e., loss ratio approach).

Under the BF approach, we estimate ultimate excess losses as follows:

$$\begin{aligned} L &= O_t \cdot LDF_t \cdot Z + E \cdot (1 - Z) \\ &= O_t + E \cdot \left(\frac{LDF_t - 1}{LDF_t} \right) \end{aligned}$$

where L = ultimate excess losses, O_t = observed excess loss at time t , LDF_t = excess age-to-ultimate development factor, $Z = \frac{1}{LDF_t}$ = credibility, and E = expected ultimate excess loss.

Advantages of the BF approach are as follows:

- We can determine liabilities either directly or indirectly
- Gives us the ability to tie into pricing estimates for recent years where excess losses have yet to emerge
- Provides more stable estimates over time

A **disadvantage** of the BF approach is that it ignores actual experience to the extent of the complement of credibility (may need to find weights that are more responsive to the actual experience).

We will look at an example in the next section.

III. Development Model

The point of this section is to show how to calculate the LDFs needed to implement the development approaches above. Siewert then caps it off with a discussion of a distributional model.

Let's start with a discussion on tail factors.

Tail Factors

Siewert computes a tail factor based on an inverse power curve to develop limited losses to ultimate. Since the inverse power curve produces age-to-age factors, they must be compounded to produce a single tail factor. Mathematically, an age-to-age factor, y , is defined as follows:

$$y = 1 + a \cdot (t + c)^{-b}$$

where a , b , and c are given parameters and t represents the beginning of the development period in years. For example, if we are calculating the age-to-age factor from 60-to-72 months, then $t = 5$ years.

Siewert's tail factor procedure is as follows:

- Fit the inverse power curve on unlimited age-to-age factors to project unlimited ultimate losses
- Select a time (ultimate age) at which the projection should stop (since the inverse power curve continues forever)
- Once the ultimate age is chosen for unlimited losses, fit the curve to each deductible limit and extend that to a common maturity

The **beauty of this method** is that it is consistent for each limit, and it produces uniformly decreasing tail factors.

One **problem with this method** is that bias exists due to extending each limit to the same maturity (lower limits should fully develop much sooner than higher limits).

Example: Using Tail Factors with the Implied Development Approach

Given the following loss triangles for a book of business as of December 31, 2023:

AY	<u>Cumulative Unlimited Reported Losses (in \$000) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	2,000	3,200	3,840
2022	1,800	2,880	
2023	2,400		

AY	<u>Cumulative Limited Reported Losses (in \$000) as of XX Months:</u>		
	12 mo.	24 mo.	36 mo.
2021	1,600	2,400	2,640
2022	1,500	2,250	
2023	1,900		

- The tail factor for unlimited losses is 1.12
- To model development in the tail for limited losses, an inverse power curve is used with the following parameters:
 - $a = 0.16$
 - $b = 1.15$
 - $c = -0.40$
- Loss development ends at 60 months for both limited and unlimited losses

First, let's calculate the age-to-age factors for the unlimited losses:

- 12-24: $\frac{3,200+2,880}{2,000+1,800} = 1.60$
- 24-36: $\frac{3,840}{3,200} = 1.20$

Second, let's calculate the ultimate unlimited losses:

- AY 2021: $3,840(1.12) = 4,301$

- AY 2022: $2,880(1.20)(1.12) = 3,871$
- AY 2023: $2,400(1.60)(1.20)(1.12) = 5,161$

Third, let's calculate the age-to-age factors for the limited losses:

- 12-24: $\frac{2,400+2,250}{1,600+1,500} = 1.50$
- 24-36: $\frac{2,640}{2,400} = 1.10$

Fourth, let's calculate the tail factor for the limited losses:

- 36-48: $1 + a \cdot (t + c)^{-b} = 1 + 0.16(3 - 0.40)^{-1.15} = 1.053$
- 48-60: $1 + 0.16(4 - 0.40)^{-1.15} = 1.037$
- Tail factor: $1.053(1.037) = 1.092$

Fifth, let's calculate the ultimate limited losses:

- AY 2021: $2,640(1.092) = 2,883$
- AY 2022: $2,250(1.10)(1.092) = 2,703$
- AY 2023: $1,900(1.50)(1.10)(1.092) = 3,423$

Finally, let's calculate the ultimate excess losses using the implied development approach:

- AY 2021: $4,301 - 2,883 = 1,418$
- AY 2022: $3,871 - 2,703 = 1,168$
- AY 2023: $5,161 - 3,423 = 1,738$

To calculate the excess IBNR by AY, we simply subtract the reported excess losses (i.e., unlimited losses – limited losses) from the ultimate excess losses.

Severity Relativities

There is a relationship between limited/excess LDFs and unlimited LDFs using loss severity relativities:

$$R_t^L = \frac{\text{Severity Limited to Limit } L \text{ at Age } t}{\text{Unlimited Severity at Age } t}$$

Severity relativities should have the following relationships:

- Severity relativity should decrease as age increases
 - This is because more losses are capped at the per-occurrence limit as age increases
- Severity relativity should be higher for a larger limit
 - This is because a higher limit means less of the loss is capped, so the relativity is higher

Consistent Development Factors

Using the severity relativities as defined above, Siewert derived the following relationships between unlimited, limited, and excess loss development factors:

$$\begin{aligned} LDF &= \frac{C}{C_t} \cdot \frac{S}{S_t} = \frac{U}{U_t} \\ LDF^L &= \frac{C}{C_t} \cdot \frac{S}{S_t} \cdot \frac{R^L}{R_t^L} = \frac{U}{U_t} \cdot \frac{R^L}{R_t^L} = LDF \cdot \Delta R^L \\ XSLDF^L &= \frac{C}{C_t} \cdot \frac{S}{S_t} \cdot \frac{1 - R^L}{1 - R_t^L} = \frac{U}{U_t} \cdot \frac{1 - R^L}{1 - R_t^L} = LDF \cdot \Delta(1 - R^L) \end{aligned}$$

where L = deductible limit, C = counts, S = severity, U = unlimited losses, R = severity relativity (limited severity divided by unlimited severity), and t = age. Note the following:

- The “LDF” without a superscript refers to the unlimited LDF
- The “ R^L ” without a subscript t refers to the severity relativity at ultimate

For the unlimited LDF, development is split between claim count development and severity development. Then, we apply the change in severity relativity (or the change in 1 minus the severity relativity) to arrive at the limited LDF (or excess LDF).

The relationships above ensure that smaller limits have lower development factors than higher limits.

Algebraically, it can also be shown that the unlimited LDF is a weighted average of the excess and limited LDFs:

$$LDF_t = R_t^L \cdot LDF_t^L + (1 - R_t^L) \cdot XSLDF_t^L$$

The nice thing about this formula is that we don't need the change in the severity relativity (or the change in 1 minus the severity relativity). We only need the severity relativity at time t .

Example: Direct Development Approach

Given the following information for a book of business:

- AY 2023 reported excess losses as of 12/31/23 = \$40,000
- R_{12}^L = severity relativity at 12 months = 0.96
- R^L = severity relativity at ultimate = 0.92
- LDF = unlimited 12-ult. factor = 2.40
- LDF^L = limited 12-ult. factor = 2.30

In order to use the direct development approach, we need to calculate an excess LDF .

Depending on the information given, this can be done in two ways:

- 1) Use $XSLDF^L = LDF \cdot \Delta(1 - R^L) = 2.40 \left(\frac{1-0.92}{1-0.96} \right) = 4.80$
- 2) Use the weighting formula $LDF_t = R_t^L \cdot LDF_t^L + (1 - R_t^L) \cdot XSLDF_t^L$. Once again,

$$XSLDF_t^L = \frac{LDF_t - R_t^L \cdot LDF_t^L}{1 - R_t^L} = \frac{2.40 - 0.96(2.30)}{1 - 0.96} = 4.80$$

The AY 2023 ultimate excess losses are calculated as $40,000(4.80) = \$192,000$.

Notice that our ultimate excess loss estimate of \$192,000 is identical to the ultimate excess loss estimate we obtained on the same data in the first implied development approach example. This is not a coincidence! The reason this happened is because the severity relativities of 0.96 and 0.92 were estimated solely on the losses for AY 2023.

If we had estimated the severity relativities using a different mechanism, then they would not be identical.

Example: BF Approach

Given the following information for a book of business:

- AY 2023 reported excess losses as of 12/31/23 = \$40,000
- $XSLDF^L = 4.80$
- Expected excess losses = \$225,000

The ultimate excess losses are equal to $O_t + E \cdot \left(\frac{LDF_t - 1}{LDF_t} \right) = 40,000 + 225,000 \cdot \left(\frac{4.80 - 1}{4.80} \right) = \$218,125$.

Example: Partitioning Development

The development factor formulas shown earlier **ensure that loss development is partitioned between limited and excess development in a consistent fashion.**

As an explicit example of partitioning, assume we are given the following for an accident year that is 36 months old (based on reported losses):

- Deductible = \$500,000
- $R_{36}^{500} = 0.93$
- $LDF_{36}^{500} = 1.45$

- $XSLDF_{36}^{500} = 3.40$

The expected future development (i.e., % unreported) can be written as follows:

$$\begin{aligned}
 1 - \frac{1}{LDF_{36}} &= \frac{LDF_{36} - 1}{LDF_{36}} \\
 &= \frac{[R_{36}^{500} \cdot LDF_{36}^{500} + (1 - R_{36}^{500}) \cdot XSLDF_{36}^{500}] - 1}{R_{36}^{500} \cdot LDF_{36}^{500} + (1 - R_{36}^{500}) \cdot XSLDF_{36}^{500}} \\
 &= \frac{R_{36}^{500} \cdot (LDF_{36}^{500} - 1) + (1 - R_{36}^{500}) \cdot (XSLDF_{36}^{500} - 1)}{R_{36}^{500} \cdot LDF_{36}^{500} + (1 - R_{36}^{500}) \cdot XSLDF_{36}^{500}} \\
 &= \frac{0.93(1.45 - 1) + (1 - 0.93)(3.40 - 1)}{0.93(1.45) + (1 - 0.93)(3.40)} \\
 &= \frac{0.4185 + 0.1680}{1.5865} \\
 &= 0.37
 \end{aligned}$$

Thus, the total future development is 37%. The 37% is comprised of $\frac{0.4185}{1.5865} = 26\%$ from limited development and $\frac{0.1680}{1.5865} = 11\%$ from excess development.

Distributional Model

Using loss data to determine development factors can lead to instances where limited development is actually greater than unlimited development. For example, if the severity relativity R^L increases from one development period to the next, limited loss development will exceed unlimited loss development. To avoid these inconsistencies, Siewert highlights a distributional approach.

The distributional approach models the development process by determining severity distribution parameters that vary over time (i.e., separate severity distributions for each development period).

Once the parameters are determined, we can calculate the severity relativities. Comparing those relativities over time results in development factors. Note that this assumes that claim count

development has already stopped. In the paper, Siewert used the distributional model for 48 months and beyond, where claim count IBNR was assumed to be zero.

A common model for Workers' Compensation (i.e., the focus of this paper) is the Weibull distribution. To estimate the parameters of the Weibull distribution, we can use one of the following approaches:

- Method of moments
- Maximum likelihood
- Siewert's approach – minimize the chi-square between actual and expected severity relativities

Advantages of a distributional model are as follows:

- Helps tie the relativities to the severities and provides consistent loss development factors
- Allows for interpolation among limits and years

To see an example of a problem using the distributional model, **take a look at the Cookbook.**

IV. Other Elements

Other elements associated with high deductible plans include aggregate limits, service revenue and ALAE.

Aggregate Limits

Recall earlier that we ignored per-aggregate excess losses for the implied development, direct development, and BF approaches. It's time to revisit them.

In general, it is difficult to determine loss development factors for losses excess of aggregate limits (for one thing, data tends to be sparse and not very credible).

Siewert discusses two approaches for estimating per-aggregate excess losses:

- 1) Collective risk modeling

2) NCCI Table M

Collective Risk Modeling

We can use collective risk modeling techniques (i.e., modeling frequency & severity separately) to determine the loss development factors, relying on the loss distributions described for deductible limits in conjunction with claims frequency distributions.

Siewert uses a Weibull distribution to model severity and a Poisson distribution to model claim counts.

Development for losses excess of aggregate limits decreases more rapidly over time with smaller deductibles than larger ones. This makes sense because most of the later development occurs in the layers of loss above the per-occurrence deductible, which is not covered by the aggregate. Only losses below the per-occurrence deductible contribute to the aggregate limit.

Due to the volatility of losses excess of aggregate limits, Siewert recommends using the BF method to determine the final estimate of ultimate aggregate excess of loss.

Example: Estimating Ultimate Aggregate Excess of Loss Using Collective Risk Modeling

This example is based on Table 8 from the paper. Given the following information for three accounts with an expected unlimited loss of \$1,000,000 and an aggregate limit of \$750,000:

Account	Per-Occ. Deductible	<u>Reptd. Losses (\$) at 48 Mo.</u>		<u>Aggregate Excess of Loss (\$)</u>		
		Per-Occ. Deductible Losses	Excess of Aggregate	Expected	LDF	Estimated Ult. Using BF
A	100,000	581,252	0	21,164	1.175	3,152
B	250,000	703,027	0	117,789	1.394	33,292
C	500,000	764,493	14,493	211,852	1.529	87,789

Let's discuss the table above:

- Each account has an expected unlimited loss of \$1,000,000 and an aggregate limit of \$750,000. When grabbing the LDFs and expected aggregate excess losses from the collective risk modeling analysis, we need to ensure the parameters come from this particular expected unlimited loss & aggregate limit combination (see page 236 of the source for an example)
- The reported excess of aggregate losses are equal to the per-occurrence deductible losses minus the aggregate limit. For example, for Account C, they are equal to $764,493 - 750,000 = 14,493$
- The expected aggregate excess losses and aggregate excess LDFs are pulled from the collective risk modeling analysis
- The estimate ultimate aggregate excess losses are based on the BF approach. For example, for Account C, they are equal to $14,493 + 211,852 \cdot \left(\frac{1.529-1}{1.529}\right) = 87,789$

NCCI Table M

A large deductible policy with an aggregate limit looks very similar to a retro policy with a maximum premium and a per loss limit. Since Table M is normally used to price retro policies, it can be used to price large deductible policies as well.

This approach is more practical than collective risk modeling and relies on using the proper insurance charge table.

The aggregate excess IBNR is determined by subtracting aggregate loss charges at different maturities.

The key to the NCCI procedure is adjusting expected losses to reflect loss limits:

$$\text{Adjustment Factor} = \frac{1 + 0.8 \cdot \chi}{1 - \chi}$$

where χ = per occurrence charge.

The intent of increasing expected losses (by the adjustment factor) for the use of a per-occurrence limit is to utilize a less dispersed loss ratio distribution, resulting in a smaller insurance charge.

There are a few important things to keep in mind regarding this method for the purposes of the exam:

- Table M is an Exam 8 topic. Outside of this one section of Siewert and briefly in Teng & Perkins, it's never mentioned again on Exam 7. For that reason alone, it feels out of scope
- The method described above is known as the ICRL method, which is actually out-of-date at this point
- That being said, there is an **example of Table M calculations in the Cookbook** in the unlikely scenario that this shows up on exam day

Service Revenue

Under a high deductible program, the insurer generally handles all of the claims, and then seeks reimbursement from the insured for losses below the per-occurrence deductible. The amount paid by the insured to the insurer for handling claims below the deductible is known as service revenue.

In general, to generate the service revenue, a factor (loss multiplier) is applied to deductible losses, limited by any applicable aggregate limit, to cover expenses that vary with those losses.

Service revenue is usually collected as losses are paid, but it could also be gathered as a function of case incurred losses.

The steps to determine the service revenue asset (i.e., service revenue “reserve”) are as follows:

- Determine the ultimate deductible losses at the account level
- Subtract ultimate losses excess of aggregate limits from ultimate deductible losses

- Apply the loss multiplier to the difference in the previous step to determine ultimate recoverables
 - Determine the total asset by subtracting any known recoveries from the ultimate recoverables, and aggregate results for all accounts
-

Example: Service Revenue Asset

Given the following information for an account:

- Ultimate deductible losses = \$1,000,000
- Ultimate losses excess of the aggregate limit = \$200,000
- Loss multiplier = 10%
- Known service revenue recoveries = \$60,000

If there was not an aggregate limit, all of the losses below the deductible would generate service revenue. However, since the insurer is responsible for the losses above the aggregate limit, those losses do not generate service revenue.

Thus, we only apply the loss multiplier to $\$1,000,000 - \$200,000 = \$800,000$. The ultimate service revenue recoverables = $\$800,000(0.10) = \$80,000$.

The service revenue asset = $\$80,000 - \$60,000 = \$20,000$.

Before we leave the discussion on service revenue, we want to note an oddity in Table 10 of the paper. In Table 10, Siewert calculates the service revenue asset for three accounts. The strange thing about the table is that the “net of aggregate” ultimate losses are *slightly larger* than the aggregate limit.

How is this possible? If the aggregate limit is hit, shouldn’t the net of aggregate losses be equal to the aggregate limit?

We actually reached out to Siewert on this seeming contradiction. The reason this is occurring is two-fold:

- 1) The LDFs for developing deductible losses were calculated separately from those used to develop aggregate excess losses
- 2) Developed losses were based on a cohort of claims, rather than the development of individual accounts

The second point is especially important. When we combine losses into a limited loss triangle and develop them using limited LDFs, some “individual” losses will develop beyond the deductible. But, in the aggregate, the limited losses should be reasonable. Since Table 10 zooms in on three individual accounts, it looks a bit odd.

Allocated Claim Expense (ALAE)

There are two ways to handle ALAE under a high deductible program:

- 1) Account manages expense itself (i.e., ALAE not covered)
 - Development patterns reflect loss only
- 2) ALAE is treated as loss and subjected to applicable limits
 - Development patterns reflect a combination of loss and expense
 - We assume expenses are equivalent to additional loss dollars (i.e., sum losses and expenses together in analysis)

Lastly, Siewert discusses the best way to split loss and ALAE for financial reporting. He recommends splitting them proportionally based upon their full coverage counterparts, since more actuarially sound approaches may not be cost justifiable.

V. Conclusion

Improvements to Siewert’s methods include:

- Obtain longer histories of experience

- Derive parameters that provide better fits to the actual data
- Determine better tail factors
- Develop more advanced approaches to index loss limits

I. Introduction

The purpose of the paper is to demonstrate the relationship between claims development, trend and claim size factors.

It provides an approach for adjusting a development pattern for any claim layer to produce a development pattern for any other layer. The approach allows for cost level adjustments and ensures that assumptions related to claim size models, claims development and trend are internally consistent.

Research Context

Currently, excess layer development is commonly estimated using the Pinto/Gogol approach, which focuses on fitting observed development factors (as a function of retentions) using industry data.

However, there are difficulties with the Pinto/Gogol approach:

- Access to industry data may be difficult
- Methodology does not use the inherent relationship between claims size models, trend and claims development patterns

Sahasrabuddhe's method aims to address these issues.

II. Background

This paper assumes we are analyzing an $n \times n$ loss triangle. Let's define some notation:

- $L(d, p)$ represents a loss layer, where d is the lower bound (ex. deductible) and p is the upper bound (ex. policy limit)

- In other words, this is the loss layer truncated from below at d and censored from above at p
- $C_{i,j}^L$ represents the cumulative losses in the layer L for exposure period i (typically an AY) and development period j
- $C_{i,\infty}^L$ represents the ultimate losses in the layer L for exposure period i

Trend Factors

In practice, trend rates have the following general characteristics:

- Refer to the annual change in cost level for a loss layer
- Tend not to vary between accident periods
- Trend that occurs in the development period or calendar period direction is often not considered
- Tend not to vary by loss layer

Rather than using annual rates of changes, Sahasrabuddhe applies cost level indices to cumulative GUU (ground-up, unlimited) claims for exposure period i as of development period j . If trend is estimated from truncated or censored data, we must adjust it to a GUU basis before applying trend.

Example: Trend Indices

Assume a cumulative loss triangle consists of five AYs and five development periods. Given the following:

- AY Trend
 - General trend = 2%
 - Between AYs 2 and 3, the trend is 5%
- CY Trend
 - General trend = 1%

- Between CYs 3 and 4, the trend is -5%

First, let's build the AY trend indices:

AY i	<u>Development Period j</u>				
	1	2	3	4	5
1	1.000	1.000	1.000	1.000	1.000
2	$1.020 = 1.000(1.02)$	1.020	1.020	1.020	1.020
3	$1.071 = 1.020(1.05)$	1.071	1.071	1.071	1.071
4	$1.092 = 1.071(1.02)$	1.092	1.092	1.092	1.092
5	$1.114 = 1.092(1.02)$	1.114	1.114	1.114	1.114

In the table above, each row has a specific trend index.

Second, let's build the CY trend indices:

AY i	<u>Development Period j</u>				
	1	2	3	4	5
1	1.000	$1.010 = 1.000(1.01)$	$1.020 = 1.01(1.01)$	0.969	0.979
2	$1.010 = 1.000(1.01)$	1.020	0.969	0.979	0.989
3	$1.020 = 1.010(1.01)$	0.969	0.979	0.989	0.998
4	$0.969 = 1.020(0.95)$	0.979	0.989	0.998	1.008
5	$0.979 = 0.969(1.01)$	0.989	0.998	1.008	1.019

In the table above, each diagonal has a specific trend index.

Finally, we can build combined trend indices by multiplying the AY trend indices by the CY trend indices (feel free to do this in a single step on the exam):

AY i	<u>Development Period j</u>				
	1	2	3	4	5
1	1.000	1.010	1.020	0.969	0.979
2	1.030	1.041	0.988	0.998	1.008
3	1.093	1.038	1.048	1.059	1.069
4	1.059	1.069	1.080	1.091	1.102

5	1.091	1.102	1.113	1.124	1.135
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Claim Size Model

To calculate development factors by layer using Saharabuddhe's method, we require limited expected values (LEVs). A claim size model underlies these limited expected values. In this paper, claim sizes are described by the exponential distribution. For an exponential distribution, the formula for an LEV at limit x is as follows:

$$LEV(x) = \theta \left(1 - e^{-\frac{x}{\theta}}\right)$$

There are **two requirements** of the claim size model:

- 1) Claim size model parameters can be adjusted for the impact of inflation
- 2) Limited expected values and unlimited means can be easily calculated

Before we can calculate LEVs for various exposure periods and development periods, we need to calculate trend-adjusted parameters for the claim size model. Since claim sizes are described by the exponential distribution, we only need to adjust a single parameter (θ).

Example: Trend-Adjusted Claim Size Parameters

Assume that we are given the following claim size parameters (θ for the exponential distribution) at the AY 5 cost level:

AY i	<u>θ at Development Period j</u>				
	1	2	3	4	5
5	28,138	84,242	133,998	182,460	204,649

To determine the claim size model parameters at other cost levels, we apply the combined trend indices to the parameters at the AY 5 cost levels as follows:

AY i	θ at Development Period j				
	1	2	3	4	5
1	$25,800 = 28,138 \left(\frac{1.000}{1.091} \right)$	77,242	122,863	157,359	176,496
2	$26,579 = 28,138 \left(\frac{1.030}{1.091} \right)$	79,574	119,054	162,112	181,826
3	$28,187 = 28,138 \left(\frac{1.093}{1.091} \right)$	79,375	126,257	171,919	192,826
4	$27,313 = 28,138 \left(\frac{1.059}{1.091} \right)$	81,772	130,070	177,111	198,650
5	28,138	84,242	133,998	182,460	204,649

At this point, we have a claim size parameter at each AY cost level and development period.

III. Main Method

In this section of the paper, Sahasrabuddhe builds out his “main” methodology. The key thing to remember is that we should only use development patterns estimated from unadjusted data if the following two things are true:

- 1) The data is provided on a GUU basis
- 2) Trend only acts in the AY direction

Since these two things are often not true in practice, we need to adjust the data before estimating development factors. Specifically, we need to adjust the data for differences in cost level and limits.

Developing the Basic Limit Loss Development Pattern

Sahasrabuddhe’s method begins by developing the basic limit loss development pattern, where the basic limit, B , represents the limit in which we have sufficient credibility to determine stable/accurate development patterns.

To do this, we start by adjusting the raw loss data to the basic limit at the latest exposure period:

$$\text{Adj. Data at } i, j = \text{Raw Data at } i, j \cdot \frac{LEV(B|\theta_{n,j})}{LEV(L|\theta_{i,j})}$$

The raw data represented a mixture of loss layer L data at various cost levels. After the adjustment above, the adjusted data represents loss data on a basic limit basis at the latest cost level. Hence, if we estimate a loss development pattern using the adjusted data, it's no longer impacted by trend differences since all of the data is on the same cost basis.

Example: Adjusting Raw Data to a Basic Limit Basis at the Latest Cost Level

Given the following cumulative losses based on policies with a limit of \$1,000,000:

AY i	<u>Cumulative Reported Losses (in \$) at Development Period j</u>				
	1	2	3	4	5
1	396,132	1,333,217	2,180,715	3,402,672	3,873,311
2	440,832	1,288,463	2,419,861	3,691,712	
3	359,840	1,421,128	2,864,498		
4	376,686	1,363,294			
5	344,014				

Suppose the selected basic limit is \$500,000.

First, using the trend-adjusted claim size parameters from the prior example, let's determine LEVs for each AY and development period at the basic limit:

AY i	<u>LEVs at Development Period j at \$500,000 Limit</u>				
	1	2	3	4	5
1	25,800	77,122	120,764	150,799	166,111
2	26,579	79,426	117,268	154,693	170,201
3	28,187	79,229	123,851	162,538	178,404
4	27,313	81,592	127,286	166,587	182,618

5	28,138	84,019	130,788	170,682	186,869
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As an **example** calculation, the LEV for AY 5 and development period 5 is 186,869 =

$$204,649 \left(1 - e^{\frac{-500,000}{204,649}} \right).$$

Second, let's determine LEVs for each AY and development at the limit of the data in the triangle (i.e., the raw data):

AY i	<u>LEVs at Development Period j at \$1,000,000 Limit</u>				
	1	2	3	4	5
1	25,800	77,241	122,827	157,086	175,885
2	26,579	79,574	119,027	161,772	181,083
3	28,187	79,375	126,211	171,407	191,748
4	27,313	81,772	130,010	176,486	197,356
5	28,138	84,241	133,921	181,700	203,104

As an **example** calculation, the LEV for AY 5 and development period 1 is 203,104 =

$$204,649 \left(1 - e^{\frac{-1,000,000}{204,649}} \right).$$

Third, we adjust the raw data to the basic limit at the latest exposure period using the adjustment formula:

AY i	<u>Adj. Cumulative Reported Losses (in \$) at Development Period j</u>				
	1	2	3	4	5
1	432,034	1,450,205	2,322,048	3,697,191	4,115,200
2	466,691	1,360,440	2,658,948	3,895,049	
3	359,215	1,504,277	2,968,365		
4	388,062	1,400,758			
5	344,014				

Let's look at a couple of example calculations:

- AY 1 at development period 1 = 432,034 = 396,132 · $\frac{LEV(B|\theta_{5,1})}{LEV(L|\theta_{1,1})} = 396,132 \cdot \left(\frac{28,138}{25,800}\right)$
- AY 2 at development 3 = 2,658,948 = 2,419,861 · $\frac{LEV(B|\theta_{5,3})}{LEV(L|\theta_{2,3})} = 2,419,861 \cdot \left(\frac{130,788}{119,027}\right)$

Once the raw data has been adjusted, we can calculate the basic limit loss development pattern at the latest cost level using the standard chain-ladder method on the adjusted loss triangle. In this case, the development factors are as follows:

	<u>Basic Limit Development Pattern at $i = 5$</u>				
	1-2	2-3	3-4	4-5	5-Ult.
Age-to-Age	3.472	1.842	1.524	1.113	1.000
Age-to-Ult	10.854	3.126	1.697	1.113	1.000

Note that the development factors only apply to losses at the **basic limit and latest cost level** (i.e., AY 5). This is because we adjusted all of the raw data to the basic limit at the latest exposure period.

Calculating Loss Development Patterns by Layer & Cost Level

Once we have the basic limit age-to-ultimate factors at the latest exposure period, we can calculate development factors for any layer X and any exposure period using the following formula:

$$F_{i,j}^X = F_{n,j}^B \cdot \frac{LEV[X|\theta_{i,\infty}]/LEV[B|\theta_{n,\infty}]}{LEV[X|\theta_{i,j}]/LEV[B|\theta_{n,j}]}$$

where F represents age-to-ultimate factors.

The formula above takes us from an age-to-ultimate factor for the basic limit at the latest cost level (i.e., latest AY) to an age-to-ultimate for the target layer at cost level i (i.e., AY i).

Example: Calculating LDFs for Other Layers & Cost Levels

Using the LEVs and basic limit development pattern we calculated earlier, let's calculate age-to-ultimate factors for the following layers:

- $L(0,1M)$: This is the layer of losses capped at \$1M. In other words, this is the layer that corresponds to the raw data
- $L(500k, 1M)$: This is the layer of losses between \$500k and \$1M

First, let's calculate the age-to-ultimate factors for the raw data layer:

AY i	<u>Age-to-Ultimate Factors for the Raw Data Layer</u>				
	1-Ult.	2-Ult.	3-Ult.	4-Ult.	5-Ult.
1	11.141	3.200	1.700	1.138	1.000
2	11.134	3.198	1.806	1.138	
3	11.118	3.395	1.804		
4	11.809	3.392			
5	11.796				

Let's look at a couple of example calculations:

- 1-Ult. Factor for AY 1

$$\circ F_{1,1}^X = F_{5,1}^B \cdot \frac{LEV[X|\theta_{1,\infty}]/LEV[B|\theta_{5,\infty}]}{LEV[X|\theta_{1,1}]/LEV[B|\theta_{5,1}]} = 10.854 \cdot \left(\frac{175,885/186,869}{25,800/28,138} \right) = 11.141$$

- 3-Ult. Factor for AY 2

$$\circ F_{2,3}^X = F_{5,3}^B \cdot \frac{LEV[X|\theta_{2,\infty}]/LEV[B|\theta_{5,\infty}]}{LEV[X|\theta_{2,3}]/LEV[B|\theta_{5,3}]} = 1.697 \cdot \left(\frac{181,083/186,869}{119,027/130,788} \right) = 1.806$$

Second, let's calculate the age-to-ultimate factors for the layer from \$500k to \$1M:

<u>Age-to-Ultimate Factors for the 500k – 1M Layer</u>					
AY i	1-Ult.	2-Ult.	3-Ult.	4-Ult.	5-Ult.
1	Large	115.330	5.625	1.580	1.000
2	Large	103.128	7.346	1.563	
3	Large	128.785	6.712		
4	Large	114.832			
5	Large				

The calculations are a bit trickier because the layer doesn't begin at 0. As an **example** calculation,

let's calculate the 3-Ult. Factor for AY 2. We know that $F_{2,3}^X = F_{5,3}^B \cdot \frac{LEV[X|\theta_{2,\infty}]/LEV[B|\theta_{5,\infty}]}{LEV[X|\theta_{2,3}]/LEV[B|\theta_{5,3}]}$.

We can calculate $LEV[X|\theta]$ as $LEV[1M|\theta] - LEV[500k|\theta]$. Thus:

$$\begin{aligned}
 F_{2,3}^X &= F_{5,3}^B \cdot \frac{(LEV[1M|\theta_{2,\infty}] - LEV[500k|\theta_{2,\infty}])/LEV[B|\theta_{5,\infty}]}{(LEV[1M|\theta_{2,3}] - LEV[500k|\theta_{2,3}])/LEV[B|\theta_{5,3}]} \\
 &= 1.697 \cdot \frac{(181,083 - 170,201)/186,869}{(119,027 - 117,268)/130,788} = 7.346
 \end{aligned}$$

The 1-Ult. factors are labeled as “Large” because they're gigantic values!

Third, let's compare the estimated ultimate losses and IBNR for the raw data layer using the unadjusted chain-ladder factors and the adjusted factors we calculated above:

AY i	<u>Adjusted</u>		<u>Unadjusted</u>	
	Ultimate (\$)	IBNR (\$)	Ultimate (\$)	IBNR (\$)
1	3,873,311	0	3,873,311	0
2	4,201,190	509,478	4,202,329	510,617
3	5,167,538	2,303,040	5,028,209	2,163,711
4	4,623,906	3,260,612	4,418,810	3,055,516
5	4,058,151	3,714,137	3,831,000	3,486,986
Total		9,787,266		9,216,829

As an **example** calculation, the adjusted ultimate losses for AY 2 are equal to $4,201,190 = 3,691,712(1.138)$, where the 3,691,712 is the value on the latest diagonal for AY 2 in the raw data triangle.

As we can see, if we had simply used the unadjusted chain-ladder factors to project ultimate losses, we would have understated the IBNR by over \$500,000.

To project ultimate losses and IBNR for the \$500k to \$1M layer, we would need to know the reported losses for that layer.

In the prior example, we illustrated that the standard chain-ladder factors understate the IBNR. It's important to note that the difference between the standard chain-ladder factors and the adjusted factors based on the paper's methodology is **primarily driven by the size of the claim size parameters**:

- Primary Layers (i.e., layers that start at 0)
 - When claim size parameters increase, a higher percentage of losses will be capped at the limit. This leads to a larger difference in the limited expected values between the basic limit and limit of the data in the triangle, which leads to a bigger deviation from the standard chain-ladder factors (see 2016 #8 for an example)
- Excess Layers (i.e., layers that do not start at 0)
 - When claim size parameters decrease, a smaller percentage of losses will be capped at the limit. This leads to a smaller difference in the expected values between the various limits. Since we have to subtract LEVs to arrive at the LEV for the layer, the difference will between the LEVs will be even smaller for the denominator, creating a leverage effect. This leads to a bigger deviation from the standard chain-ladder factors (see 2019 #4 for an example)

To convince yourself of the two facts above, take a look at the examples in the paper, along with the problems referenced above. Honestly, the best way to learn is to create your own example, start adjusting the claim size parameters up and down, and see what happens!

The difference between the standard chain-ladder factors and the adjusted factors **will also increase** *with higher trend rates* (whether highly negative or highly positive) and *longer development patterns* (i.e., long-tailed lines).

The prior examples lay out the calculations for Sahasrabuddhe's "main" method. Let's summarize the process:

- Calculate trend indices
- Develop claim size model parameters at the latest cost level
- Calculate trend-adjusted claim size model parameters using the trend indices
- Calculate LEVs for the basic limit and the limit in the raw data triangle
- Calculate the adjusted cumulative loss triangle at the basic limit and latest cost level
- Calculate the basic limit development pattern at the latest cost level
- Calculate the development pattern for any desired layer and cost level based on the basic limit development pattern at the latest cost level
- Use the adjusted development patterns to calculate ultimate losses and reserves for any desired layer and cost level

IV. Practical Issues

In practice, we are typically only provided with a development pattern (i.e., cost level not stated). Although we have shown that development patterns for limited claims vary with cost level, we can ignore this fact for simplicity. Instead, we simply **assume that that the development pattern is at the latest cost level**.

In addition, we typically only have claim size models *at ultimate* (i.e., no claim size models at earlier age).

Based on the two practical issues above, we restate our development factors for any layer X as follows:

$$F_{i,j}^X = F_{n,j}^B \cdot \frac{LEV[X|\theta_{i,\infty}]/LEV[B|\theta_{i,\infty}]}{R_j(X,B)}$$

where $R_j(X,B) = LEV[X|\theta_{i,j}]/LEV[B|\theta_{i,j}]$. This is known as the *simplified model*.

Let's dissect the formula above:

- As stated earlier, the $F_{n,j}^B$ is often provided to us with no cost level stated. For simplicity, we assume that this development pattern is at the cost level of the latest exposure period. If a triangle is provided instead of a development pattern, then we would proceed by calculating the unadjusted chain-ladder development factors. Again, we would assume that these factors are at the latest cost level
- The author points out that the impact of trend is immaterial when calculating ratios of LEVs. Thus, the cost level for each LEV (the first subscript) is i . That being said, trend assumptions would still need to be used to determine the required LEVs if they are not given to us
- Since we are assuming claim size models are not available prior to ultimate, the $R_j(X,B)$ term must be estimated (or, given)

Important Notes on $R_j(X,B)$

Sahasrabuddhe assumes that $R_j(X,B) < 1$. This implies that X is smaller than the basic limit B . This is different from the “main” method shown earlier where X was the layer with the higher limit and B was the layer with the lower limit (in which case, $R_j(X,B) > 1$). The point is that Sahasrabuddhe frames his “simplified” model in terms of going from a higher limit B to a lower limit X . Keep this mind as you review the exhibits in the paper.

The assumption that $R_j(X, B) < 1$ leads to the following three properties:

- 1) $R_a > R_b$, where $a < b$
 - At early maturities (such as a), there will be less development in the excess layer, resulting in an R ratio close to 1
- 2) $R_a \geq U$, where $U = \lim_{a \rightarrow \infty} R_a$
 - U is simply R_j at ultimate
 - Similar to the first property, $R_a \geq U$ because there is more development associated with the denominator of R (losses in layer B) than the numerator of R (claims in layer X)
- 3) If $B = GUU$ and if all development in the unlimited layer occurs above X , then the maximum value for R is calculated as U times the unlimited loss development factor

Note that the three conditions above will be violated if there is negative development or if we assume that an excess layer may develop more quickly than a working layer.

Estimating $R_j(X, B)$

Sahasrabuddhe uses the following decay model to estimate $R_j(X, B)$:

$$R_j(X, B) = U + (1 - U) \cdot \text{Decay Factor}$$

where the decay factors are determined/calibrated using the observed ratios at each development period.

The decay model ensures that R_j is “high” at early maturities (closer to 1) and “low” at later maturities (further from 1). At ultimate, the decay factor is 0 and $R_j = U$.

As an example, suppose that we have the following observed ratios:

AY i	Dev. Period j	Cumulative Losses at \$1M	Cumulative Losses at \$500k	Observed Ratio
1	5	3,873,311	3,670,631	$0.948 = \frac{3,670,631}{3,873,311}$
2	4	3,691,712	2,750,008	0.745
3	3	2,864,498	1,771,896	0.619
4	2	1,363,294	1,363,294	1.000
5	1	344,014	344,014	1000

Using the observed ratios as a guideline, we select a decay pattern that best reproduces them.

On the exam, if you are required to apply a decay model, the decay pattern would need to be given to you.

Note also that the decay model above only applies to primary layers. It doesn't work for an excess layer because excess layers don't "decay." Instead, R_j starts "low" and ends "high." Thus, in the event we need to deal with an excess layer, we have to be clever. Let's look at an example.

Example: Simplified Model

Given the following information for a book of business as of December 31, 2023:

AY	Cumulative Incurred Losses (in \$000) as of XX Months:			
	12 mo.	24 mo.	36 mo.	48 mo.
2020	4,000	12,000	19,200	24,000
2021	4,800	14,000	22,400	
2022	4,200	13,000		
2023	4,500			

The ultimate limited expected values at \$2,500,000 and \$5,000,000 are as follows:

AY	LEV(2.5M) (in \$000)	LEV(5M) (in \$000)
2020	1,191	1,404
2021	1,217	1,446

2022	1,241	1,488
2023	1,265	1,530

The following decay factors are used to select $R_j(X, B)$, where X is the layer $L(0, 2.5M)$ and B is the layer $L(0, 5M)$:

j	Decay Factor
1	0.9
2	0.6
3	0.4

- Limit of the data in the triangle = \$5M
- Basic limit = \$5M

First, let's calculate U and R_j for each AY:

AY	$U(X, B) = R_\infty(X, B)$	$R_j(X, B)$
2020	$0.848 = \frac{1,191}{1,404}$	0.848
2021	$0.842 = \frac{1,217}{1,446}$	$0.905 = 0.842 + (1 - 0.842) \cdot (0.4)$
2022	$0.834 = \frac{1,241}{1,488}$	$0.934 = 0.834 + (1 - 0.834) \cdot (0.6)$
2023	$0.827 = \frac{1,265}{1,530}$	$0.983 = 0.827 + (1 - 0.827) \cdot (0.9)$

Notice that $U = R_j(X, B)$ for AY 2020. This is because AY 2020 is assumed to be at ultimate.

Second, let's calculate the standard chain-ladder factors using the raw data:

	<u>Standard Chain-Ladder Factors</u>		
	12-24	24-36	36-48
Age-to-Age	3.000	1.600	1.250
Age-to-Ult.	6.000	2.000	1.250

Since the data in the triangle is at the basic limit of \$5M, these are the factors we want to use to calculate development patterns for other layers.

Third, let's calculate the adjusted development patterns for the layer $L(0, 2.5M)$:

- $F_{i,j}^X = F_{n,j}^B \cdot \frac{LEV[X|\theta_{i,\infty}]/LEV[B|\theta_{i,\infty}]}{R_j(X,B)} = F_{n,j}^B \cdot \frac{U(X,B)}{R_j(X,B)}$
- $F_{2020,4}^X = 1.000$ (already at ultimate)
- $F_{2021,3}^X = 1.250 \cdot \left(\frac{0.842}{0.905}\right) = 1.163$
- $F_{2022,2}^X = 2.000 \cdot \left(\frac{0.834}{0.934}\right) = 1.787$
- $F_{2023,1}^X = 6.000 \cdot \left(\frac{0.827}{0.983}\right) = 5.048$

In the work steps above, the layer X was defined as the layer from 0 to \$2,500,000. Let's change that up. Suppose X is the layer from \$2,500,000 to \$5,000,000. How does this change the calculations?

If X is the layer from \$2,500,000 to \$5,000,000, then we have the following:

$$\begin{aligned}
 F_{i,j}^X &= F_{n,j}^B \cdot \frac{LEV[X|\theta_{i,\infty}]/LEV[B|\theta_{i,\infty}]}{R_j(X,B)} \\
 &= F_{n,j}^B \cdot \frac{(LEV[5M|\theta_{i,\infty}] - LEV[2.5M|\theta_{i,\infty}])/LEV[5M|\theta_{i,\infty}]}{(LEV[5M|\theta_{i,j}] - LEV[2.5M|\theta_{i,j}])/LEV[5M|\theta_{i,j}]} \\
 &= F_{n,j}^B \cdot \frac{1 - (LEV[2.5M|\theta_{i,\infty}]/LEV[5M|\theta_{i,\infty}])}{1 - (LEV[2.5M|\theta_{i,j}]/LEV[5M|\theta_{i,j}])} \\
 &= F_{n,j}^B \cdot \frac{1 - U((0 - 2.5M), (0 - 5M))}{1 - R_j((0 - 2.5M), (0 - 5M))}
 \end{aligned}$$

Since we already calculated these U and R_j values in the prior steps, these factors are simple to compute:

- $F_{2020,4}^X = 1.000$ (already at ultimate)
- $F_{2021,3}^X = 1.250 \cdot \left(\frac{1-0.842}{1-0.905}\right) = 2.083$

- $F_{2022,2}^X = 2.000 \cdot \left(\frac{1-0.834}{1-0.934} \right) = 5.000$
- $F_{2023,1}^X = 6.000 \cdot \left(\frac{1-0.827}{1-0.983} \right) = 60.000$

And just like that, we have the development patterns for the layer that runs from \$2.5M to \$5M.

V. Model Assumptions

Sahasrabuddhe describes two different sets of assumptions needed to use the procedure outlined in the paper. The first set describes assumptions that are **simple to implement**. The second set describes assumptions that are **difficult to implement**.

Simple to Implement Assumptions

- 1) The procedures require us to select a basic limit
 - This is not always an explicit selection. When analyzing a loss triangle, actuaries make an implicit assumption that the limit associated with the triangle is sufficiently credible
- 2) The procedure requires the use of a claim size model
 - Typically, claim size information (i.e., increased limit factors) or a claim size model is readily available
 - It's important to note that the claim size model does not necessarily need to be 100% accurate. Since the primary goal of the claim size model is to obtain reasonable ratios of limited expectations to unlimited means at higher values, it is less important that the absolute limited expected values are completely accurate
 - This means a simpler claims size model may be sufficient
- 3) The procedure requires that the data triangle be adjusted to a basic limit and common cost level
 - Given claim size and trend information, this is fairly simple to implement

Difficult to Implement Assumptions

- 1) The procedure requires claim size models at maturities prior to ultimate
 - These are generally not available and have limited application outside of the approach described in this paper
 - A benefit of creating claim size models over time is that it would provide an increased understanding of excess layer development
 - With enough data, an algorithm could be programmed to calculate these claim size models
 - Since only the ratio of expected values is required to adjust development patterns from one layer to another, a robust claim size model is not necessarily required
- 2) The procedure requires a triangle of trend indices
 - The trend indices specify the cost level associated with cumulative claims, which is an issue because trend typically occurs on an incremental basis
 - In addition, the impact of trend on reported incurred claims is difficult to quantify (in particular, the timing of the effect of trend on case reserves)
 - We can avoid these issues if we assume that development only occurs in the exposure period direction (i.e. no calendar year trend)
 - One benefit of directly confronting and documenting trend assumptions is that it produces greater transparency and better informs the consumer of actuarial information

VI. Appendix

The appendix provides the proof behind the third R_j property. Recall that when $B = GUU$ and all development in the unlimited layer occurs above X , the maximum value for R is calculated as U times the unlimited claims development factor.

Using the formula that converts from layer to another, we have the following:

$$F_{i,j}^X = F_{n,j}^B \cdot \frac{U}{R_j}$$

Re-arranging terms, we find that $F_{i,j}^X \cdot R_j = F_{n,j}^B \cdot U$. Since all development in the unlimited layer occurs above X , it must be the case that $F_{i,j}^X = 1$. Assuming there is no negative development, this is as small as $F_{i,j}^X$ can get.

$$\text{Thus, } R_j = \frac{F_{n,j}^B \cdot U}{F_{i,j}^X} = \frac{F_{n,j}^B \cdot U}{1} = F_{n,j}^B \cdot U = R_{max}.$$

I. Introduction

Under a retrospectively rated (retro) policy, the premium paid by the insured reflects its actual loss experience during the policy period. In general, the process is as follows:

- A deposit premium is paid at the start of the policy period based on expected losses
- A few months after the policy has expired (six months is common), a premium adjustment is made. If losses are less than expected, premium is returned to the insured. If losses are more than expected, additional premium is collected from the insured
- Additional premium adjustments often occur annually thereafter, and stop after a pre-determined number of adjustments

At each adjustment, the premium paid by the insured is subject to maximum and minimum premium amounts. Without these, a retro policy would not be insurance. The insured would simply be paying all of its own losses.

In addition, retro policies often include a per-occurrence limit that caps individual large losses before they are included in the premium calculation.

This paper focuses on the premium asset, which is equal to the ultimate premium that the insurer expects to collect based on the expected ultimate loss experience, less the premium that the insurer has already booked.

Retro policies are **popular** for a number of reasons:

- They encourage loss control and loss management by returning premium to the insured for good loss experience. This is great for the insurer as well because it attracts preferred customers

- They offer a cash flow advantage to insureds by allowing them to pay premium as losses are reported or paid
- Since premium varies directly with the insured's actual loss experience, it shifts a large portion of the risk to the insured. This is great for the insurer due to the increasing difficulty in predicting the cost of insurance

In this paper, Teng & Perkins present a method for calculating the premium asset based on the **premium development to loss development (PDL D)** ratio. In his review of the paper, Feldblum explains why the PDL D method is preferred to historical approaches for calculating the premium asset. He also offers up his own enhancement to the PDL D method.

II. The Formula Approach to Calculating PDL D Ratios

A common formula for the premium under a retro policy is as follows:

$$P_n = [BP + (CL_n \times LCF)] \times TM$$

where:

- P_n is the premium at the n^{th} adjustment
- BP is the basic premium
 - Covers fixed expenses, the net insurance charge, and the per-occurrence excess loss charge (described more in Feldblum's review)
- CL_n is the capped loss at the n^{th} adjustment
 - Individual losses may be subject to the per-occurrence limit if present on the policy
 - Aggregate losses are subject to the maximum & minimum loss amounts that correspond to the maximum & minimum premiums mentioned earlier
- LCF is the loss conversion factor
 - Covers LAE
- TM is the tax multiplier

To build out the PDL ratios, let's define L_n as the unlimited loss at the n^{th} adjustment.

The First PDL Ratio

The first PDL ratio is as follows:

$$\begin{aligned}
 PDL_1 &= \frac{P_1}{L_1} \\
 &= \frac{[BP + (CL_1 \times LCF)] \times TM}{L_1} \\
 &= \left[\frac{BP}{L_1} \times TM \right] + \left[\frac{CL_1}{L_1} \times LCF \times TM \right]
 \end{aligned}$$

The first term above, $\left[\frac{BP}{L_1} \times TM \right]$, can be approximated as follows:

$$\left[\frac{BP}{L_1} \times TM \right] \approx \frac{BP \times TM}{SP \times ELR \times \%Loss_1}$$

where SP is the standard premium, ELR is the unlimited expected loss ratio, and $\%Loss_1$ is the expected percentage of loss emerged for the first adjustment.

We can re-write the approximation above as follows:

$$\left[\frac{BP}{L_1} \times TM \right] \approx \frac{\frac{BP}{SP} \times TM}{ELR \times \%Loss_1}$$

In this form, $\frac{BP}{SP}$ represents the **basic premium factor** and $ELR \times \%Loss_1$ represents the **expected loss ratio for the first adjustment**.

Capped Losses

Notice the $\frac{CL_1}{L_1}$ ratio in the second term of PDL_1 . This is known as the cumulative loss capping ratio at the first adjustment. We can generalize this to the n^{th} adjustment as $\frac{CL_n}{L_n}$.

Capped losses refer to losses that contribute to additional premium. As mentioned earlier, the difference between the capped loss and the uncapped loss can be viewed as the portion of loss outside the boundaries of the retro minimum loss and maximum loss. On plans with per accident loss limits, the capped losses would exclude the losses exceeding this limit as well.

The loss capping ratio decreases as the data becomes more mature, since an increasing portion of the loss development occurs outside of loss limitations.

We can define the cumulative loss capping ratio mathematically using the net insurance charge, along with a loss elimination ratio (LER) for the per-accident limit:

$$\frac{CL_n}{L_n} = 1 - \text{Net Insurance Charge} - LER$$

where the net insurance charge = Table M insurance charge at the retro maximum – Table M insurance savings at the retro minimum.

The formula above is introduced in Section 5 of the paper, which is off-syllabus. However, since it's used in the exhibits that are on the syllabus, we are including it here. The net insurance charge is described in more detail in Feldblum's review

The Second PDL Ratio

The second PDL ratio refers to the incremental premiums developed between the first and second retro adjustments divided by the incremental losses developed between these two adjustments (i.e., the change in premium divided by the change in loss):

$$\begin{aligned} PDL_2 &= \frac{P_2 - P_1}{L_2 - L_1} \\ &= \frac{(CL_2 - CL_1) \times LCF \times TM}{L_2 - L_1} \\ &= \frac{CL_2 - CL_1}{L_2 - L_1} \times LCF \times TM \end{aligned}$$

where $\frac{CL_2 - CL_1}{L_2 - L_1}$ is known as the *incremental loss capping ratio*. Once again, this can be generalized to other adjustment periods.

Subsequent PDL ratios are calculated in the same way as the second PDL ratio.

Advantages & Disadvantages of the Formula Approach

An **advantage** of using the retro formula to estimate the PDL ratio is that it responds to changes in the retro rating parameters that are sold. If retro parameters change significantly over time, more weight should be given to PDL ratios derived from the formula than those derived from historical data.

A **disadvantage** of using the retro formula to estimate the PDL ratio is that potential bias exists since the formula approach uses the average parameters for the LCF, tax multiplier, maximum, minimum, and per accident limitation. We should retrospectively test PDL ratios against actual emergence to check for bias.

Example: Formula Approach

Given the following for a set of retrospectively rated workers' compensation (WC) policies:

- Basic premium = \$25,000
- Standard premium = \$100,000
- ELR = 0.80
- LCF = 1.25
- Tax multiplier = 1.04

There will be three retro adjustments, with the following estimated factors:

Retro Adj. Period	% Loss Emerged	Cumulative Loss Capping Ratio
First	60%	0.95
Second	30%	0.80
Third	10%	0.75

First, let's calculate $PDL D_1$:

- $L_1 = SP(ELR)(\%Loss_1) = 100,000(0.80)(0.60) = 48,000$
- $CL_1 = \frac{CL_1}{L_1} \cdot L_1 = 0.95(48,000) = 45,600$
- $PDL D_1 = \frac{[BP + (CL_1 \times LCF)] \times TM}{L_1} = \frac{[25,000 + 45,600(1.25)](1.04)}{48,000} = 1.777$

Second, let's calculate $PDL D_2$:

- $L_2 = SP(ELR)(\%Loss_2) = 100,000(0.80)(0.60 + 0.30) = 72,000$
- $CL_2 = \frac{CL_2}{L_2} \cdot L_2 = 0.80(72,000) = 57,600$
- $PDL D_2 = \frac{(CL_2 - CL_1) \times LCF \times TM}{L_2 - L_1} = \frac{(57,600 - 45,600)(1.25)(1.04)}{72,000 - 48,000} = 0.650$

Third, let's calculate $PDL D_3$:

- $L_3 = SP(ELR)(\%Loss_3) = 100,000(0.80)(0.60 + 0.30 + 0.10) = 80,000$
- $CL_3 = \frac{CL_3}{L_3} \cdot L_3 = 0.75(80,000) = 60,000$
- $PDL D_3 = \frac{(CL_3 - CL_2) \times LCF \times TM}{L_3 - L_2} = \frac{(60,000 - 57,600)(1.25)(1.04)}{80,000 - 72,000} = 0.390$

The expected incremental premiums at each adjustment are represented by the numerators of each of the PDL D ratios above. Hence, the expected incremental premiums at each adjustment are as follows:

- First Adjustment: \$85,280
- Second Adjustment: \$15,600
- Third Adjustment: \$3,120

III. The Empirical Approach to Calculating PDL D Ratios

Two types of data are needed for the empirical approach:

- 1) Booked premium development

2) Reported loss development

Data should be segregated into homogeneous groups by:

- Size of account
- Type of rating plan sold

In the paper, Teng & Perkins present an example of the approach where policies are grouped based on the calendar quarter in which they became effective (i.e., policy effective quarters). On the exam, you may see them grouped differently (ex. many past exam questions grouped policies by policy year).

Empirical PDLR Ratios

The first retro premium computation is usually based on losses developed through 18 months. Due to processing and recording delays, premiums for the first retro adjustment are assumed to be booked at 27 months.

Thus, under the empirical approach, the first PDLR ratio is defined as follows:

$$PDLR_1^{emp} = \frac{\text{premiums booked through 27 months}}{\text{losses reported through 18 months}}$$

The second retro premium booking would occur a year later (i.e., at 39 months) based on losses reported through 30 months. Thus, under the empirical approach, the second PDLR ratio is defined as follows:

$$PDLR_2^{emp} = \frac{\text{prems. booked at 39 mos.} - \text{prems. booked at 27 mos.}}{\text{losses reptd. at 30 mos.} - \text{losses reptd. at 18 mos.}}$$

Subsequent PDLR ratios are calculated in a similar fashion.

Example: Empirical Approach

Given the following information:

PY	<u>Cumulative Reported Losses (in \$000) as of XX Months:</u>			
	18 mo.	30 mo.	42 mo.	54 mo.
2020	2,000	2,300	2,438	2,463
2021	2,500	2,850	2,993	
2022	3,000	3,480		

PY	<u>Cumulative Booked Premium (in \$000) as of XX Months:</u>		
	27 mo.	39 mo.	51 mo.
2020	2,800	3,000	3,050
2021	3,750	4,000	
2022	4,800		

Let's calculate and select the PDL ratios at each retro adjustment:

PY	$PDL D_1^{emp}$	$PDL D_2^{emp}$	$PDL D_3^{emp}$
2020	$1.400 = \frac{2,800}{2,000}$	$0.667 = \frac{3,000 - 2,800}{2,300 - 2,000}$	0.362
2021	1.500	0.714	
2022	1.600		
Selected (Average)	1.500	0.690	0.362

Upward Trend in the PDLR Ratios

An upward trend in the PDLR ratios may indicate the following:

- More liberal retro rating parameters (i.e., change in retro parameters being sold), such as a higher maximum, minimum, or per accident limitation
- Improvement in loss experience, resulting in a larger portion of loss being within the boundaries of the retro maximum and the per accident limitation. This drives additional premium per dollar of unlimited loss

Fluctuation in Historical PDLR Ratios

Historical PDLR ratios may **fluctuate significantly** after the first retro adjustment:

- Premium and loss development on a few policies can drive total incremental development on quarterly data
- Negative PDLR ratios are possible – upward development in high loss layers (resulting in no additional premium) and downward development in layers within loss limitations (resulting in return premium)
- If large fluctuation exists, average as many historical points as possible OR use the formula approach

Historical vs. Formula PDLR Ratios

Historical and formula PDLR ratios **could diverge** for a number of reasons. For example:

- Worse (better) than expected loss experience may have caused a larger portion of the loss to be outside (inside) the boundaries of the retro maximum and the per accident limitation than the formula approach predicted
- Average retro parameters may be changing over time

IV. Cumulative PDL Ratios

Since the PDL ratios are incremental factors, we must calculate the **cumulative PDL (CPDL)** ratios to estimate the premium asset.

A CPDL ratio is the average of the PDL ratios in all subsequent retro periods (including the current adjustment period), weighted by the percentage of losses to emerge in each period. It tells an insurer how much premium it can expect to collect for a dollar of loss that has yet to emerge.

For example, if CPDL ratio is 1.63, then each dollar of loss emerged provides the insurer one dollar and 63 cents of premium.

The CPDL ratio at the first retro adjustment is **normally greater than 1.0** for the following reasons:

- First retro premium computation includes the basic premium
- Only a small portion of loss is limited at this point
- The application of the loss conversion factor and the tax multiplier results in more than a dollar of premium per dollar of loss

At subsequent adjustments, the CPDL ratio should be less than 1.0 due to the retro maximum and the per accident limitation.

Premium Asset

Once the CPDL ratios have been calculated, we multiply them by the expected future loss emergence to determine the expected future premiums.

Adding future premiums to the booked premiums from the prior adjustment gives an estimate for ultimate premium.

The final step is to subtract current (or some other specified date) booked premium from the estimated ultimate premium to get the premium asset.

Example: Premium Asset

Given the following information for a book of business as of December 31, 2023:

Retro Adjustment Period	PDL Ratio	Development Period	Age-to-Ult. Factor
First	1.50	18-Ult.	1.80
Second	0.95	30-Ult.	1.25
Third	0.40	42-Ult.	1.10
Fourth	0.10	54-Ult.	1.05
Subsequent	0.00	66-Ult.	1.00

- ELR = 80%
- Standard premium = \$10,000,000
- Total current booked premium = \$7,000,000
- All policies were written on January 1, 2023
- None of the policies has yet had its first retro adjustment
- Retro premium is booked 27 months after policy effective dates. Retro premium recordings are assumed to lag losses by 9 months

First, let's convert the age-to-ultimate factors into an emergence pattern:

Retro Adjustment Period	Age-to-Ult. Factor	Cumulative % Emerged	Incremental % Emerged
First	1.80	$0.556 = \frac{1}{1.80}$	0.556
Second	1.25	0.800	0.244
Third	1.10	0.909	0.109
Fourth	1.05	0.952	0.043
Subsequent	1.00	1.000	0.048

Based on the information given, the first retro premium booking occurs at 27 months and is based on losses reported through 18 months. Hence, the 18-ult. factor goes with the first adjustment, the 30-ult. factor goes with the second adjustment, etc.

Second, let's calculate the CPDLD ratios by weighting future PDL ratios with the incremental emergence:

- $CPDLD_1 = \frac{1.50(0.556)+0.95(0.244)+0.40(0.109)+0.10(0.043)+0(0.048)}{0.556+0.244+0.109+0.043+0.048} = 1.114$
- $CPDLD_2 = \frac{0.95(0.244)+0.40(0.109)+0.10(0.043)+0(0.048)}{0.244+0.109+0.043+0.048} = 0.630$
- $CPDLD_3 = \frac{0.40(0.109)+0.10(0.043)+0(0.048)}{0.109+0.043+0.048} = 0.240$
- $CPDLD_4 = \frac{0.10(0.043)+0(0.048)}{0.043+0.048} = 0.047$
- $CPDLD_5 = \frac{0(0.048)}{0.048} = 0.000$

Third, let's calculate the premium asset:

- Since the first retro adjustment has not occurred yet for these policies, the expected future loss is the total future loss. Since we are not provided with an ultimate loss, we calculate the total future loss as $SP(ELR) = \$10,000,000(0.80) = \$8,000,000$
- We need to use $CPDLD_1$ to calculate the expected future premium for this group of policies (again, because we have not had a retro adjustment yet). Thus, the expected future premium is $\$8,000,000(1.114) = \$8,908,167$
- The ultimate premium is the booked premium from prior adjustments plus the expected future premium. Since this is the first adjustment, there is no premium booked from prior adjustments. Thus, the ultimate premium is $\$8,908,167$
- However, it appears that premium was booked between policy issuance and the first retro adjustment (we are given $\$7,000,000$ of currently booked premium in the problem).
Thus, the premium asset is $\$8,908,167 - \$7,000,000 = \$1,908,167$

Currently Booked Premium vs. Premium Booked from the Prior Retro Adjustment

In this example, there was a difference between the “currently booked premium” and the “premium booked from the prior retro adjustment.” The authors state this occurs due to the following:

- Differences in the timing of retro adjustments
- Minor premium adjustments
- Interim premium booking that occurs between scheduled retro adjustments

Policy Quarters

This example is based on policy years. However, the paper shows a nice example based on policy quarters. We highly recommend reviewing the paper’s example and recreating it on your own.

V. Further Issues

The authors mention a few other issues to consider:

- Handling ALAE
 - The definition of loss may include ALAE
 - Make sure the data used in computing the PDL ratios is consistent with that used in the rating plan
- Mix of Business
 - Changes in the mix of business by state, industry group, or geographical region may change the PDL ratio
 - This can alter average rating parameters and the underlying claim frequency and severity
- Collectibility of premium
 - If a portion of the premium asset is not secured, then a provision should be made to anticipate bad debt

This covers the material from the Teng & Perkins paper. Next, we will dive into Feldblum's review of the paper.

VI. Feldblum Review

The PDL method has **several advantages** over other methods for estimating the premium asset:

- It is modeled directly on the retro rating formula, making it easy to explain
- Its focus on premium sensitivity is similar to how loss sensitive contracts are handled under Risk-Based Capital
- It may be useful when changes to retro rating parameters distorts indications from other methods

The PDL method is based on the relationship between premium and losses. **Why not estimate the accrued retro premium asset using a chain-ladder approach** on billed or collected premium?

There are a few reasons for this:

- Due to the lag in processing and recording retro premium adjustments, a chain-ladder estimate of the premium asset is not available until at least nine months after the policy expiration, and it can be updated only annually thereafter
- An initial estimate of the premium asset can be produced soon after the policy expires, using the known loss information and the relationships between incurred losses and retro premium
- The premium asset estimate can also be updated each quarter as new loss data becomes available

Retrospective Premium Determination

Recall that the retro premium is $P = [BP + (CL_n \times LCF)] \times TM$. The **two major components** of the retro premium are as follows:

1) Ratable Losses & Associated LAE

- This is the $LCF \times CL \times TM$ portion of the retro premium
- We say “ratable losses” because of the retro max, retro min, and per-accident limit

2) Basic Premium

- This is the $BP \times TM$ portion of the retro premium
- This covers non-LAE expenses, the net insurance charge, and the per-occurrence excess loss charge

Since the ratable losses are capped or floored by the retro max, retro min, and per-accident limit, we must load in expected amounts to cover these amounts. These expected amounts are represented by the net insurance charge and the per-occurrence excess loss charge:

- The net insurance charge is the difference between:
 - The expected loss to the insurer caused by the maximum retro premium (or equivalently, the maximum ratable loss)
 - The expected gain to the insurer caused by the minimum retro premium (or equivalently, the minimum ratable loss)
- The per-occurrence excess loss charge is expected losses excess of the per-accident limit

Fitzgibbon's Method

The retro premium $P = [BP + (CL_n \times LCF)] \times TM$ serves as the basis for Fitzgibbon's method. Fitzgibbon assumes that premium is a **linear function** of the (unlimited) incurred losses:

$$\boxed{\text{Retro Premium} = C + B \times \text{Losses}}$$

Then, he makes two modifications to this equation:

- Uses ratios to standard premium
- Subtracts 1 from both sides to obtain a retro adjustment

With these two modifications, the formula above is re-written as follows:

$$\text{Retro Adj.} = A + B \times \text{Standard Loss Ratio}$$

where $A = \frac{C}{\text{Standard Prem.}} - 1$.

Estimating the Parameters in Fitzgibbon's Method

Fitzgibbon uses a linear regression to estimate the parameters A and B . Feldblum points out something odd here. Since the parameters A and B clearly depend on the retro parameters being sold, why should we use a regression to estimate them? Wouldn't it be simpler to ask the pricing actuary what parameters are used in the retro rating plan?

The reason for this is simple. While actuarial reserves (in this case, the premium asset) are estimated on an aggregate basis, retro plan parameters vary from:

- Year to year
 - The basic premium may be low when interest rates are high, and the insurer expects high earnings on investments
- State to state
 - Premium taxes differ by state
- Plan to plan
 - A smaller insured may prefer a plan with a low maximum premium

Hence, a regression may be more accurate than collecting the actual parameters being sold. For instance, although the pricing actuary may recommend a basic premium charge of 30% of standard premium, the underwriter might reduce the basic premium to 25% of standard premium when selling the policy. The regression does a better job of capturing what is actually happening at policy issuance.

Difficulties with the regression are as follows:

- Regression performed on historical data may not apply to current policies due to changes in rating plan factors and aggregate loss ratios
- Premium on individual plans is not a simple linear function of total incurred losses (remember that losses are capped!)

Before leaving this section, you may be wondering how Teng & Perkins are able to deploy the formula approach version of the PDL method. Doesn't it rely on the retro parameters being sold? Yes, it does! Feldblum points out in the footnote on the bottom of page 282 that estimating PDL ratios using the retro formula is very difficult.

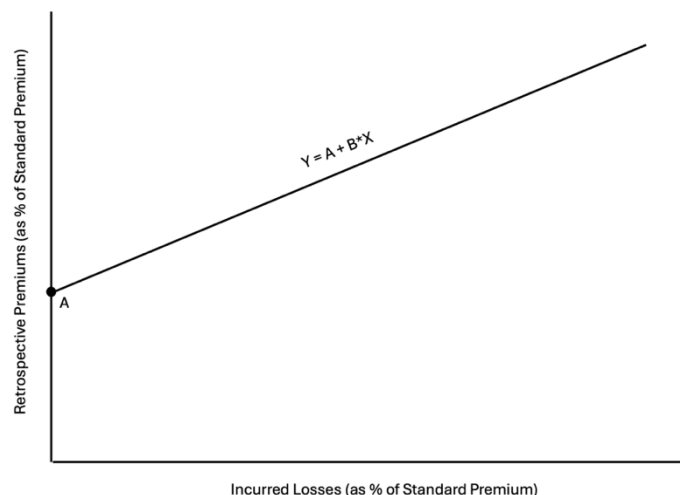
Graphical Representations

Fitzgibbon's method plots net earned premium as a % of standard premium against incurred losses as a % of standard premium. Net earned premium refers to earned premium net of retro adjustments (i.e., after refunds or additional premium).

This plot results in a straight line, $Y = A + B \cdot X$, where:

- A = basic premium percentage (net premium/standard premium)
- B = premium responsiveness

Here is a graphical example of Fitzgibbon's method:



As losses are incurred, the loss ratio increases, we move up the line, and the retro premium as a percentage of standard premium increases.

Note the following on the plot above:

- A and B are averages since they regressions across a book of business
- Feldblum uses the general labels “Incurred Losses” and “Retrospective Premiums” in his version of the plot above (page 283 of the text). Technically, everything should be as a percentage of standard premium for A and B to have the correct interpretation

The slope B is not exactly equal to 1 for the following reasons:

- Loss limitations (i.e., retro max and per-accident limit)
- Minimum premiums might exceed the basic premium
- The LCF and TM are applied to the incurred losses, which changes the slope

The combined impact of the three things above depends on the swing of the plan:

- For plans with narrow swing (i.e., small accounts with low loss limits and low maximum premiums), $B < 1$
- For plans with wide swing (i.e., large accounts with high loss limits and high maximum premiums), $B > 1$

Projections vs. Reality

Technically, Fitzgibbon’s graph is meant to relate the ultimate retrospective premium ratio to the ultimate loss ratio. Hence, it fails to consider the emerging experience of the book of business itself. By “emerging experience,” we are referring to the **emerging premium responsiveness** of the book, not the emerging losses.

The emerging premium responsiveness may differ from what is expected from the plot for a few reasons:

- A and B are estimates
 - They may vary from year to year
- Premium responsiveness may be less than expected due to large losses on individual policies
 - The effect of large losses on the retrospective premium may differ due to policy attributes such as per-accident limit and maximum premiums

If the true B is less than estimate of B shown in the graph, then this method will overstate the retro premium and the premium asset. The point is that this method has no way to “get back on track” if the actual premium responsiveness differs from the expected premium responsiveness.

Berry's Method

Similar to Fitzgibbon, Berry assumes a linear relationship between the loss ratio and the retrospective premium ratio. However, he modifies Fitzgibbon's method by gradually discarding the method and giving more weight to the actual experience of the book of business.

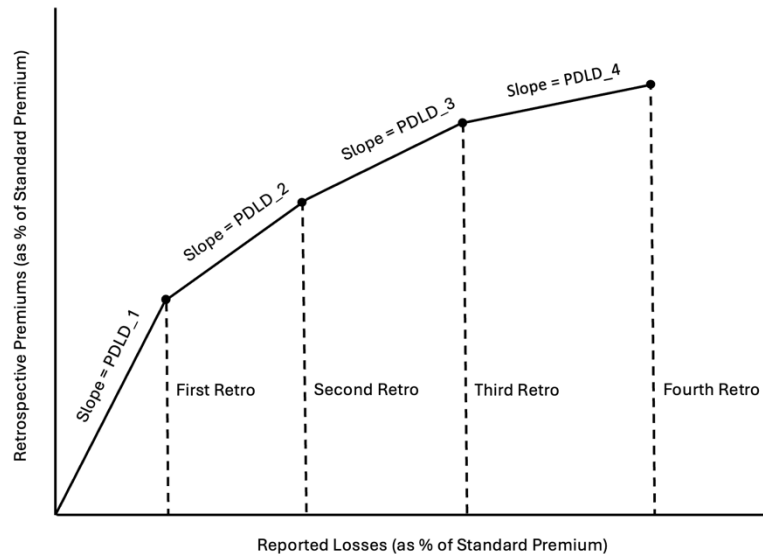
The Teng & Perkins Solution

There are two ways to interpret the Fitzgibbon (& Berry) graph of $Y = A + B \cdot X$:

- 1) Relates the ultimate loss ratio and the ultimate retro premium ratio among different books of business or different years (intention of Fitzgibbon & Berry)
- 2) Relates the reported loss ratio and the net earned premium (i.e., the retro premium) at different points of time for a single book of business (intention of Teng & Perkins)

Decreasing Slopes

As shown above, the Fitzgibbon (& Berry) method can be plotted as a straight line. The Teng & Perkins solution is plotted as a set of line segments with decreasing slope as losses are reported and move to the right:



The decreasing slopes are caused by loss limits and maximum premiums. After policy issuance, most of the losses are ratable. As more losses are reported and large losses develop, a higher portion of losses fall outside of the loss boundaries, resulting in less premium per dollar of loss.

Although this paper argues for decreasing slopes, some actuaries argue that the slopes of the early line segments should be the flattest. Their reasoning is that high-cost claims are often immediately recognized, well before the first retro adjustment. Since high-costs claims are the ones most likely to be truncated by loss limits, it seems to indicate flatter slopes early on. But, it's important to keep in mind that these types of claims are rare.

Feldblum states the following general rules regarding premium responsiveness:

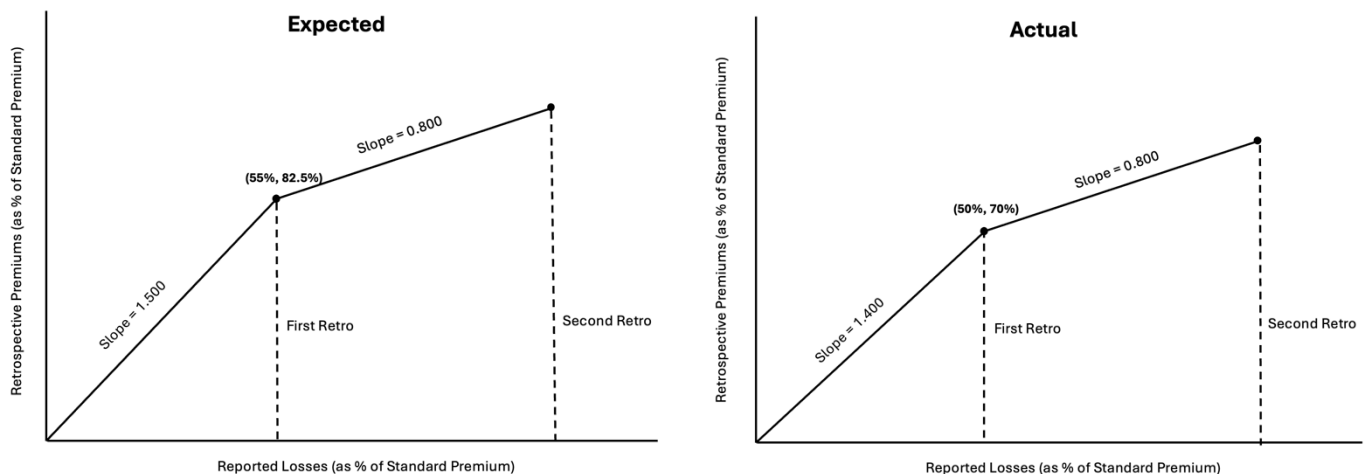
- 1) As a book of business matures, premium responsiveness on loss-sensitive contracts declines
 - As policies mature, a greater percentage of policies are excluded from retro rating by the maximum premium and by the loss limit
- 2) At higher loss ratios, premium responsiveness on loss-sensitive contracts declines
 - This refers to the overall loss ratio, not to specific types of claims
 - At higher overall loss ratios, more policyholders have reached their maximum premiums. This means even a small claim will result in no additional premium

Actual vs. Expected Experience

At the first adjustment, actual experience may differ from expected experience in two ways:

- Actual reported losses may differ from projected reported losses at policy inception
 - Ex. At policy inception, the projected reported loss ratio (losses divided by standard premium) at 18 months is 55%. However, the actual reported loss ratio at 18 months is 50%
- The relationship between reported losses and retro premium may differ from what was projected at policy inception
 - Ex. The projection suggests that an actual reported loss ratio of 50% should correspond to a retrospective premium ratio of 75%. However, the actual retrospective premium ratio is only 70%

The following plot illustrates both of the “deviations from expected” above:



- Expected Plot
 - The expected reported loss ratio at the first retro adjustment is 55%
 - Based on an expected slope of $PDL D_1 = 1.500$, the expected retrospective premium ratio at the first retro adjustment is $1.500(0.55) = 82.5\%$
- Actual Plot
 - The actual reported loss ratio at the first retro adjustment is 50%

- Based on an actual slope of $PDDL_1 = 1.400$, the actual retrospective premium ratio at the first retro adjustment is $1.400(0.50) = 70\%$

The Teng & Perkins Assumptions

Two assumptions underlie the PDL method:

- 1) The premium responsiveness during subsequent adjustments is independent of the premium responsiveness during preceding adjustments
 - Even if the slope of the first line segment is different from what is expected, we do not change our expectations for the slope of the second line
- 2) The slope of the line segment depends on the time period, not on the beginning loss ratio or the beginning retro premium ratio
 - This assumption relates to when we change from one line to another
 - Ex. When moving from the first line segment to the second line segment, we change at the first retro adjustment, regardless of where the loss ratio ended up

We can see both of these assumptions in action in the prior plots of expected and actual:

- **Application of Assumption 1:** In both plots, the subsequent PDL ratio (i.e., $PDDL_2$) is assumed to be 0.800, despite the fact that the actual slope of the first line segment was lower than expected
- **Application of Assumption 2:** In both plots, the change in slope occurs at the first retro adjustment, despite the fact that the actual reported loss ratio was lower than expected

These two assumptions fix the issues in Fitzgibbon's method:

- Fitzgibbon relates the ultimate loss ratio to the ultimate retro premium ratio. If the actual premium experience differs from what is expected, we have no way to get back on track
- The PDL ratio relates the reported loss ratio to the retro premium ratio. If the actual premium experience deviates from what is expected, the next line segment begins at a starting point that corresponds to actual experience

Feldblum's Enhancement

As shown in the prior plots, Teng & Perkins assume that the first line segment passes through the origin. In doing so, they are combining two separate items into the slope of the first line segment:

- 1) Basic premium ratio
- 2) True slope of the first line segment (i.e., $PDLD_1$)

One **problem with combining these items** is that we cannot tell how much each item contributes to the total slope of the first line segment.

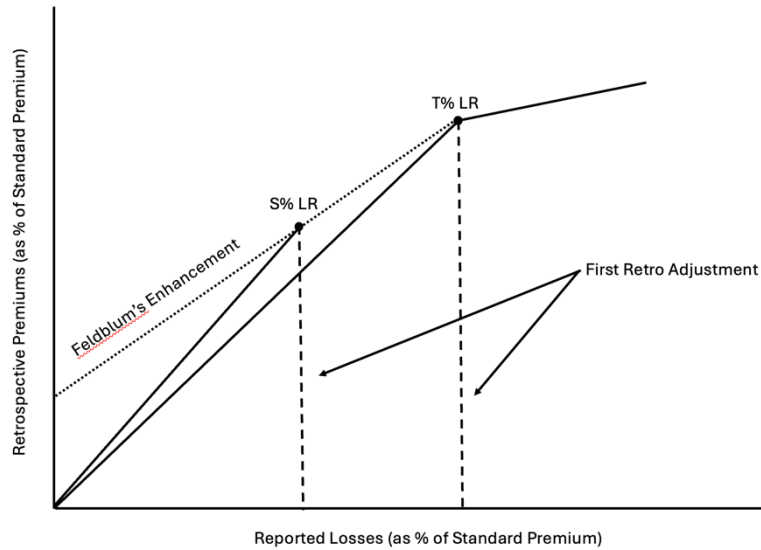
Another **problem** is that combining these items leads to confusing interpretations. For example:

- We normally interpret a CPDLD ratio of 1.492 at policy inception to the first retro adjustment to mean that each dollar of loss emerged provides the insurer one dollar and 49 cents of premium
- However, this is wrong because part of that 1.492 comes from the basic premium charge, which does not represent emerged losses

The failure to distinguish these items also makes it harder to analyze changes over time:

- Assume the slope of the first line segment has risen steadily over the past 30 years
- Is this due to a change in the average basic premium ratio, a change in premium responsiveness during the first period, or a lengthening of the loss reporting pattern?

As an example of how a lengthening of the loss reporting pattern can lead to confusing interpretations, consider the following plot:



In the plot above, we originally expected the reported loss ratio at the first retro adjustment to be T%. However, due to a lengthening of the loss reporting pattern, we now expect the reported loss ratio to be S% at the first retro adjustment (note that the expected ultimate loss ratio remains the same).

What happened when we moved from T% to S%? The slope of the first line segment changed even though premium responsiveness didn't change! It was caused by a change in the loss reporting pattern.

To fix this issue, Feldblum suggests that we subtract the average basic premium charge as a ratio to the standard loss ratio (i.e., $\frac{BP/SP}{ELR}$) from the first CPDLD ratio:

- We could also subtract the average basic premium charge as a ratio to the standard loss ratio multiplied by the incremental loss emergence at the first adjustment (i.e., $\frac{BP/SP}{ELR \times \%Loss_1}$) from the first PDL ratio
- In both cases, we are ignoring the tax multiplier
- To properly account for the tax multiplier, we would need to multiply the pieces being subtracted by the tax multiplier

- The plot would then have a positive y -intercept and the slope of the first line would be flatter
- In addition, the S% loss ratio and T% loss ratio would live on the same line because premium responsiveness has not changed

Example: Feldblum's Enhancement

Given the following information as of December 31, 2023 (in \$000):

PY	Ultimate Losses	Losses Reported at Prior Adjustment	Premium Booked at Prior Adjustment	Currently Booked Premium	Standard Premium
2020	1875	1,725	2,375	2,377	2,500
2021	1880	1,400	2,650	2,800	2,750
2022	1750	0	0	2,000	2,800
2023	2100	0	0	2,070	2,400

	<u>Retrospective Adjustments</u>				
	First	Second	Third	Fourth	Subsequent
$PDLL_n$	1.55	0.85	0.65	0.45	0.00
% Emerged	72%	12%	8%	6%	2%

- Average basic premium factor (including tax) = 0.15
- ELR = 70%
- Policy years 2022-2023 have not had a retro adjustment

First, let's calculate the "enhanced" versions of the CPDLD ratios:

PY	Teng & Perkins CPDLD Ratios	Feldblum's Enhanced CPDLD Ratios
2020	0.494	0.494
2021	0.646	0.646
2022	1.297	1.083
2023	1.297	1.083

Feldblum's enhancement only impacts the first CPDLD ratio:

- $CPDLD_1^{TP} = \frac{1.55(0.72)+0.85(0.12)+0.65(0.08)+0.45(0.06)+0(0.02)}{0.72+0.12+0.08+0.06+0.02} = 1.297$
- $CPDLD_1^{Feld} = CPDLD_1 - \left(\frac{BP/SP}{ELR}\right)TM = 1.297 - \frac{0.15}{0.70} = 1.083$

Second, let's calculate the premium asset based on the enhanced CPDLD ratios:

PY	Expected Future Loss Development	Expected Future Premium from Loss	Basic Premium Portion for First Adjustment	Ultimate Premium	Premium Asset
2020	150	74	0	2,449	72
2021	480	310	0	2,960	160
2022	1,750	1,895	420	2,315	315
2023	2,100	2,275	360	2,634	564
Total				10,358	1,111

Let's discuss the table above, using PY 2023 as an example:

- Expected Future Loss Development
 - Ultimate Losses – Losses Reported at Prior Adjustment
 - PY 2023: 2,100 – 0 = 2,100
- Expected Future Premium from Loss
 - Expected Future Loss Development × Enhanced CPDLD Ratio
 - PY 2023: 2,100(1.083) = 2,274
- Basic Premium Portion for First Adjustment
 - Average Basic Premium Factor × Standard Premium
 - PY 2023: 0.15(2,400) = 360
- Ultimate Premium
 - Expected Future Premium from Loss + Basic Premium Portion + Premium Booked at Prior Adjustment
 - PY 2023: 2,274 + 360 + 0 = 2,634

- Premium Asset
 - Ultimate Premium – Currently Booked Premium
 - PY 2023: $2,634 - 2,070 = 564$

You may be wondering why we bother splitting the expected future premium into “loss” and “basic premium” components. In the end, don’t we just end up in the same place? Not necessarily. If the expected future loss emergence *differs* from the a priori expected loss for the first retro adjustment (i.e., $ELR \times SP$), then the Teng & Perkins approach will give a different answer than Feldblum’s enhancement. See the Cookbook recipe for a good example of this.

I. Introduction

In this paper, Meyers uses Schedule P data from 1997 to validate the Mack model and the bootstrap ODP model. He then searches for better models based on Bayesian MCMC methods.

When models do not accurately predict the distribution of outcomes for the test data, there are three explanations:

- 1) The insurance process is too dynamic to be captured in a single model (black swan type events will never be modeled well)
- 2) There could be other models that better fit the data
- 3) The data used to calibrate the model is missing crucial information needed to make a reliable prediction (such as changes in claim processes, changes in the way the underlying business is conducted, etc.)

He ends the paper by exploring dependencies between lines and risk margins.

II. The CAS Loss Reserve Database

As stated earlier, this paper relies on Schedule P data from 1997 to fit various loss reserve models to a set of 200 insurer loss triangles. Data beyond 1997 was also used so that the model predictions could be compared to actual outcomes.

For all of the models mentioned in the paper, the underlying assumption is that there have not been any substantial changes in the insurers' operations. Although this is not a realistic assumption, Meyers created an algorithm to search for loss triangles that showed overall stability.

III. Validating the Mack Model

To test the Mack model, Meyers selected 200 incurred loss triangles (50 each from four different LOBs – commercial auto (CA), personal auto (PA), workers' compensation (WC), and other liability (OL)).

For each triangle, he used the Mack model to calculate the mean and standard deviation (i.e., standard error). Then, Meyers fit a lognormal distribution with a mean and standard deviation that matched those produced by the Mack model. Finally, the actual outcome was converted into a percentile of the lognormal distribution.

As an **example**, consider the following output from the Mack model for a single insurer's incurred losses:

w	$\hat{C}_{w,10}$	SD	$C_{w,10}$	Percentile
1	3,917	0	3,917	
2	2,538	0	2,532	
3	4,167	3	4,279	
4	4,367	37	4,341	
5	3,597	34	3,587	
6	3,236	40	3,268	
7	5,358	146	5,684	
8	3,765	225	4,128	
9	4,013	412	4,144	
10	3,955	878	4,181	
Total	38,914	1,057	40,061	86.03

In the table above, w represents the accident year, $\hat{C}_{w,10}$ represents the estimated ultimate loss for each accident year, and $C_{w,10}$ represents the actual ultimate loss for each accident year. In this example, the mean and standard deviation of the total ultimate losses from the Mack model are 38,914 and 1,057, respectively. These two values were used to parameterize a lognormal distribution. The actual total ultimate loss is 40061. It falls in the 86th percentile of the lognormal distribution.

Once we build the predictive distributions for all 200 insurers using the Mack model, we can test the validity of the model using a variety of tests. Each of the tests leverages the fact that the percentiles from each insurer should be uniformly distributed. This fact can be used because the 200 data points form a large enough sample to test for uniformity.

Test 1: Histogram

If the percentiles are uniformly distributed, the height of the bars should be equal. For a small sample, the heights will not be perfectly level.

Test 2: $p - p$ Plot & Kolmogorov-Smirnov (K-S) Test

Test 2 is a formal test for the statistical significance of uniformity.

Suppose we have a sample of n predicted percentiles ranging from 0 to 100. We first sort them into increasing order. The expected value of these percentiles is given by $\{e_i\} = 100 \cdot \{1/(n + 1), 2/(n + 1), \dots, n/(n + 1)\}$. We then plot the expected percentiles on the x -axis and the sorted predicted percentiles on the y -axis. If these predicted percentiles are uniformly distributed, we expect this plot to lie along a 45-degree line.

We can test the significance of the *percentiles using the K-S test*. With this test, we can reject the hypothesis that a set of percentiles is uniform at the 5% level if the K-S statistic, D , is greater than its critical value. The K-S statistic, D , is defined as follows:

$$D = \max |p_i - f_i|$$

where $\{p_i\}$ represents the set of percentiles and $\{f_i\} = 100 \cdot \{1/n, 2/n, \dots, n/n\}$. The critical value is $\frac{136}{\sqrt{n}}$.

On a $p - p$ plot, these critical values appear as 45-degree bands that run parallel to the line $y = x$. We reject the hypothesis of uniformity if the $p - p$ plot lies outside the bands.

In this paper, Meyers deems a model “validated” if it passes the K-S test.

Example: K-S Test

An actuary is validating a model using the K-S test on loss development data for 20 companies. Suppose we are given the following percentiles for each company's actual loss development within the distribution derived with the model, in ascending order:

4	11	19	57
6	14	21	66
8	15	28	78
9	16	32	91
10	18	42	95

To determine whether the model passes the K-S test, we need to calculate $D = \max|p_i - f_i|$. If D is greater than the critical value, then the model fails the K-S test. In this example, the critical value is $\frac{136}{\sqrt{n}} = \frac{136}{\sqrt{20}} = 30.4$.

Let's calculate the absolute difference between p and f for each company:

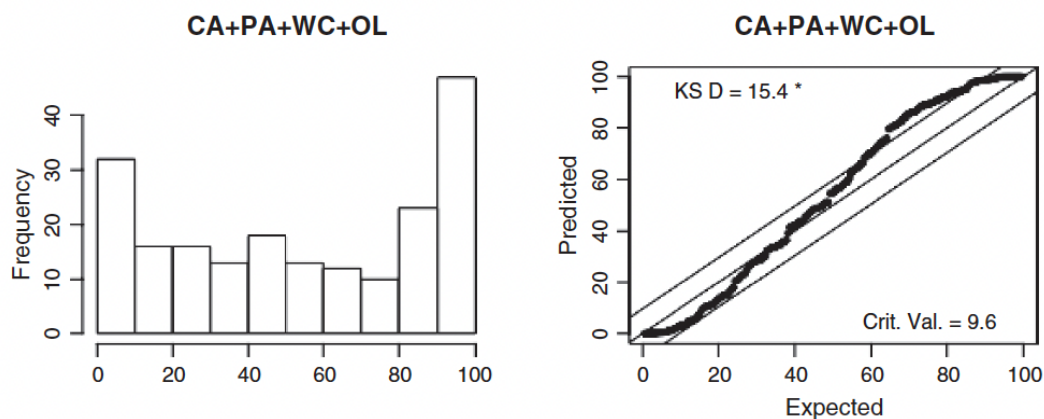
Company	p	f	Abs. Diff.
1	4	$1/20 \cdot 100 = 5$	$1 = 5 - 4 $
2	6	10	4
3	8	15	7
4	9	20	11
5	10	25	15
6	11	30	19
7	14	35	21
8	15	40	25
9	16	45	29
10	18	50	32
11	19	55	36
12	21	60	39
13	28	65	37

14	32	70	38
15	42	75	33
16	57	80	23
17	66	85	19
18	78	90	12
19	91	95	4
20	95	100	5

The maximum absolute difference is $D = 39$. Since $D > 30.4$, the model is NOT validated.

Mack Model Plots – Incurred Losses

Moving back to the 200 insurers used by Meyers, here are the plots for the Mack model for all insurers (and all lines combined) on incurred loss triangles:



Histogram

Looking at the histogram, we see that the percentiles show little uniformity. In addition, since the actual outcomes are falling into the smaller and larger percentiles of the distributions produced by the Mack model more often than the middle percentiles, we can conclude that the Mack model produces a distribution that is light-tailed. If the Mack model produced larger tails, the outcomes currently falling in the largest percentile would shift towards the middle percentiles.

p – p Plot

The $p - p$ plot produces the same interpretation. As we can see in the graph, the predicted percentiles (based on where the **actual outcomes fall** within the distributions produced by the Mack model) form an “S” shape. This means that the model is light-tailed because the actual outcomes are falling into percentiles that are lower than expected in the left tail and higher than expected in the right tail.

For example, when the expected percentile is 20%, the predicted percentile appears to be closer to 15%. If the tails were heavier, the predicted percentile would shift towards 20%. Note that if a backward “S” shape was present, it would imply that the Mack model was heavy-tailed.

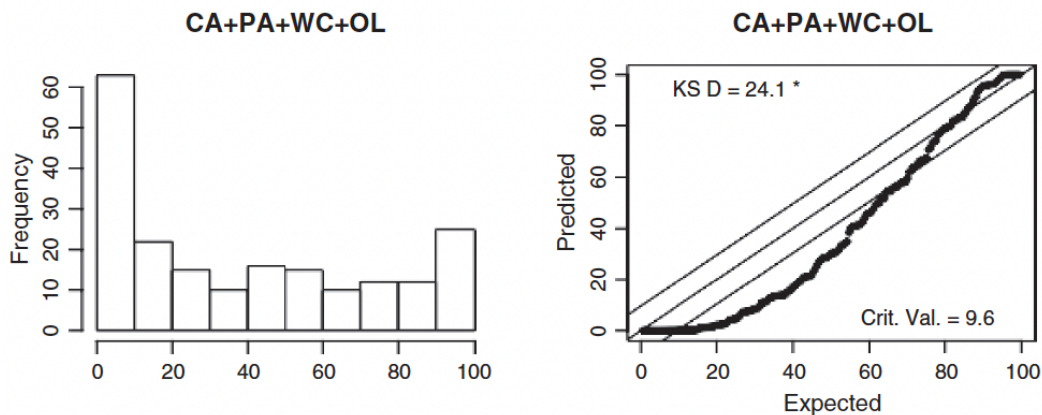
K-S Statistic

Lastly, we see that the K-S D statistic is 15.6 and the critical value is $9.6 = \frac{136}{\sqrt{200}}$. Since D is greater than the critical value, we reject the hypothesis of uniformity at the 5% level.

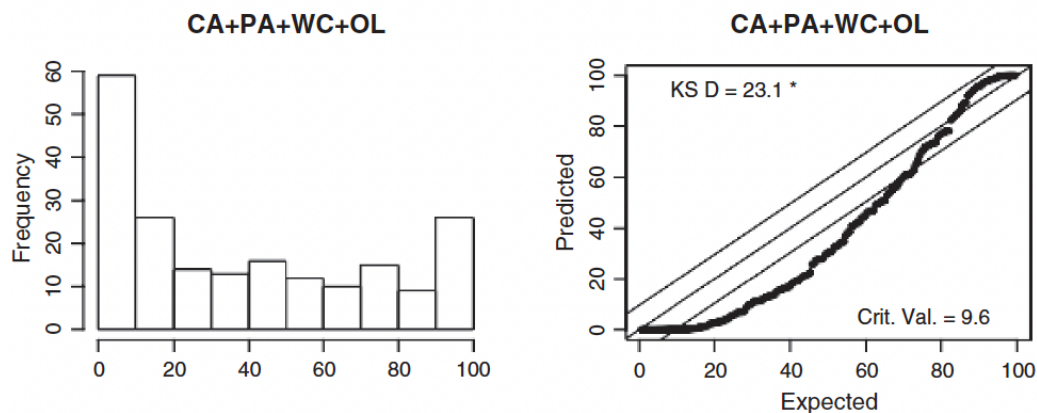
IV. Validating the Bootstrap ODP Model

To test the bootstrap ODP model, Meyers selected 200 paid loss triangles. He also tested the Mack model on 200 paid loss triangles.

Here are the plots for the bootstrap ODP model for all insurers (and all lines combined) on paid loss triangles:



Here are the plots for the Mack model for all insurers (and all lines combined) on paid loss triangles:



Interpreting the Plots

Looking at all four plots, we see that the actual outcomes are occurring in the lower percentiles of the model distributions more often. This implies that both models produce expected loss estimates that are biased high when modeling paid losses. This means that more of the actual outcomes will fall in lower percentiles because the model distributions are shifted too far to the right. If they shifted back to the left, then there would be fewer outcomes falling into the left tails.

In addition, the K-S D statistics are greater than the critical value, implying that uniformity is not present in either model.

The next few sections of this paper focus on various Bayesian MCMC reserving models that could potentially address the deficiencies in the Mack and ODP models. The first one we will look at is the cross classified model.

V. The Cross Classified Model

The cross classified (CRC) model covered here is similar to the one presented in Taylor & McGuire. The main difference is that the model specification explicitly includes premium and the expected loss ratio. As a result, the interpretation of the parameters changes somewhat.

The model specifications for a 10×10 triangle are as follows:

- $\log elr \sim \text{normal}(-0.4, \sqrt{10})$
- $\alpha_w \sim \text{normal}(0, \sqrt{10})$ for $w = 2, \dots, 10$. Set $\alpha_1 = 0$
- $\beta_d \sim \text{normal}(0, \sqrt{10})$ for $d = 1, \dots, 9$. Set $\beta_{10} = 0$
- $a_i \sim \text{uniform}(0,1)$ for $i = 1, \dots, 10$
- Set $\sigma_d^2 = \sum_{i=d}^{10} a_i$ for $d = 1, \dots, 10$. This ensures that $\sigma_1^2 > \dots > \sigma_{10}^2$
- Set $\mu_{w,d} = \log(\text{Premium}_w) + \log elr + \alpha_w + \beta_d$
- Then $C_{w,d} \sim \text{lognormal}(\mu_{w,d}, \sigma_d)$

Let's discuss the specification above:

- Since this is a Bayesian model, each parameter is assigned a prior distribution. For example, the natural log of the expected loss ratio is assigned a normal prior with mean -0.4 and standard deviation $\sqrt{10}$
- Since $C_{w,d}$ is lognormally distributed with parameters $\mu_{w,d}$ and σ_d , expected cumulative losses for cell (w, d) is $e^{\mu_{w,d} + \frac{\sigma_d^2}{2}}$. Hence, when $d = 10$, the expected ultimate losses for AY w are $e^{\mu_{w,10} + \frac{\sigma_{10}^2}{2}}$
- When $w = 1$ and $d = 10$:
 - $\alpha_1 + \beta_{10} = 0 + 0 = 0$
 - $\mu_{1,10} = \log(\text{Premium}_1) + \log elr + \alpha_1 + \beta_{10} = \log(\text{Premium}_1) + \log elr$
 - The expected ultimate losses are $e^{\mu_{1,10} + \frac{\sigma_{10}^2}{2}} = e^{\log(\text{Premium}_1) + \log elr + \frac{\sigma_{10}^2}{2}} = \text{Premium}_1 \times elr \times e^{\frac{\sigma_{10}^2}{2}}$
 - Since σ_{10} is typically small, $\text{Premium}_1 \times elr \times e^{\frac{\sigma_{10}^2}{2}} \approx \text{Premium}_1 \times elr$
 - Thus, $e^{\log elr}$ can be approximately interpreted as the expected ultimate loss ratio
- The magnitude of α_w s gives an indication of how much the expected ultimate loss ratio varies from year to year

- The prior distributions are wide (i.e., large variances). Thus, the posterior distributions of each parameter will be heavily influenced by the data
- In total, there are 29 parameters in the CRC model

Producing Predictive Distributions

The goal is to obtain the predictive distribution of ultimate losses for each AY and in total. To accomplish this, MCMC simulation is used to produce a sample size of 10,000 from the posterior distribution of the model. In other words, we produce 10,000 parameter sets, where each set contains 29 parameter values (one draw from the posterior distribution).

For each of the 10,000 parameter sets, we do the following:

- Set $\mu_{w,d} = \log(\text{Premium}_w) + \log elr + \alpha_w + \beta_d$ for $w = 2, \dots, 10$
- Simulate $\tilde{C}_{w,10} \sim \text{lognormal}(\mu_{w,10}, \sigma_{10})$ for $w = 2, \dots, 10$
- Calculate $\tilde{C}_{Tot,10} = C_{1,10} + \sum_{w=2}^{10} \tilde{C}_{w,10}$

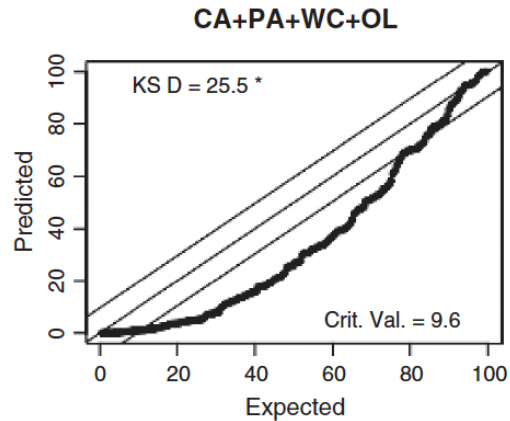
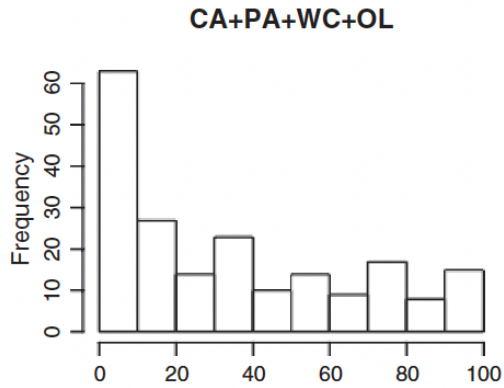
At this point, we have 10,000 simulated ultimate loss outcomes by AY and in total. These are the predictive distributions.

For each predictive distribution, the sample mean and sample standard deviation of the 10,000 simulated ultimate losses are referred to as the estimate and standard error, respectively. We obtain the coefficient of variation (CV) by dividing the standard error by the estimate.

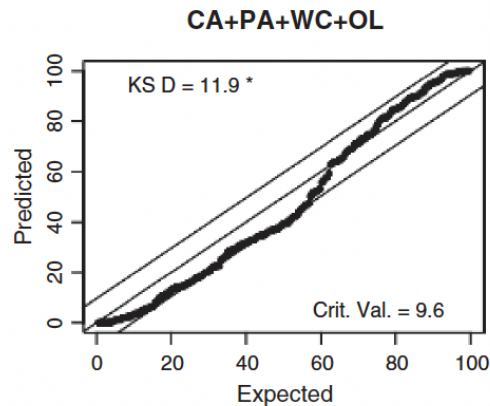
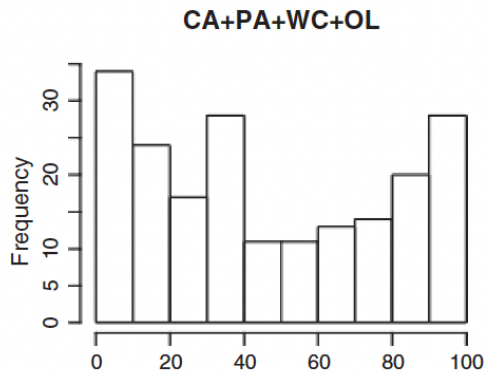
To obtain the predicted percentile of the outcome for all AYs combined, we compute the proportion of $\tilde{C}_{Tot,10}$ s that are less than or equal to the actual outcome.

CRC Model Histograms & $p - p$ Plots

The histogram and $p - p$ plot for the CRC model applied to the 200 selected paid triangles are as follows:



The histogram and $p - p$ plot for the CRC model applied to the 200 selected incurred triangles are as follows:



The CRC model performs similarly to the Mack and ODP models:

- Paid losses – the CRC model tends to overpredict losses
- Incurred losses – the CRC model tends to understate the variability of the predicted losses

Meyers refers to the plots above as indicators of the reputation of the model, where reputation means the result of a **retrospective analysis** based on the set of 200 loss triangles (i.e., how well the model predicts the holdout ultimate losses in the lower right portion of the triangle). In this case, the CRC model has a reputation of producing light tails on incurred data and overstating losses on paid data.

Although a model's reputation is helpful, a reserving actuary should understand how the model performs on the data being analyzed. To do this, Meyers uses residual plots, which act as a **prospective diagnostic** of model performance.

CRC Model Residual Box Plots

When fitting a stochastic reserving model to a loss triangle, the loss triangle data is the training data. Unlike a frequentist model, which produces a single set of parameter estimates, a Bayesian MCMC model produces j realizations of each parameter.

To produce residuals, we take a random sample of the j MCMC draws (e.g., 100) and calculate the standardized residuals, $r_{w,d}^j$, for the log of all losses in the training data. For **example**, since a 10×10 triangle has 55 observed cells, we would compute 5,500 standardized residuals (100 for each cell). The residuals are computed as follows:

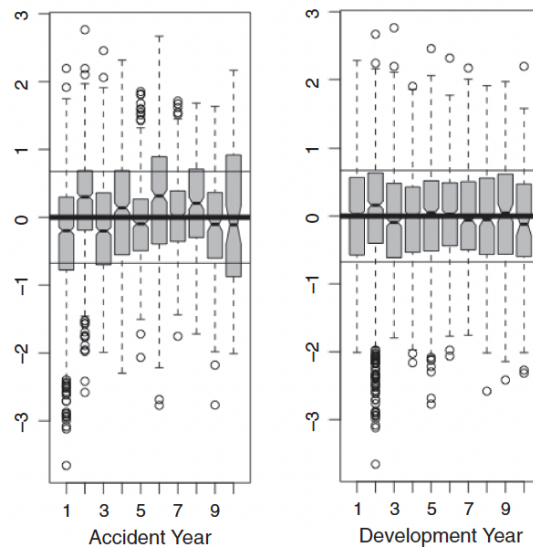
$$r_{w,d}^j = \frac{\log(C_{w,d}) - \mu_{w,d}^j}{\sigma_d^j}$$

As expected, the standardized residuals are found by subtracting the model-implied mean, $\mu_{w,d}^j$, from $\log(C_{w,d})$ and dividing by the model-implied standard deviation, σ_d^j .

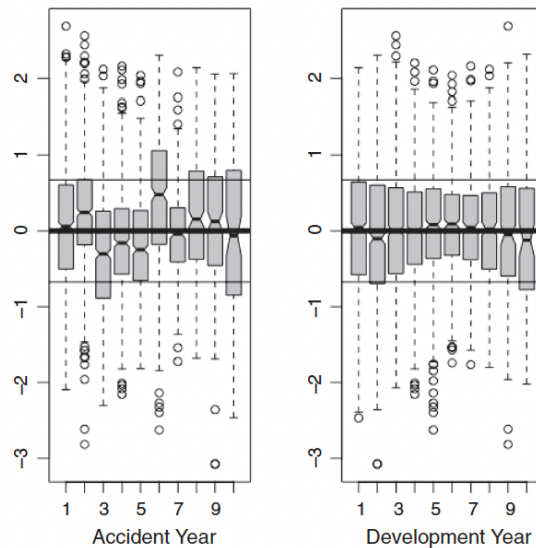
Then, box plots are created for the residuals by accident year and development year. Recall the following characteristics of a box plot:

- Plots the interquartile range (IQR), which runs from the 25th percentile to 75th percentile
- Plots whiskers that extend from the 25th and 75th percentiles. Each whisker extends 1.5 times the IQR
- Plots individual points located beyond the whiskers (generally interpreted as outliers)

The residual box plots for the CRC model applied to paid losses are as follows:



The residual box plots for the CRC model applied to incurred losses are as follows:



The thin black lines represent the expected IQR, while the thick black line represents the expected median of 0. The plots appear fairly reasonable, especially since each box plot contains 0 within its IQR. However, we will gain additional perspective on these plots as we move to better-fitting models.

VI. The Stochastic Cape Cod Model

Suppose we have a 10×10 loss triangle. Under the Cape Cod model, the projected ultimate losses for AY w are as follows:

$$\hat{C}_{w,11-w} = C_{w,11-w} + Premium_w \times elr \times V_{11-w}$$

where elr is estimated from the data and V_{11-w} is the estimated proportion of losses that remain unreported at the end of the most recent development period $11 - w$.

The Stochastic Cape Cod (SCC) model is specified as follows:

- $logelr \sim \text{normal}(-0.4, \sqrt{10})$
- $\beta_d \sim \text{normal}(0, \sqrt{10})$ for $d = 1, \dots, 9$. Set $\beta_{10} = 0$
- $a_i \sim \text{uniform}(0,1)$ for $i = 1, \dots, 10$
- Set $\sigma_d^2 = \sum_{i=d}^{10} a_i$ for $d = 1, \dots, 10$. This ensures that $\sigma_1^2 > \dots > \sigma_{10}^2$
- Set $\mu_{w,d} = \log(Premium_w) + logelr + \beta_d$
- Then $C_{w,d} \sim \text{lognormal}(\mu_{w,d}, \sigma_d)$

Producing Predictive Distributions

For each of the 10,000 parameter sets, we do the following:

- Set $\mu_{w,10} = \log(Premium_w) + logelr$ for $w = 2, \dots, 10$
Recall that $\beta_{10} = 0$
- Simulate $\tilde{C}_{w,10} \sim \text{lognormal}(\mu_{w,10}, \sigma_{10}) - e^{\mu_{w,11-w} + \sigma_{11-w}^2/2} + C_{w,11-w}$ for $w = 2, \dots, 10$
- Calculate $\tilde{C}_{Tot,10} = C_{1,10} + \sum_{w=2}^{10} \tilde{C}_{w,10}$

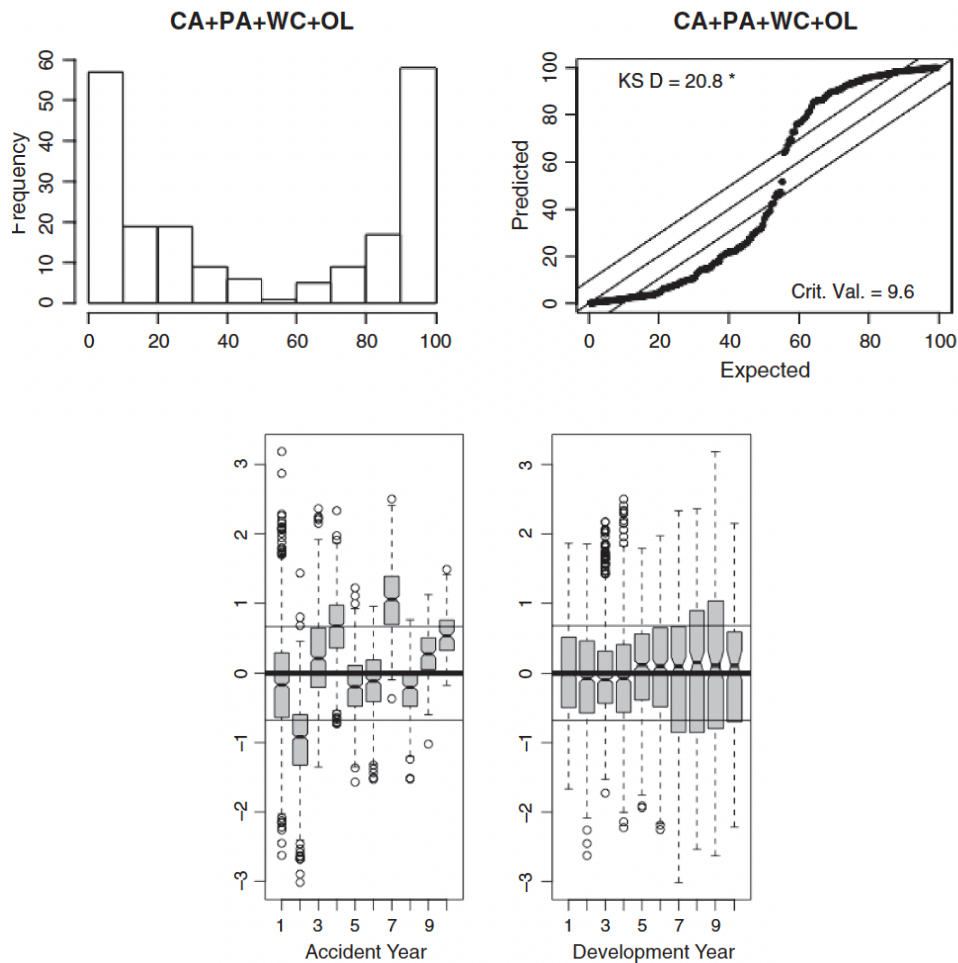
Notice that the second bullet above differs considerably from the corresponding step in the CRC model. Under the SCC model, the ultimate loss is equal to the current reported losses plus the Cape Cod reserve. We mirror this in the model by doing the following:

- Simulate an ultimate loss from a lognormal distribution

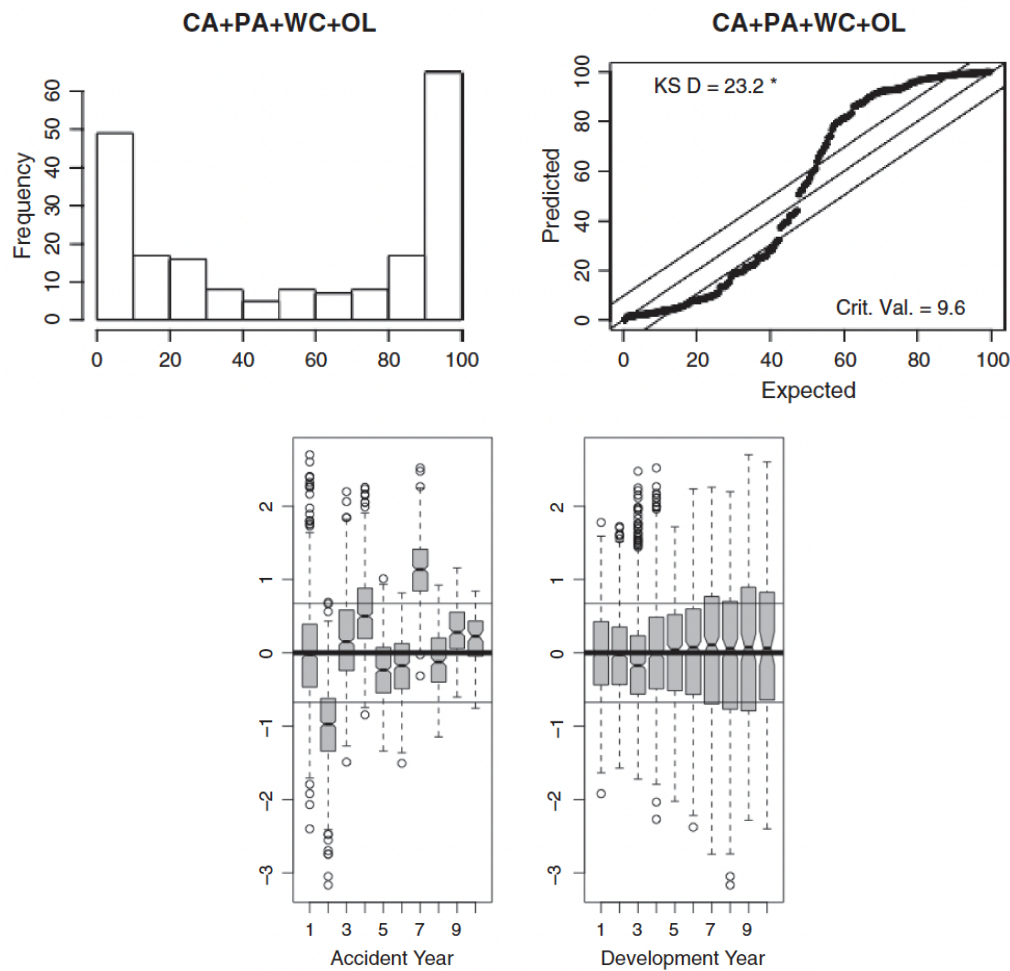
- Subtract the SCC model's expected current reported loss to obtain the Cape Cod reserve
- Add the actual current reported loss to the Cape Cod reserve to obtain the Cape Cod ultimate loss

SCC Model Results

The histogram, $p - p$ plot, and residual box plots for the SCC model applied to the 200 selected paid triangles are as follows:



The histogram, $p - p$ plot, and residual box plots for the SCC model applied to the 200 selected incurred triangles are as follows:



In both cases, the SCC model has a reputation for understating the variability of the predictive losses. In addition, the model shows worse performance than the CRC model. The residual box plots by AY are noticeably worse than those of the CRC model, suggesting that the assumption of a fixed expected loss ratio across AYs may be driving the poor model performance.

Example: Running the SCC Model

An SCC model is fit to the following incurred loss triangle as of 12/31/2025:

Accident Year	Cumulative Incurred Losses at Month XX (in \$000)					Premium
	12	24	36	48	60	
2021	69,975	78,590	80,805	82,400	83,805	116,010
2022	82,600	90,865	93,850	94,900		128,230
2023	91,215	99,065	101,170			137,225
2024	93,395	103,025				149,104
2025	100,050					157,710

The following parameters were sampled from the posterior distribution for a single simulation from the SCC model:

Parameter	Sampled Value
$\log elr$	-0.3005
β_1	-0.2050
β_2	-0.1150
...	...
β_5	0.0000
σ_1	0.2632
σ_2	0.1511
...	...
σ_5	0.0128

An actuary wishes to simulate the ultimate loss outcome for accident year 2025 based on the SCC model parameters above.

Recall that $\tilde{C}_{w,5} \sim \text{lognormal}(\mu_{w,5}, \sigma_5) - e^{\mu_{w,6-w} + \sigma_6^2 - w/2} + C_{w,6-w}$. For accident year 2025, $w = 5$. Thus, $\tilde{C}_{5,5} \sim \text{lognormal}(\mu_{5,5}, \sigma_5) - e^{\mu_{5,1} + \sigma_1^2/2} + C_{5,1}$.

First, let's determine the log means:

- $\mu_{5,5} = \log(\text{Premium}_5) + \log \text{elr} = \log(157,710) - 0.3005 = 11.6680$
- $\mu_{5,1} = \log(\text{Premium}_5) + \log \text{elr} + \beta_1 = \log(157,710) - 0.3005 - 0.2050 = 11.4630$

Second, let's simulate an ultimate loss for accident year 2025:

- $\tilde{C}_{5,5} \sim \text{lognormal}(11.6680, 0.0128) - e^{11.4630+0.2632^2/2} + 100,050$
- Using Excel, we do "LOGNORM.INV(rand(), 11.6680, 0.0128) – exp(11.4630+(0.2632^2)/2)+100500"
- For this specific random seed, this generates a simulated ultimate loss of 116,265

Comparing Models Using Akaike Information Criteria (AIC)

The general formula for AIC is as follows:

$$AIC = 2 \times p - 2 \times L(\text{data}|\hat{\theta})$$

where $\hat{\theta}$ is the parameter vector that maximizes the log-likelihood, L , of the data. AIC rewards a model for having a high log-likelihood, but it penalizes the model for having more parameters. Thus, models with low AIC values are preferred.

Note that although this formula appears to differ from the one in Shapland, the two are equivalent. Shapland has simply substituted a particular log-likelihood into the general AIC formula.

AIC poses a couple of problems under a Bayesian framework:

- 1) There is no single maximum likelihood estimate of the parameter vector θ . Instead, there is an entire sample of parameter vectors drawn from the model's posterior distribution
- 2) The penalty for the number of parameters should be smaller in the presence of strong prior information

To address these problems, Meyers uses the expected log predictive density (elpd). The elpd measures how a model predicts observations that were not used to fit the model.

For each observation x_i , the estimated log predictive density (lpd) is defined as follows:

$$\widehat{lpd}_i = \log \left(\frac{1}{J} \sum_{j=1}^J p(x_i | \theta_j) \right)$$

where:

- J is the number of MCMC simulations
- θ_j is the j th parameter vector from the posterior distribution, $j = 1, \dots, J$
- $p(x_i | \theta_j)$ is the probability density of observing x_i under parameter vector θ_j

In words, we calculate the probability of observing x_i under each simulated parameter vector, average those probabilities, and then take the log.

Next, we sum over all I observations to obtain $\widehat{lpd}$:

$$\widehat{lpd} = \sum_{i=1}^I \log \left(\frac{1}{J} \sum_{j=1}^J p(x_i | \theta_j) \right)$$

When $\widehat{lpd}$ is computed on holdout data (the lower right portion of the loss triangle), it is an unbiased estimate of the elpd. However, when $\widehat{lpd}$ is computed on the training data (the upper left portion of the loss triangle) used to generate the parameters, it is a biased estimate of the elpd. The amount of bias is called the effective number of parameters.

Thus, Meyers uses leave-one-out cross-validation, or LOO. The idea is to leave out the observation x_i , fit the model using all the other observations, and then evaluate how well the model predicts the omitted observation. Repeating this for each observation gives:

$$\widehat{elpd}_{loo} = \sum_i^I \log \left(\frac{1}{J} \sum_{j=1}^J p(x_i | \theta_j^{(-i)}) \right)$$

where $\theta_j^{(-i)}$ represents parameter vectors from a model fit without observation x_i .

In practice, Meyers does not refit the model separately for each omitted observation because that would be computationally expensive. Instead, software packages have methods for efficiently computing $\widehat{elpd}_{loo}$.

Models with higher $\widehat{elpd}_{loo}$ values are preferred. However, we can restate model results in a manner similar to AIC as follows:

$$LOOIC = -2 \times \widehat{elpd}_{loo} = 2 \times p_{loo} - 2 \times \widehat{lpd}$$

where p_{loo} is the effective number of parameters. In this case, models with lower LOOIC values are preferred.

The table below summarizes the quantities above for the CRC and SCC models:

Model	$\widehat{elpd}_{loo}$	p_{loo}	LOOIC
CRC-Paid	47.80	14.97	-95.60 = -2(47.80)
SCC-Paid	-5.14	8.75	10.28
CRC-Incurred	70.97	15.07	-141.93
SCC-Incurred	-2.85	9.13	5.69

As seen above, the CRC models have higher $\widehat{elpd}_{loo}$ values and lower LOOIC values. Hence, they are preferred over the SCC models.

The poor SCC results may be partly attributable to the lack of premium adjustments. In general, under the Cape Cod method, premiums should be adjusted to a level consistent with expected losses. Meyers points out that he did not make these adjustments.

VII. The Changing Settlement Rate Model

The changing settlement rate (CSR) model is intended to correct for the tendency of the CRC model to overestimate ultimate losses when applied to paid data. It does this by allowing for changes in the claim settlement rate.

The model is specified as follows:

- $\log elr \sim \text{normal}(-0.4, \sqrt{10})$
- $\alpha_w \sim \text{normal}(0, \sqrt{10})$ for $w = 2, \dots, 10$. Set $\alpha_1 = 0$
- $\beta_d \sim \text{normal}(0, \sqrt{10})$ for $d = 1, \dots, 9$. Set $\beta_{10} = 0$
- $\gamma \sim \text{normal}(0, 0.05)$
- $a_i \sim \text{uniform}(0, 1)$ for $i = 1, \dots, 10$
- Set $\sigma_d^2 = \sum_{i=d}^{10} a_i$ for $d = 1, \dots, 10$. This ensures that $\sigma_1^2 > \dots > \sigma_{10}^2$
- Set $\mu_{w,d} = \log(\text{Premium}_w) + \log elr + \alpha_w + \beta_d(1 - \gamma)^{w-1}$
- Then $C_{w,d} \sim \text{lognormal}(\mu_{w,d}, \sigma_d)$

Due to the model specification above, β_d is typically negative. However, it generally becomes less negative as d increases, resulting in higher cumulative paid losses at later development ages.

A positive value for γ causes $\beta_d(1 - \gamma)^{(w-1)}$ to increase with w . Since β_d is typically negative, this causes $\mu_{w,d}$ to increase for more recent accident years. This pattern is consistent with a speedup in claims settlement.

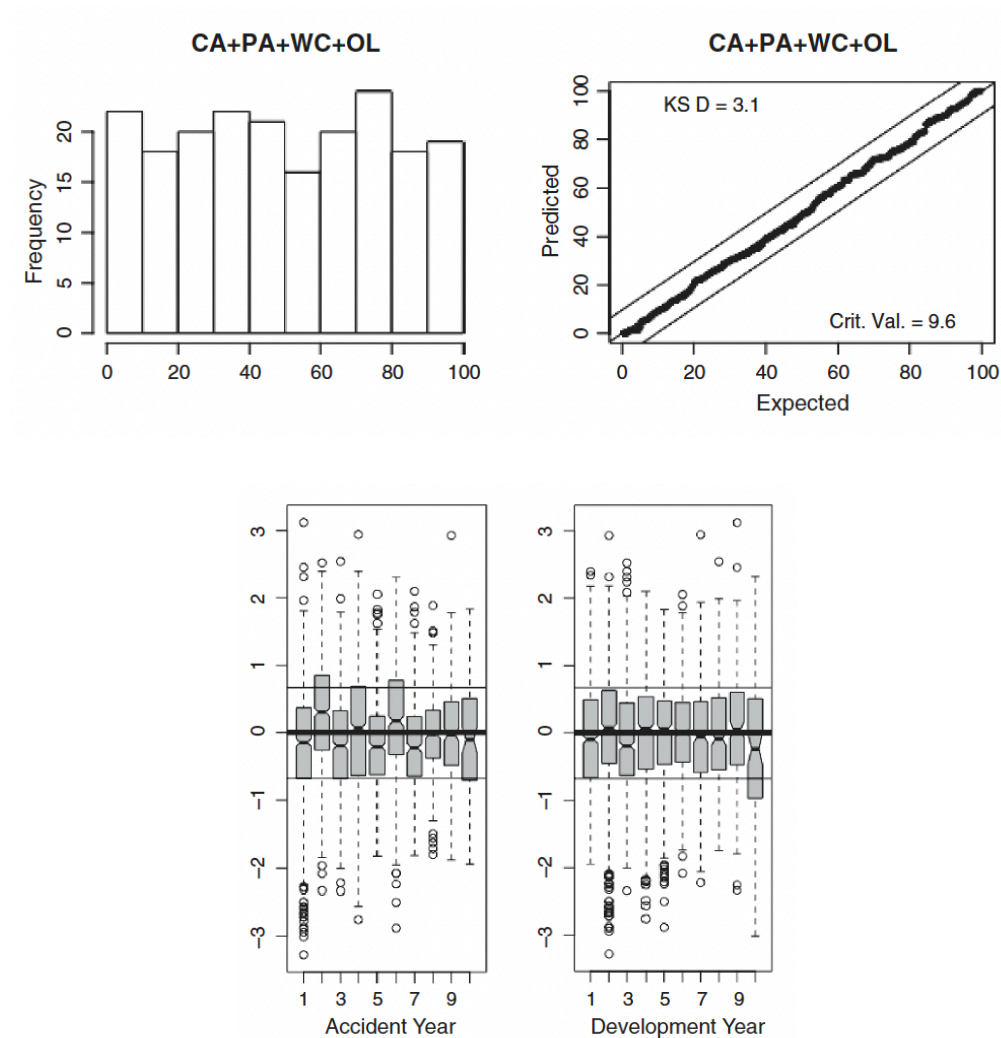
When Meyers fit the model, the posterior distribution for γ was primarily positive, indicating a speedup in claims settlement.

Producing Predictive Distributions

The process for producing predictive distributions for the CSR model matches that of the CRC model.

CSR Model Results

The histogram, $p - p$ plot, and residual box plots for the CSR model applied to the 200 selected paid triangles are as follows:



Based on the histogram and $p - p$ plot, the model has a strong reputation for producing uniformly distributed predictive percentiles. The residual box plots look reasonable as well.

The table below compares the CSR and CRC models using $\widehat{elpd}_{loo}$ and LOOIC:

Model	$\widehat{elpd}_{loo}$	p_{loo}	LOOIC
CSR-Paid	49.76	15.09	-99.53
CRC-Paid	47.80	14.97	-95.60

The quantities above clearly favor the CSR model. Combined with its strong reputation, including a settlement rate parameter appears to improve the tendency of the CRC model to overestimate ultimate losses.

Using the elpd Statistic on Test Data

So far, we have computed $\widehat{elpd}_{loo}$, which is a leave-one-out cross-validation estimate of the elpd based on the training data.

We now introduce $\widehat{elpd}_{test}$, which refers to the elpd statistic computed on test/holdout data (i.e., lower right triangle data). Since we only see lower right triangle data “after the fact,” this is a retrospective test statistic that informs a model’s reputation.

The statistic is computed as follows:

$$\widehat{elpd}_{test} = \sum_{i=1}^N \log \left(\frac{1}{J} \sum_{j=1}^J p(x_i | \theta_j) \right)$$

In this case, we are summing over the N individual observations in the lower right triangle.

When a changing settlement rate is present, we would expect the CSR model to outperform the CRC model. When settlement rates are relatively stable, we would expect the CRC model to perform as well as or better than the CSR model.

The following table counts the number of times that the elpd statistic favors the CSR model for both the “loo” (upper left triangle) and the “test” (lower right triangle) statistics:

Line	CSR > CRC (loo)	CSR > CRC (test)
CA	27	27
PA	26	30
WC	27	30
OL	26	32
Total	106	119

For both loo and test data, the CSR model outperformed the CRC model in more than half of the 200 paid triangles. This suggests that changes in claim settlement rates occur fairly often across all four lines of insurance.

$p - p$ Plot vs. $\widehat{elpd}_{test}$

The $p - p$ plot and $\widehat{elpd}_{test}$ evaluate different aspects of predictive performance:

- **$p - p$ plot**
 - Examines the sum of the last column of the lower right triangle
 - Assesses whether the predictive distribution of the total ultimate loss is well calibrated by comparing the realized ultimate outcome to its predicted percentile
- **$\widehat{elpd}_{test}$**
 - Evaluates the predictive density assigned to each individual observation in the lower right triangle
 - As a result, $\widehat{elpd}_{test}$ incorporates substantially more information than the $p - p$ plot and can detect improvements in cell-level predictive accuracy even when two models produce similar distributions for total ultimate losses
 - This makes $\widehat{elpd}_{test}$ a more sensitive tool for comparing competing models

VIII. Correlated Accident Year Model

- The correlated accident year (CAY) model is intended to correct for the tendency of the CRC model to underestimate the variability in predicted losses when applied to incurred data. It does this by allowing for correlation between accident years.

Suppose we have a 10×10 loss triangle. The model is specified as follows:

- $\log elr \sim \text{normal}(-0.4, \sqrt{10})$
- $\alpha_w \sim \text{normal}(0, \sqrt{10})$ for $w = 2, \dots, 10$. Set $\alpha_1 = 0$
- $\beta_d \sim \text{normal}(0, \sqrt{10})$ for $d = 1, \dots, 9$. Set $\beta_{10} = 0$
- Set $\rho = 2 \times \rho_{pos} - 1$, where $\rho_{pos} \sim \text{beta}(2, 2)$. This allows ρ to take on any value in the interval $(-1, 1)$
- $a_i \sim \text{uniform}(0, 1)$ for $i = 1, \dots, 10$
- Set $\sigma_d^2 = \sum_{i=d}^{10} a_i$ for $d = 1, \dots, 10$. This ensures that $\sigma_1^2 > \dots > \sigma_{10}^2$
- Set $\mu_{1,d} = \log(\text{Premium}_1) + \log elr + \beta_d$
- Set $\mu_{w,d} = \log(\text{Premium}_w) + \log elr + \alpha_w + \beta_d + \rho(\log(C_{w-1,d}) - \mu_{w-1,d})$, $w > 1$
- Then $C_{w,d} \sim \text{lognormal}(\mu_{w,d}, \sigma_d)$

Regarding the correlation term above, there are a couple important notes:

- When $\rho = 0$, the CAY model reduces to the CRC model
- $\text{Corr}[\log(C_{w,d}), \log(C_{w-k,d})] = \frac{\rho}{1+\rho^2}$ when $k = 1$ and 0 when $k > 1$

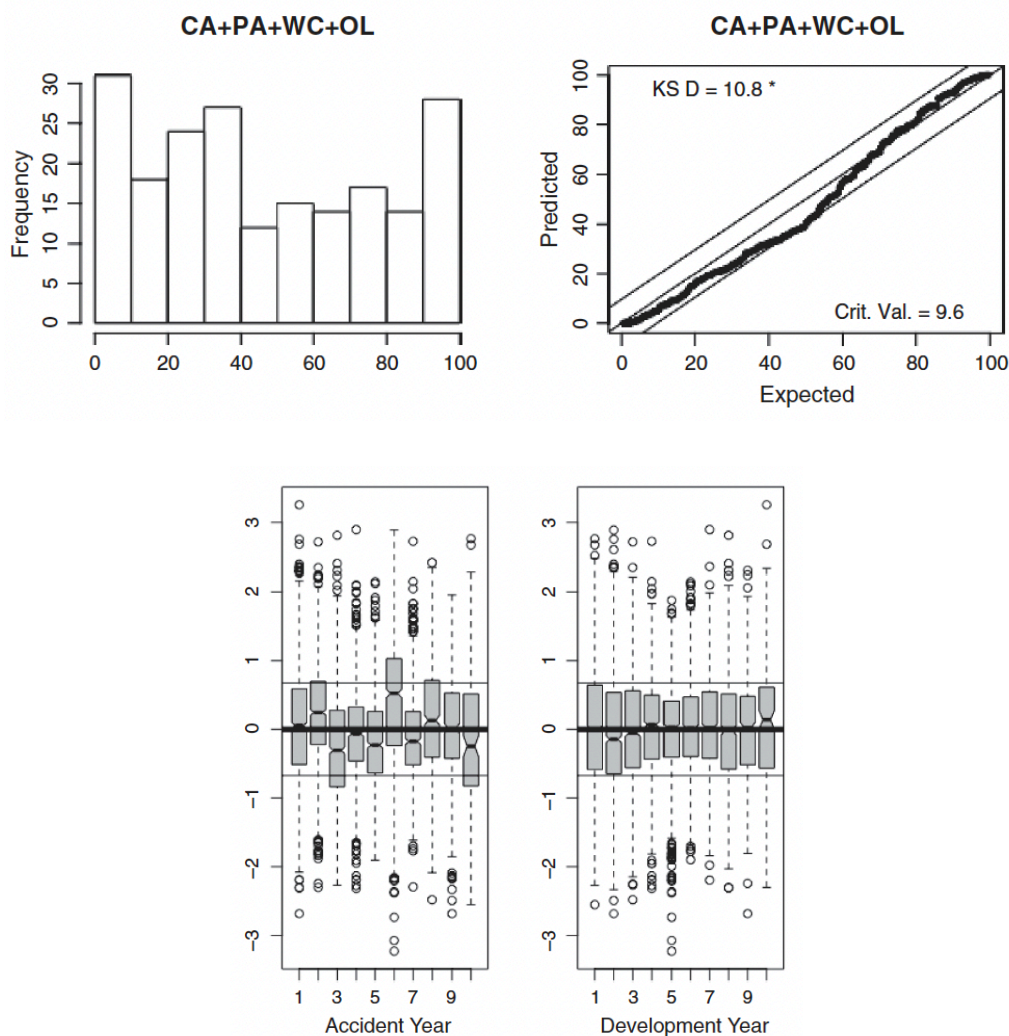
Producing Predictive Distributions

For each of the 10,000 parameter sets, we do the following:

- Set $\mu_{1,10} = \log(\text{Premium}_1) + \log elr$
 - Recall that $\alpha_1 = \beta_{10} = 0$
- Set $\mu_{2,10} = \log(\text{Premium}_2) + \log elr + \alpha_2 + \rho(\log(C_{1,10}) - \mu_{1,10})$
- Simulate $\tilde{C}_{2,10} \sim \text{lognormal}(\mu_{2,10}, \sigma_{10})$
- Set $\mu_{w,10} = \log(\text{Premium}_w) + \log elr + \alpha_w + \rho(\log(\tilde{C}_{w-1,10}) - \mu_{w-1,10})$ for $w = 3, \dots, 10$
- Simulate $\tilde{C}_{w,10} \sim \text{lognormal}(\mu_{w,10}, \sigma_{10})$ for $w = 3, \dots, 10$
- Calculate $\tilde{C}_{Tot,10} = C_{1,10} + \sum_{w=2}^{10} \tilde{C}_{w,10}$

CAY Model Results

The histogram, $p - p$ plot, and residual box plots for the CAY model applied to the 200 selected incurred triangles are as follows:



The histogram and $p - p$ plot show that the CAY model still has a reputation for producing a predictive distribution with light tails. However, the plots look better than those of the CRC model, and the K-S statistic is smaller. Similar to the CRC model, the residual box plots look reasonable. In addition, the posterior distribution of ρ is primarily positive, indicating positive correlation between adjacent accident year losses. Hence, the standard error of the total ultimate loss across accident years is larger than that produced by the CRC model.

Regarding the elpd statistics, the following table counts the number of times that the CAY model is favored using both the “loo” and “test” data:

Line	CAY > CRC (loo)	CAY > CRC (test)
CA	5	31
PA	7	28
WC	11	38
OL	3	24
Total	26	121

Although the CAY model outperforms the CRC model on the test data (which aligns with the $p - p$ plot), the loo statistic favors the CRC model. The posterior distribution of the correlation parameter ρ is fairly wide, indicating substantial uncertainty regarding the strength of the correlation. This suggests that the training data may not contain enough information to clearly distinguish between the CAY and CRC models.

Example: Running the CAY Model

A CAY model is fit to a 10×10 loss triangle to estimate a distribution of ultimate losses at age 10. During the MCMC process, 10,000 parameter sets were simulated to fill out the posterior distribution of the parameters.

Simulation #5 contained the following parameters:

Parameter	Sampled Value
<i>logelr</i>	-0.3945
α_1	0.0000
α_2	-0.2619
...	...
α_{10}	0.1779
β_1	-0.5976
...	...
β_{10}	0.0000

σ_1	0.2632
σ_2	0.1511
$\dots$	$\dots$
σ_{10}	0.0128
ρ	0.1709

Suppose that the cumulative losses for accident year 1 as of development period 10 are equal to \$3,900. Further suppose that the premiums for accident years 1 and 2 are \$5,812 and \$4,908, respectively.

Let's calculate the log mean for accident year 2:

- The log mean for accident year 2 is $\mu_{2,10} = \log(\text{Premium}_2) + \text{logelr} + \alpha_2 + \rho(\log(C_{1,10}) - \mu_{1,10})$
- We have everything we need except for $\mu_{1,10}$. To calculate $\mu_{1,10}$, we simply do $\mu_{1,10} = \log(\text{Premium}_1) + \text{logelr} = \log(5,812) - 0.3945 = 8.2732$
- Thus, $\mu_{2,10} = \log(4,908) - 0.3945 - 0.2619 + 0.1709(\log(3,900) - 8.2732) = 7.8415$

At this point, we can simulate an ultimate loss for accident year 2 by drawing from a lognormal distribution with log mean $\mu_{2,10} = 7.8415$ and log standard deviation $\sigma_{10} = 0.0128$.

For this specific simulation, the expected ultimate loss for accident year 2 is $e^{\mu_{2,10} + \frac{\sigma_{10}^2}{2}} = \$2,544$.

IX. Combining the CAY & CSR Models

As shown in the prior sections, the CSR model appears to perform best on paid data, while the CAY model appears to perform best on incurred data. In both models, the *logelr* and α_w parameters have the same interpretation, and possibly the same values.

In this section, Meyers combines aspects of each model to produce the **Integrated Paid and Incurred (IPI) model**. The key idea is that we may be able to produce more precise estimates of

the $\log elr$ and α_w parameters by using information from both the paid and incurred losses when estimating them. In doing so, we have less parameter variance and lower ultimate loss standard error.

When specifying the model, some parameters have “P” prefixes and others have “I” prefixes. These indicate that the parameters are specific to the paid and incurred data, respectively.

The model is specified as follows:

- $\log elr \sim \text{normal}(-0.4, \sqrt{10})$
- $\alpha_w \sim \text{normal}(0, \sqrt{10})$ for $w = 2, \dots, 10$. Set $\alpha_1 = 0$
- ${}_P\beta_d \sim \text{normal}(0, \sqrt{10})$ for $d = 1, \dots, 10$
 - Notice that we do not set ${}_P\beta_{10} = 0$
 - This reduces any distortion caused by the $\log elr$ parameter for the paid loss triangle being materially different from the $\log elr$ parameter for the incurred loss triangle
- $\gamma \sim \text{normal}(0, 0.05)$
- ${}_Pa_i \sim \text{normal}(0, 1)$ for $i = 1, \dots, 10$
- Set ${}_P\sigma_d^2 = \sum_{i=d}^{10} {}_Pa_i$ for $d = 1, \dots, 10$. This ensures that ${}_P\sigma_1^2 > \dots > {}_P\sigma_{10}^2$
- Set ${}_P\mu_{w,d} = \log(\text{Premium}_w) + \log elr + \alpha_w + {}_P\beta_d(1 - \gamma)^{w-1}$
- Then ${}_PC_{w,d} \sim \text{lognormal}({}_P\mu_{w,d}, {}_P\sigma_d)$
- ${}_I\beta_d \sim \text{normal}(0, \sqrt{10})$ for $d = 1, \dots, 9$. Set ${}_I\beta_{10} = 0$
- Set $\rho = 2 \times \rho_{pos} - 1$, where $\rho_{pos} \sim \text{beta}(2, 2)$. This allows ρ to take on any value in the interval $(-1, 1)$
- ${}_Ia_i \sim \text{normal}(0, 1)$ for $i = 1, \dots, 10$
- Set ${}_I\sigma_d^2 = \sum_{i=d}^{10} {}_Ia_i$ for $d = 1, \dots, 10$. This ensures that ${}_I\sigma_1^2 > \dots > {}_I\sigma_{10}^2$
- Set ${}_I\mu_{1,d} = \log(\text{Premium}_1) + \log elr + {}_I\beta_d$
- Set ${}_I\mu_{w,d} = \log(\text{Premium}_w) + \log elr + \alpha_w + {}_I\beta_d + \rho(\log({}_IC_{w-1,d} - {}_I\mu_{w-1,d}))$ for $w > 1$
- Then ${}_IC_{w,d} \sim \text{lognormal}({}_I\mu_{w,d}, {}_I\sigma_d)$

What do we notice in the specification above?

- 1) The *logelr* and α_w parameters have no prefixes and are shared parameters. Hence, their posterior distributions are informed by both the paid data and the incurred data
- 2) The next set of parameters include “P” prefixes (with the exception of γ) and are identical to those found in the CSR model. Their posterior distributions are informed primarily by the paid data
- 3) The final set of parameters include “I” prefixes (with the exception of ρ) and are identical to those found in the CAY model. Their posterior distributions are informed primarily by the incurred data
- 4) The ${}_PC_{w,d}$ values are simulated from a lognormal distribution whose mean is determined by the shared parameters and the paid-specific parameters
- 5) The ${}_IC_{w,d}$ values are simulated from a lognormal distribution whose mean is impacted by the shared parameters and the incurred-specific parameters

Producing Predictive Distributions

Similar to the other models, the Bayesian process begins by estimating the joint posterior distribution of all model parameters. Unlike the other models, the estimation process for the IPI model requires both paid and incurred data because the *logelr* and α_w are shared.

Once the joint posterior distribution is estimated, 10,000 parameter sets are drawn and applied separately to the paid and incurred portions of the model to produce paid and incurred predictive distributions.

The paid predictive distribution is based on the shared parameters and the paid-specific parameters. It follows the predictive distribution process shown earlier for the CSR model.

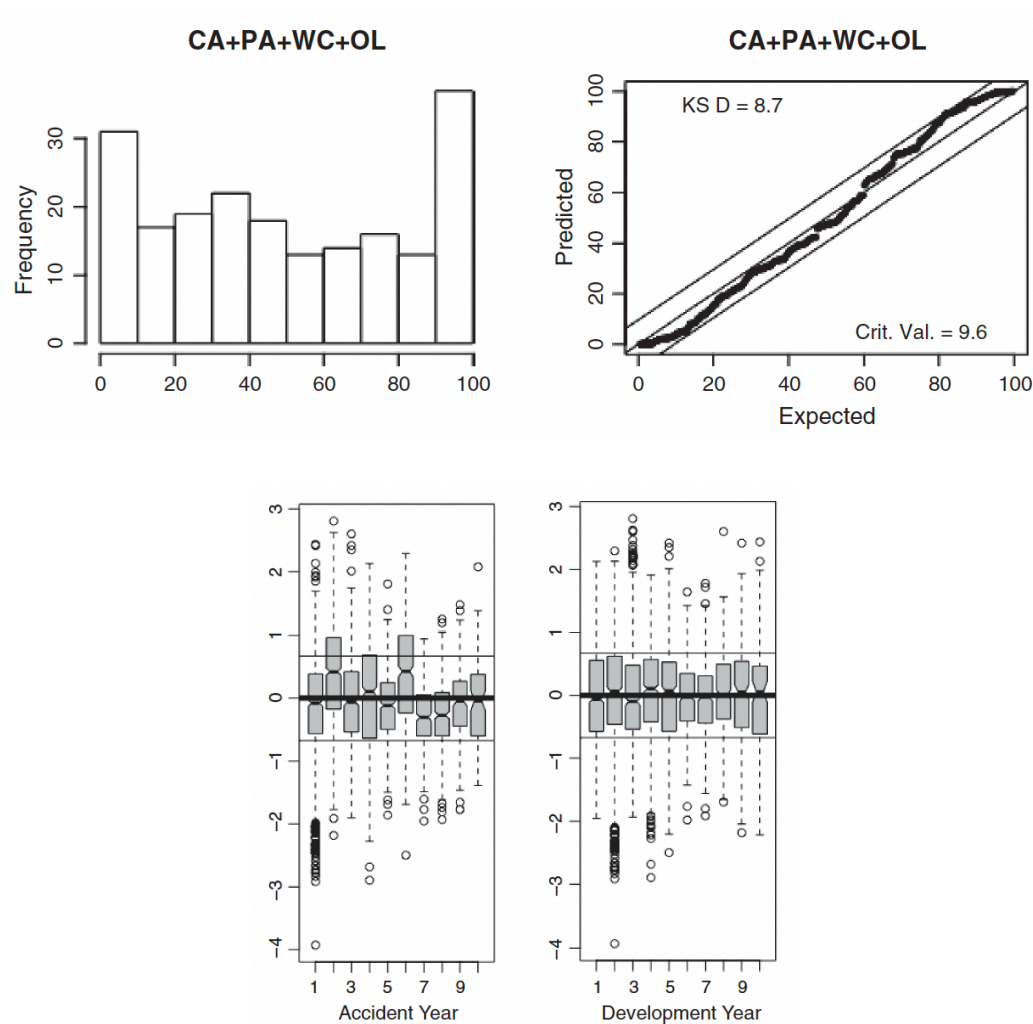
The incurred predictive distribution is based on the shared parameters and the incurred-specific parameters. It follows the predictive distribution process shown earlier for the CAY model.

IPI Model Results

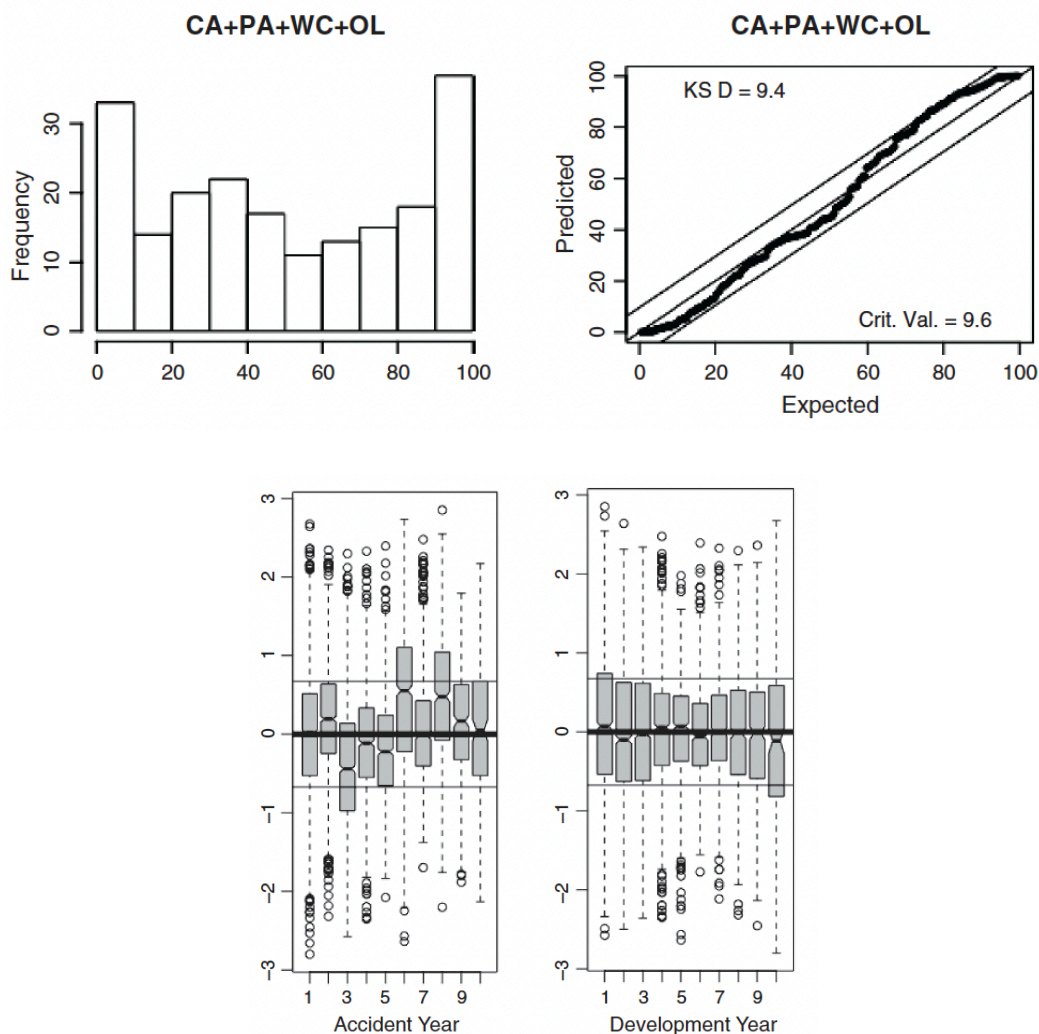
For earlier accident years, the posterior means of the *logelr* and α_w parameters are very similar to those of the CSR and CAY models. However, there is some divergence in the later accident years.

A more interesting result is that the posterior standard deviations of the *logelr* and α_w parameters are significantly smaller than those of the other models. As a result, the ultimate loss estimates produced by the IPI model have smaller standard errors than those of the other models.

The histogram, $p - p$ plot, and residual box plots for the IPI model applied to the 200 selected paid triangles are as follows:



The histogram, $p - p$ plot, and residual box plots for the IPI model applied to the 200 selected incurred triangles are as follows:



In both cases, the $p - p$ plots remain within the bounds. Of course, they are on an all lines combined basis. If we examine the $p - p$ plots by line, we find that the IPI model appears to understate the variability in predicted losses for the WC line. This is not completely surprising, as WC is a long-tailed line in which significant differences between paid and incurred losses may remain even after 10 years.

Although the residual plots are slightly worse than the corresponding plots for the CSR and CAY models, they still appear reasonable.

Looking at the $\widehat{elpd}_{loo}$ and $\widehat{elpd}_{test}$ statistics, there are several instances where the IPI model outperforms the corresponding CSR and CAY models. This is especially true for the paid data. This result is consistent with the key idea behind the IPI model that information from the incurred losses can improve estimation of the shared parameters that drive the paid predictive distribution.

X. Dependencies Between Lines of Insurance

For the reserving actuary, the ultimate objective is to estimate the insurer's aggregate loss reserve liability. For insurers writing multiple lines of business, this requires combining reserve estimates across lines into a single aggregate reserve distribution.

To accomplish this, one must consider the correlations that remain between lines after models have been fit to the data. The idea is that many apparent correlations can be explained through appropriate modeling methodology. Our goal is to capture any remaining correlation.

Recall that the models for our lines of business take the following form:

$$\log(C_{w,d}) \sim \text{normal}(\mu_{w,d}^j, \sigma_d^j)$$

where j represents one of the J MCMC simulations. The $\mu_{w,d}^j$ and σ_d^j parameters can come from any of the paid SCC, CSR, or IPI models.

For a given simulation j , we use the following bivariate normal model for lines of business X and Y :

$$\begin{pmatrix} \log(C_{w,d}^X) \\ \log(C_{w,d}^Y) \end{pmatrix} \sim \text{normal} \left(\begin{pmatrix} \mu_{w,d}^j \\ \mu_{w,d}^j \end{pmatrix}, \begin{pmatrix} (\sigma_d^j)^2 & \rho^j \cdot \sigma_d^j \cdot \sigma_d^j \\ \rho^j \cdot \sigma_d^j \cdot \sigma_d^j & (\sigma_d^j)^2 \end{pmatrix} \right)$$

where ρ^j represents the correlation between the two lines of business for simulation j .

Now, we need a large number of posterior parameter sets (e.g., 10,000) that contain the quantities above. This posterior distribution is obtained as follows:

- 1) Use Bayesian MCMC to obtain the following sample:

$$\{\mu_{w,a}^j, \sigma_{d,a}^j, \mu_{w,d}^j, \sigma_{d,d}^j\}_{j=1}^{10,000}$$

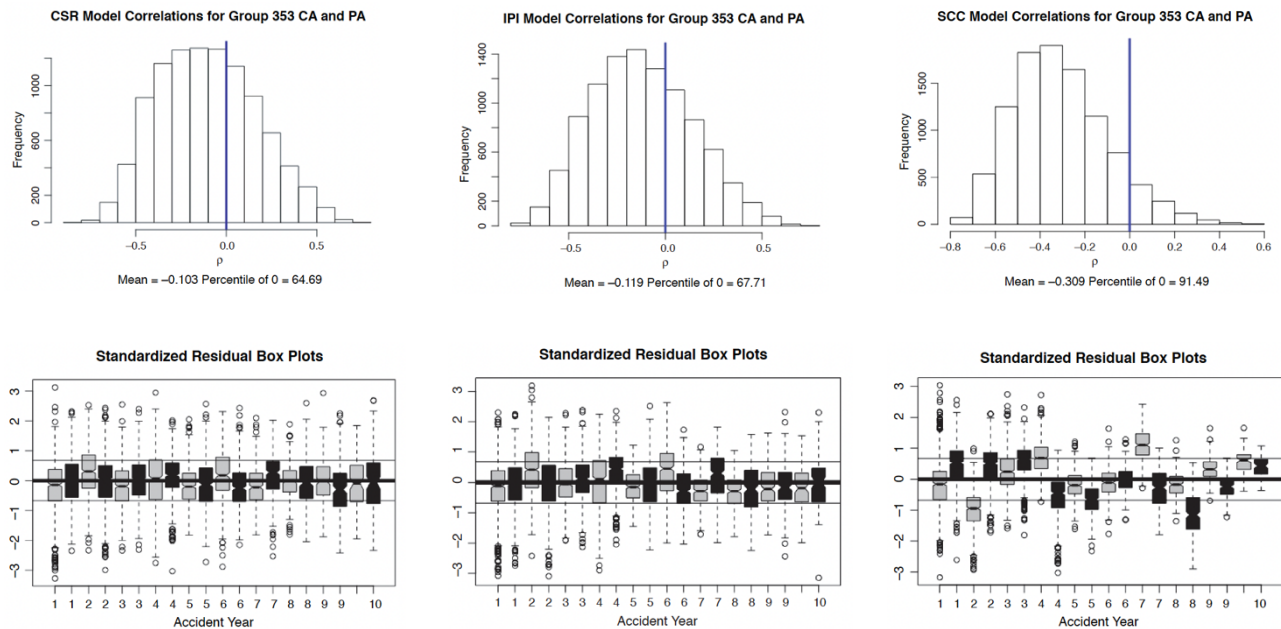
- 2) For each j , use Bayesian MCMC to take a sample of size one from the posterior distribution of:

$$\rho^j | \mu_{w,a}^j, \sigma_{d,a}^j, \mu_{w,d}^j, \sigma_{d,d}^j$$

In other words, we first obtain 10,000 sets of posterior μ and σ parameters from the individual model runs. Then, we obtain a single correlation coefficient for each of the 10,000 parameter sets by drawing from the corresponding conditional posterior distribution.

Bivariate Normal Model Results

The posterior distributions of ρ and AY residual box plots for CA and PA for the CSR, IPI, and SCC models are as follows:



Note: In the box plots, the “gray” boxes represent CA and the “black” boxes represent PA.

The key takeaways from the plots above are as follows:

- The posterior distribution of ρ is fairly wide (more uncertainty), and the point $\rho = 0$ is not close to the tails of the distributions. This suggests little evidence of dependency between the lines
- As shown in the box plots, the ability to capture AY effects decreases as we move from the CSR model to the SCC model
 - For the SCC model, 6 of the CA and PA medians lie on opposite sides of 0 within the same AY, while the remaining 4 are close to 0. For this reason, the posterior mean of ρ is far more negative than for the other models (i.e., -0.309)

Independence vs. Dependence

Next, Meyers compared the dependence and independence assumptions. He created independence by setting $\rho = 0$ in the bivariate normal model.

Comparing the assumptions using the $\widehat{elpd}_{loo}$ statistic, Meyers found that the independence assumption was favored in all cases for the CSR and IPI models, and nearly all cases for the SCC model.

Meyers analyzed a specific insurer for CA and OL where the independence assumption was barely favored. He noted the following important results:

- For this insurer, the posterior mean of ρ was positive. The strength of the positive correlation decreased as the model’s ability to capture the AY effect decreased. In other words, $\rho_{SCC} < \rho_{IPI} < \rho_{CSR}$
- For the SCC model, the estimated correlation actually flipped from positive to negative due to its inability to adequately capture AY effects. As a result, Meyers dropped the SCC model

Finally, Meyers compared the CSR and IPI model using the $\widehat{elpd}_{\text{test}}$ statistic. In this case, the dependence assumption was favored in more cases. This suggests that there are factors affecting the lower right triangle data that were not captured by the CSR and IPI models when fit to the upper left triangle data.

Since the dependence assumption performed better using the $\widehat{elpd}_{\text{test}}$ statistic, the evidence favors retaining the dependence parameter rather than setting $\rho = 0$.

Sensitivity of the Dependence Assumption

Meyers finished this chapter by testing the sensitivity of the dependence assumption on the standard error of the estimated ultimate loss. He does this by comparing the ratio of the standard error under dependence to the standard error under independence for each value of ρ in its posterior distribution.

Under the CSR and IPI models, he finds that the ratio is close to 1 in most cases, suggesting little risk of materially understating the standard error under an independence assumption.

Correlation Conclusion

In conclusion:

- It is difficult to detect dependencies between lines using only upper left triangle data
- The main source of dependency identified is the failure to adequately recognize AY effects

XI. Risk Margin

Under IFRS 17 and Solvency II, the reserve liability reported on an insurer's balance sheet consists of the expected present value of the unpaid claims, plus a risk margin.

In this section, Meyers specifically focuses on Solvency II. Under Solvency II, the reported reserve liability is referred to as technical provisions. Then:

- The value of the technical provisions equals the sum of the best estimate and risk margin
- The best estimate is the probability-weighted average of future cash flows, discounted using risk-free rates
- The risk margin is the cost of providing an amount of eligible own funds equal to the Solvency Capital Requirement needed to support the insurance obligations over their lifetime

Cash Flow Assumptions

Under a Bayesian MCMC framework, we have a large number (e.g., 10,000) of potential ultimate loss scenarios. For each scenario, we can compute the future cash flows underlying the technical provisions.

For simplicity, Meyers makes several simplifying assumptions about non-loss related parameters, such as the solvency criteria, interest rates, and payment timing.

Cash flow assumptions can be summarized as follows:

- Time $t = 0$
 - Time $t = 0$ is the end of the current CY
 - The insurer posts its best estimate of the liability, along with the amount of capital, K_0 , needed to cover the uncertainty in the best estimate
 - The insurer invests K_0 in a fund that earns income at the risk-free interest rate i
- Time $t = 1$
 - Time $t = 1$ is the end of the next CY
 - The insurer posts its updated best estimate of the liability, along with the capital, K_1 , needed to cover the uncertainty in the best estimate
 - The difference between $K_0(1 + i)$ and K_1 is returned to the investor. If that difference is negative, the investor is expected to contribute an amount to make up that difference
 - The insurer invests K_1 in a fund that earns income at the risk-free interest rate i

- Future CYs
 - The process continues, with the amount, $K_{t-1}(1 + i) - K_t$, being returned to (or being contributed by) the investor
- Ultimate
 - At some time $t = u$, the loss is at ultimate and we set $K_t = 0$ for $t > u$
 - For a 10×10 triangle, $u = 9$

Computing the Risk Margin

The present value, discounted at the risky rate r , of the amount returned to the investor is equal to the following:

$$\sum_{t=1}^{u+1} \frac{K_{t-1}(1 + i) - K_t}{(1 + r)^t}$$

Since $r > i$, the present value of the future capital releases will be less than the initial capital investment K_0 . This is problematic because the investor is not being adequately compensated for providing capital to support the reserve risk.

To address this, we increase the posted liability by adding an additional amount that compensates the investor for this shortfall. This amount is known as the cost of capital risk margin and is denoted R_{COC} . Mathematically:

$$\begin{aligned} R_{COC} &= K_0 - \sum_{t=1}^{u+1} \frac{K_{t-1}(1 + i) - K_t}{(1 + r)^t} \\ &= (r - i) \cdot \sum_{t=0}^u \frac{K_t}{(1 + r)^{t+1}} \end{aligned}$$

The final risk margin posted to the balance sheet is the arithmetic average of the risk margins calculated across all MCMC simulations.

Computing the Best Estimate

For the remainder of this section, we focus on the CSR and IPI models and assume they have been applied to a 10×10 triangle.

Assuming losses are lognormally distributed, the expected payment during development year d for simulation j is as follows:

$$P_{w,d}^j = e^{\mu_{w,d}^j + (\sigma_d^j)^2 / 2} - e^{\mu_{w,d-1}^j + (\sigma_{d-1}^j)^2 / 2}$$

for $d = 2, \dots, 10$.

Now, Meyers assumes that the expected incurred loss at $d = 10$ differs from the expected paid loss. For that reason, he assumes that the final expected payment for the IPI model will occur at $d = 11$. It is equal to the difference between the expected ultimate losses for the incurred and paid IPI models:

$$P_{w,11}^j = e^{\mu_{w,10}^j + (\sigma_{10}^2)/2} - e^{\mu_{w,d}^j + (\sigma_{10}^2)/2}$$

The final expected payment for the CSR model occurs at $d = 10$.

For simulation j , the present value of the liability for the CSR (IPI) model is equal to the following:

$$E_{best}^j = \sum_{w=2(1)}^{10} \sum_{d=12-w}^{10(11)} \frac{P_{w,d}^j}{(1+i)^{w+d-11.5}}$$

Regarding the formula above:

- The IPI model sums over AYs 1 through 10 (i.e., the first sum starts at $w = 1$) because the final expected payment is assumed to occur at $d = 11$
- Losses are assumed to be paid halfway through each future CY (hence, the $w + d - 11.5$ exponent)

Since all simulations are equally likely, we simply average E_{best}^j over all simulations to obtain the final best estimate:

$$E_{best} = \frac{1}{10,000} \sum_{j=1}^{10,000} E_{best}^j$$

Selected Ultimate Loss

For simulation j , the selected ultimate loss, U_j , is the sum of the expected ultimate losses across all AYs. Mathematically:

$$U_j = \sum_{w=1}^{10} e^{\mu_{w,10}^j + (\sigma_{10}^j)^2 / 2}$$

Future Loss Trapezoids

For simulation j , the simulated loss trapezoid for future calendar year t includes the upper left loss triangle, T_0 , and the first t simulated diagonals from the lower right loss triangle.

Mathematically:

$$T_t^j = \begin{cases} C_{w,d} & \text{for } w = 1, \dots, 10 \text{ and } d = 1, \dots, 11 - w \\ \tilde{C}_{w,d}^j & \text{for } w = t + 1, \dots, 10 \text{ and } d = 12 - w, \dots, \min(11 - w + t, 10) \end{cases}$$

The observed loss trapezoid for future calendar year t is defined as T_t .

Sampling Probabilities

Each MCMC simulation j represents a state of the world described by a set of parameters.

Initially, all MCMC simulations are assumed to be equally likely, so $P(J = j) = \frac{1}{10,000}$.

However, as we move further out into the future, some states become more probable while others become less probable. These updated probabilities are referred to as sampling probabilities and are computed using Bayes' Theorem:

$$P(J = j|T_t)$$

The formula for this probability is fairly technical, so we have decided to leave it out of the outline.

Ultimate Loss Estimate

Given T_t , the ultimate loss estimate, denoted E_t , is computed as follows:

$$\begin{aligned} E_t &= E \left[\sum_{w=1}^{10} C_{w,10} |T_t \right] \\ &= \sum_{j=1}^{10,000} P(J = j|T_t) \cdot U_j \end{aligned}$$

In words, the ultimate loss estimate is the expected value of the total ultimate loss across AYs given the observed trapezoid at time t . We compute it as a probability-weighted average of the selected ultimate losses, where the weights are the updated sampling probabilities after observing new data.

Algorithm 1 – Calculate Capital Simulations

Before we dive into Algorithm 1, we need to define a few terms:

- **Tail Value-at-Risk at the α level ($TVaR@\alpha$)** – the average of the worst $(1 - \alpha)$ proportion of ultimate loss outcomes
- **Total Required Capital (i.e., K_t)** – equal to the difference between $TVaR@\alpha$ and the ultimate loss estimate E_t (i.e., $TVaR@\alpha - E_t$)

Algorithm 1 computes the ultimate loss estimate and total required capital at each future time period:

- 1) **for** $k = 1, \dots, 10,000$ **do**
- 2) **for** $t = 0, \dots, 9$ **do**

- 3) Simulate cash flows $\{T_t^k\}$ using the $\{(\mu_{w,d}^k, \sigma_d^k)\}$
- 4) Compute $P(J = j|T_t^k)$ for each $j = 1, \dots, 10,000$
- 5) Take a random sample of size 10,000 with replacement, denoted $\{S_t^k\}$, from $\{U_j\}_{j=1}^{10,000}$ with sampling probabilities $P(J = j|T_t^k)$
- 6) Set E_t^k equal to the arithmetic average of $\{S_t^k\}$
- 7) Set K_t^k equal to the arithmetic average of the highest $(1 - \alpha) \cdot 10,000$ highest values of $\{S_t^k\}$, minus E_t^k
- 8) **end for**
- 9) **end for**

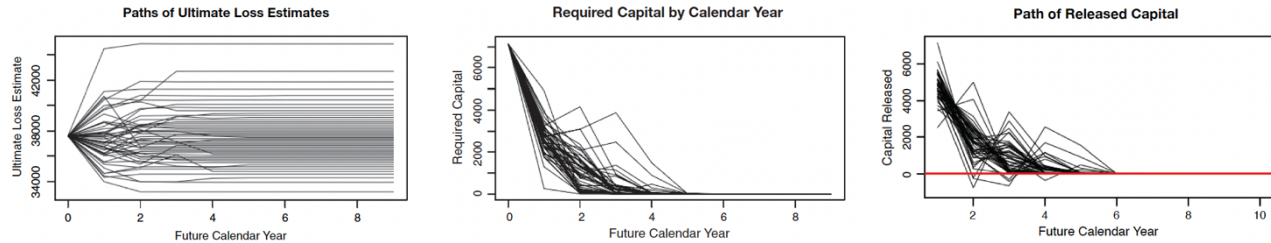
Let's discuss what each step of the algorithm is doing:

- Step 1 initializes the outer for loop
 - The $k = 1, \dots, 10,000$ means that we will run these capital calculations 10,000 times, producing 10,000 possible future capital paths
- Step 2 initializes the inner for loop
 - The $t = 0, \dots, 9$ means that we will complete the following steps for each future calendar year in the run-off period
- Step 3 simulates future loss trapezoids based on the posterior parameters from the model
- Step 4 computes updated sampling probabilities based on each future loss trapezoid
- Step 5 randomly samples 10,000 selected ultimate losses with replacement, where the probability of each selected ultimate loss corresponds to the updated sampling probabilities
- Step 6 computes the expected ultimate loss estimate at each time period
- Step 7 computes the total required capital for each time period

Plotting Pathways

After running Algorithm 1, we end up with 10,000 possible paths for E_t^j and K_t^j . For *example*, E_t^{275} yields the 275th path that the ultimate loss estimate can take from $t = 0$ to $t = 9$.

The following plots show the paths for the ultimate loss estimates, total required capital amounts, and released capital amounts (i.e., $K_{t-1}^j - K_t^j$) for the CSR model:



Key takeaways from the plots above are as follows:

- As time increases, the ultimate loss estimates become more certain and the paths converge toward their final values
- The total required capital amounts tend to decrease over time
- Most of the capital gets released early on, and it's occasionally necessary to add capital

If we compare the ultimate loss estimate pathways for the IPI and CSR models, we find that the IPI model produces a much narrower distribution of ultimate loss estimates. This indicates substantially less uncertainty regarding the ultimate loss estimate. As a result, the required capital is significantly reduced because the difference between $TVaR@α$ and E_t is much smaller.

Example: Computing the Best Estimate

A paid CSR Bayesian MCMC model is fit to the following 5×5 triangle as of 12/31/2025. The model's expected future losses for one simulation have been provided as well (i.e., lower right portion of triangle):

Accident Year	<u>Cumulative Paid Losses at Month XX (in \$000)</u>				
	12	24	36	48	60
2021	69,975	78,590	80,805	82,400	83,805
2022	82,600	90,865	93,850	94,900	95,980
2023	91,215	99,065	101,170	102,542	103,687
2024	93,395	103,025	105,980	107,111	108,000
2025	100,050	108,920	111,564	113,120	114,023

An actuary wants to compute the discounted best estimate of the total unpaid loss liability for this simulation. The risk-free rate is 3%.

First, let's compute the expected incremental loss payments for each future cell after the latest triangle diagonal:

Accident Year	<u>Expected Incremental Paid Losses at Month XX (in \$000)</u>				
	12	24	36	48	60
2021					
2022					1,080
2023				1,372	1,145
2024			2,955	1,131	889
2025		8,870	2,644	1,556	903

Second, let's aggregate the future loss payments by calendar year t (i.e., future diagonals) and discount them to time zero:

CY (t)	1	2	3	4
Expected Paid	14,277	4,920	2,445	903
Present Value	14,068	4,707	2,271	814

Using CY 1 as an example:

- Expected Paid = $8,870 + 2,955 + 1,372 + 1,080 = 14,277$
- Present Value = $\frac{14,277}{1.03^{0.5}} = 14,068$

Note that we discount future incremental loss payments from the middle of each year. This differs from the risk margin where capital amounts are discounted from the end of each year.

For this simulation, the discounted best estimate of the total unpaid loss liability is $14,068 + 4,707 + 2,271 + 814 = 21,859$.

The overall best estimate is found by averaging the best estimates across all simulations.

Example: Computing the Risk Margin

A paid CSR Bayesian MCMC model is fit to a loss triangle for a line of business using 10,000 simulations. For one simulation, the model's expected ultimate loss estimate and the $TVaR@98\%$ of the ultimate loss estimate, as estimated at each calendar year t using updated probabilities, are below:

CY (t)	0	1	2	3	4
Ult. Loss Est. (E_t)	350,627	352,168	353,879	354,021	354,021
$TVaR@98\%$ (of Ult.)	532,953	422,602	380,420	354,375	354,021

An actuary wants to compute the risk margin for this simulation. The risk-free rate is 3% and the risky rate is 8%.

First, let's calculate the required capital, K_t , at each CY as $TVaR@98\% - E_t$:

CY (t)	0	1	2	3	4
Ult. Loss Est. (E_t)	350,627	352,168	353,879	354,021	354,021
$TVaR@98\%$ (of Ult.)	532,953	422,602	380,420	354,375	354,021
Required Capital (K_t)	182,326	70,434	26,541	354	0

Second, let's compute the risk margin:

- Recall that $R_{COC} = (r - i) \cdot \sum_{t=0}^u \frac{K_t}{(1+r)^{t+1}}$
- Thus, $R_{COC} = (0.08 - 0.03) \cdot \left(\frac{182,326}{1.08^1} + \frac{70,434}{1.08^2} + \frac{26,541}{1.08^3} + \frac{354}{1.08^4} \right) = 12,527$

Note that we could also compute the risk margin using $R_{COC} = K_0 - \sum_{t=1}^{u+1} \frac{K_{t-1}(1+i) - K_t}{(1+r)^t}$. An example of this calculation can be found in the Cookbook.

The overall risk margin is found by averaging the risk margins across all simulations.

Effect of Insurer Size & Line of Business on Risk Margins

Meyers computed the risk margin and best estimate for all 200 loss triangles in his study and noted the following:

- There are significant differences in risk margins by line of business
- There is an approximate linear relationship between the log of the risk margin and the log of the best estimate

We can re-write the second conclusion above mathematically:

$$\log(R_{COC}) = a + b \cdot \log(E_{best})$$

Rearranging this formula, we obtain the following risk margin to best estimate ratio:

$$\boxed{\frac{R_{COC}}{E_{best}} = e^a \cdot (E_{best})^{b-1}}$$

For each line of business (CA, PA, WC, OL), $b < 1$. This implies that the risk margin to best estimate ratio decreases as the best estimate increases.

Effect of Diversification by Line of Business

Under Solvency II, the total risk margin for a multi-line insurer is the sum of the risk margins across lines of business. This assumes perfect dependence between the lines and fails to consider diversification.

While this is the most prudent approach, some recognition of diversification is warranted. To account for diversification, Meyers proposes Algorithm 2.

Algorithm 2 is defined as follows:

- 1) **for** $k = 1, \dots, 10,000$ **do**
- 2) **for** $t = 1, \dots, 9$ **do**
- 3) Simulate an L -tuple vector $\{p_l^k\}_{l=1}^L$ of uniform(0,1) numbers from the normal copula with correlation coefficient, ρ
- 4) For each line of business, l , select ${}_lQ_t^k$ to be the $\{p_l^k\}_{l=1}^L \cdot 10,000$ highest value of ${}_lS_t^k$
- 5) **end for**
- 6) Set the total ultimate loss estimate ${}_TS_t^k = {}_1Q_t^k + \dots + {}_LQ_t^k$
- 7) **end for**

Let's discuss what each step of the algorithm is doing:

- Step 1 initializes the outer for loop
 - The $k = 1, \dots, 10,000$ means that we will run these dependency calculations 10,000 times
- Step 2 initializes the inner for loop
 - The $t = 1, \dots, 9$ means that we will complete the following steps for 9 future time periods
- Step 3 generates correlated random percentiles for each line of business l using a normal copula

- Step 4 selects the corresponding percentile from each line's ultimate loss distribution
- Step 6 sums the selected losses across all lines to obtain a combined insurer-level loss

In the end, Algorithm 2 produces a combined ultimate loss distribution consisting of 10,000 total ultimate loss estimates. We use this combined loss distribution as the input to Algorithm 1 to calculate the diversified risk margin.

Meyers found that the diversification credit depends on the mix of business. The dominant line tends to receive the smallest credit, whereas smaller lines receive larger credits.

Example: Diversification Credit

Suppose an insurer's best estimates and stand-alone risk margins are as follows:

Line	Best Estimate	Risk Margin
CA	336,301	6,985
PA	10,280,570	215,633
WC	238,395	5,676
OL	909,123	86,791
Total	11,764,389	315,085

Now, suppose the insurer runs Algorithms 2 and 1 and obtains a diversified risk margin of 236,532. The insurer also ran Algorithms 2 and 1 for each three-line combination and obtained the following diversified risk margins:

- PA, WC, & OL – 235,872
- CA, WC, & OL – 88,966
- CA, PA, & OL – 236,418
- CA, PA, & WC – 216,617

The insurer wishes to compute the diversification credit by both line of business and in total.

Diversification credit is computed as follows:

$$\text{Diversification Credit} = 1 - \frac{\text{Allocated Risk Margin}}{\text{Stand-Alone Risk Margin}}$$

First, we compute the marginal risk margins for each line by subtracting the combination excluding that line from the total diversified risk margin:

Line	Marginal Risk Margin
CA	660 = 236,532 – 235,872
PA	147,566 = 236,532 – 88,966
WC	114
OL	19,915
Total	168,255

Second, we compute the allocated risk margins for each line by allocating the total diversified risk margin to each line in proportion to the marginal risk margins:

Line	Allocated Risk Margin
CA	928 = 236,532(660/168,255)
PA	207,448 = 236,532(147,566/168,255)
WC	160
OL	27,996
Total	236,532

Third, we compute the diversification credit by line and in total:

Line	Diversification Credit
CA	86.7% = 1 – 928/6,985
PA	3.8% = 1 – 207,448/215,633
WC	97.2%
OL	67.7%
Total	24.9%

As expected, the WC line (the smallest line) has the largest credit.

One-Year Time Horizon

In the prior sections, required capital was based on the *distribution of ultimate loss estimates* at the *ultimate time horizon* (i.e., the U_j).

Under Solvency II, however, required capital is determined using a one-year time horizon by considering the *distribution of ultimate loss estimates* that would be obtained *one year after the valuation date*.

To accomplish this, for each MCMC simulation, Meyers simulates multiple possible realizations of the next year's loss experience. For each realization, the additional information is used to update the sampling probabilities and calculate a revised ultimate loss estimate. These estimates are then averaged to produce an expected ultimate loss estimate one year from now for that simulation.

By simulating multiple realizations of next year's loss experience, we recognize that many scenarios will still have material positive probability one year from now. Different realizations of loss emergence can therefore produce different ultimate loss estimates. Averaging across these realizations produces a more stable estimate of the ultimate loss estimate that would be held one year after the valuation date.

Algorithm 3 produces the one-year-ahead ultimate loss estimates and is defined as follows:

- 1) **for** $m = 1, \dots, M$ **do**
- 2) **for** $j = 1, \dots, 10,000$ **do**
- 3) **for** $t = 1, \dots, 9$ **do**
- 4) Simulate T_t using the parameters $\{\mu_{w,d}^j, \sigma_d^j\}$
- 5) Calculate $\{P(J = j|T_t)\}_{j=1}^{10,000}$
- 6) Calculate the ultimate loss estimate, $O_{t,j}^m$
- 7) **end for**

```

8)      Set  $O_{10,j}^m = O_{9,j}^m$ 
9)      end for
10)     end for
11)     for  $j = 1, \dots, 10,000$  do
12)       for  $t = 1, \dots, 10$  do
13)         Set  $O_{t,j}^m = \text{mean}(O_{t,j}^m)$ 
14)       end for
15)     end for

```

Let's discuss what each step of the algorithm is doing:

- Step 1 initializes the outer for loop
 - The $m = 1, \dots, M$ means that Meyers repeats the simulation procedure M times
 - This is the key addition of Algorithm 3. Meyers simulates M possible realizations of future loss experience for each MCMC scenario and averages the resulting ultimate loss estimates to obtain an average ultimate loss estimate $O_{t,j}$
- Step 2 initializes the next for loop
 - The $j = 1, \dots, 10,000$ means that the procedure is applied to each MCMC scenario
- Step 3 initializes the next for loop
 - The $t = 1, \dots, 9$ means that the future loss emergence is evaluated at each future calendar year
- Step 4 simulates future loss trapezoids conditional on MCMC scenario j
- Step 5 updates the sampling probabilities based on the simulated loss trapezoid
- Step 6 calculates the updated ultimate loss estimates based on the updated sampling probabilities. Note that we use $O_{t,j}^m = \sum_{k=1}^{10,000} P(J = k | T_t^m) \cdot U_k$ here
- Step 8 handles the final year by setting the year 10 estimate equal to the year 9 estimate (since run-off is effectively complete)

- Steps 11-15 average across the M ultimate loss estimates $O_{t,j}^m$ to obtain the final one-year-ahead ultimate loss estimates $O_{t,j}$ for MCMC scenario j

Finally, Meyers introduces Algorithm 4 to calculate the required capital based on a one-year time horizon.

Algorithm 4 is defined as follows:

- 1) **for** $k = 1, \dots, 10,000$ **do**
- 2) **for** $t = 0, \dots, 9$ **do**
- 3) Simulate cash flows $\{T_t^k\}$ using the $\{(\mu_{w,d}^k, \sigma_d^k)\}$
- 4) Compute $P(J = j|T_t^k)$ for each $j = 1, \dots, 10,000$
- 5) Take a random sample of size 10,000 with replacement, denoted $\{S_t^k\}$, from $\{O_{t+1,j}\}_{j=1}^{10,000}$ with sampling probabilities $P(J = j|T_t^k)$
- 6) Set E_t^k equal to the arithmetic average of $\{S_t^k\}$
- 7) Set K_t^k equal to the arithmetic average of the highest $(1 - \alpha) \cdot 10,000$ highest values of $\{S_t^k\}$, minus E_t^k
- 8) **end for**
- 9) **end for**

Notice that algorithm 4 is identical to algorithm 1 except for Step 5. In Step 5, we replaced U_j with the ultimate loss estimate, $O_{t+1,j}$, from Algorithm 3. The difference between these two estimates is as follows:

- U_j is the ultimate loss estimate we compute today under posterior scenario j
- $O_{t+1,j}$ is the ultimate loss estimate that would be obtained one year from now under posterior scenario j , after allowing for another year of loss emergence and Bayesian updating

Using the output from Algorithm 4 (i.e., required capital amounts over time), we can compute risk margins for each simulation j using the R_{COC} formula. As before, the final posted risk margin is the average of the risk margins over the simulations.

Meyers shows that the initial capital and risk under the IPI model are higher for the one-year time horizon than for the ultimate time horizon. Therefore, we cannot automatically assume that the one-year time horizon will produce lower risk margins.

XII. Summary & Conclusions

In this section, Meyers broadly reviews the main topics discussed in the monograph.

Retrospective Tests

Retrospective tests inform a model's *reputation* and are based on future losses in the lower-right portion of the loss triangle (i.e., the holdout data). The tests are as follows:

- **Test #1:** Evaluate a model's predicted percentiles for uniformity using histograms, $p - p$ plots, and the $K - S$ test
- **Test #2:** Compare competing models by using the expected log posterior density, $\widehat{elpd}_{test}$

Prospective Tests

Prospective tests are based on the observed losses in the upper-left portion of the loss triangle (i.e., the training data). The tests are as follows:

- **Test #1:** Review standardized residual box plots constructed using parameter samples drawn from the posterior distribution
- **Test #2:** Compare competing models by using the “leave-one-out” cross-validation test statistic, $\widehat{elpd}_{loo}$. Alternatively, the equivalent LOOIC statistic may be used

Model Performance

The following table summarizes the performance of each model discussed in the monograph based on the 200 loss triangles analyzed by Meyers:

Model	Performance
Mack	• <u>Incurred</u> Losses – Underpredicts the variability of the ultimate loss outcomes • <u>Paid</u> Losses – Overpredicts expected ultimate losses
ODP	• <u>Paid</u> Losses – Overpredicts expected ultimate losses
CRC	• <u>Incurred</u> Losses – Underpredicts the variability of the ultimate loss outcomes • <u>Paid</u> Losses – Overpredicts expected ultimate losses
SCC	• <u>Incurred & Paid</u> Losses – Underpredicts the variability of the ultimate loss outcomes • Residual box plot indicates problems with a constant ELR assumption
CSR	• <u>Paid</u> Losses – Passes $K - S$ test • Performed better than the CRC model in over half of the cases according to the $\widehat{elpd}_{loo}$ and $\widehat{elpd}_{test}$ statistics. This indicates that the settlement rate was changing
CAY	• <u>Incurred</u> Losses – Underpredicts the variability of the ultimate loss outcomes • Tends to generate a positive correlation between the losses of adjacent AYs • The $\widehat{elpd}_{loo}$ statistic rarely favors the CAY model over the CRC model • The $\widehat{elpd}_{test}$ statistic favors the CAY model over the CRC model in more than half of the cases
IPI	• <u>Incurred & Paid</u> Losses – Passes $K - S$ test • The $\widehat{elpd}_{loo}$ and $\widehat{elpd}_{test}$ statistics favor the IPI model over the corresponding CSR and CAY models in more than half of the cases • Produces lower standard errors by incorporating information from both paid and incurred losses • Works best for the Auto Liability LOB where the paid and incurred losses are almost equal at 10 years • Works less well for Workers' Compensation where a large portion of the losses remain unpaid after 10 years

I. Introduction

The purpose of this paper is to showcase various stochastic models where the **chain-ladder reserve happens to be the maximum likelihood forecast of the true loss reserve**.

All of the stochastic models outlined in this paper are generalized linear models (GLMs).

II. Framework & Notation

Before diving into the actual meat of the paper, we have to define some notation. Similar to other papers on the syllabus, understanding the notation is useful when learning the models described in the paper. The notation is as follows:

- Accident years are represented by rows and labeled $k = 1, 2, \dots, K$
- Development periods are represented by columns and labeled $j = 1, 2, \dots, J$, where $J \leq K$
- Experience periods (i.e. calendar periods) are represented by diagonals and labeled $d = 1, 2, \dots, K$. Each observation on diagonal d has $d = k + j - 1$
- Y_{kj} represents incremental losses for accident year k and development period j
- $X_{kj} = \sum_{i=1}^j Y_{ki}$ represents cumulative losses for accident year k and development period j
- D_K represents the **trapezoid of past Y_{kj} observations**. In general, actuaries tend to think of past observations as a data triangle (i.e. the special case where $K = J$). For this paper, the authors have adopted the more general framework where $J \leq K$
- D_K^c represents the **triangle of future Y_{kj} observations**
- D_K^+ represents the **combined set of past and future Y_{kj} observations**. Thus, $D_K^+ = D_K \cup D_K^c$

- $R_k = \sum_{j=K-k+2}^J Y_{kj} = X_{kJ} - X_{k,K-k+1}$ represents the sum of the future observations for accident period k (i.e., the true accident period k reserve)
- R represents the sum of the future observations for all accident periods (i.e., the true total reserve)
- $\hat{f}_j$ represents the all-year volume-weighted average age-to-age factor that projects accident period k cumulative losses from development period j to $j + 1$. This is the same chain-ladder factor we have been discussing throughout the syllabus
- $\hat{R}_k$ represents the chain-ladder reserve for accident period k
- $\hat{R}$ represents the total chain-ladder reserve
- $\hat{R}_k^{(B)} = B_k \left[1 - \frac{1}{\hat{f}_{K-k+1} \dots \hat{f}_{J-1}} \right]$ represents the estimate of the reserve for accident period k based on an a priori estimate of ultimate losses. In this case B_k represents the a priori estimate of ultimate losses and is commonly based on the following **two methods**:
 - 1) Bornhuetter-Ferguson (BF) method
 - 2) Cape Cod method

BF Method

Under the BF method, the a priori loss, B_k , is defined as follows:

$$B_k = P_k \pi_k$$

where P_k denotes earned premium for accident period k and π_k denotes the expected loss ratio for accident period k . In this case, B_k uses a separate a priori loss ratio (i.e. π_k) for each accident year.

Cape Cod Method

Under the Cape Cod method, the a priori loss, B_k , is defined as follows:

$$B_k = P_k \left(\frac{\sum_{i=1}^K P_i \omega_i [(X_{i,K-i+1} + \hat{R}_i)/P_i]}{\sum_{i=1}^K P_i \omega_i} \right)$$

where ω_i = percent emerged for accident year i . In this case, B_k uses the same a priori loss ratio for each accident year. The overall loss ratio (i.e. the part to the right of the P_k) is simply the weighted average of each accident year's chain-ladder loss ratio, where the weights are the $P_i\omega_i$.

Example: Cape Cod Expected Loss

To better understand the notation above, let's look at a simple example (note that the non-bolded figures below represent the past observations, and the bolded figures represent the future observations that will eventually occur):

AY	Premium (\$000)	Inc. losses in Dev. Year (\$000)			
		1	2	3	4
1	20,000	10,250	4,750	2,250	875
2	20,500	10,500	4,625	2,000	1,150
3	21,000	10,375	5,000	1,610	1,000
4	20,375	10,765	4,220	1,930	995
5	20,750	10,550	4,675	1,970	1,015

First, let's look at the quantities related to the fully developed data rectangle:

- Accident years are represented by $k = 1, 2, \dots, 5$. Thus, $K = 5$
- Development periods are represented by $j = 1, 2, \dots, 4$. Thus, $J = 4$
- Experience periods are represented by $d = 1, 2, \dots, 5$. Experience period 1 has a single observation (i.e., 10,250). Experience period 5 has four observations (i.e., 10,550, 4,220, 1,610, and 1,150)
- $Y_{4,3} = 1,930$ (i.e., the incremental loss for accident period 4 that comes out between 2 and 3 years of development)
- $X_{4,3} = 10,765 + 4,220 + 1,930 = 16,915$ (i.e., the cumulative losses for accident period 4 as of 3 years of development)
- D_5 is represented by the trapezoid of past observations shown in the data trapezoid above (i.e., the non-bolded data)

- D_5^c is represented by the triangle of future observations (i.e., the bolded data)
- D_5^+ is represented by the full rectangle of observations (i.e., the union of the non-bolded data and bolded data)
- $R_4 = 1,930 + 995 = 2,925$ (i.e., the sum of the future observations for accident period 4)
- $R = 1,000 + 1,930 + 995 + 4,675 + 1,970 + 1,015 = 11,585$ (i.e., the sum of all future observations)

Second, let's look at how B_k is calculated for $k = 3$ using the Cape Cod method:

- The chain-ladder factors ($\hat{f}_1, \hat{f}_2, \hat{f}_3$) are 1.444, 1.129, and 1.059, respectively
- Assuming development stops at $J = 4$, the emergence patterns (i.e., the ω_i) and chain-ladder reserves by accident period are as follows:

Accident Period i	ω_i	$\hat{R}_i$ (in \$000)
1	100%	0
2	100%	0
3	94.4%	1,001
4	83.7%	2,926
5	57.9%	7,658

- The Cape Cod (CC) loss ratio is as follows:

$$CC LR = \frac{20(1)\left(\frac{18.125}{20}\right) + \dots + 20.75(0.579)\left(\frac{10.55 + 7.658}{20.75}\right)}{20(1) + 20.5(1) + \dots + 20.75(0.579)} = 0.883$$

- Thus, $B_3 = P_3(0.883) = 21,000(0.883) = \$18,538$

III. The Exponential Dispersion Family (EDF)

The EDF is the family of distributions with probability density function (pdf) $\pi(y; \theta, \phi)$ of the form:

$$\ln(\pi(y; \theta, \phi)) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)$$

where:

- y is the value of an observation Y
- θ is a location parameter called the **canonical parameter**
- ϕ is a dispersion parameter called the **scale parameter**
- $b(\theta)$ is the **cumulant function**, which determines the shape of the distribution
- $e^{c(y, \phi)}$ is a **normalizing factor** producing unit total mass for the distribution

Some **important assumptions** regarding the functions above are as follows:

- a , b , and c are continuous
- b is a one-to-one function and twice differentiable
- The first derivative of b is a one-to-one function
- For the purposes of this paper, $a(\phi) = \frac{\phi}{w}$ for some constant w . In general, we will assume $w = 1$ and $a(\phi) = \phi$

The expected value and variance of an EDF distribution are as follows:

- $E[Y] = \mu = b'(\theta)$. This implies that $\theta = (b')^{-1}(\mu)$
- $Var(Y) = a(\phi)b''(\theta) = a(\phi)b''((b')^{-1}(\mu)) = a(\phi)V(\mu)$
- Notice that $Var(Y)$ has been decomposed into factors that depend on the mean and the dispersion parameter respectively. The **variance function** $V(\mu)$ depends on the mean while $a(\phi)$ depends on the dispersion parameter

Examples of EDF distributions include the Poisson, binomial, and gamma distributions. The Poisson and binomial distributions are useful for modeling counts. The gamma distribution is useful for modeling amounts. Let's take a look at how the Poisson distribution fits into the EDF:

- Let $b(\theta) = e^\theta$, $a(\phi) = 1$, and $c(y, \phi) = -\ln(y!)$
- In this case, $\ln(\pi(y; \theta, \phi)) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) = \frac{y\theta - e^\theta}{1} - \ln(y!)$
- If we exponentiate both sides, we find that $\pi(y; \theta, \phi) = \frac{e^{y\theta} e^{-e^\theta}}{y!}$
- If we let $e^\theta = \mu$, then $\pi(y; \mu, \phi) = \frac{\mu^y e^{-\mu}}{y!}$
- Thus, $Y \sim \text{Pois}(\mu)$
- Using the expected value formula above, $E[Y] = b'(\theta) = e^\theta$. This matches what we just derived in the prior bullet
- Using the variance formula above (and the fact that $a(\phi) = 1$), we find that $\text{Var}(Y) = a(\phi)V(\mu) = V(\mu) = b''((b')^{-1}(\mu)) = b''(\theta) = e^\theta$. This matches the known fact that the mean and variance of a Poisson distribution are identical

The Tweedie Sub-Family

The Tweedie sub-family results from restricting the variance function to the following:

$$V(\mu) = \mu^p, p \leq 0 \text{ or } p \geq 1$$

where $\mu = [(1 - p)\theta]^{\frac{1}{1-p}}$. Note that this formula for μ holds for $p \neq 1$. When $p = 1$, $\mu = e^\theta$ as shown earlier for the Poisson distribution.

Assuming $a(\phi) = \phi$, this means that $\text{Var}(Y) = \phi\mu^p$. In words, this means that the variance is proportional to a power of the mean. Note that several well-known distributions are special cases of the Tweedie sub-family:

p	Distribution
0	Normal
1	Over-dispersed Poisson
2	Gamma
3	Inverse Gaussian
[1, 2)	Compound Poisson with a Gamma Severity Distribution

The choice of p is somewhat informed by the heaviness of the tail indicated by the data (the tail heaviness of Tweedie distributions increases as p increases). If a model based on a specific p generates more widely dispersed residuals than are consistent with that model, then an increase in p might be warranted.

It's common to model insurance losses as a compound distribution. Hence, for pure premium modeling, $1 \leq p < 2$.

The Over-Dispersed Poisson Sub-Family

As mentioned in the previous section, Y is an over-dispersed Poisson (ODP) distribution when $p = 1$. Let's go ahead and demonstrate this by deriving the mean and variance of Y :

- For the Tweedie sub-family, $p = 1$ results in $b(\theta) = e^\theta$. Furthermore, let $a(\phi) = \phi$ and $c(y, \phi) = \ln(\phi^{-y/\phi} [(y/\phi)!]^{-1})$
- In this case, $\ln(\pi(y; \theta, \phi)) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) = \frac{y\theta - e^\theta}{\phi} + \ln(\phi^{-y/\phi} [(y/\phi)!]^{-1})$
- If we exponentiate both sides, we find that $\pi(y; \theta, \phi) = \frac{\phi^{(-y/\phi)} e^{y\theta/\phi} e^{-e^\theta/\phi}}{(y/\phi)!}$
- If we let $e^\theta = \mu$, then $\pi(y; \mu, \phi) = \frac{\phi^{(-y/\phi)} \mu^{y/\phi} e^{-\mu/\phi}}{(y/\phi)!} = \frac{(\mu/\phi)^{y/\phi} e^{-\mu/\phi}}{(y/\phi)!}$. This is the pdf of an ODP distribution with parameters μ and ϕ
- This also shows that $Y/\phi \sim \text{Pois}(\mu/\phi)$. We can use this fact to calculate the expected value and variance of Y :
 - $E[Y] = \phi E[Y/\phi] = \phi(\mu/\phi) = \mu$
 - $\text{Var}(Y) = \phi^2 \text{Var}(Y/\phi) = \phi^2(\mu/\phi) = \phi\mu$
- As expected for the ODP distribution, the variance is proportional to the mean. Notice that the standard Poisson distribution occurs when $a(\phi) = 1$

The ODP distribution is **useful when little is known of the subject distribution**.

Generalized Linear Models (GLMs)

Consider a sample of observations Y_i , where $i = 1, 2, \dots, n$. Suppose that each Y_i is associated with the following q predictors: $x_{i1}, x_{i2}, \dots, x_{iq}$. As an example related to auto insurance, x_{i1} might be the operator age of the i^{th} observation.

Further suppose that the observations satisfy the following **three conditions**:

- 1) $Y_i \sim \pi(\cdot; \mu_i, \phi_i)$, with μ_i being unknown parameters (we will talk about the ϕ_i down below). This just means that each observation is a member of the EDF
- 2) $h(\mu_i) = x_i^T \beta$, where $h(\cdot)$ is known as the **link function** and β is a vector of unknown parameters. For simplicity, just think about $x_i^T \beta$ as a sumproduct of x s and β s
- 3) The observations Y_i are stochastically independent

The three conditions above **define the structure of a GLM**. The variate Y_i is called the **response** and the linear expression $x_i^T \beta$ is called the **linear response**. The intent of the link function is to **transform the mean of each observation into a linear function of the parameter vector β** .

For example, suppose the mean of Y_1 is $e^{\beta_0 + x_{11}\beta_1 + x_{12}\beta_2}$. Using a log link function, the mean is transformed to $\ln(e^{\beta_0 + x_{11}\beta_1 + x_{12}\beta_2}) = \beta_0 + x_{11}\beta_1 + x_{12}\beta_2$, which is linear.

Connection to Weighted Linear Regression

For the dispersion parameters ϕ_i , we typically assume that $\phi_i = \frac{\phi}{w_i}$, where ϕ is unknown and w_i is the **weight** of the i^{th} observation. This assumption leads to the following connection to weighted linear regression:

- Instead of the EDF, assume that Y_i is normally distributed with parameters μ_i and ϕ_i . Further assume that the link function h is the identity function (i.e., $h(x) = x$)
- In this case, $Y_i = x_i^T \beta + \epsilon_i$, where $\epsilon_i \sim N(0, \phi_i)$. This is a **weighted linear regression model**. In other words, it's a standard linear regression where the errors are normally distributed with **unequal variances**

- If we generalize the weighted linear regression further by allowing a non-linear relationship between observations and predictors (i.e., a link function other than the identity function) AND non-normal errors, we end up with a GLM

Matrix Notation

When defining the structure of a GLM above, the second condition was $h(\mu_i) = x_i^T \beta$. It's useful to rewrite this condition in matrix notation as follows:

$$\mu = h^{-1}(X\beta)$$

where:

- μ is the matrix of means where the i^{th} component is μ_i
- X is the **design matrix** where the i^{th} row is x_i^T
- β is the matrix of parameters θ

Using our definition from earlier, $b'(\theta_i) = E[Y_i] = \mu_i$ is the i^{th} component of $h^{-1}(X\beta)$.

Selection of a GLM

The selection of a GLM consists of the following:

- Selection of a cumulant function, controlling the model's assumed error distribution
- Selection of an index p , controlling the relationship between the model's mean and variance
- Selection of the covariates x_i^T (i.e., the variables that explain μ_i)
- Selection of a link function, controlling the relationship between the mean μ_i and the associated covariates

Parameters of a GLM are often estimated using **maximum likelihood estimation (MLE)**.

Categorical Covariates

Categorical covariates are predictors with discrete levels. Suppose we have a categorical covariate with m possible values (also known as **levels**). In a GLM, each level of the categorical covariate is treated as a dummy variable with possible values of 0 or 1.

As an **example**, assume we are fitting a model on a dataset comprised of the following three states: Alaska, Texas, and New York. Each record on the dataset corresponds to a policyholder living in one of these three states. If the policyholder lives in Texas, then the covariate for Texas is assigned a 1 and the covariates for the other two states are assigned 0s

In practice, one level from each categorical covariate is considered the "base level" and would not be explicitly represented in the model as a covariate. Instead, its effect would be captured in the intercept. Excluding one level from each categorical variable ensures that parameter values calculated by the model are unique. We will discuss parameter redundancy in more detail later.

Continuous Covariates

Continuous covariates are predictors with continuous levels (ex. age). These covariates are typically represented as a polynomial (ex. x, x^2, x^3 , etc...) in the model.

In addition, it's common to transform continuous variables (ex. apply a log transformation) before placing them in the model. Transformations also include piecewise functions.

As an **example**, consider the following function:

$$L_{mM}(x) = \min[M - m, \max(0, x - m)] \text{ with } m < M$$

When x is between m and M , the function above is linear. When $x < m$ or $x > M$, the function above is constant. We could incorporate **linear splines** by incorporating a number of these L functions (known as **basis functions**) into our model.

Goodness-of-Fit and Deviance

Earlier, we showed that $\mu = h^{-1}(X\beta)$. Let $\hat{\beta}$ denote the MLE of β . The vector:

$$\hat{Y} = h^{-1}(X\hat{\beta})$$

is the MLE of μ and is referred to as the vector of **fitted values**.

For GLMs, goodness-of-fit is generally measured using **scaled deviance**. The scaled deviance is defined as follows:

$$D(Y, \hat{Y}) = 2 \sum_{i=1}^n [\ln \pi(Y_i; \hat{\theta}^{(s)}, \phi) - \ln \pi(Y_i; \hat{\theta}, \phi)]$$

where $\hat{\theta}$ is the MLE of θ and $\hat{\theta}^{(s)}$ is the estimate of θ in the **saturated model** (i.e., a model with a parameter for every observation so that $\hat{Y} = Y$).

The main takeaway from the formula above is that we are finding the difference between the saturated model and the actual model. When this difference is small, then the fitted values are close to the actual values. Thus, we want to determine parameters that minimize the scaled deviance.

Another thing to notice is that the scaled deviance formula requires ϕ . It turns out that minimizing the scaled deviance does not rely on ϕ (it factors out of the minimization procedure). Thus, it's **common to minimize the unscaled deviance instead**. This is simply known as the **deviance** and has the following formula:

$$D^*(Y, \hat{Y}) = 2 \sum_{i=1}^n [\ln \pi(Y_i; \hat{\theta}^{(s)}, 1) - \ln \pi(Y_i; \hat{\theta}, 1)]$$

The deviance can be used to estimate the scale parameter ϕ as follows:

$$\hat{\phi} = \frac{D^*(Y, \hat{Y})}{n - p}$$

where n is the number of observations and p is the number of parameters.

Pearson Residuals

The **standardized Pearson residuals** are defined as follows:

$$R_i^P = \frac{Y_i - \hat{Y}_i}{\hat{\sigma}_i}$$

where $\hat{\sigma}_i^2$ is an estimator of $\sigma_i^2 = \text{Var}(Y_i)$.

Similar to the Shapland paper, we would like our residuals to exhibit unbiasedness (i.e., revolve around $y = 0$) and homoscedasticity (i.e., constant variance) when plotted against covariates.

For **example**, if we plot 100,000 standardized residuals against the “age” covariate, we want those residuals scattered evenly about $y = 0$ and exhibiting uniform dispersion across the ages.

Deviance Residuals

One consequence of Pearson residuals is that they will reproduce any non-normality that exists in the observations (ex. skewed loss data). Since normally distributed residuals are more useful for model assessment, it is common to look at **standardized deviance residuals** when assessing a GLM (since they are much closer to normal than the Pearson residuals). The formula is as follows:

$$R_i^D = \text{sgn}(Y_i - \hat{Y}_i)(d_i/\hat{\phi})^{0.5}$$

where d_i is the contribution of the i^{th} observation to the deviance $D^*(Y, \hat{Y})$. Note that the $\text{sgn}(\cdot)$ function produces a -1 if the quantity is negative, 0 if the quantity is 0 , and 1 if the quantity is positive.

Use of Weights

If a residual plot exhibits heteroscedasticity, then weights can be used to correct it. The **general rule** is that observations should be assigned weights that are inversely proportional to the variance of the residuals.

Example: Removing Heteroscedasticity in the Residuals

Suppose a GLM is formulated assuming homoscedasticity is present. In other words, we are assuming that $Var(Y_i) = \phi V(\mu_i)$, where ϕ is independent of i .

Using this model, the standardized Pearson residuals are defined as $R_i^P = \frac{Y_i - \hat{Y}_i}{[\hat{\phi} V(\hat{\mu})]^{0.5}}$ (assuming a common variance function $V(\mu)$ across all i).

Further suppose that the standardized Pearson residuals are plotted by age and appear to show that the residuals above age 55 have double the standard deviation of those below age 55. This means that the residuals above age 55 have four times the variance of those below age 55.

Earlier, we assumed that $\phi_i = \frac{\phi}{w_i}$ with w_i typically equal to 1. Instead of assuming $w_i = 1$ for observations with an age above 55, we can assume that $w_i = \frac{1}{4}$. In summary, we are setting the weights as follows:

$$\begin{aligned} w_i &= 1 \text{ if observation } i \text{ is below age 55} \\ &= 1/4 \text{ if observation } i \text{ is above age 55} \end{aligned}$$

In doing so, observations with larger variance receive less weight. This recognizes that ϕ_i is NOT the same for each observation i . In addition, it should remove the heteroscedasticity in the residuals because we are dividing by the square root of $\phi_i = \frac{\phi}{w_i}$ in the denominator.

Outliers

Residual plots might identify isolated observations with large residuals. These are known as **outliers**. Since these observations can heavily influence the regression by **shifting the fitted values away from the main body of observations in favor of the outliers**, they may need to be dealt with. We can remove the influence of outliers by assigning them weights of 0 in the model.

If we remove an outlier caused by a major catastrophe, we need to ensure that the cost of those events is being captured somewhere. Otherwise, the model fails to recognize the possible impact of such an event.

Forecasts

Unlike the data used to create the model, data used in forecasting represents future observations. We will notate these future observations as Y_i^* with corresponding covariates x_i^* . Then, a reasonable forecast of Y^* is as follows:

$$\hat{Y}^* = \hat{\mu}^* = h^{-1}(X^* \hat{\beta})$$

where X^* is known as the **forecast design matrix**.

IV. Stochastic Models Supporting the Chain Ladder

Non-Parametric Mack Model

This is the same model detailed in the Mack Chain-Ladder paper. It satisfies the following conditions:

- (M1): Accident years are stochastically independent (similar to assumption 2 from the Mack Chain-Ladder paper)
- (M2): For each $k = 1, 2, \dots, K$, the X_{kj} form a Markov chain (means that X_{kj} is only dependent on $X_{k,j-1}$)
- (M3a): For each $k = 1, 2, \dots, K$ and $j = 1, 2, \dots, J - 1$, $E[X_{k,j+1}|X_{kj}] = f_j X_{kj}$ for some parameter $f_j > 0$ (similar to assumption 1 from the Mack Chain-Ladder paper)
- (M3b): For each $k = 1, 2, \dots, K$ and $j = 1, 2, \dots, J - 1$, $Var[X_{k,j+1}|X_{kj}] = \sigma_j^2 X_{kj}$ for some parameter $\sigma_j > 0$ (similar to assumption 3 from the Mack Chain-Ladder paper)

Mack derived the following results from this model:

- **Result 1:** The conventional chain-ladder estimators $\hat{f}_j$ are unbiased AND minimum variance estimators among estimators that are unbiased linear combinations of the $\hat{f}_{kj}$
- **Result 2:** The conventional chain-ladder estimate $\hat{R}_k$ is unbiased

The Mack model is stochastic because it considers the means AND variances of observations. It is non-parametric because it does not consider the distribution of the observations.

Parametric Mack Models

As mentioned above, the Mack model does not consider the distributions of the observations. In this section, we will begin assigning distributions to the **incremental** observations Y_{kj} . We will start with the **EDF Mack model**.

EDF Mack Model

This model has the exact same conditions (M1 - M3a) as the non-parametric Mack model, with one exception: the variance assumption in M3b is replaced with the following:

$$Y_{k,j+1}|X_{kj} \sim EDF(\theta_{kj}, \phi_{kj}; a, b, c)$$

Notice that this model does not tell us much about the variance of the observations. It simply tells us that the data comes from the EDF.

Tweedie Mack Model

Similar to the EDF Mack model, this model shares all conditions with the standard Mack model, excluding the variance assumption. In this case, $Y_{k,j+1}|X_{kj}$ is Tweedie distributed

ODP Mack Model

Similar to the EDF Mack model, this model shares all conditions with the standard Mack model, excluding the variance assumption. In this case, $Y_{k,j+1}|X_{kj}$ is ODP distributed.

Theorem 3.1

The **EDF Mack model** results in the following assuming the **data array is a triangle** ($J = K$):

- If **M3b holds** (i.e., $\text{Var}[X_{k,j+1}|X_{kj}] = \sigma_j^2 X_{kj}$), then the maximum likelihood estimators (MLEs) of the f_i are the conventional chain-ladder estimators (which are unbiased)
- If we are in the special case of the ODP Mack model **AND** the dispersion parameters ϕ_{kj} are just column dependent ($\phi_{kj} = \phi_j$), then the conventional chain-ladder estimators are minimum variance unbiased estimators (MVUEs). In addition, the cumulative loss estimates $\hat{X}_{kj}$ and the reserve estimates $\hat{R}_k$ are also MVUEs

Recall that one of the results from the Mack model (Result 1) was that the chain-ladder estimators were MVUEs among all linear combinations of the $\hat{f}_{kj}$. The MVUE result in the prior bullet is *much stronger* because the estimators here are the minimum variance out of **all unbiased estimators**, not just out of the linear combinations of the $\hat{f}_{kj}$.

Cross-Classified Models

The assumptions for the **EDF cross-classified model** are as follows:

- The random variables Y_{kj} are stochastically independent
- For each $k = 1, 2, \dots, K$ and $j = 1, 2, \dots, J$:
 - $Y_{kj} \sim \text{EDF}(\theta_{kj}, \phi_{kj}, a, b, c)$
 - $E[Y_{kj}] = \alpha_k \beta_j$, for $\alpha_k, \beta_j > 0$
 - $\sum_{j=1}^J \beta_j = 1$

Notice that the EDF cross-classified model includes explicit row (α_k) and column (β_j) parameters whereas the Mack model only includes explicit column (f_j) parameters. Although this appears to imply that the cross-classified model is more general, the Mack model implicitly includes row parameters by conditioning on the accumulated experience to date (i.e., $X_{k,J-k+1}$).

Theorem 3.2

Given the assumptions from the EDF cross-classified model above, as well as the following **ODP cross-classified model** assumptions:

- Y_{kj} is restricted to an ODP distribution
- The dispersion parameters ϕ_{kj} are identical for all cells (i.e., $\phi_{kj} = \phi$)

Then, the MLE fitted values and forecasts $\hat{Y}_{kj}$ are the same as those given by the conventional chain-ladder method. Note that we are specifically referring to *future* incremental losses here.

Theorem 3.3

In general, the MLEs $\hat{Y}_{kj}$ will not be unbiased. However, if we assume that the ODP cross-classified model assumptions apply (i.e., Theorem 3.2) AND that the fitted values and forecasts $\hat{Y}_{kj}$ and $\hat{R}_k$ are corrected for bias, then they are MVUEs of Y_{kj} and R_k .

Comparing the Theorems

Theorems 3.2 and 3.3 are *more remarkable* than Theorem 3.1 for the following reasons:

- They state that the forecasts from the ODP Mack and ODP cross-classified models are identical and the same as those from the conventional chain-ladder method despite the different formulations
- Forecasts can be obtained from the ODP cross-classified model without any explicit consideration of its parameters by working as if the model were the ODP Mack model

Example: Determining ODP Cross-Classified Parameters (Non-GLM)

Given the following information as of December 31, 2017:

AY	<u>Cumulative Losses (in \$000) as of XX Months:</u>			
	12 mo.	24 mo.	36 mo.	48 mo.
2014	2,000	4,000	5,000	5,250
2015	1,800	3,900	5,200	
2016	2,200	4,100		
2017	1,900			

- $\hat{\phi}$ (in 000s) = 40
- The contribution (in \$000) of the AY 2016 incremental paid loss between 0 and 12 months to the total deviance is 2,500

The process for determining the parameters is iterative:

- Begin by setting $\hat{\alpha}_1$ equal to the upper right **cumulative** loss
- Determine subsequent $\hat{\alpha}_k$ parameters using the following formula:

$$\hat{\alpha}_k = \frac{X_{k,j}}{1 - \sum_{\text{Remaining}} \hat{\beta}_j}$$

- Determine the $\hat{\beta}_j$ parameters using the following formula:

$$\hat{\beta}_j = \frac{\sum_k^{\text{Column}} Y_{kj}}{\sum_k^{\text{Column}} \hat{\alpha}_k}$$

First, let's create the incremental loss triangle:

AY	<u>Incremental Losses (in \$000) as of XX Months:</u>			
	12 mo.	24 mo.	36 mo.	48 mo.
2014	2,000	2,000	1,000	250
2015	1,800	2,100	1,300	
2016	2,200	1,900		
2017	1,900			

Second, let's calculate the $\hat{\alpha}_k$ and $\hat{\beta}_j$ parameters:

- $\hat{\alpha}_1 = 5,250$
- $\hat{\beta}_4 = \frac{250}{5,250} = 0.048$
- $\hat{\alpha}_2 = \frac{5,200}{1-0.048} = 5,460$
- $\hat{\beta}_3 = \frac{1,000+1,300}{5,250+5,460} = 0.215$
- $\hat{\alpha}_3 = \frac{4,100}{1-0.048-0.215} = 5,558$
- $\hat{\beta}_2 = \frac{2,000+2,100+1,900}{5,250+5,460+5,558} = 0.369$
- $\hat{\alpha}_4 = \frac{1,900}{1-0.048-0.215-0.369} = 5,152$
- $\hat{\beta}_1 = \frac{2,000+1,800+2,200+1,900}{5,250+5,460+5,558+5,152} = 0.369$

Third, let's calculate the forecasted AY 2016 incremental losses to emerge between 24 and 36 months:

- $\hat{\alpha}_3\hat{\beta}_3 = 5,558(0.215) = 1,194$
- This is identical to the chain-ladder estimate of $4,100(1.291 - 1) = 1,194$

Fourth, let's calculate the coefficient of variation (CV) of the forecasted AY 2016 incremental losses to emerge between 24 and 36 months:

- For the ODP distribution, the variance is ϕ times the mean, which is $40(1,194) = 47,747$
- Thus, the CV is $\frac{\sqrt{47,747}}{1,194} = 18.3\%$

Fifth, let's calculate the standardized Pearson residual for the incremental paid losses between 0 and 12 months for accident year 2016:

- The fitted value for the incremental paid loss between 0 and 12 months for AY 2016 is $\hat{\alpha}_3\hat{\beta}_1 = 5,558(0.369) = 2,050$
- The standardized Pearson residual is $R_i^P = \frac{Y_i - \hat{Y}_i}{\hat{\sigma}_i} = \frac{2,200 - 2,050}{\sqrt{40(2,050)}} = 0.524$

Sixth, let's calculate the standardized deviance residual for the incremental paid losses between 0 and 12 months for accident year 2016:

- The standardized deviance residual is $R_i^D = \text{sgn}(Y_i - \hat{Y}_i)(d_i/\hat{\phi})^{0.5} = \text{sgn}(2,200 - 2,050)(2,500/40)^{0.5} = (+1)(2,500/40)^{0.5} = 7.906$

GLM Representation of Chain-Ladder Models

In this section, we will look at the GLM representation of the ODP Mack model and ODP Cross-Classified model for a 4 x 4 loss triangle.

ODP Mack Model

Under the ODP Mack model, $Y_{k,j+1}|X_{kj} \sim ODP((f_j - 1)X_{kj}, \phi_{kj})$. We can re-formulate the model so that we are modeling age-to-age factors.

Assuming $\phi_{kj} = \phi_j$, the paper re-formulates the model so that $\hat{f}_{kj} - 1|X_{kj} \sim ODP(f_j - 1, \frac{\phi_j}{X_{kj}})$.

We will translate this to a GLM in matrix form where the $\hat{f}_{kj} - 1$ are the responses being modeled:

$$Y = \begin{bmatrix} \hat{f}_{1,1} - 1 \\ \hat{f}_{1,2} - 1 \\ \hat{f}_{1,3} - 1 \\ \hat{f}_{2,1} - 1 \\ \hat{f}_{2,2} - 1 \\ \hat{f}_{3,1} - 1 \end{bmatrix} \quad X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \beta = \begin{bmatrix} f_1 - 1 \\ f_2 - 1 \\ f_3 - 1 \end{bmatrix}$$

Recall that $E[Y] = \mu = h^{-1}(X\beta)$. For the ODP Mack model, h is the identity function. Thus $\mu = X\beta$. The weights underlying the variance function are $w_{k,j} = X_{k,j}$.

ODP Cross-Classified Model

Under the ODP Cross-Classified model, $Y_{kj} \sim \text{ODP}(\alpha_k \beta_j, \phi)$. We will translate this to a GLM in matrix form where the Y_{kj} are the responses being modeled:

$$Y = \begin{bmatrix} Y_{1,1} \\ Y_{1,2} \\ Y_{1,3} \\ Y_{1,4} \\ Y_{2,1} \\ Y_{2,2} \\ Y_{2,3} \\ Y_{3,1} \\ Y_{3,2} \\ Y_{4,1} \end{bmatrix} \quad X = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \end{bmatrix} \quad \beta = \begin{bmatrix} \ln(\alpha_1) \\ \ln(\alpha_2) \\ \ln(\alpha_3) \\ \ln(\alpha_4) \\ \ln(\beta_1) \\ \ln(\beta_2) \\ \ln(\beta_3) \\ \ln(\beta_4) \end{bmatrix}$$

Recall that $E[Y] = \mu = h^{-1}(X\beta)$. For the ODP cross-classified model, h is the log link function. Thus $\mu = \exp(X\beta)$ and $\mu_{i,j} = \exp(\ln\alpha_i + \ln\beta_j)$.

As an example, $E[Y_{2,4}] = \mu_{2,4} = \exp(\ln\alpha_2 + \ln\beta_4) = \alpha_2\beta_4$. The weights underlying the variance function are $w_{k,j} = 1$.

This particular setup leads to parameter redundancy because the model is over-specified. In this case, the parameters are said to be **aliased**. GLM software will set one of the parameters to zero to remove the redundancy. In doing so, the parameter estimates will no longer match the parameters produced under the non-GLM version of the ODP cross-classified model (see prior example). However, we can force the parameters to match by re-normalizing the parameters.

Note that the GLM setup above is different from how Shapland formulates it. For the purposes of the exam, we recommend viewing them as two different formulations of the same GLM.

Forecast Design Matrix

In the ODP cross-classified section above, we looked at the calculation of $E[Y_{2,4}]$, which is the expected future incremental loss for the second accident year in the fourth development period. Using a forecast design matrix, we can calculate all of the expected future incremental losses at once using $Y^* = h^{-1}(X^*\hat{\beta})$. The required matrices are as follows:

$$Y^* = \begin{bmatrix} Y_{2,4}^* \\ Y_{3,3}^* \\ Y_{3,4}^* \\ Y_{4,2}^* \\ Y_{4,3}^* \\ Y_{4,4}^* \end{bmatrix} \quad X^* = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \hat{\beta} = \begin{bmatrix} \ln(\hat{\alpha}_1) \\ \ln(\hat{\alpha}_2) \\ \ln(\hat{\alpha}_3) \\ \ln(\hat{\alpha}_4) \\ \ln(\hat{\beta}_1) \\ \ln(\hat{\beta}_2) \\ \ln(\hat{\beta}_3) \\ \ln(\hat{\beta}_4) \end{bmatrix}$$

Reliance On Only Recent Experience Years

In practice, it is common to use only the most recent m experience years to determine development factors. Thus, all observations **prior** to the most recent m experience years would be given zero weight in the GLM.

Outlier Observations

As mentioned earlier, if an actuary wants to exclude outlier observations in a GLM, the outlier should be given zero weight.

Example: Determining ODP Cross-Classified Parameters (GLM)

Given the following output from an ODP cross-classified GLM:

j or k	$\ln(\hat{\alpha}_k)$	$\ln(\hat{\beta}_j)$
1	4.561	2.231
2	4.787	2.463

3	5.102	1.987
4	5.315	0.000

Let's exponentiate the parameters to produce $\hat{\alpha}_k$ and $\hat{\beta}_j$ parameters:

j or k	$\hat{\alpha}_k$	$\hat{\beta}_j$
1	$95.679 = e^{4.561}$	9.309
2	119.941	11.740
3	164.350	7.294
4	203.365	1.000
Total		29.343

At this point, we can calculate expected values using the $\hat{\alpha}_k$ and $\hat{\beta}_j$ parameters above and they will match the chain-ladder method. However, the $\hat{\beta}_j$ parameters above do NOT sum to 1 like the non-GLM version of the ODP cross-classified model. To compensate for this, we divide each $\hat{\beta}_j$ parameter by the sum of the $\hat{\beta}_j$ parameters AND multiply each $\hat{\alpha}_k$ by the sum of the $\hat{\beta}_j$ parameters. This ensures the expected values remain the sum while also forcing the $\hat{\beta}_j$ parameters to sum to 1.

The re-normalized parameters are as follows:

j or k	$\hat{\alpha}_k$	$\hat{\beta}_j$
1	$2,807 = 95.679(29.343)$	$0.317 = \frac{9.309}{29.343}$
2	3,519	0.400
3	4,822	0.249
4	5,967	0.034
Total		1.000

To demonstrate that the parameters produce equivalent results, let's look at $E[Y_{2,4}]$:

- Using the parameters produced by the GLM, $E[Y_{2,4}] = \hat{\alpha}_2 \hat{\beta}_4 = 119.941(1) = 119.941$

- Using the re-normalized parameters (which match the non-GLM parameters for the ODP cross-classified model), we have $E[Y_{2,4}] = \hat{\alpha}_2 \hat{\beta}_4 = 3519(0.034) = 119.941$

V. Prediction Error

This chapter focuses on the error contained in a forecast (known as prediction or forecast error). It can be decomposed into three components:

- 1) Parameter Error
- 2) Process Error
- 3) Model Error

Parameter Error & Process Error – Individual Observations

In this chapter, the authors use the following *general model* to forecast outstanding losses:

$$Y_{kj} = u(k, j; \theta) + \varepsilon_{kj} \text{ for } Y_{kj} \in D_K^+$$

for some function u , dependent on a parameter vector θ , and *centered* stochastic error ε_{kj} (i.e., $E[\varepsilon_{kj}] = 0$). More simply, each observation Y_{kj} can be thought of a mean plus a random error.

As one might expect, we assume that the model has been constructed using the data set D_K , yielding parameter estimates $\hat{\theta}$. Using these parameter estimates, we define the following values:

$$\hat{Y}_{kj} = u(k, j; \hat{\theta}) \text{ for } Y_{kj} \in D_K^+$$

The $\hat{Y}_{kj}$ associated with the past observations (i.e., $Y_{kj} \in D_K$) are *fitted values*. The $\hat{Y}_{kj}$ associated with the future observations (i.e., $Y_{kj} \in D_K^c$) are forecasts.

The *prediction error* associated with the forecast $\hat{Y}_{kj}$ is as follows:

$$e_{kj} = Y_{kj} - \hat{Y}_{kj} = [u(k, j; \theta) - u(k, j; \hat{\theta})] + \varepsilon_{kj}$$

Since $E[\varepsilon_{kj}] = 0$, it must be the case that $E[Y_{kj}] = \mu_{kj} = u(k, j; \theta)$. Hence, we can re-write the prediction error above as follows:

$$e_{kj} = Y_{kj} - \hat{Y}_{kj} = [\mu_{kj} - \hat{Y}_{kj}] + \varepsilon_{kj}$$

The prediction error above can be thought of as follows:

$$\begin{aligned} e_{kj} &= [\mu_{kj} - \hat{Y}_{kj}] + \varepsilon_{kj} \\ &\quad \downarrow \quad \searrow \\ &= \text{Parameter Error} + \text{Process Error} \end{aligned}$$

It is often the case that the parameter error and process error are stochastically independent. If the model is formulated such that the past Y_{kj} and future ε_{kj} are independent, then the parameter and process errors are also independent. For example, the EDF cross-classified model is formulated in this manner since it assumes all observations are independent.

There is one caveat to the prediction error decomposition above. It assumes that the model u has been correctly specified. In reality, there are no perfect models. The difference between $E[Y_{kj}] = u(k, j; \theta)$ and its correct specification is known as model error. If the model is incorrectly specified, this discrepancy is not separately identifiable in the decomposition and is therefore absorbed into the process error component.

Parameter Error & Process Error – Loss Reserves

At this point, the authors move from individual observations to loss reserves. To make the notation less cumbersome, they introduce vector terms (we saw some of these earlier in the outline):

- Y is the vector of past observations $Y_{kj} \in D_K$
- Y^* is the vector of future observations $Y_{kj} \in D_K^c$
- μ^* is the vector of future true means μ_{kj}
- $\hat{Y}^*$ is the vector of forecasts $\hat{Y}_{kj}$

- e^* is the vector of prediction errors e_{kj} associated with future observations
- ε^* is the vector of centered stochastic errors ε_{kj} associated with future observations

Note that the “stars” are simply meant to refer to future observations. Using this notation, we can re-write the prediction error formula as follows:

$$e^* = [\mu^* - \hat{Y}^*] + \varepsilon^*$$

Next, the authors introduce a general column vector r . This vector is used to produce linear combinations of the other vectors. For *example*, suppose the vector r is comprised of 1s. Then, based on standard matrix multiplication, $r^T \mu^*$ (note that the superscript T means “transpose”) is the **expected value of total outstanding losses**.

The prediction error associated with the total outstanding losses $r^T Y^*$ is defined as follows:

$$\begin{aligned} e_{(r)}^* &= r^T e^* \\ &= [r^T \mu^* - r^T \hat{Y}^*] + r^T \varepsilon^* \end{aligned}$$

where:

- $r^T \mu^*$ is the expected value of total outstanding losses
- $r^T \hat{Y}^*$ is the forecast of total outstanding losses
- $r^T \varepsilon^*$ is the process error associated with total outstanding losses

As a *simple example*, suppose we are forecasting losses for a 3 x 3 triangle, where the first AY is already at ultimate. The r and μ^* vectors are defined as follows:

$$r = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \mu^* = \begin{bmatrix} \mu_{2,3}^* \\ \mu_{3,2}^* \\ \mu_{3,3}^* \end{bmatrix}$$

Then, we compute $r^T \mu^*$ as follows:

$$\begin{aligned} r^T \mu^* &= [1 \quad 1 \quad 1] \begin{bmatrix} \mu_{2,3}^* \\ \mu_{3,2}^* \\ \mu_{3,3}^* \end{bmatrix} \\ &= \mu_{2,3}^* + \mu_{3,2}^* + \mu_{3,3}^* \end{aligned}$$

As we see above, $r^T \mu^*$ sums together the expected value of the future observations, resulting in the expected value of the total outstanding losses.

Mean Square Error of Prediction (MSEP)

The mean square error of prediction, denoted as $MSEP[e_{(r)}^*]$, is used to summarize the magnitude of the prediction error. It is defined as follows:

$$MSEP[e_{(r)}^*] = E \left\{ [e_{(r)}^*]^2 \right\}$$

If the parameter and process errors are independent, the MSEP can be re-written as follows:

$$MSEP[e_{(r)}^*] = E \left\{ [e_{(r)param}^*]^2 \right\} + E \left\{ [e_{(r)process}^*]^2 \right\}$$

where:

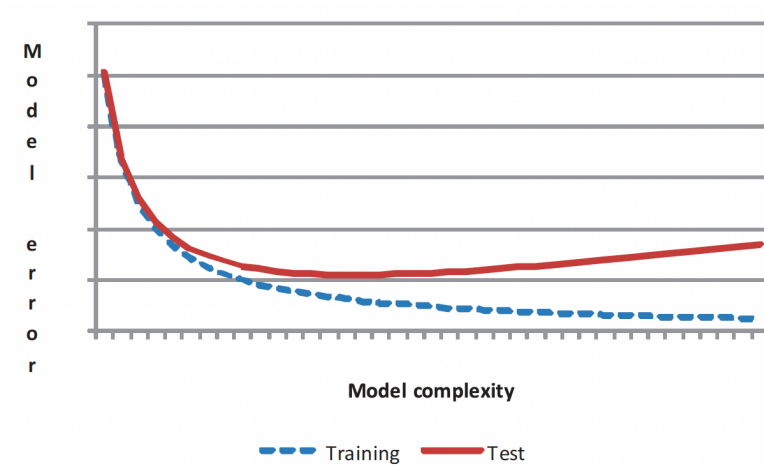
- $e_{(r)param}^* = r^T \mu^* - r^T \hat{Y}^* = \text{parameter error}$
- $e_{(r)process}^* = r^T \varepsilon^* = \text{process error}$

Goodness-of-Fit & Prediction Error

Since the MSEP estimates the tightness of a forecast around its target, a model that produces smaller MSEP is generally preferred over a model generating a larger MSEP.

A common misconception is that improving a model's goodness-of-fit automatically decreases the MSEP. However, that is not always the case. Models with an excessive number of parameters can overfit and lead to a deterioration in the MSEP when applied to test data (i.e., holdout data).

Consider Figure 4.1 from the text:



Since the model is built using the training data, increasing the model complexity (i.e., adding predictors, interactions, etc.) always lowers the MSEP on the training data. However, at some point, the increased model complexity leads to deterioration in performance on unseen, holdout data. The reversal in model error represents the point at which the model is beginning to overfit the training data.

Information Criteria

Information criteria (IC) refer to statistics that function as proxies for measurement of prediction error relative to test data. In general, they take the following form:

$$IC = \text{model fit error (on training data)} + \text{penalty for number of parameters}$$

By including a penalty for the number of parameters, information criteria discourage unnecessary model complexity. Although adding parameters typically improves model fit on training data, the information criterion will increase unless that improvement is sufficient to offset the added penalty (similar to what we saw earlier).

For a GLM, a specific information criterion can be written as follows:

$$IC(Y, \hat{Y}) = D(Y, \hat{Y}) + f(p)$$

where:

- $IC(Y, \hat{Y})$ denotes the information criterion
- $D(Y, \hat{Y})$ denotes the scaled deviance
- p is the number of model parameters
- $f(\cdot)$ is a monotonically increasing function

Three common information criteria are as follows:

Information Criterion	Function $f(p)$
Akaike Information Criterion (AIC)	$2p$
Modified AIC (AICc)	$2p \left[1 + \frac{p+1}{n-p-1} \right]$
Bayes Information Criterion (BIC)	$p \ln(n)$

In the table above, n is the number of observations. In general, BIC applies a heavier penalty than AIC or AICc. When evaluating information criterion, models with lower values are preferred.

Generalized Cross-Validation

Cross-validation is a method used to estimate prediction error. In k -fold cross-validation, the data is split into k equal-sized parts, with the model fitted on $k - 1$ parts and tested on the final k^{th} part. This produces k estimates of the prediction error. The average of the k estimates is the cross-validation error.

If we let $k = n$, then one point is left out of the fit for each of the n iterations of the model fit. This is known as leave-one-out cross-validation.

When a fitted value can be expressed as $\hat{y} = Hy$ (such as in a linear regression), the leave-one-out cross-validation error can be approximated using the generalized cross-validation (GCV) measure:

$$GCV = \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n[1 - \text{trace}(H)/n]^2}$$

where:

- Y_i is the i^{th} observed value
- $\hat{Y}_i$ is the i^{th} fitted value
- $\text{trace}(H)$ is the sum of the diagonal elements of the hat matrix H . This actually gives you the effective number of parameters in a model

At this point in the text, the authors have not defined the hat matrix. For a linear regression, the hat matrix is defined as $X(X^T X)^{-1} X^T$, where X is the design matrix. We call it the “hat” matrix because applying it to the observation vector y produces fitted values.

Similar to AIC and BIC, the GCV penalizes overfitting.

Model Error

At the beginning of this section, we proposed the following general model for $Y_{kj} \in D_K^+$:

$$Y_{kj} = u(k, j; \theta) + \varepsilon_{kj}$$

Then, we assumed that the fitted values took the form $u(k, j; \hat{\theta})$, where the unknown parameters θ were replaced by the estimates $\hat{\theta}$. The issue with this model is that it assumes we identify the function $u(k, j; \theta)$. In reality, this function is unknown. As a result, it's more appropriate to use the following model form:

$$Y_{kj} = v(k, j; \xi) + \varepsilon_{kj} \text{ for } Y_{kj} \in D_K^+$$

where $v(\cdot)$ is an approximation function with a different parameter set ξ . Under this model form, the fitted values are as follows:

$$\hat{Y}_{kj} = v(k, j; \hat{\xi}) \text{ for } Y_{kj} \in D_K^+$$

The prediction error is decomposed as follows:

$$\begin{aligned}
 e_{kj} &= Y_{kj} - \hat{Y}_{kj} \\
 &= [v(k, j; \xi) - v(k, j; \hat{\xi})] + \varepsilon_{kj} + [u(k, j; \theta) - v(k, j; \xi)] \\
 &\quad \downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow \\
 &= \text{Parameter Error} + \text{Process Error} + \text{Model Error}
 \end{aligned}$$

In the equation above, model error is the difference between the parametric form assumed for the model and the true but unknown parametric form. Parameter error is the difference between the assumed parametric model evaluated at the true parameter values and the final fitted version of that model.

Although model error clearly exists, there is no known procedure for estimating it. Hence, it's typically absorbed into process error.

VI. The Bootstrap

In this section, we investigate two approaches for estimating the prediction error associated with outstanding losses:

- 1) The delta method
- 2) The bootstrap

Although the delta method is simple to implement, its accuracy is questionable and it does not provide any distributional information (e.g., quantiles) about the MSEP. Hence, the bootstrap is preferred for these reasons.

The Delta Method – Uni-Dimensional

The delta method is used to compute the variance of a transformed random variable when the variance of the untransformed random variable is known.

Let X be a random variable with $E[X] = \mu$ and $Var(X) = \sigma^2$. Further suppose that f is a differentiable one-to-one transformation of X . Then, we can approximate the variance of $f(x)$ using the delta method as follows:

$$\begin{aligned} Var(f(X)) &= E[(x - \mu)^2 [f'(\mu)]^2 + \dots] \\ &= \sigma^2 [f'(\mu)]^2 + \dots \end{aligned}$$

The resulting expression provides a second-order accurate approximation to $Var(f(X))$, derived from a second-order Taylor expansion of $f(X)$ about μ .

The Delta Method – Multi-Dimensional

Now, let's extend the delta method from a single-dimensional variate to a multi-dimensional vector. Let $Y = f(X)$, where X is a column vector with n rows. The components of X and Y are denoted as X_i and Y_i , respectively.

The variance of Y is as follows:

$$Var(Y) = DVar(X)D$$

where:

- $Var(Y)$ represents the variance-covariance matrix of vector Y
- $Var(X)$ represents the variance-covariance matrix of vector X
- $D = diag(f'(\mu_1), \dots, f'(\mu_n))$ is a diagonal matrix whose entries are $f'(\mu_1), \dots, f'(\mu_n)$ along the diagonal and 0 elsewhere

The Delta Method – Application to Loss Reserving

As we saw earlier, when dealing with total outstanding losses, we use column vectors. Hence, we can take the general formula for $Var(Y)$ above and swap in the values related to the reserving model:

General Quantity	Loss Reserving Replacement
Y	$\hat{Y}^*$
X	$X^* \hat{\beta}$
μ	$X^* \beta$
f	h^{-1}

where the loss reserving replacements are defined as we saw earlier when investigating the GLM representations of various reserving models.

Using these replacements, the general formula for $Var(Y)$ becomes the following:

$$\begin{aligned} Var_{param}[\hat{Y}^*] &= DVar(X^* \hat{\beta})D \\ &= DX^*Var(\hat{\beta})(X^*)^T D \end{aligned}$$

Regarding the formula above:

- The subscript “param” has been added to reflect the fact that only parameter error is being estimated
- $Var(\hat{\beta})$ is estimated by the GLM software (i.e., the GLM software provides the variance-covariance matrix for the estimated coefficients)
- $D = diag \left[(h^{-1})' \left(h(\hat{Y}_{2,K}^*) \right), \dots, (h^{-1})' \left(h(\hat{Y}_{K,K}^*) \right) \right]$. This assumes we have a $K \times K$ loss triangle

To arrive at the full prediction error of $\hat{Y}^*$ (ignoring model error), we must add a process error term. Assuming incremental losses follow the ODP distribution, the process error term is written as follows:

$$Var_{proc}(\hat{Y}^*) = \hat{\phi} diag(\hat{Y}^*)$$

where $\hat{\phi}$ is an estimate of the scale parameter obtained from the GLM software and $diag(\hat{Y}^*)$ is a diagonal matrix whose entries are the forecasted values along the diagonal and 0 elsewhere.

Putting it all together, we obtain the delta method version of the MSEP of the forecasted loss vector $\hat{Y}^*$:

$$\begin{aligned} MSEP[\hat{Y}^*] &= Var_{param}[\hat{Y}^*] + Var_{proc}(\hat{Y}^*) \\ &= DX^*Var(\hat{\beta})(X^*)^T D + \hat{\phi}diag(\hat{Y}^*) \end{aligned}$$

Note that $MSEP[\hat{Y}^*]$ represents a variance-covariance matrix of size $m \times m$, where m represents the number of future observations. To help understand this, let's break the individual matrices out piece by piece:

- D is $m \times m$
- X^* is $m \times p$, where p is the number of parameters
- $Var(\hat{\beta})$ is $p \times p$
- $(X^*)^T$ is $p \times m$
- $diag(\hat{Y}^*)$ is $m \times m$

Using matrix multiplication and addition, we obtain an $m \times m$ matrix. For a 10×10 loss triangle with 10 accident year parameters and 9 development year parameters (omitting one for redundancy), $m = 45$ and $p = 19$.

In order to produce the MSEP of total reserve estimate $\hat{R}$, we first note that $\hat{R}$ can be obtained by multiplying the $\hat{Y}^*$ by a column vector consisting of 1's (this simply adds up the forecasted losses to produce a total reserve estimate).

When working in matrix notation, the variance of the linear combination $a^T X$ is $Var(a^T X) = a^T Var(X)a$, where a is a fixed vector. Thus, the MSEP of the total reserve estimate is computed as follows:

$$MSEP[\hat{R}] = 1^T MSEP[\hat{Y}^*] 1$$

where "1" is a column vector consisting of 1s.

To compute the MSEP of an individual AY reserve estimate $\hat{R}_k$, we use a similar approach:

$$\boxed{MSEP[\hat{R}_k] = 1_k^T MSEP[\hat{Y}^*] 1_k}$$

where 1_k is a column vector consisting of a 1 in the entries corresponding to $\hat{Y}_{k,J-k+2}^*, \dots, \hat{Y}_{kJ}^*$, and 0 elsewhere.

As an *alternative* to computing $MSEP[\hat{R}]$ and $MSEP[\hat{R}_k]$ as shown above, we can split the calculations into separate parameter variance and process variance portions:

- $Var_{param}[\hat{R}_k] = 1_k^T Var_{param}[\hat{Y}^*] 1_k$
- $Var_{process}[\hat{R}_k] = 1_k^T Var_{process}[\hat{Y}^*] 1_k = \hat{\phi} \hat{R}_k$
- For the total reserve estimate $\hat{R}$, simply swap in $\hat{R}$, 1^T , and 1 into the prior two equations

The nice thing about splitting the MSEP into two pieces is that the process variance portion simplifies to a simple calculation. We simply multiply each reserve estimate by the scale parameter to compute the process variance for that AY (or in total).

See the Cookbook for a detailed example of the Delta Method.

The Bootstrap

As mentioned earlier, the delta method has a couple of problems:

- Since it relies on a second-order Taylor series expansion, its accuracy can deteriorate when higher-order derivatives are large
- It provides little distributional information about the outstanding losses

The issues above are *particularly problematic* when we need to estimate quantiles of our estimated outstanding losses. For *example*, a regulator might require the loss reserves to be equal to the estimated amount of outstanding losses with 100p% ($p > 50$) *probability of adequacy* (PoA).

If we denote this amount as $\hat{R}_p$, then:

$$P(R < \hat{R}_p) = p$$

Although a distribution can be assumed to estimate this amount, there is no guarantee that the assumed distribution will perform well. A better solution is the bootstrap.

The text describes two versions of the bootstrap:

- 1) Semi-Parametric Bootstrap
- 2) Parametric Bootstrap

Semi-Parametric Bootstrap

The bootstrap is useful when it is challenging to estimate the properties of a statistic in an analytical manner. With the increase in computing power, the bootstrap became a viable way forward. The bootstrap involves re-sampling from the available data.

Given that the bootstrap is a process, we will skip most of the mathematics in this section and focus on how the bootstrap is implemented.

For the semi-parametric bootstrap described in the text, our goal is to resample from the residuals and construct pseudo datasets using the resampled datasets and the fitted values (similar to what is done in Shapland).

Taylor and McGuire point out that the standardized residuals may be standardized Pearson residuals, standardized deviance residuals, or any other standardized residuals in which it is possible to “back-solve” for the observation.

For example, as shown earlier in the paper, the standardized Pearson residuals are defined as follows:

$$R_i^P = \frac{Y_i - \hat{Y}_i}{\hat{\sigma}_i}$$

In this case, we can easily solve for Y_i , resulting in $Y_i = R_i^P \hat{\sigma}_i + \hat{Y}_i$.

Once we have determined which standardized residuals we will use, the bootstrap process continues as follows:

- 1) Draw a random n -sample (always sample the same number of observations as used in the original model), with or without replacement
 - Generally, it's with replacement
- 2) For each residual in the random n -sample, compute pseudo observations by “back-solving”
 - This produces a pseudo-dataset $\tilde{Y}$ with the same stochastic properties as the original one
- 3) Re-run steps 1 and 2 a large number of times (let's say r times), producing r pseudo-data sets, denoted $\tilde{Y}_{(1)}, \tilde{Y}_{(2)}, \dots, \tilde{Y}_{(r)}$
- 4) Model the r pseudo-data sets using the exact same model form used for the original fitted values
 - The same model form means the same model structure. Since the data being used to fit the model is different from the data used to fit the original model, the parameter values will change
- 5) For each of the r pseudo-data sets, form the same forecasts (known as pseudo-forecasts) that were formed for the original data set. We denote the pseudo-forecasts as $\hat{Y}_{(j)}^*$, where j represents the j th replicate among the r total replicates
 - For example, if we used the ODP cross-classified to model the original data set and forecast outstanding losses, we would produce the same forecasts for the pseudo data sets based on the ODP cross-classified parameter values from step 4

The variability in the pseudo-forecasts reflects parameter error. To account for process error, we use a separate re-sampling process. Before doing so, let the process error associated with the i th component of Y^* be denoted as follows:

$$\varepsilon_i^* = Y_i^* - E[Y_i^*]$$

In other words, the future process error is the difference between the actual future outcome and the expected future outcome (assuming we know the true parameter values).

Now, the process error re-sampling steps are as follows:

- 1) For the j th pseudo-forecast $\hat{Y}_{(j)}^*$, treat each component as an estimate of $E[Y_i^*]$
- 2) Draw a second random sample of standardized residuals, $\tilde{S}_{proc}$, using the same procedure used earlier to generate the pseudo-data sets
- 3) Compute a process error vector ε_{proc}^* , where component i of the vector is $\varepsilon_{proc,i}^* = \hat{\sigma}_i^* \tilde{S}_{proc,i}$
- 4) Repeat steps 2-4 above r times (once for each replicate) to obtain r process error replicates
- 5) Add the process error replicate to the corresponding pseudo-forecast to obtain

$$\left(\hat{Y}_{(j)}^*\right)^+ = \hat{Y}_{(j)}^* + \varepsilon_{proc(j)}^*$$

In this case, $\left(\hat{Y}_{(j)}^*\right)^+$ is a pseudo-forecast, augmented to include process error. Using $\left(\hat{Y}_{(j)}^*\right)^+$, we can easily obtain pseudo-forecasts of the reserves including process error, denoted $R\left(\left(\hat{Y}_{(j)}^*\right)^+\right)$. At this point, the r replicates form an empirical distribution of $R(Y^*)$. Based on this empirical distribution, we can estimate quantities such as MSEP.

The bootstrap above is known as the semi-parametric bootstrap because the generation of the pseudo-data sets using resampling makes no distributional assumption. However, the original fitted values used to compute the original residuals is based on an assumed model.

Finally, it's worth noting that any outliers excluded from the original model fit will clearly reduce any bootstrap estimate of variability. Adjustments may be needed if the actuary feels that variability is understated.

Differences Between Shapland & Taylor Bootstrapping

The semi-parametric bootstrap process described in Taylor & McGuire is very similar to the one shown in Shapland. In fact, the parameter error steps are identical between the two papers.

However, the process error steps differ.

In Shapland, we draw from a gamma distribution to add process error to each future cell. In Taylor & McGuire, we use a second empirical residual resampling to add process error.

See the Cookbook for an example of the semi-parametric bootstrap as described in Taylor & McGuire.

Parametric Bootstrap

The parametric bootstrap is similar to the semi-parametric bootstrap. Traditionally, the parametric bootstrap uses theoretical residuals to perform the bootstrap process. Taylor uses an alternative approach based on the multivariate normal distribution.

The bootstrap process is as follows:

- 1) Fit the GLM to the original data and obtain:
 - The parameter estimates, $\hat{\beta}$
 - The estimated variance-covariance matrix for $\hat{\beta}$, denoted as $\hat{C}$
 - Assume that $\hat{\beta} \sim N(\beta, \hat{C})$. In other words, the estimated parameters follow a multivariate normal distribution with mean vector β and variance-covariance matrix $\hat{C}$
 - This assumes n is large enough to invoke asymptotic normality of the MLE
- 2) Apply a linear transformation, M , to the parameter estimates to produce uncorrelated parameter estimates
 - Independent parameter estimates simplify the sampling process
- 3) Sample each transformed parameter independently from its corresponding normal distribution

- 4) Combine the sampled values into a random vector, γ
- 5) Apply the inversion of M to the γ vector to take the transformed parameter estimates back to their original parameter space
 - This produces a single bootstrap replicate of the parameter vector, $\hat{\beta}_{(j)} = M^{-1}\gamma$
- 6) Repeat steps 4-6 r times to obtain r parameter estimate replicates
- 7) Use each parameter replicate to generate forecasts

Similar to the semi-parametric bootstrap, the process above reflects parameter error.

Under the parametric bootstrap, process error is added directly from the distribution assumed by the GLM. The process is as follows:

- 1) For each parameter replicate, generating the corresponding forecast of future losses
- 2) Treat the forecasted values as the means of the future observations
- 3) Draw random observations directly from the distribution assumed by the GLM
 - For example, under an ODP model, draw from ODP distribution with a mean equal to the forecasted value $\tilde{Y}_i$, and a scale parameter equal to $\hat{\phi} \backslash w_i$
- 4) The resulting random draw is a pseudo-forecast and reflects both parameter and process error

Based on the pseudo-forecasts of future observations, we can produce pseudo-forecasts of the reserves. The collection of reserve estimates forms an empirical distribution of possible reserve outcomes, from which quantities such as the MSEP may be estimated.

As shown above, the parametric bootstrap assumes a multivariate normal distribution for the parameter error, and the GLM's chosen distributional form for process error.

Advantages of the parametric bootstrap over the semi-parametric bootstrap are as follows:

- Simpler
- Less computationally intensive

Disadvantages of the parametric bootstrap over the semi-parametric bootstrap are as follows:

- For smaller n , the asymptotic normality of the MLE may not be appropriate
- Relies heavily on the error structure of the GLM being a good representation of the data

Example – Delta Method

The authors applied the delta method to estimate the forecast error associated with the ODP cross-classified model forecasts for a 10×10 triangle.

Since the ODP cross-classified model uses a log link function, $h(x) = \ln(x)$ and $h^{-1}(x) = e^x$. As a result, $(h^{-1})'(x) = e^x$ and $(h^{-1})'(h(x)) = x$.

Given this, the diagonal matrix D simplifies to $D = \text{diag}(\hat{\mu}_{2,10}^*, \hat{\mu}_{3,9}^*, \dots, \hat{\mu}_{10,10}^*)$. This is used to compute the delta method MSEF.

The results are as follows:

AY	<u>Outstanding Losses</u>		
	Forecast (\$000)	RMSEP (\$000)	CVP (%)
2	3,398	924	27.2
3	8,155	1,363	16.7
4	14,579	1,775	12.2
5	22,645	2,169	9.6
6	31,865	2,523	7.9
7	45,753	3,036	6.6
8	60,093	3,577	6.0
9	80,983	4,538	5.6
10	105,874	6,786	6.4
Total	373,346	14,076	3.8

Regarding the table above:

- CVP stands for the coefficient of variation of prediction. It is defined as $CVP = \frac{RMSEP}{Forecast}$

- RMSEP is the root mean square error of prediction. It is equal to the square root of the MSEP
- If we square the RMSEP column to obtain the MSEP figures, we notice that the total MSEP is greater than the sum of the AY MSEPs. This indicates positive correlation among at least some of the AYs. If the AYs were independent, then the total MSEP would be equal to the sum of the AY MSEPs

Example – Parametric Bootstrap

Next, the authors applied the parametric bootstrap to estimate the forecast error associated with the ODP cross-classified model forecasts for the same triangle:

- The forecasts were computed by taking the simple average of the 10,000 replicates of the forecast
- The RMSEPs were computed by taking the square root of the sample variance of the 10,000 replicates

The results are as follows:

AY	<u>Outstanding Losses</u>		
	Forecast (\$000)	RMSEP (\$000)	CVP (%)
2	3,476	937	27.0
3	8,269	1,366	16.5
4	14,738	1,794	12.2
5	22,776	2,186	9.6
6	32,043	2,525	7.9
7	45,963	3,057	6.7
8	60,273	3,608	6.0
9	81,249	4,589	5.6
10	106,204	6,831	6.4
Total	374,992	14,286	3.8

Overall, the results are very similar to those from the Delta Method. The small differences in forecasts are due to sampling error arising from the limited number of bootstrap replicates.

VII. Model Validation

Model validation is the “process of examining whether the fitted model (i.e., the distributional assumption and fitted parameter effects) are acceptable and adequate descriptions of the data being modeled.”

There are three aspects to model validation:

- Analysis of the distributional assumptions
- Analysis of the goodness-of-fit of the model
- Analysis of the model’s predictive performance on data beyond those used in the model estimation

For a loss triangle, the third item above is typically not possible since the entire triangle is usually used to build the model. For that reason, we focus on the first two items.

The authors use the following approach for model fitting and validation:

- Select the appropriate link function
- Select a reasonable distribution
 - Example: Use the ODP distribution for a cross-classified model
- Fit the main effects in the model and any obvious interactions
- Check the residual diagnostics for any significant violations of the distributional assumptions and make changes if needed
- Continue with model fitting using goodness-of-fit tests until all individual cell means look satisfactory
 - Interactions may be needed to ensure strong goodness-of-fit
- Review the distributional diagnostics in detail and adjust the model as needed
 - After adjustments, re-check the goodness-of-fit and adjust the model as needed

- Continue until a satisfactory model is obtained

In general, the estimated mean of the distribution is insensitive to the distributional choice. In contrast, the estimated variance is sensitive to the distributional choice.

Summary of Assumptions & Tests

This section summarizes the various assumptions and tests we will consider throughout this chapter.

Regarding *distributional assumptions*:

- **Appropriateness of the Link Function**
 - Largely determined by modeled quantity. In other words, if a multiplicative model is needed, use log link function
 - The link function is *validated* if the model passes the other diagnostic tests *without requiring a large number of interactions*
- **Appropriateness of the Distribution**
 - Review probability-probability (P-P) plot
 - Review residual plots by accident, development, and calendar year periods
 - Review histograms or kernel density plots of the residuals

Regarding *goodness-of-fit*:

- **Model Fits Well by Accident, Development, and Calendar Year Periods**
 - Review plots by accident, development, and calendar year periods of *actual* and *expected* in one of the following forms:
 - Actual vs. expected
 - Log(actual) vs. log(expected)
 - Actual/expected
 - Review residual plots by accident, development, and calendar year periods
- **All Significant Interactions Have Been Identified**
 - Use a triangular heat map of actual/expected

- Use actual and expected plots for specific parts of the experience

Diagnostic Graphs

In this section, we cover several diagnostic graphs. All diagnostic graphs involve the comparison of actual and expected quantities, including the following:

- Pearson residuals – both raw and standardized
- Deviance residuals – both raw and standardized
- Actual values including sums across rows, columns, and diagonals
- Expected values including sums across rows, columns, and diagonals
- Actual/expected values in each cell of the triangle
- Actual/expected marginal values by row, column, and diagonal (i.e., dividing sums)

As discussed earlier, deviance residuals display a greater degree of normality when the underlying distribution is not normal. For that reason, the authors use **standardized deviance residuals** for the rest of this section.

We will cover the following diagnostic plots:

- Scatterplot of residuals
- Spread plot of residuals
- Actual and expected comparison plots
- Actual and expected ratio plots
- Actual and expected ratio 2-D heat map
- P-P plot
- Histogram of residuals

Scatterplot of Residuals

A scatterplot of residuals plots residuals against expected value, accident period, development period, or calendar period.

We desire a random scattering around zero (i.e., random and homoscedastic residuals). Otherwise, it suggests problems with the model:

- A trend in the residuals indicates possible goodness-of-fit issues
- Heteroscedasticity indicates possible goodness-of-fit issues or inappropriate dispersion assumptions

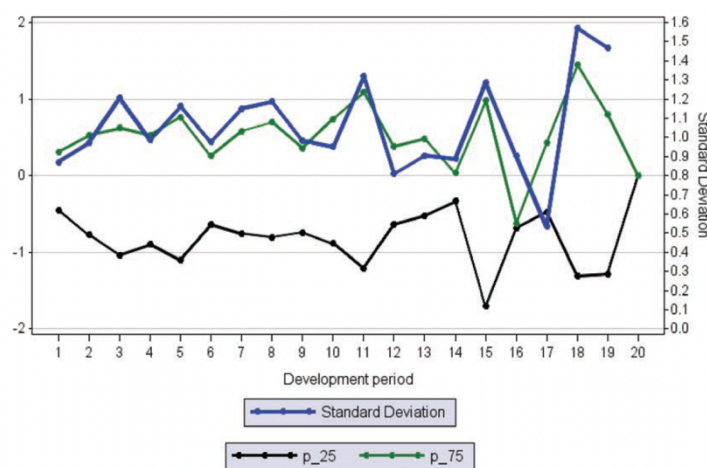
Spread Plot of Residuals

A spread plot of residuals plots summary statistics (e.g., 25th and 75th percentiles, standard deviation) for the residuals against expected value, accident period, development period, or calendar period.

A spread plot is useful for detecting heteroscedasticity of the residuals:

- If the inter-quartile range (i.e., the range from the 25th percentile to the 75th percentile) widens or narrows, then heteroscedasticity is indicated
- If there are significant changes in standard deviation, then heteroscedasticity is indicated

As an example, consider Figure 6-2 from the monograph:



Overall, the spread plot is fairly stable and does not indicate material heteroscedasticity:

- The inter-quartile range is reasonable stable over time

- The standard deviation varies randomly about 1, which is what we would expect for standardized residuals
- There is some volatility in the statistics at later development periods, but this is most likely due to the limited data at this point in the triangle

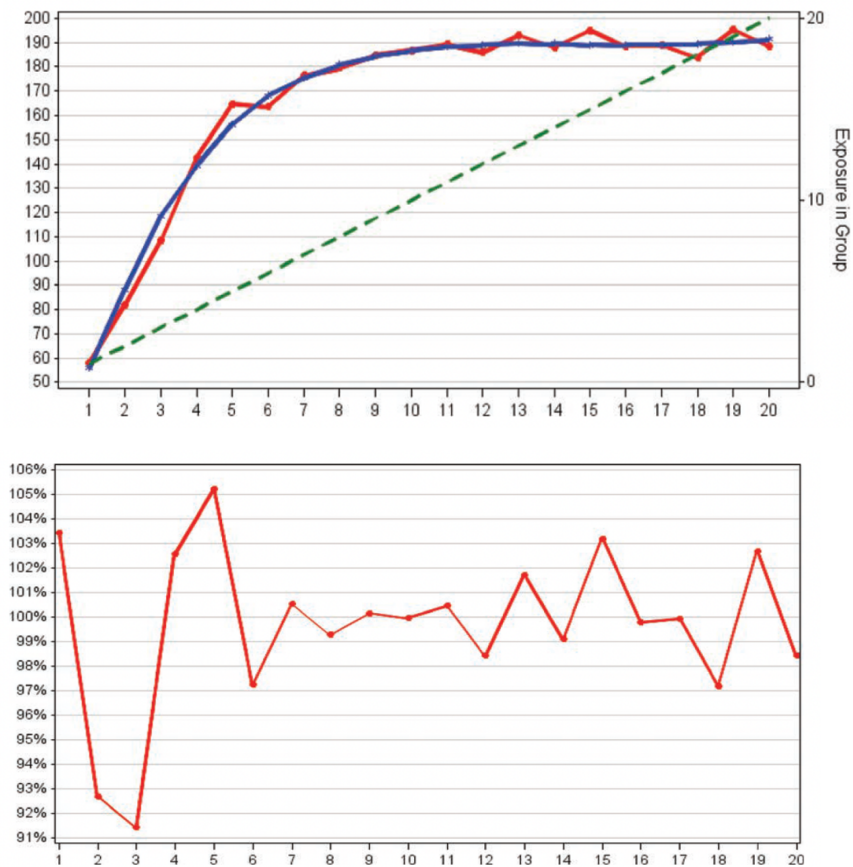
Actual and Expected Comparison Plots

These plot the actual and expected totals (i.e., sums) by accident, development, or calendar period. Areas of *poor fit* correspond to *consistent differences* in the actual and expected values.

Actual and Expected Ratio Plots

These plot the actual/expected ratio (based on sums) by accident, development, or calendar period. *Systemic deviations from 100%* indicate regions of *poor fit*.

Figures 6-3 (top) and 6-4 (bottom) from the monograph illustrate the two plots above:



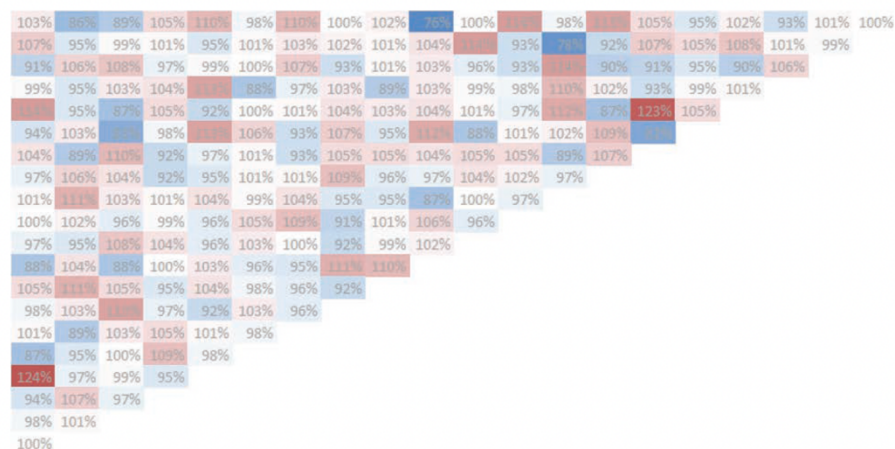
We summarize the plots as follows:

- Actual and Expected Comparison Plot (Top)
 - The x -axis is the calendar period (i.e., diagonal)
 - The actual and expected values are the red and blue lines, respectively
 - The green line shows the number of data points at each level
 - The red and blue lines are very similar, indicating adequate fit
- Actual and Expected Ratio Plot (Bottom)
 - The x -axis is the calendar period (i.e., diagonal)
 - The red line is the actual/expected ratio for each diagonal
 - Overall, the ratio randomly varies about 1, indicating adequate fit
 - There is some volatility in earlier calendar periods, but this is most likely due to limited data in that part of the triangle (i.e., the upper left)

Actual and Expected Ratio 2-D Heat Map

Unlike the prior two plots that compare actual and expected by period, this examines goodness-of-fit across the entire triangle. Specifically, it calculates the actual/expected ratio in each cell of the triangle and applies a conditional formatting based on the deviation of the ratio from 100%.

Figure 6-5 from the monograph displays a heat map:



In the plot above, pink values indicate ratios larger than 100% and blue values indicate ratios smaller than 100%. As we move further away from 100%, the colors get darker.

A random scattering of colors suggests an adequate fit. If we saw clumps of colors, then additional terms may be needed. For example, suppose a model includes accident period and development period terms. If we see color clumps, then we should consider adding calendar period terms or interactions between accident period and development period terms.

The plot above shows a random scattering, which indicates an adequate fit.

P-P Plot

A P-P plot compares two probability distributions by plotting their cumulative distribution functions (cdfs) against one another. The distributions are denoted F and G , and are each evaluated at a common value u .

For a given value of u , the point $(F(u), G(u))$ is plotted. Since a cdf gives the probability that a random variable is less than or equal to u , each point represents the cumulative probabilities assigned to u under the two distributions being compared.

For our purposes, the two distributions are as follows:

- **Theoretical Distribution**
 - For each observed cell Y_{kj} in the triangle, compute its percentile u_{kj} under the fitted ODP distribution with mean $\hat{Y}_{kj}$ and variance $\hat{\phi}\hat{Y}_{kj}$
- **Empirical Distribution**
 - Sort the u_{kj} values in ascending order
 - Let $\hat{u}_{kj}$ be the proportion of data points $\leq u_{kj}$

If the model provides a good fit to the data, the two cdfs will be similar, and the plotted points will lie close to the reference line $y = x$.

A related diagnostic is the **quantile-quantile (Q-Q) plot**, which compares two distributions by plotting their quantiles against each other. Quantiles are obtained by applying the inverse of each cdf. For example, suppose $F(3,398) = 0.25$. Then, $F^{-1}(0.25) = 3,398$. In other words, 3,398 is the quantile associated with the 25th percentile. Whereas a P-P plot compares cumulative probabilities, a Q-Q plot compares the values associated with corresponding percentiles.

Histogram of Residuals

As discussed earlier in the monograph, a histogram of *standardized deviance residuals* should be approximately normally distributed with mean 0 and standard deviation 1. Material departures from this pattern may indicate that the model is inappropriate.

Simulated Data Sets & Fitted Models

The authors use three simulated data sets to illustrate the model diagnostics introduced in this chapter. All three data sets were generated assuming a Poisson distribution. However, different model structures were required depending on the characteristics of the simulated data:

- **Simulated Dataset #1**
 - Accident period effects and development period effects that are common across all periods
 - Constant scale parameter
 - Fitted using a standard cross-classified model
- **Simulated Dataset #2**
 - Accident period effects and development period effects that are common across all periods
 - Scale parameter varies by development period
 - Fitted using a cross-classified model with development-period-specific weights
- **Simulated Dataset #3**
 - Accident period effects that are common across all periods
 - Development period effects that vary by accident period (i.e., accident-development interactions are present)

- Fitted using an interaction model, consisting of a modified cross-classified model that allows accident period and development period effects to interact

Analysis of the Goodness-of-Fit

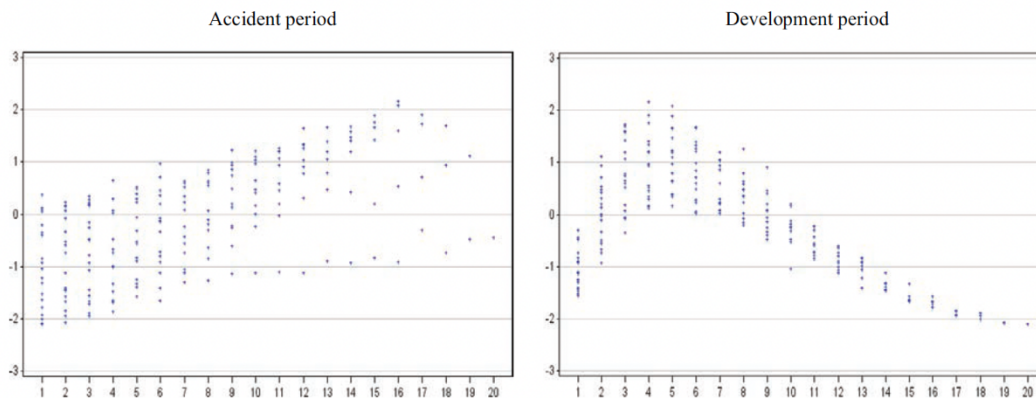
In general, analyzing goodness-of-fit means that we examine the data to ensure all significant drivers of the target value have been identified.

Applying this concept to claims reserving, it means that we review diagnostics by accident, development, and calendar period to determine if there are any un-modeled trends in the data. We do this by examining the following plots:

- Residual Plots
 - Actual & Expected Comparison Plots
 - Actual & Expected Ratio Plots
-

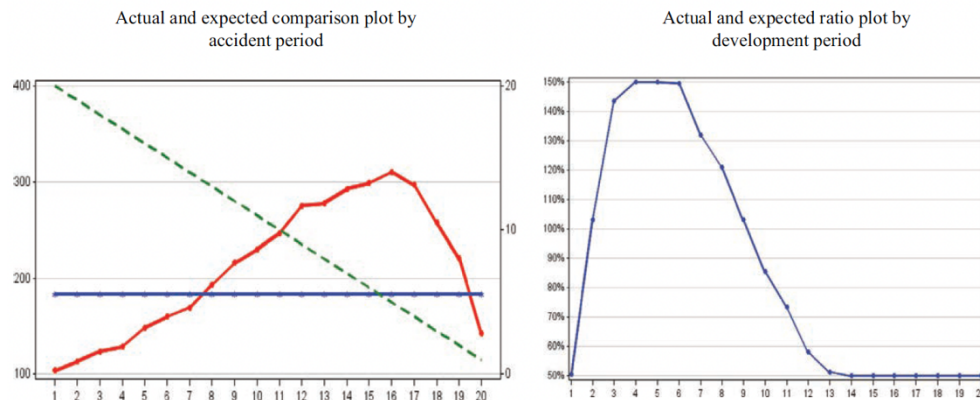
Example: Analyzing Goodness-of-Fit Using Plots

An actuary fits a mean model (i.e., intercept only) to Simulated Dataset #1. The standardized deviance residual plots by accident period and development period are as follows:



For residual plots, we desired a random scattering around zero. In this case, there are clear trends in both plots above. This is because a mean model does not include accident period or development period effects.

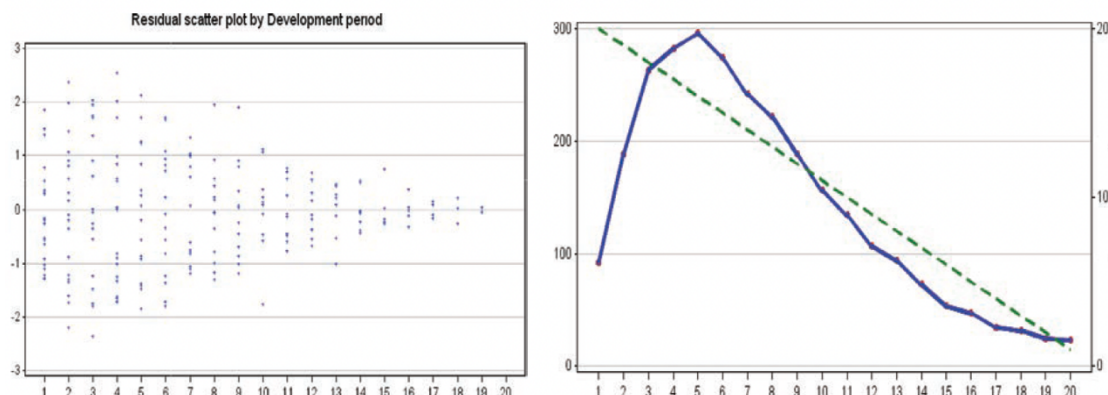
The actual and expected comparison plot and actual and expected ratio plot are as follows:



The two plots above tell the same story:

- **Comparison Plot**
 - The red line represents actual values and the blue line represents expected values
 - The two lines are clearly different, indicating that the model does not adequately capture accident period trends
- **Ratio Plot**
 - There is a clear non-random deviation from 1.0 across development periods, indicating that the model does not adequately capture development period trends

Next, the actuary fits a development model (i.e., development period effects only) to Simulated Dataset #1. We review the residual plot and actual and expected comparison plot by development period:



In this case, the plots tell slightly different stories:

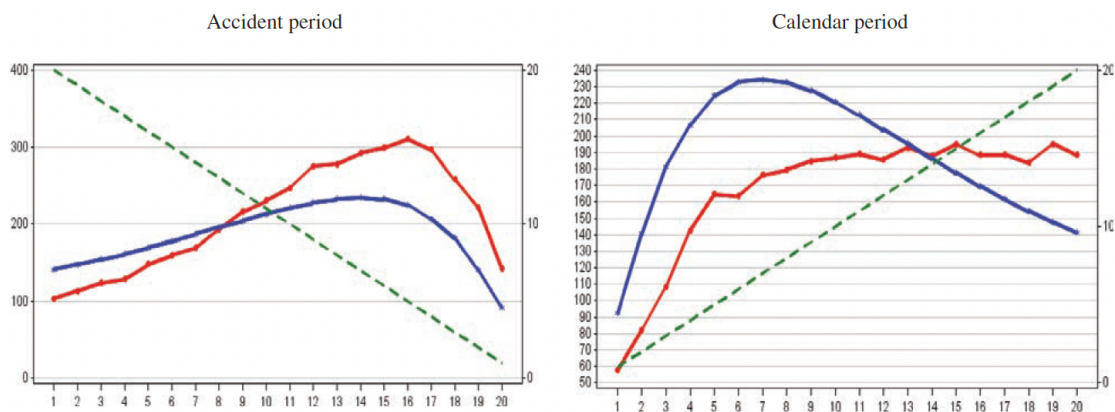
- **Comparison Plot**

- The two lines are identical (the blue sits on top of the red line), indicating that the model adequately captures development period trends

- **Residual Plot**

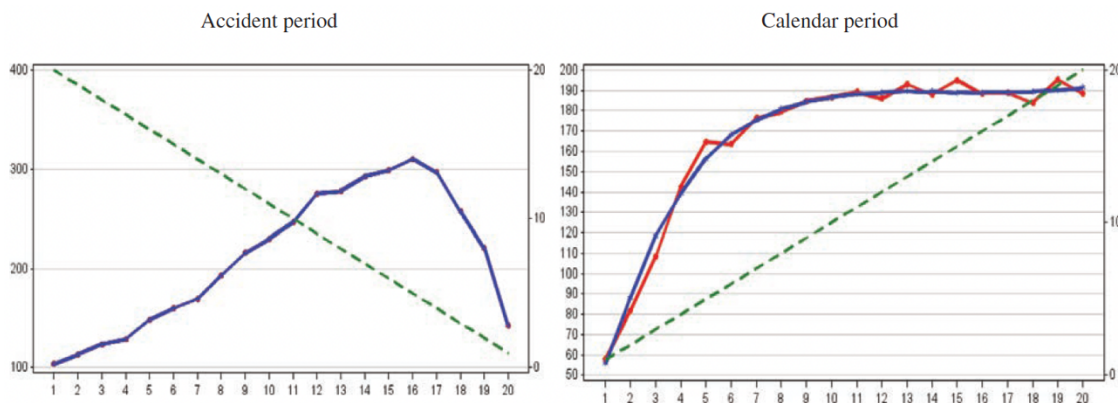
- There is no upward or downward trend in the residuals, indicating that the model adequately captures development period trends
- There is a clear fan shape in the residuals, indicating heteroscedasticity

The plots above illustrate a limitation of the actual and expected comparison plot. Since the model includes development period effects, the red and blue lines will always be identical when plotted by development period (for the ODP model, the actual and expected marginal sums associated with each model parameter are the same). As a result, the plot provides little information beyond confirming that development period effects were included in the model. The residual plot, however, tells a different story. Although the development period trend has been removed, there is still structure remaining in the residuals. To investigate further, we review the actual and expected comparison plots by accident period and calendar period:



As expected, in each plot, the red and blue lines differ markedly. This makes sense because the model only includes development period effects.

Finally, the actuary fits a full cross-classified model to the data, including accident period and development period effects:



Now, the comparison plots by accident period and calendar period look strong. The close agreement between the actual and expected values indicates that the model adequately fits the data. In addition, the calendar period plot shows a good fit despite the fact that the model does not include calendar period effects. This suggests that there are no material calendar period effects present in the simulated data.

Identifying Interactions

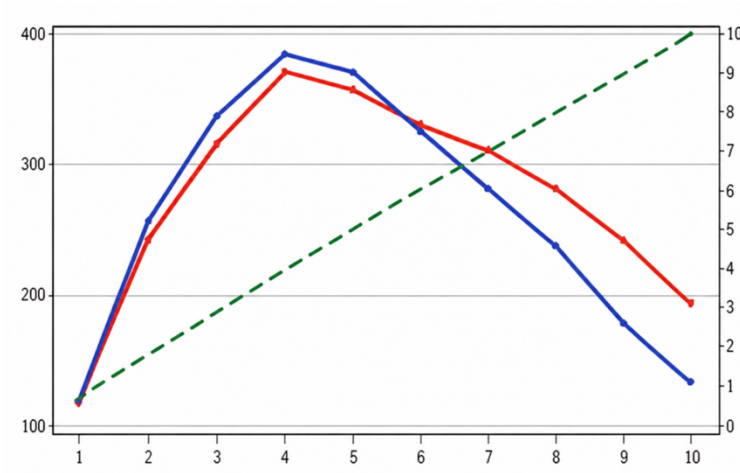
The same plots used to analyze goodness-of-fit can also be used to identify interactions. In the prior example, we fit models to Simulated Dataset #1. To illustrate the presence of interactions, we now fit models to Simulated Dataset #3. Recall that this dataset assumes development period effects vary by accident period.

Example: Identifying Interactions

Suppose an actuary fits a standard cross-classified model with accident period and development period effects to Simulated Dataset #3. The actual and expected comparison plots are not useful when viewed by accident period or development period because the marginal sums will always match between actual and expected for effects explicitly included in the model. However, since

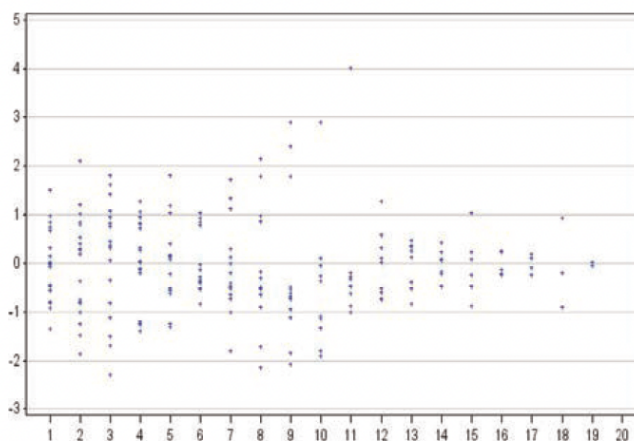
calendar period effects are not included in the model, the actual and expected comparison plot by calendar period can provide useful insights.

The actual and expected comparison plot is as follows:



The plot above shows a clear deviation between the two lines, particularly in the later calendar periods.

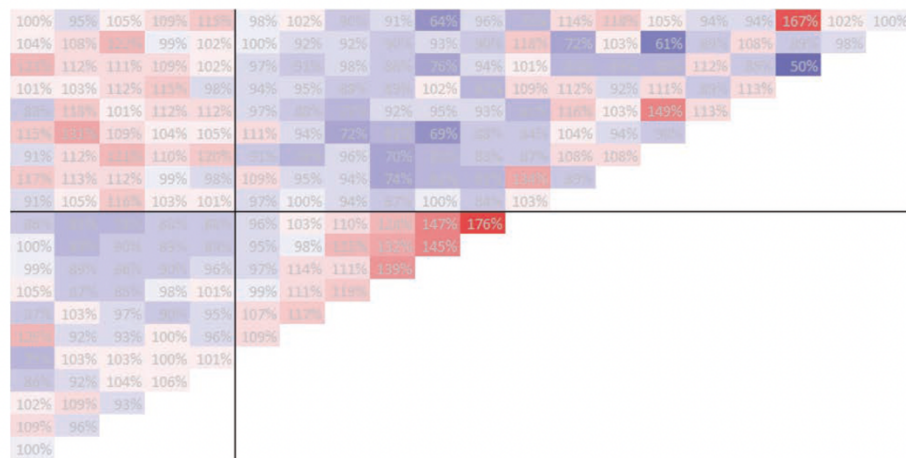
A review of the residual plot by development period is also useful:



The plot exhibits a clear non-random pattern, indicating an inadequate model fit. At this point, we have verified that there is an issue with model fit. Since accident period and development

period effects are included in the model, we conclude that the model is either missing a calendar period effect or an interaction between accident period and development period (or both).

To determine if it's a missing interaction, we can utilize the following actual and expected ratio 2-D heat map:



As expected, the distribution of actual/expected ratios is non-random, with clusters of ratios above 100% and below 100%. However, the clusters tend to be confined to specific accident period and development period combinations rather than extending along entire calendar period diagonals. For example, the latest diagonal contains both red and blue clusters. This suggests a missing interaction rather than missing calendar period effects.

Analysis of the Distribution Assumptions

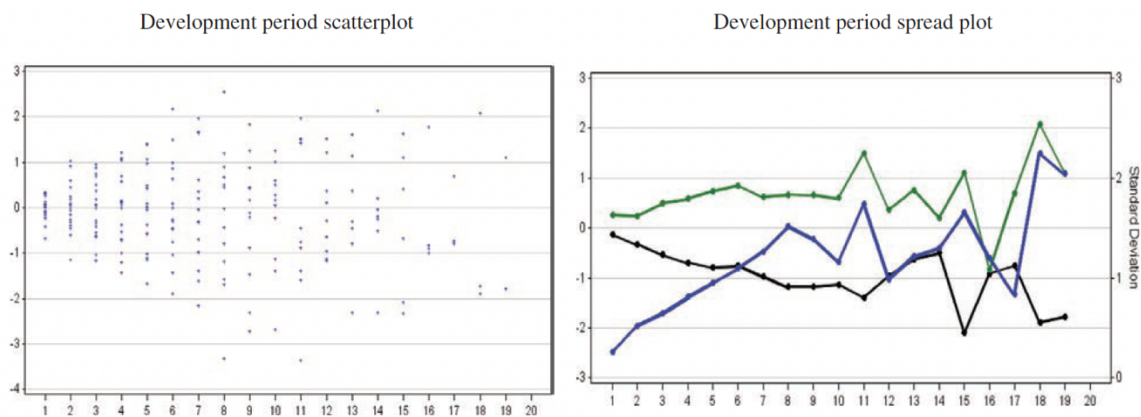
Once model goodness-of-fit has been addressed, we should complete a final review of the distributional assumptions. Since the model structure has already been refined through goodness-of-fit diagnostics, any remaining issues are more likely attributable to the distributional assumptions. The following plots are useful for reviewing distributional assumptions:

- Residual plots
- P-P plots

Example: Reviewing Distributional Assumptions

An actuary fits a standard cross-classified model with a constant scale parameter to Simulated Dataset #2.

The actuary produces the following residual plots by development period:



The scatterplot shows a clear fan shape, indicating heteroscedasticity. This is further supported by the spread plot, which shows the standard deviation of the residuals (the blue line) increasing as development period increases.

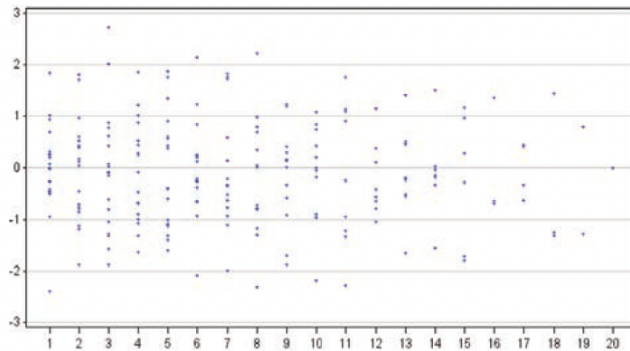
Since the model structure has already been addressed, the heteroscedasticity is unlikely to be caused by missing effects. Instead, it suggests an issue with the distributional assumptions.

To address the problem, the actuary refits the model with a non-constant scale parameter. Recall that the scale parameter for each cell in the triangle is defined as follows:

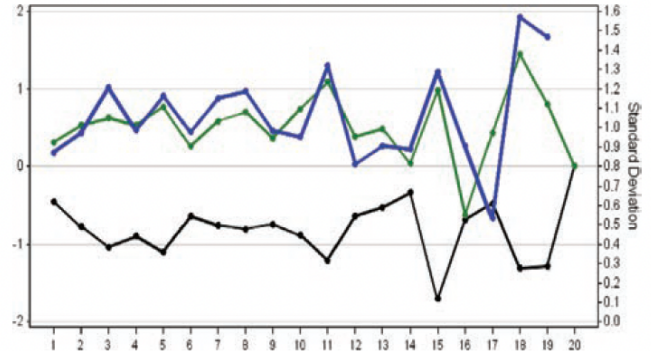
$$\phi_{kj} = \frac{\phi}{w_{kj}}$$

where w_{kj} is the weight. The actuary varies the weight by development period and refits the model. The residual plots are as follows:

Scatterplot by development year

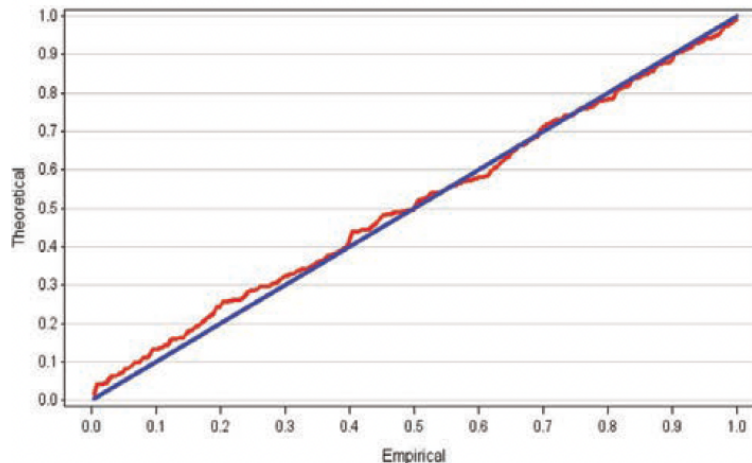


Spread plot by development year



The plots above show a clear improvement over the original model. The scatterplot no longer displays a fan shape, and the spread plot shows the standard deviation fluctuating around 1.

A P-P plot for the updated model is as follows:



The theoretical distribution (blue line) and empirical distribution (red line) appear very similar, further supporting the conclusions drawn from the residual plots.

Note that if revising the weights does not adequately correct the model, the actuary should consider whether a different distribution (e.g., gamma) may be more appropriate for the data.

Regarding the link function, there is no formal statistical test for determining whether the assumed link function is appropriate. Instead, the link function should be selected based on the application. For example, if a multiplicative model structure is desired, a log link function should be used. If an additive model structure is desired, an identity link function should be used.

Model Validation for Real Data

In this section, the authors fit a standard cross-classified model with accident period and development period effects to real data rather than a simulated dataset. The model validation process is identical to that used in the prior examples.

Initial Check of Distributional Assumptions

When applied to the real data, an initial inspection of the P-P plot and residual plots revealed the following:

- The P-P plot (Figure 6-18) showed some deviation from the line $y = x$, but overall provided reasonable support for the Poisson distribution
- The residual scatterplot and spread plot by development period (Figure 6-19) showed greater dispersion at ages 1 and 2, suggesting that a constant scale parameter may not be appropriate
- The residual scatterplot by calendar period exhibited a clear non-random pattern, indicating potential goodness-of-fit issues that warranted further investigation

Goodness-of-Fit

To investigate these fit issues, the authors reviewed an actual and expected comparison plot by calendar year and an actual and expected ratio heat map.

While the comparison plot by calendar period indicated an adequate fit, the heat map revealed evidence of missing interactions:

Accident	Development									
	0	1	2	3	4	5	6	7	8	9
1	98%	100%	100%	104%	113%	87%	96%	92%	100%	100%
2	99%	99%	106%	103%	95%	95%	99%	102%	100%	
3	96%	108%	107%	91%	90%	102%	92%	104%		
4	97%	103%	96%	97%	103%	111%	111%			
5	95%	107%	100%	100%	97%	100%				
6	98%	105%	93%	101%	104%					
7	109%	91%	95%	104%						
8	106%	90%	105%							
9	103%	97%								
10	100%									

As we see in the heat map above, early AYs tended to be overestimated at age 0 and underestimated at age 1, while more recent AYs exhibited the opposite pattern. These clusters of systematic deviation across specific accident period/development period combinations suggest the presence of missing interaction effects.

This example illustrates why heat maps should always be reviewed when validating reserving models. Although aggregate diagnostics may indicate an acceptable fit, the heat map can reveal localized patterns that point to missing interactions or other structural deficiencies.

The authors concluded that the model should be refit with interaction terms. Once refit, the modeler should run a final check of distributional assumptions and adjust as needed.

I. Introduction

In recent years, stochastic reserving models have become increasingly popular. One aspect of reserving that has been missing from the literature on stochastic reserving is the idea of adjusting the results based on actuarial judgment or other expert knowledge.

Examples of situations where results may need to be adjusted include:

- A change in the payment pattern due to a change in company policy
- The legislature has enacted benefit limitations that restrict the potential for loss development

The problem with adjusting model results based on expert opinion is that it disrupts the assumptions underlying the stochastic model:

- Suppose you decide to adjust a residual that appears out of line before applying the bootstrap procedure
- At this point, the assumptions underlying the stochastic model are violated and calculated prediction errors may no longer be valid

The **Bayesian approach** provides a way to incorporate expert knowledge into the reserving process.

Verrall concentrates on two areas where expert knowledge is applied:

- 1) The Bornhuetter/Ferguson (BF) method
- 2) The insertion of prior knowledge about individual development factors in the chain-ladder method

Bayesian models have **two important properties**:

- 1) Can incorporate expert knowledge
- 2) Can be easily implemented

The ease of implementation is due to developments in Bayesian methodology based on “Markov chain Monte Carlo” (MCMC) methods:

- MCMC methods simulate the posterior distribution by breaking the process down into a number of simulations that are easy to carry out
- A common problem with Bayesian methods is that it can be difficult to derive the posterior distribution, which may be multidimensional
- MCMC methods use the conditional distribution of each parameter, given all the others, making the simulation a univariate distribution and much easier to work with
- Considering each parameter in turn creates a Markov chain. Since the method is simulation-based, we get the name, “Markov chain Monte Carlo”

II. Notation & Basic Methods

Assume we have a loss triangle consisting of the set of incremental claims data C_{ij} , where i refers to the row (the accident year) and j refers to the column (the age in years). Note that $i = 1, \dots, n$ and $j = 1, \dots, n - i + 1$.

Cumulative claims are then defined as:

$$D_{ij} = \sum_{k=1}^j C_{ik}$$

Development factors based on the standard chain-ladder method can be estimated as follows:

$$\hat{\lambda}_j = \frac{\sum_{i=1}^{n-j+1} D_{ij}}{\sum_{i=1}^{n-j+1} D_{i,j-1}}$$

The **chain-ladder (CL) method** estimates the outstanding losses as follows:

$$D_{i,n-i+1}(\hat{\lambda}_{n-i+2}\hat{\lambda}_{n-i+3} \dots \hat{\lambda}_n - 1)$$

The **BF method** incorporates knowledge about the level of each row by replacing the CL estimate of outstanding losses with the following:

$$M_i \frac{1}{\hat{\lambda}_{n-i+2}\hat{\lambda}_{n-i+3} \dots \hat{\lambda}_n} (\hat{\lambda}_{n-i+2}\hat{\lambda}_{n-i+3} \dots \hat{\lambda}_n - 1)$$

where M_i denotes a value for the ultimate losses in accident year i , obtained using expert knowledge about the losses.

The difference between the two methods is that the BF method uses an external estimate for the level of each row, while the CL method uses the data in each row.

III. Stochastic Models for the Chain-Ladder Technique

This section highlights several stochastic versions of the CL method.

Mack's Model

As we saw in the Mack Chain-Ladder paper, Mack's model specifies only the first two moments for the cumulative losses:

$$\begin{aligned} E[D_{i,j}] &= \lambda_j D_{i,j-1} \\ \text{Var}(D_{i,j}) &= \sigma_j^2 D_{i,j-1} \end{aligned}$$

An **advantage** of this method is that the parameter estimates and prediction errors can be obtained using a spreadsheet (i.e., it's simple).

Disadvantages include the following:

- There is no predictive distribution
- Additional parameters must be estimated in order to calculate the variance

Over-Dispersed Poisson (ODP) Model

The ODP model is a model for *incremental losses*. Using a generalized linear model (GLM) approach, we can apply a log link function to make it easier to calculate reserve estimates:

$$\ln(m_{ij}) = c + \alpha_i + \beta_j$$

where m_{ij} represents the mean incremental loss for AY i and age j .

An **alternative** way to specify the ODP model is as follows:

$$\begin{aligned} E[C_{ij}] &= x_i y_j \\ \text{Var}(C_{ij}) &= \varphi x_i y_j \end{aligned}$$

where x_i represents the expected ultimate losses in each accident year, y_j represents the proportion of ultimate losses that have emerged in each development year, and φ represents the dispersion factor.

Advantages of the ODP model are as follows:

- It doesn't necessarily break down if there are some negative incremental values (only requires the column sums to be positive)
- It gives the same reserve estimate as the CL method
- It's more stable than the log-normal model of Kremer

Disadvantages of the ODP model are as follows:

- The connection to the CL method is not immediately apparent

Over-Dispersed Negative Binomial (ODNB) Model

The ODNB model can be applied to both incremental and cumulative claims.

For incremental claims, the mean and variance are as follows:

$$\begin{aligned} E[C_{ij}] &= (\lambda_j - 1)D_{i,j-1} \\ \text{Var}(C_{ij}) &= \phi \lambda_j (\lambda_j - 1) D_{i,j-1} \end{aligned}$$

It's clear from this formulation that the sum of incremental claims down columns must be positive. Otherwise, $\lambda_j < 1$ and the variance would be negative.

For cumulative claims, the mean and variance are as follows:

$$\begin{aligned} E[D_{ij}] &= \lambda_j D_{i,j-1} \\ \text{Var}(D_{i,j}) &= \phi \lambda_j (\lambda_j - 1) D_{i,j-1} \end{aligned}$$

Advantages of the ODNB model are as follows:

- It doesn't necessarily break down if there are some negative incremental values (only requires the column sums to be positive)
- It gives the same reserve estimate and has the same form as the chain ladder method

The **disadvantage** is that it cannot handle negative development (the column sums must be positive).

Normal Approximation to the Negative Binomial Model

This model replaces the negative binomial with a normal distribution whose mean is unchanged, but whose variance is altered to allow for negative incremental claims.

For incremental claims, the mean and variance are as follows:

$$\begin{aligned} E[C_{ij}] &= (\lambda_j - 1)D_{i,j-1} \\ \text{Var}(C_{ij}) &= \phi_j D_{i,j-1} \end{aligned}$$

For cumulative claims, the mean and variance are as follows:

$$\begin{aligned} E[D_{ij}] &= \lambda_j D_{i,j-1} \\ \text{Var}(D_{i,j}) &= \varphi_j D_{i,j-1} \end{aligned}$$

In both variance formulas above, the constant dispersion parameter φ has been replaced by a column-specific parameter φ_j . This is **disadvantageous** since additional parameters must be estimated in order to calculate the variance (similar to Mack's model).

The **advantage** of this model is that it can handle negative development (i.e., a negative column sum of incremental values) because the variance is no longer tied to λ_j .

Mean Squared Error of Prediction (MSEP)

For each of the stochastic methods above, the MSEP (i.e., prediction variance) can be obtained, allowing for the calculation of prediction intervals. After a bit of derivation, the MSEP can be written as follows:

$$\text{MSEP (prediction variance)} = \text{process variance} + \text{estimation variance}$$

The prediction error is the square root of the MSEP. It is known as the root mean squared error of prediction (RMSEP)

$$\text{RMSEP (prediction error)} = \sqrt{\text{process variance} + \text{estimation variance}}$$

Difference Between Standard Error & Prediction Error

The prediction error accounts for uncertainty in the parameter estimation (estimation variance) and the variability of future losses (process variance). Standard error only accounts for estimation variance.

- prediction error = $\sqrt{\text{process variance} + \text{estimation variance}}$
- standard error = $\sqrt{\text{estimation variance}}$

The difficulty in calculating the prediction error highlights a **few advantages** of Bayesian methods:

- The full predictive distribution can be found using simulation methods
- The RMSEP can be obtained directly by calculating the standard deviation of the distribution

IV. Incorporating Expert Opinion About the Development Factors

This section covers two cases of the actuary intervening in the estimation of the development factors for the chain-ladder method:

- 1) A development factor is changed in some rows due to external information
- 2) Development factors are based on a five-year volume weighted average rather than all of the available data in the triangle

Suppose that our model for the data is the ODNB model described earlier, modified to allow separate LDFs for each AY and development period:

$$E[C_{ij}] = (\lambda_{i,j} - 1)D_{i,j-1}$$

$$Var(C_{ij}) = \phi \lambda_{i,j} (\lambda_{i,j} - 1) D_{i,j-1}$$

Case 1: Development Factor is Changed in Some Rows Due to External Information

Given a 10 x 10 loss triangle, suppose that there is information that implies the second development factor (from column 2 to column 3) should be 1.5 for rows 8, 9, and 10. Assume that all other parameters remain unchanged.

To specify the development factors, do the following:

- $\lambda_{i,j} = \lambda_j$ for $i = 1, 2, \dots, n - j + 1$ and $j = 2, 4, 5, \dots, n$
- $\lambda_{i,3} = \lambda_3$ for $i = 1, 2, \dots, 7$
- $\lambda_{8,3} = \lambda_{9,3} = \lambda_{10,3}$

The means and variances of the prior distributions of the development factors reflect the expert opinion:

- $\lambda_{8,3}$ has mean 1.5 and variance W , where W depends on the strength of the opinion
- λ_j have prior distributions with large variances (known as “vague” priors)
- If W is large, the posterior mean of the development factor will be pulled closer to the CL development factor and the reserve will closely resemble the CL reserve
- If W is small, the posterior mean of the development factor will be pulled closer to the prior mean and the reserve will move away from the CL reserve
- The remaining development factors will have posterior means equal to those implied by the CL model due to their vague prior distributions with large variances. When vague priors are used, the posterior distribution will be completely driven by the data

Case 2: Development Factors are Based on a Five-Year Weighted Average

In this case, we are not using expert opinion to set a specific prior mean for the development factors. All of the development factors will be given vague prior distributions so that the posterior distributions will be informed by the data.

However, we are limiting the data used in the estimation process, which means the final development factors will not resemble CL all-year volume-weighted averages. Instead, they will resemble CL five-year volume-weighted averages.

To specify the development factors, do the following:

- $\lambda_{i,j} = \lambda_j$ for the most recent n calendar year diagonals
- $\lambda_{i,j} = \lambda_j^*$ for all diagonals *prior* to the latest n calendar year diagonals

We should specify prior distributions with large variances for λ_j and λ_j^* to ensure that they estimated from the data.

Note that in the later development years where there are less than five rows, the columns have just one development factor.

V. A Bayesian Model for the BF Method

Assume that the data is described by the ODP model.

Since the BF method assumes that there is expert opinion about the level of each row, we must specify a prior distribution for x_i . Verrall uses a gamma distribution for the prior:

$$x_i \sim \text{Gamma}(\alpha_i, \beta_i)$$

where the mean and variance are as follows:

$$\begin{aligned} E[x_i] &= \frac{\alpha_i}{\beta_i} = M_i \\ \text{Var}(x_i) &= \frac{\alpha_i}{\beta_i^2} = \frac{M_i}{\beta_i} \end{aligned}$$

For a given choice of M_i , the variance can be altered by changing the value β_i . Choosing a larger value of β_i implies we are more certain about the value of M_i .

Once the row parameters have been estimated using a standard prior/posterior analysis, it can be shown that the mean of the incremental claims is as follows:

$$E[C_{ij}] = Z_{ij}(\lambda_j - 1)D_{i,j-1} + (1 - Z_{ij})(\lambda_j - 1)M_i \frac{1}{\lambda_j \lambda_{j+1} \dots \lambda_n}$$

where $Z_{ij} = \frac{\sum_{k=1}^{j-1} y_k}{\beta_i \varphi + \sum_{k=1}^{j-1} y_k}$. This is a *credibility formula*, where the Z_{ij} control the trade-off between the prior mean (BF) and the data (CL).

Example: Bayesian Model for the BF Method

An actuary is modeling incremental losses using a Bayesian model for the BF method. She assumes that incremental paid losses follow an ODP distribution with $\varphi = 15$. Given the following loss information as of December 31, 2023:

<u>Incremental Paid Losses (in \$000) as of XX Months:</u>				
AY	12 mo.	24 mo.	36 mo.	48 mo.
2020	2,000	1,000	600	200
2021	2,400	1,250	400	
2022	2,200	1,400		
2023	2,500			

The actuary sets gamma priors for the level of each AY. Here is a subset of the prior parameters:

- $\alpha_3 = 46$
- $\beta_3 = 0.01$

First, let's create the cumulative loss triangle and calculate the age-to-age factors:

<u>Cumulative Paid Losses (in \$000) as of XX Months:</u>				
AY	12 mo.	24 mo.	36 mo.	48 mo.
2020	2,000	3,000	3,600	3,800
2021	2,400	3,650	4,050	
2022	2,200	3,600		
2023	2,500			
Age-to-Age		1.553	1.150	1.056

Second, let's calculate $Z_{3,3}$:

- Recall that $Z_{ij} = \frac{\sum_{k=1}^{j-1} y_k}{\beta_i \varphi + \sum_{k=1}^{j-1} y_k}$
- The numerator, $\sum_{k=1}^{j-1} y_k$, is simply the cumulative percentage of losses emerged for AY i .
For AY 3 (i.e., 2022), $\sum_{k=1}^{j-1} y_k = \frac{1}{1.150(1.056)} = 0.824$
- $Z_{3,3} = \frac{0.824}{0.01(15) + 0.824} = 0.846$

Third, let's calculate $E[C_{3,3}]$:

- Recall that $E[C_{ij}] = Z_{ij}(\lambda_j - 1)D_{i,j-1} + (1 - Z_{ij})(\lambda_j - 1)M_i \frac{1}{\lambda_j \lambda_{j+1} \dots \lambda_n}$

- $M_3 = \frac{\alpha_3}{\beta_3} = \frac{46}{0.01} = 4,600$
- $E[C_{3,3}] = 0.846(1.150 - 1)(3,600) + (1 - 0.846)(1.150 - 1)(4,600)(0.824) = 546$
- Thus, the expected incremental losses to emerge between 24 and 36 months for AY 2022 are equal to \$546,000

Impact of β on the Variance of the Prior Distribution

As described earlier, β_i can be used to alter the variance of x_i :

- If we choose prior distributions with large variances (small β s), we have low confidence (no prior knowledge) in our prior mean and the result will be close to the CL method
- If we choose prior distributions with small variances (large β s), we have high confidence (prior knowledge) in our prior mean and the result will be close to the BF method

Fully Stochastic Bayesian Model for the BF Model

At this point in the paper, we have only discussed the estimation of the row parameters (i.e., x_i). To account for all variability, we also need to estimate the column parameters (i.e., y_j). There are two options:

- 1) Use plug-in estimates from the traditional CL method with no variability (this is what was done in the prior example)
- 2) Define prior distributions for the column parameters, and estimate the column parameters first, before applying prior distributions for the row parameters and estimating these

Although both of these options will reproduce the BF method, the second option is the preferred option since it allows us to consider the fact that the column parameters have been estimated when calculating the prediction errors, predictive distribution, etc. In other words, the second option provides a fully stochastic version of the BF method.

To implement the fully stochastic version of the BF method, Verrall defines improper prior distributions for the column parameters. The improper priors serve as non-informative priors since we don't have any information regarding the column parameters.

Once the column parameters are estimated using a standard prior/posterior analysis, we obtain an ODNB model with the following mean:

$$E[C_{ij}] = (\gamma_i - 1) \sum_{m=1}^{i-1} C_{m,j}$$

Notice that there are no x_i s in the formula above. Instead, the model has been re-parameterized in terms of γ_i . For $i > 1$, the γ_i s are defined as follows:

$$\gamma_i = 1 + \frac{\text{BF Reserve for AY } i}{\text{Inc. Losses for Prior AYs in Future Dev. Periods for AY } i}$$

We assume that $\gamma_1 = 1$.

Example: Fully Stochastic Bayesian Model for the BF Method

An actuary is modeling incremental losses using a Bayesian model for the BF method. He sets prior distributions for the row and column parameters. The columns priors are improper. Given the following loss information as of December 31, 2023:

AY	Incremental Paid Losses (in \$000) as of XX Months:			
	12 mo.	24 mo.	36 mo.	48 mo.
2020	2,000	1,000	600	200
2021	2,400	1,250	400	
2022	2,200	1,400		
2023	2,500			

- $x_2 = \$4,500,000$
- $x_3 = \$4,600,000$

- $x_4 = \$4,800,000$

First, let's calculate the cumulative percentage emerged using the standard CL method:

AY	Cumulative Paid Losses (in \$000) as of XX Months:			
	12 mo.	24 mo.	36 mo.	48 mo.
2020	2,000	3,000	3,600	3,800
2021	2,400	3,650	4,050	
2022	2,200	3,600		
2023	2,500			
Age-to-Age		1.553	1.150	1.056
Age-to-Ult.		1.886	1.214	1.056
% Emerged		53.0%	82.4%	94.7%

Second, let's calculate the γ_i s and expected future incremental losses for each AY i :

- Recall that $\gamma_i = 1 + \frac{\text{BF Reserve for AY } i}{\text{Inc. Losses for Prior AYs in Future Dev. Periods for AY } i}$
- Recall that $E[C_{ij}] = (\gamma_i - 1) \sum_{m=1}^{i-1} C_{m,j}$
- $\gamma_1 = 1$
- $\gamma_2 = 1 + \frac{4,500(1-0.947)}{200} = 2.184$
 - $E[C_{2,4}] = (2.184 - 1)(200) = 237$
- $\gamma_3 = 1 + \frac{4,600(1-0.824)}{600+400+200+237} = 1.565$
 - $E[C_{3,3}] = (1.565 - 1)(600 + 400) = 565$
 - $E[C_{3,4}] = (1.565 - 1)(200 + 237) = 247$
- $\gamma_4 = 1 + \frac{4,800(1-0.530)}{1,000+1,250+1,400+600+400+565+200+237+247} = 1.382$
 - $E[C_{4,2}] = (1.382 - 1)(1,000 + 1,250 + 1,400) = 1,395$
 - $E[C_{4,3}] = (1.382 - 1)(600 + 400 + 565) = 598$
 - $E[C_{4,4}] = (1.382 - 1)(200 + 237 + 247) = 261$

As we can see from above, calculating the γ_i s and expected future incremental losses is an iterative process. Also, you can think of calculating the expected future incremental losses as running a reverse CL method, where the CL factors are the γ_i s.

The complete triangle under the fully stochastic BF model is as follows:

AY	Incremental Paid Losses (in \$000) as of XX Months:			
	12 mo.	24 mo.	36 mo.	48 mo.
2020	2,000	1,000	600	200
2021	2,400	1,250	400	237
2022	2,200	1,400	565	247
2023	2,500	1,395	598	261

If we sum up the expected future incremental losses for each AY, we obtain the BF reserves.

Notice that we do not have a nice analytical credibility type formula for the mean when running the fully stochastic version of the BF method. That only happens when we use plug-in estimates from the traditional CL method.

The steps for defining a fully stochastic version of the BF method are as follows:

- Estimate the column parameters
 - Since we are using improper gamma distributions, their estimates will be those implied by the CL method (remember, improper means “no information”)
- Incorporate prior information into the distributions for the parameters x_i
- Use x_i to determine γ_i
- Calculate the expected future incremental losses using the γ_i s
- The sum of the expected future incremental losses will be the BF reserve

VI. Implementation

In this section, we will take a look at a couple of examples illustrating how varying degrees of expert opinion affect model results.

Example: Overriding a Link Ratio

An actuary is comparing two Bayesian models with the deterministic chain-ladder method. In each model, the actuary is overriding the 12-24 LDF with a prior mean of 1.50. The CL LDFs and model results are as follows:

LDFs			
12-24	24-36	36-48	48-60
1.40	1.20	1.10	1.05

AY	Model Results (in \$000)		
	CL Reserve	Model 1 Reserve	Model 2 Reserve
2014	1,800	1,850	1,900

In this case, the Model 1 results are closer to the CL results than the Model 2 results. This implies that the variance of the 12-24 LDF in Model 1 is large (i.e., low confidence in the prior mean of 1.50).

On the other hand, the variance of the 12-24 LDF in Model 2 is small (i.e., high confidence in the prior mean of 1.50). This high confidence drove the estimate of the 12-24 LDF closer to 1.50, leading to an increase in the ultimate losses and reserves.

Example: BF Model Implementation

An actuary is implementing a Bayesian model for incremental losses where prior information is known about the ultimate losses for each accident year.

Incremental losses follow an ODP distribution with mean $x_i y_j$. The model results, as well as the CL and BF reserves, are given below for accident year 2014:

AY	<u>Model Results (in \$000)</u>		
	CL Reserve	Bayesian Reserve	BF Reserve
2014	1,500	1,550	1,800

In this case, the model results (i.e., the Bayesian reserve) are closer to the CL reserves than the BF reserves. This implies that the variance of the prior distribution for the AY 2014 row parameter is large (i.e., low confidence in the prior estimate of the row parameter). If the variance of the row parameter was small, the Bayesian reserve would be closer to the BF reserve.

I. Introduction

This paper provides a framework for assessing risk margins for insurance liabilities. Risk margins are intended to provide an extra bit of cushion above the central estimate (i.e., the mean).

Current Approaches to Assessing Risk Margins

In general, insurers use the following methodology when assessing risk margins:

- **Coefficients of variation (CoVs)** are determined for individual valuation portfolios or groupings of portfolios, where the groupings are fairly homogeneous
- A **correlation matrix** is populated with correlation coefficients that reflect expected correlations between valuation portfolios or groupings of portfolios
- CoVs and correlation coefficients are determined separately for outstanding claim liabilities (OSC) and premium liabilities (PL). Assumptions are made about the correlation *between these two types of liabilities* as well
- A **distribution** is selected and combined with the CoVs and correlation coefficients to determine the **aggregate risk margin** at a particular probability of adequacy

Determining COVs

The authors mention two approaches for determining COVs:

- 1) *Less sophisticated* methods ignore the individual characteristics of the valuation portfolio being assessed. Instead, they are based on “model” portfolios analyzed in two famous papers on risk margin analysis
- 2) *More sophisticated* methods combine quantitative analysis with qualitative analysis of the sources of uncertainty not captured by quantitative techniques

Populating the Correlation Matrix

The *most common* approach for populating the correlation matrix relies heavily on actuarial judgment. The key risks that are believed to cause correlation between valuation portfolios are categorized as high, medium or low. Each category is assigned a correlation coefficient value.

Although more rigorous, quantitative approaches are used as well, they tend to be the exception due to the amount of time and effort required to provide credible results.

Selecting the Distribution

The *most common* distribution used to determine the aggregate risk margin is the *lognormal distribution*.

The normal distribution is used as well, especially at lower probabilities of adequacy where it can generate a higher risk margin than the heavy-tailed lognormal distribution.

The Bolt-On Approach

When determining risk margins, most actuaries use the “bolt-on” approach, which means separate analyses are used to determine the central estimate and the risk margin. Hence, the risk margin is “bolted on” to the central estimate.

In general, the term “bolt-on” refers to any approach that does not involve a single unified distribution of the entire distribution of possible future claim costs.

II. The Proposed Framework

Before diving into this section, let’s define a few terms:

- Claims portfolio – the aggregate portfolio for which the risk margins must be estimated (ex. XYZ Insurance)
- Valuation classes – the portfolios that are considered individually as part of the risk margin analysis (ex. Property vs. Auto)
- Claim group – a group of claims with common risk characteristics

Sources of Uncertainty

At the highest level, there are two sources of uncertainty:

- 1) Systemic risk
- 2) Independent risk

Systemic Risk

Systemic risk represents risks that are common across valuation classes or claim groups. There are **two sources of systemic risk**:

- 1) **Internal systemic risk** – risks internal to the insurance liability valuation/modeling process
 - Examples include model structure, model parameterization and data accuracy
 - This risk is also referred to as **model specification risk**
- 2) **External systemic risk** – risks external to the insurance liability valuation/modeling process
 - An example is future trends in claim cost outcomes (ex. change in inflation) that may cause actual experience to differ from what is expected based on the current environment and trends

Independent Risk

Independent risk represents risks that occur due to the randomness inherent in the insurance process. There are **two sources of independent risk**:

- 1) Random component of **parameter risk** – uncertainty in selecting appropriate parameters due to the randomness of insurance
- 2) Random component of **process risk** – pure effect of randomness due to the insurance process

Quantitative vs. Qualitative Analysis

Quantitative analysis can only reflect uncertainty in historical experience and cannot adequately reflect all possible sources of future uncertainty.

For this reason, traditional quantitative modeling techniques (ex. bootstrapping, stochastic chain-ladder, GLMs, Bayesian methods, etc.) are **best suited** for measuring the following:

- Independent risk
- Past episodes of external systemic risk

To measure internal systemic risk or external systemic risk that differs from historical experience, traditional quantitative techniques should be combined with other quantitative and qualitative analyses.

Preparing the Claims Portfolio for Analysis

As defined earlier, the claims portfolio represents the aggregate insurance entity. The claims portfolio should be split into valuation classes based on a number of considerations:

- Should the valuation classes be aligned to the valuation portfolios analyzed for central estimate purposes?
 - This may be preferable since it allows the risk margin analysis to be conducted in the context of the central estimate analysis and it allows quantitative and qualitative analysis to be aligned with the central estimate valuation
- It may not be possible to conduct quantitative analysis at the same granularity used for central estimate purposes
 - This may be due to credibility concerns, implementation costs, etc.
 - In these cases, quantitative analysis may be conducted on aggregated valuation classes and allocated down to more granular valuation classes
- In the end, an approach should be used that balances practicality and granularity and retains as much consistency as possible between the central estimate analysis and the risk margin analysis

Once the claims portfolio has been divided into valuation classes, we must determine whether or not specific classes should be divided further into claim groups.

For **example**, consider a home portfolio:

- Development patterns tend to differ significantly between cat and non-cat claims
- In addition, home liability claims tend to behave differently from other home claims
- Hence, it may make sense to analyze these groups separately
- As always, any class divisions should be practical

Analyzing Independent Risk Sources

Although commonly used, stochastic modeling techniques **fail to capture** all sources of uncertainty for the following reasons:

- **Good stochastic** models effectively “fit away” past episodes of external systemic risk. If a model does not fully remove these past risks, consideration must be given to whether we should expect similar risks to continue in the future
 - **If we don’t expect these external risk episodes to continue**, then the model results that “failed to fit them away” are not good representations of the true uncertainty
- Stochastic models do not capture uncertainty arising from internal systemic risk

Analyzing Internal Systemic Risk

As stated earlier, internal systemic risk refers to the risks internal to the modeling process. **Three sources** of internal systemic risk are as follows:

- 1) **Specification error** – the error that arises because the model cannot perfectly model the insurance process
- 2) **Parameter selection error** – the error that arises because the model cannot adequately measure all the predictors of claim cost outcomes and their trends
- 3) **Data error** – the error that arises due to poor or unavailable data that’s needed for the reserve analysis

On the surface, parameter selection error might seem like a subset of parameter risk, but they represent different sources of uncertainty. **Parameter selection error** refers to the risk that the model does not capture all relevant predictors of claim costs or their trends. In other words, we may have chosen the wrong variables or omitted important ones. **Parameter risk**, on the other hand, is the uncertainty in the estimated values of the parameters that were selected.

The authors recommend using the **balanced scorecard** approach to analyze internal systemic risk:

- A balanced scorecard is developed to objectively assess the model specification against a set of criteria
- For each of the sources of internal systemic risk, risk indicators are developed and scored against the criteria
 - For example, a risk indicator for specification error might be “number of independent models used.” More independent models would lead to a higher score
 - See page 33 in the paper for a full list of potential risk indicators
- The scores are aggregated for each valuation class and mapped to a CoV

Although the goal is to make the balanced scorecard approach as objective as possible, some subjective decisions must be made. These include the risk indicators, the measurement and scoring criteria, the weight given to each risk indicator, and the CoVs that map to each score. We will see an example of the balanced scorecard approach later on.

Analyzing External Systemic Risk

Since models analyze historical claim experience, they may not fully capture future external systemic risk. This is especially true for valuation classes where future external systemic risk is significantly different from past episodes.

We can identify key potential sources of systemic risk through discussions with business experts as part of the valuation process. These discussions should consider:

- Underwriting and risk selection

- Pricing
- Claims management
- Expense management
- Emerging portfolio trends
- The environment in which the portfolio operates

The main external systemic risks for any valuation class belong to specific **risk categories**. These categories include the following:

- **Economic and social risks** – the uncertainty associated with inflation, social trends, etc.
- **Legislative, political and claim inflation risks** – the uncertainty associated with changes in the political landscape, shifts/trends in the level of claim settlements, etc.
- **Claim management process change risk** – the uncertainty associated with changes in claim reporting, payment, estimation, etc.
- **Expense risk** – the uncertainty associated with the cost of managing the run-off of the insurance liabilities or the cost of maintaining the unexpired risk until the date of loss
- **Event risk** – the uncertainty associated with claim costs arising from events (i.e. cats), either natural or man-made
- **Latent claim risk** – the uncertainty associated with claims that arise from sources that are not currently covered (i.e., asbestos)
- **Recovery risk** – the uncertainty associated with recoveries, either reinsurance or non-reinsurance

CoVs for each external systemic risk category should be selected using a mix of quantitative analysis and qualitative judgment. We will go into more detail on each of these risk categories later on.

For most valuation classes, a *small number of risk categories account* for most of the uncertainty:

- Property classes – event risk dominates the volatility of the premium liabilities

- Long-tailed classes – legislative, political and claim inflation risks dominate the volatility of both the outstanding claim liabilities and the premium liabilities

It's useful to rank each of the risk categories in descending from most impactful on liability uncertainty to least impactful on liability uncertainty. Then, we can spend more time quantifying the more critical categories.

Correlation Effects

Once we have derived CoVs for independent risk, internal systemic risk, and for each source of external systemic risk, we must account for the fact that these sources of risk are not fully correlated within valuation classes OR between valuation classes.

The **paper does not discuss quantitative methods** to assess correlation effects for the following reasons:

- The techniques tend to be very complex and require large amounts of data
- The techniques produce correlations that are heavily influenced by the correlations displayed in the past data. This is a problem for external systemic risk where future correlations may differ from past correlations
- It is difficult to separate the past correlation effects between independent risk and systemic risk OR to identify the pure effect of each past systemic risk
- Internal systemic risk cannot be modeled using standard correlation modeling techniques

The authors provide several **rules of thumb** regarding correlation. We will look at those now.

Independent Risk

Independent risks are uncorrelated with other sources of uncertainty both within and between valuation classes.

Internal Systemic Risks

Internal systemic risks are **uncorrelated** with independent risk and external systemic risk sources.

Internal systemic risks are **correlated** between classes or between outstanding claim and premium liabilities due to the following:

- **Same actuary effect** – causes correlation between classes due to common valuation approaches
- Link between premium liability methodology and outcomes from outstanding claim valuation cause correlation between the two liability types

External Systemic Risks

External systemic risks are **uncorrelated** with independent risk, internal systemic risk, and (in general) with other risk categories.

External systemic risks are **correlated** between classes or between outstanding claim and premium liabilities in the same risk category. In addition, it's possible that correlation may exist between risk categories within or between valuation classes. If this occurs, correlated risk categories can be aggregated into broader categories that are not correlated.

The paper defines five correlation bands:

- Nil – 0% correlation
- Low – 25% correlation
- Medium – 50% correlation
- High – 75% correlation
- Full – 100% correlation

Consolidation of Analysis Into Risk Margin Calculation

Once we have the necessary CoV and correlation inputs, we can aggregate them to produce a risk margin for the assessed liability.

As mentioned earlier, two distributions are commonly used to represent insurance liabilities:

- 1) **Normal** distribution – produces a higher risk margin at lower probabilities of adequacy, irrespective of the size of the consolidated CoV. The risk margin providing an X% probability of reserve adequacy is calculated as follows:

$$\text{Risk Margin} = Z \cdot CoV$$

where Z is the z -value for the X^{th} percentile of the normal distribution.

- 2) **Lognormal** distribution – produces a higher risk margin at higher probabilities of adequacy as long as the consolidated CoV is not too high. For extremely high CoVs, the risk margin can actually reduce as a percentage of the CoV. The risk margin providing an X% probability of reserve adequacy is calculated as follows:

$$\text{Risk Margin} = e^{Z\sigma - \frac{\sigma^2}{2}} - 1$$

where $\sigma^2 = \ln(1 + CoV^2)$

The risk margins above can be converted into dollars by multiplying them by the central estimates for the liabilities in question.

Example: Calculating Risk Margins

Given the following for a company that writes auto and home insurance:

LOB	<u>Proportion of Insurance Liabilities</u>	
	OSC	PL
Auto	30%	40%
Home	15%	15%

LOB	<u>Independent Risk CoVs</u>	
	OSC	PL
Auto	8%	4%
Home	5%	4%

	<u>Internal Systemic Risk CoVs</u>		
	OSC	PL	Total
Auto	6.5%	4.0%	4.7%
Home	3.0%	5.0%	?
Total	4.9%	3.8%	4.1%

	<u>Internal Systemic Risk Correlation Matrix</u>			
	Auto OSC	Auto PL	Home OSC	Home PL
Auto OSC	100%	75%	50%	50%
Auto PL	75%	100%	50%	50%
Home OSC	50%	50%	100%	75%
Home PL	50%	50%	75%	100%

	<u>External Systemic Risk CoVs</u>		
	OSC	PL	Total
Auto	4.0%	6.8%	5.6%
Home	3.4%	15.5%	12.3%
Total	3.0%	8.5%	6.4%

- The z-value for the 90th percentile of the normal distribution is 1.28

First, let's calculate the independent risk CoV for both lines combined:

- Let w_i represent the weight for the i^{th} liability
- Let CoV_i represent the CoV for the i^{th} liability
- Let $i = 1, 2, 3, 4$ represent Auto OSC, Home OSC, Auto PL, and Home PL, respectively
- The total independent risk CoV for both lines combined is as follows:

$$\begin{aligned}
 CoV_{ind} &= \frac{\sqrt{[w_1 \cdot CoV_1]^2 + [w_2 \cdot CoV_2]^2 + [w_3 \cdot CoV_3]^2 + [w_4 \cdot CoV_4]^2}}{w_1 + w_2 + w_3 + w_4} \\
 &= \frac{\sqrt{[0.30(0.08)]^2 + [0.15(0.05)]^2 + [0.40(0.04)]^2 + [0.15(0.04)]^2}}{0.30 + 0.15 + 0.40 + 0.15} \\
 &= 3.0\%
 \end{aligned}$$

The formula above comes from the variance formula for the sum of independent random variables. Each $[w_i \cdot CoV_i]^2$ term represents the variance of the i^{th} liability. If we sum up the variance terms under the square root, we obtain the aggregate standard deviation for independent risk. Then, when we divide by the sum of the weights, we obtain the aggregate CoV for independent risk.

Second, let's calculate the aggregate CoV for internal systemic risk for the Home line:

$$\begin{aligned} CoV_{Home} &= \frac{\sqrt{[w_2 \cdot CoV_2]^2 + [w_4 \cdot CoV_4]^2 + 2 \cdot w_2 \cdot w_4 \cdot CoV_2 \cdot CoV_4 \cdot \rho}}{w_2 + w_4} \\ &= \frac{\sqrt{[0.15(0.03)]^2 + [0.15(0.05)]^2 + 2(0.15)(0.15)(0.03)(0.05)(0.75)}}{0.15 + 0.15} \\ &= 3.8\% \end{aligned}$$

The formula above comes from the variance formula for the sum of dependent random variables. When calculating the variance of the sum of dependent random variables, we must consider the covariance between the variables. This is represented by the $2 \cdot w_i \cdot w_j \cdot CoV_i \cdot CoV_j \cdot \rho$ term. If we had three lines of business, there would be three separate covariance terms (see the Marshall exhibits in the Online Course for an example of this).

Third, let's calculate the total insurance liabilities CoV for all sources of risk combined:

$$\begin{aligned} CoV_{Total} &= \sqrt{(CoV_{ind})^2 + (CoV_{int})^2 + (CoV_{ext})^2} \\ &= \sqrt{0.03^2 + 0.041^2 + 0.064^2} \\ &= 8.2\% \end{aligned}$$

Notice that we simply take the square root of the sum of the squared CoVs for each source of risk since each overall source of risk (i.e., independent, internal systemic, external systemic) is assumed to be uncorrelated.

Fourth, let's calculate the risk margin for the overall book of business that provides a 90% probability of reserve adequacy, assuming liabilities are normally distributed:

$$\begin{aligned}
 \text{Risk Margin} &= z \cdot CoV_{Total} \\
 &= 1.28(0.082) \\
 &= 10.5\%
 \end{aligned}$$

To calculate the risk margin in raw dollars, we would multiply 0.105 by the total central estimate (OSC + PL).

Fifth, let's calculate the risk margin for the Auto OSC that provides a 90% probability of reserve adequacy, assuming the Auto OSC are lognormally distributed:

- $CoV_{Auto\ OSC} = \sqrt{(CoV_{ind\ Auto\ OSC})^2 + (CoV_{int\ Auto\ OSC})^2 + (CoV_{ext\ Auto\ OSC})^2} = \sqrt{0.08^2 + 0.065^2 + 0.04^2} = 0.111$
- $\sigma_{Auto\ OSC}^2 = \ln(1 + CoV_{Auto\ OSC}^2) = \ln(1 + 0.111^2) = 0.0122$
- $\text{Risk Margin} = e^{z\sigma - \frac{\sigma^2}{2}} - 1 = e^{1.28\sqrt{0.0122} - \frac{0.0122}{2}} - 1 = 14.5\%$

To calculate the risk margin in raw dollars, we would multiply 0.145 by the central estimate for Auto OSC.

Sensitivity Testing

The **goal** of sensitivity testing is to gain insights about the sensitivity of the final risk margin to key assumptions.

We do this by varying each key assumption (i.e., the CoVs and correlations) and monitor the impact on the risk margin. We should review key assumptions that have significant impact on the risk margin.

Scenario Testing

The **goal** of scenario testing is to tie the risk margin to a set of valuation outcomes. We adjust the assumptions used for the central estimate in order to align the central estimate with the assessed provisions including the risk margin.

Scenario testing asks the question: “What scenario (ex. higher frequency or severity) would result in the central estimate increasing to the current central estimate plus the risk margin?”

Internal Benchmarking

The goal of internal benchmarking is to compare the proposed CoVs to one another to check for internal consistency and reasonableness.

Independent Risk CoV Selection

For independent risk, internal benchmarking should be done for two main dimensions:

- 1) **Portfolio size** – *larger portfolios* should have *lower CoVs* due to lower volatility from random effects
- 2) **Length of claim run-off** – *longer-tailed lines* should have *higher CoVs* due to more time for random effects to have an impact

The two dimensions above have the following implications:

- **Long-tailed portfolios:** PL CoV should be *higher* than OSC CoV since the PL for a long-tailed line is smaller than the OSC for a long-tailed line
- **Short-tailed portfolios:** PL CoV should be *lower* than the OSC CoV since the PL for a short-tailed line is larger than the OSC for a short-tailed line

Internal Systemic Risk CoV Selection

If template models are used for similar valuation classes, we would expect similar CoVs.

If a similar valuation methodology is used on both short and long-tail classes, then we would expect a higher CoV for the long-tail class since they tend to be more complicated classes.

External Systemic Risk CoV Selection

Long-tailed portfolios should have higher CoVs than short-tailed portfolios in most cases (exceptions include event risk and liability risk for home classes).

External Benchmarking

The **goal** of external benchmarking is to compare the reasonableness of CoVs and risk margins to those from external sources.

External benchmarking should not replace analysis. Instead, it should serve as a reasonability check. It is especially useful when little information is available for analysis.

Hindsight Analysis

The **goal** of hindsight analysis is to compare the past reserve estimates of liabilities against the latest view of the same liabilities to review the actual volatility in the past.

We should be careful when interpreting hindsight analysis for the following reasons:

- Models may have improved significantly since previous valuations
- Future external sources of systemic risk may be significantly different from past episodes, which could make it appear as though estimates were off

Hindsight analysis is *more useful* for *short-tailed portfolios* where serial correlation between consecutive valuations is less significant.

The authors highlight a specific type of hindsight analysis, known as **mechanical hindsight analysis**. It works as follows:

- Apply the chain ladder method to data as of the valuation date to calculate a current unpaid estimate
- Iteratively, remove a diagonal of data and apply the same method to calculate unpaid estimates for prior valuation dates
- Compare the past estimates to the current estimate for the same accident years. The relevant payments made between the past valuation dates and the current valuation date should be added to the current unpaid estimate

Mechanical hindsight analysis may be used to analyze the following:

- Independent risk, by focusing on the periods with stable experience
- Internal systemic risk, by applying this technique using a number of actuarial methods and observing the differences in volatility
- All past sources of uncertainty, by applying the approach across all past periods

Regularity & Review

It is not necessary to complete each step of the framework described in the paper each time an actuary produces a central estimate of insurance liabilities. At the very least, a full application of the framework should be applied every three years.

A less comprehensive review of the key assumptions should be completed when central estimate valuations are conducted. These key assumptions include:

- Any emerging trends
- Emerging systemic risks
- Changes to valuation methodologies

These regular reviews can be thought of as monitoring exercises where key assumptions from the previous full application are compared against emerging experience.

III. Independent Risk Assessment

When using statistical methods to assess independent risk (ex. Mack method, bootstrapping, GLM, etc.), we are attempting to fit a model to past systemic episodes and trends. Once the model is fit, any residual volatility should reflect random effects only (i.e., the independent risk).

The choice of the statistical method depends on the following factors:

- The valuation class being assessed
- The materiality of the independent risk for that class relative to the overall claims portfolio risk margin

- The cost and effort associated with applying the method

When the cost or effort outweighs the benefit, a simpler approach should be considered (such as benchmarking).

Independent Risk Assessment for OSC

Regardless of the statistical method used, it is important to align the methodology used for central estimate purposes with that used to analyze uncertainty. This ensures internal consistency.

The paper highlights two of the techniques:

- GLMs – for reserving, GLMs are used to identify the key factors that have contributed to past claim costs. Thus, GLMs are a good choice for analyzing independent risk
- Bootstrapping – although less flexible than GLMs, bootstrapping is useful when we are able to identify past episodes of stable experience. We can bootstrap residuals *specifically from these periods* to measure uncertainty

Independent Risk Assessment for PL

Bootstrapping, GLMs, and Bayesian approaches may be used to support the analysis of premium liabilities.

The authors highlight a simpler approach as well:

- Consider a valuation class where the central estimate of the claim cost component of the PL is calculated by combining a projected claim frequency and average claim size
- For both claim frequencies and average claim sizes, past systemic episodes (such as seasonality and inflation) can be removed. A CoV can then be derived for the residual volatility

When a loss ratio approach is used to project premium liabilities, systemic shifts in premium levels should be considered as well.

Note that it may be necessary to analyze large claims separately.

IV. Systemic Risk Assessment

Internal Systemic Risk

As mentioned earlier, the authors recommend a balanced scorecard approach when assessing a model's infrastructure and its ability to predict the insurance process.

Let's look at an example. Afterwards, we will discuss some of the underpinnings in more detail.

Example: Balanced Scorecard Approach

Given the following information from a balanced scorecard for outstanding claim liabilities (OSC):

<u>Balanced Scorecard</u>				
Potential Risk Indicator	Auto Score	Auto Weight	Home Score	Home Weight
Specification Error				
Number of independent models used	4	7	4	7
Range of results produced by models	5	3	4	3
Parameter Selection Error				
Value of predictors used	4	5	4	5
Data Error				
Knowledge of past processes affecting predictors	4	8	4	8
Reconciliations & quality controls	3	7	5	6

<u>Internal Systemic Risk CoV Scale</u>		
Balanced Scorecard Score	Auto CoV	Home CoV
1-2	13%	15%
2-3	10%	11%
3-4	6%	8%

4-5

5%

6%

First, let's calculate the weighted average score by line of business:

- Auto Score = $\frac{4(7)+5(3)+4(5)+4(8)+3(7)}{7+3+5+8+7} = 3.9$
- Home Score = $\frac{4(7)+4(3)+4(5)+4(8)+5(6)}{7+3+5+8+6} = 4.2$

Second, let's map each weighted average score to the appropriate CoV:

- Auto CoV = 6%
- Home CoV = 6%

These are the internal systemic risk CoVs that will go into the risk margin calculation.

When developing a balanced scorecard, actuaries should include risk indicators that are more relevant to the classes within their own claims portfolio.

Additional risk indicators may be required for PL:

- Examples include whether OSC are used as an input to the PL assessment OR whether credible portfolio level pricing analysis is used as an input to the PL assessment
- If used in PL assessment, both of these examples would drive the risk score up and the CoV down (less systemic risk)

For short-tail portfolios, some risk indicators may be less relevant for PL:

- For example, a large variance in the OSC may only affect the most recent accident periods and have a small impact on projected claim frequencies and average claim sizes

CoV scales should represent the full range of possible scorecard outcomes, ranging from worst practice to best practice.

Properties of CoV scales are as follows:

- The minimum CoV for best practice is unlikely to be less than 5%.
- The maximum CoV for worst practice could be greater than 20% (ex. single, aggregated model with limited data)
- The scale should not be linear – model improvements show diminishing returns
 - Improvement from a poor to fair model is greater than from a fair to good model
- CoVs for long-tail lines should be higher than short-tail CoVs for the same score
 - It's more difficult to model the underlying process of a long-tail line and predictors are less stable
- It's reasonable to use the same scale for outstanding claim liabilities and premium liabilities

External Systemic Risk

In this section of the paper, the authors re-introduce external systemic risk and then dive into individual external risk categories. This is where we will focus our attention.

Economic & Social Risks

This category includes normal inflation (ex. CPI) and general economic conditions (ex. GDP growth, unemployment, etc.).

Some of these risks impact both OSC and PL, while others only impact PL (ex. uncertainty around driving conditions is less relevant for Auto OSC than it is for Auto PL).

Legislative, Political, & Claims Inflation Risk

These risks are often **correlated** and more significant for long-tailed lines. Examples include:

- Impact of recent legislation or changes in court interpretation
- Potential future legislation that impacts losses retrospectively
- Precedent-setting in courts

- Changes in medical technology costs or legal costs
- Systemic shifts in large claim frequency or severity

Claim Management Process Change Risk

Changes to the claim management process can impact all valuation classes. However, they are more relevant for OSC than for PL. For example, OSC for short-tailed classes are materially impacted since this risk impacts the emergence pattern for credible claim estimates.

It's important to discuss current and future process changes with claims management and how changes could impact reporting/payment patterns, closing and reopening rates, and case reserves.

Expense Risk

Expense risk is typically a small contributor to total external systemic risk.

In general, the actuary should spend time with product and claim management to better understand the key drivers of policy maintenance and claim handling expenses. Understanding these drivers allows the actuary to identify the key sources of variation relative to the central estimate assumptions.

Event Risk

Event risk relates to single events that cause a large number of claims (i.e., catastrophes).

Assessing event risk is significantly different for OSC and PL:

- OSC
 - Focus is on material outstanding events
 - Discussions with claim management should help set expectations on final claim cost outcomes
 - The range of development patterns on past events may influence the view on uncertainty as well
- PL
 - A number of things can be used to quantify event risk

- These include past experience for event claims, output from proprietary cat models, and output from models for perils not covered by cat models
- When analyzing past experience to help quantify event risk for PL, it is important to allow for changes in portfolio size, geographical spread, inflation, reinsurance arrangements, etc.

Latent Claim Risk

Latent claim risk is negligible for most valuation classes. However, it may be material for workers' compensation and liability classes.

This type of risk is difficult to quantify because it has extremely low frequency and extremely high severity. Thus, a large degree of judgment is required to quantify it. Discussions with underwriters can help gain insights on potential sources of this risk, along with their likelihoods and potential impacts.

Recovery Risk

Recovery risk is negligible for most valuation classes. However, it may be material for auto classes due to third-party recoveries (ex. subrogation).

When assessing this risk, the goal is to identify systemic events that may lead to different recovery outcomes than those used for central estimate purposes.

For **non-reinsurance** recoveries, a range of systemic outcomes can be formed by analyzing past experience and talking to claim management about current/future trends in recovery levels.

For **reinsurance** recoveries, a range of systemic outcomes can be formed by determining how shifts in reinsurance market conditions (such as a high number of catastrophes) impact the ability of the reinsurer to pay.